



OCCUPATIONAL AUDIOMETRY

MONITORING AND
PROTECTING HEARING
AT WORK

Maryanne Maltby

Occupational Audiometry

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Foreword

Occupational Audiometry: Monitoring and Protecting Hearing at Work by Maryanne Maltby provides relevant reading material for students in training, experienced practitioners, managers and others who are interested in the field. The text contains all the essential elements that are required to run a comprehensive programme of care and management within the workplace with added information on the pathological and physiological effects that occur with excessive noise exposure.

This publication has come at a very opportune moment of time with the revision of the Health and Safety Regulations on Noise Exposure (2005) that lay down the statutory action requirements.

Maryanne Maltby, with a wealth of experience in both training and practical application in different fields of Audiology, has managed to capture the subject matter in an accessible style that makes this book sit well on the shelf both as reading material for education and training courses and as a reference book.

Gillian Booth MSc
Lecturer in Acoustics and Occupational Audiology
Former Chair of the British Association of Audiology Technicians

Preface

Over the past few years, my work has involved me in training many nurses, doctors and health and safety professionals in Occupational Audiometry, in accordance with the requirements of the Health and Safety Executive, and I am constantly being asked to recommend a book for reference. This book was written as there did not appear to be any one book available that fitted the bill. It is intended to be a practical and readable book but it is not intended to reduce the need for a course of study, which together with experience is required to become competent. Each chapter is intended to stand on its own so that it is possible to dip in and out for information and guidance.

The changes in the legislation in 2005 regarding control of noise at work have had a profound effect on industry. Many occupations which were not previously covered will now come under the regulations and all professionals and managers working with employees exposed to noise levels at or above 80 dBA need to be aware of the legislation and the duty to comply.

This book provides a comprehensive guide to the theory and practice of Occupational Audiometry and covers assessment (including case history, otoscopy and hearing tests), record keeping, noise regulations, personal protective equipment and hearing conservation, as well as the necessary background science to understand the subject. There are simple but accurate instructions for testing hearing and for undertaking otoscopic examination. The style is formal but readable and the information is explained as simply as possible whilst providing the depth required for practice. There are many simple diagrams to aid understanding together with a wide variety of examples of types of audiograms that might be found when carrying out hearing tests. This book will be useful to all those involved in testing programmes including for those professionals who only undertake audiometry from time to time and need a book to which they can refer to remind themselves of the methods and techniques. It should also be helpful to those who are involved in the management of conservation programmes.

Maryanne Maltby

Acknowledgements

I would like to thank all those who helped me to complete this book, especially: The Health and Safety Executive (HSE) for their helpful replies to my queries; Gillian Booth (audiological scientist) who amazingly managed to read and comment on the entire book; David Gaszczyk (audiology manager) who helped me to produce the figures and the index; Clare Bowling (HSE) for looking at the final manuscript; Joanne Williams (British Library) who read and advised me on some of the chapters; Matthew Tate (I.T. consultant) who gave me encouragement and help in producing many of the figures and tables; Stuart Russell (P.C. Werth Ltd) for locating the photographs; John Irwin (audiological physician) for his help with references; John Shuttleworth (Amplivox Ltd) for his encouragement and all those occupational health professionals who shared their ideas and experiences with me.

I

Noise Induced Hearing Loss

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Noise induced hearing loss

Introduction

It is thought that at least one in ten people in the United Kingdom has a hearing loss that affects their ability to hear and understand normal speech. The two most common causes of hearing loss in adults are generally accepted as being:

1. The effects of ageing
2. Noise induced hearing loss (NIHL, the effects of excessive noise exposure).

Noise induced hearing loss is totally preventable but cannot be reversed. Occupational noise is the most common cause of noise induced hearing loss. It is estimated that 1.1 million people are exposed to excessive noise at work and of these 170 000 will suffer significant ear damage as a direct result of the noise. A constant barrage of noise from machinery will impair hearing over time, the degree of loss depending on the intensity of the noise, the hours exposed per day and the number of years of exposure. Some noises, for example explosions, shots and hammers, which are experienced only for a short period can have the same effect. In fact, the characteristics of impulse noise make it more intrusive than the sound level would suggest (South, 2004). A single episode of exposure to very loud noise can create hearing damage and may also perforate the eardrum and possibly dislocate the bones in the middle ear.

Occupations at risk are many, for example: firemen, armed police, police motorcyclists, soldiers, construction and factory workers, printers, foundry workers, couriers and despatch riders, musicians, farmers, lorry drivers and many others. As early as the 1900s, it was recognised that certain occupations caused hearing loss and terms such as 'boilermakers' deafness' and 'weavers' deafness' were used. However, there were no noise guidelines in the United Kingdom until the Noise in Factories Guidelines of 1963 published by the Ministry of Labour as 'Noise and the Worker'.

4 Occupational Audiometry

The National Physics Laboratory carried out research into the effects of noise on hearing in the 1970s and a Code of Practice was introduced in 1972, which was the basis of the Health and Safety at Work Regulations of 1974. This was followed by the Protection of Hearing at Work Regulations of 1981 and then by the Noise at Work Regulations 1989 and the Control of Noise at Work Regulations 2005.

A temporary partial loss of hearing, known as ‘temporary threshold shift’ (TTS), often occurs in the early stages of being exposed to excessive noise. The person may notice that their hearing is temporarily dulled and may experience temporary tinnitus but, after a rest away from the noise, there is usually full recovery. Individuals are often so used to high levels of noise that they are not even aware that they may be damaging their hearing. However, if the exposure to noise is repeated sufficiently often, or if it occurs again before recovery is complete, the hearing damage may become permanent. Removal from the noise will not then produce recovery from deafness although it will prevent *further* damage. The noise induced permanent threshold shift will not progress once there is no further noise exposure but, in later life, changes in hearing due to ageing, known as ‘presbycusis’, will add to any existing hearing loss and the individual is likely to suffer from a greater degree of deafness than that experienced by others of their age.

The risk of noise damage

Factors affecting noise risk

The effect of excessive noise on hearing depends upon a number of factors. These include:

- Noise level
- Duration of exposure
- Frequency of the sound
- Individual susceptibility
- Vulnerability due to environmental factors
- Vulnerability due to biological factors.

Sound above a certain level may cause hearing loss. The longer that someone is exposed to loud sound, and the louder it is, the more likely it is that it will cause damage. Sound becomes louder the closer one moves towards the sound source. Sound close to the ear is considerably louder for that individual than for someone else even a short distance away. The duration of the exposure refers to the length of time spent in the noise but is not just the length of a single exposure to noise. It is an accumulation of time spent in excessive noise, thus a noise induced hearing loss will be the result of the total noise exposure over that person’s lifetime.

Excessive noise of any frequency can cause hearing loss and the level at which hearing loss starts to occur is dependent only to a small extent on the frequency of the noise. High frequency noise is probably the most damaging but all excessive noise causes hearing damage. The A-weighting scale is intended to approximate the contribution of different frequencies to hearing loss.

The concept of equal energy

In general, equal amounts of acoustic energy are thought to cause equal amounts of hearing damage. This is the concept of equal energy. In other words, a person could be exposed to the same amount of sound energy by hearing intense noise for a relatively short period of time or less intense noise for a longer period. This is known as an 'equivalent continuous noise level' (L_{eq}). The amount of sound energy to which the worker has been exposed over the day ($L_{EP,d}$ also known as $L_{EX,8h}$) or over the week is expressed as an equivalent continuous noise level in dBA. The L_{eq} is used in the prediction of levels of noise likely to cause hearing damage. It is generally accepted that 70 dBA is a safe level of sound that should not cause hearing damage, although most (95 per cent) of the population will be safe at levels greater than this, possibly up to 85 dBA.

Vulnerability

Some workers may be especially vulnerable to noise damage and require special consideration. These include:

- Those with a pre-existing hearing problem.
- Those with a history of genetic hearing loss, military service or noisy leisure hobbies.
- Those who smoke. In general, smokers are 1.69 times more vulnerable and the risk increases with the intensity and duration of exposure to cigarette smoke. Passive smokers are also at increased risk and non-smokers living with a smoker have been found to be 1.94 times more likely to suffer a hearing loss than those who do not live with one (Cruickshank *et al.*, 1998).
- Pregnant women.
- Children and young people.
- Individuals who show a hearing loss greater than would normally be expected for the level of noise to which they have been exposed. This small group of people has to be found through audiometric testing.
- Individuals working with certain chemicals, for example solvents, such as:
 - toluene (used in printing and leather manufacture)
 - styrene (used in the plastics industry)
 - mixed xylene (used in the plastics industry)
 - trichloroethylene (used for cleaning metal parts).

There are no reliable methods to assess the interaction between different chemicals and noise. Chemical exposure should be considered in the work history and estimations of exposure should be made by monitoring and from other data.

- Individuals affected by vibration.
- Divers. Professional divers are likely to develop hearing loss at an early age. Anyone diving regularly has an increased risk of high frequency hearing loss. The hearing loss is greatest over the frequencies 4, 6 and 8 kHz and is probably due to exposure to major changes in pressure as well as possible noise damage from the equipment used (Zulkaflay *et al.*, 1996). The pressure change

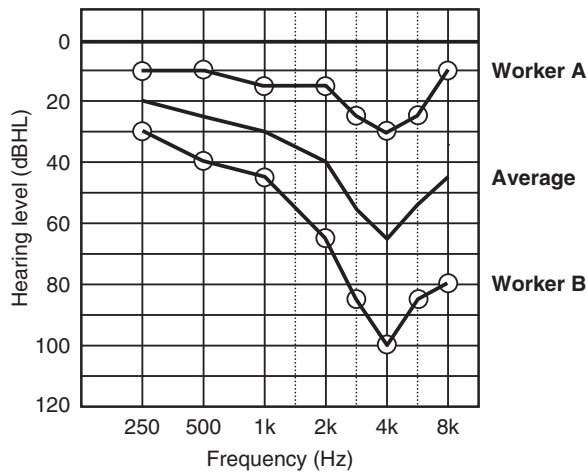


Figure 1.1 Individual susceptibility. The right ear hearing threshold levels from the audiograms of two workers of the same age and occupation reported to have had the same noise exposure. Worker A is an individual with 'strong' ears, whilst worker B is an individual with 'tender' ears.

experienced can also weaken and rupture the round window of the cochlea, causing sudden dizziness and a flat sensorineural hearing loss. This may occur immediately after the event or some weeks, months or even years later.

- Individuals with certain medical conditions, such as high blood pressure, elevated cholesterol, circulation problems or diabetes (Pykkhö *et al.*, 1998).
- Individuals on certain medications, such as painkillers.

Figure 1.1 shows an example of two workers, who are reported to be of the same age and to have held the same type of job as each other for an equal number of years. Worker A is an individual with 'strong' ears, whilst worker B is an individual with 'tender' ears. The average effect on the hearing levels lies somewhere between the two. Noise levels that appear to be safe may not be so for susceptible individuals and these individuals need to be made subject to increased audiometric testing, provided with adequate ear protection and, in extreme cases, removed from the noise.

The effect of noise on hearing

Cochlear hair cell damage

Excessive noise primarily damages the cochlear hair cells; damage may be confined to the outer hair cells (OHCs) (Figures 1.2 and 1.3) but if noise exposure continues (or in many cases of noise trauma) the damage may also involve the inner hair cells (IHCs) (Figure 1.4). In the severest cases, there may be total destruction of the cells in the organ of Corti. The area of greatest damage is

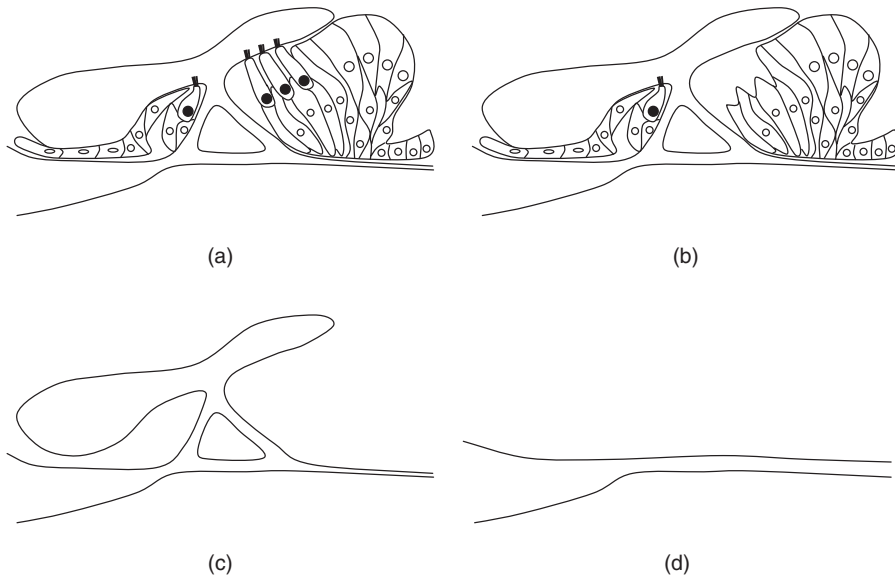


Figure 1.2 Damage to the organ of Corti due to excessive noise: (a) Normal organ of Corti; (b) Outer hair cells are missing; (c) Outer hair cells and inner hair cells are missing and supporting structures have collapsed and (d) The whole organ of Corti has collapsed.

usually about 10 to 30 mm from the round window. This is where the frequencies between 3 and 6 kHz are received, which may explain the existence of the 4 kHz notch that is a common feature of noise induced hearing loss.

In the earlier stages of noise induced hearing loss, damage is restricted to outer hair cell damage. Damage to the outer hair cells not only results in an inability to hear quieter sounds but, in general, tends to cause:

1. Reduced sensitivity for quiet sounds. (Speaking louder and turning the television up may be enough to compensate for a mild to moderate loss of hearing sensitivity.)
2. Some loss of frequency resolution, that is the ability to distinguish one frequency sound from another, especially in the presence of background noise. This can occur even before the audiogram indicates a hearing loss and it is particularly noticeable if the hearing in one ear is worse than the hearing in the other.
3. Discomfort with loud sounds ('recruitment'). Damaged hair cells become less sensitive and less specific and can no longer react to quiet sounds. As the sound level rises an increasing number of neighbouring hair cells will also start reacting, with the result that the person hears nothing but then suddenly hears something that rapidly becomes too loud.
4. Over-reaction to some sounds ('hyperacusis'). This is where the brain increases the volume of sounds, which it inappropriately perceives to be important or dangerous. This may include normal levels of noise, for example the alarm on the microwave, which then become difficult to tolerate.

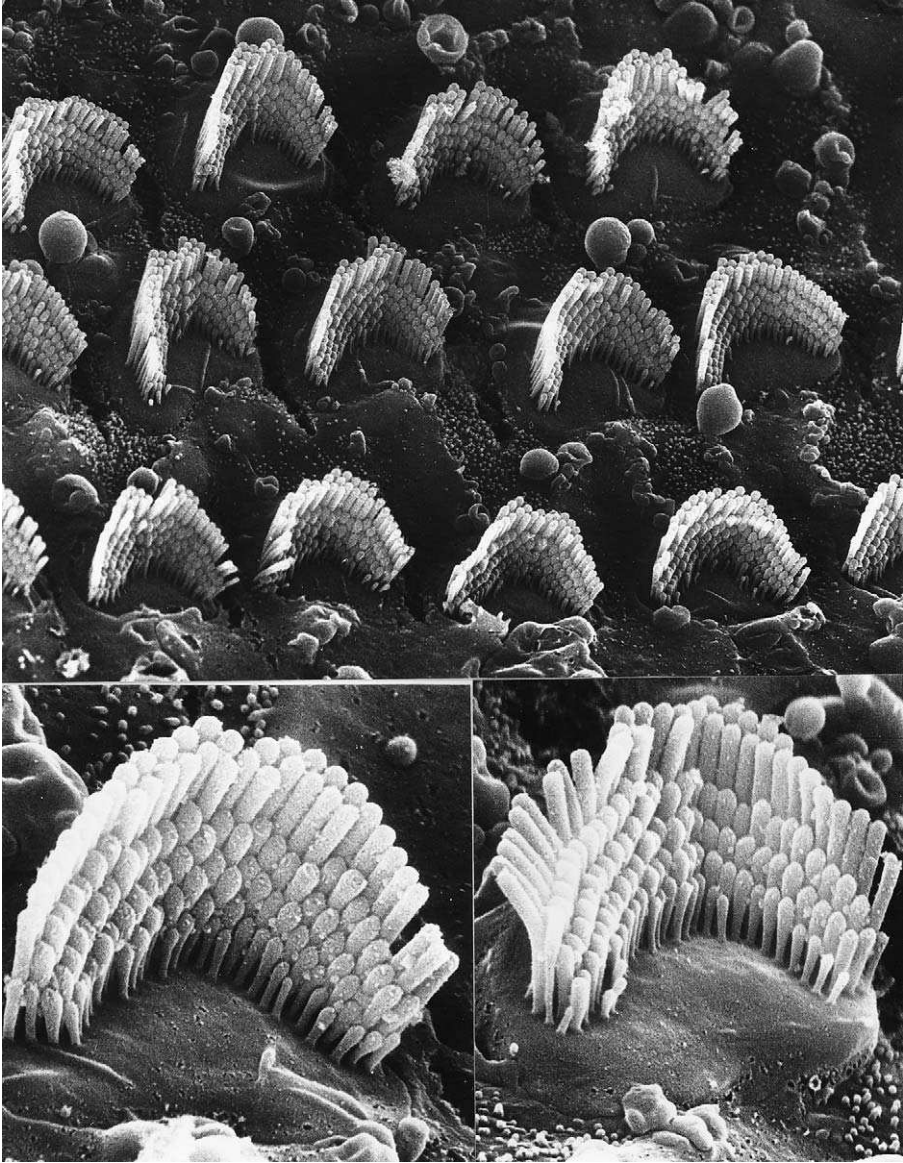


Figure 1.3 Human outer hair cells with only very minor damage. (Photograph courtesy of Widex/Engström.)

More severe damage may also include inner hair cells, in which case information from some areas of the cochlea may be incomplete, distorted or missing. Commonly, loss of hearing for high frequency consonants makes it particularly difficult to understand conversation. A few individuals also experience one tone as a different sound in each ear ('diplacusis').

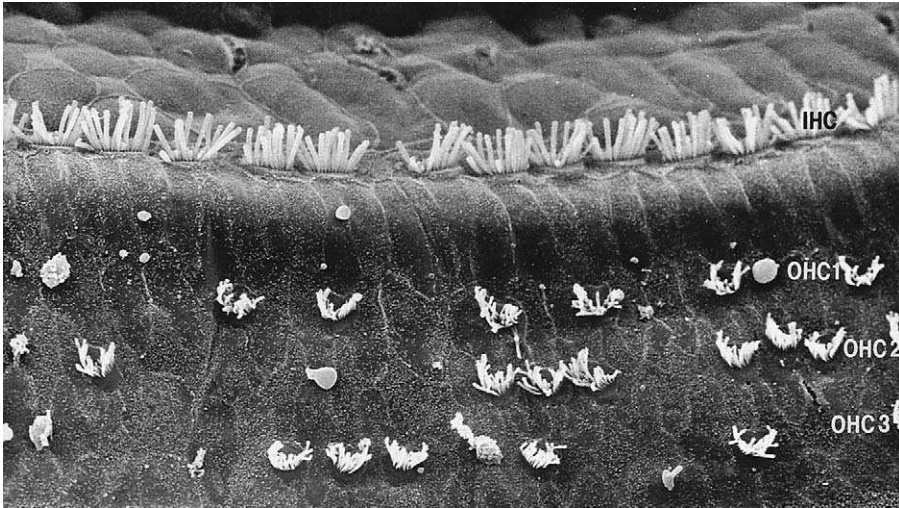


Figure 1.4 Human cochlear hair cells showing extensive damage to outer hair cells and considerable damage to inner hair cells. (Photograph courtesy of Widex/Engström.)

The audiogram and repeated noise exposure

On an audiogram, noise induced hearing loss will usually be seen first as a slight loss of hearing in the 4 kHz region. This dip in hearing is known as a 'notch' in the audiogram. A '4 kHz notch' is a common characteristic of noise induced hearing loss (Figure 1.5). Less commonly, a noise notch may occur at 3 or 6 kHz.

Further noise exposure causes further deterioration in hearing levels and also widening of the frequency range affected (see Figure 1.5). Noise induced hearing loss is generally sensorineural in nature, its onset may be quite rapid and its rate of increase is gradually progressive. The loss affects high frequencies more than low frequencies and tinnitus (ringing in the ears) is often present. Removal from the noise will prevent noise induced hearing loss from worsening but further deterioration in hearing will usually occur in old age due to the effects of presbycusis, making the overall problem worse. The effect of noise and age combined is not simply additive, it is such that the effect of one is reduced in proportion to the other but the combined effect is still very significant. In many cases of noise induced hearing loss, there is an element of presbycusis present and it is difficult to separate them, although tables are available for estimating the degree to which a hearing loss is likely to be due to age or to noise.

As long as the hearing loss affects only the higher frequencies (approximately 3 kHz and above), most people manage very well, particularly in quiet conditions. In noisy conditions, however, speech may become difficult to discriminate. When the hearing loss affects lower frequencies (2 kHz and below) in addition to the higher frequencies, an individual may be unable to hear well even in quiet conditions.

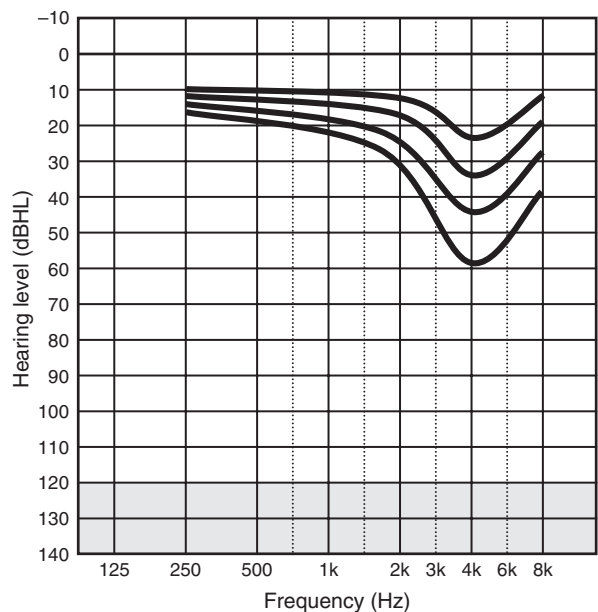


Figure 1.5 Progression of damage in noise induced hearing loss. In fact, the effects of noise exposure join with hearing loss due to presbycusis to give an effective hearing loss over time that can be much worse than that shown.

Many industries have an obvious noise problem, for example coalmining, engineering plants, steelworks, packing plants and bottling plants; but there are less obvious cases of dangerous noise levels, in jobs such as those of waiters and barpersons, call centre operators and musicians. Sound levels in an orchestra, for example, can reach 112 dBA and in a rock group may reach as much as 130 dBA. Over half of all classical musicians suffer from noise induced hearing loss (Einhorn, 1999).

Call centre operators may be subject to acoustic shock, which is a sudden and unexpected noise burst through the headset. The noise burst is usually a high frequency screech and could be caused by interference or misdirected faxes, or noise at the caller's end, for example an alarm, a television, or someone screaming or shouting. Call centre operators' headsets are normally limited to a maximum output of 118 dBA and an operator is likely to remove the headset immediately when exposed to such unexpected loud sounds. Exposure is therefore only for a very short time (5 to 15 seconds) and the exposure level is below the action levels referring to impulse noise. It is therefore thought unlikely that the exposure is sufficient to cause a hearing loss as assessed by conventional methods (Lawton, 2003). However, it is possible that the middle ear muscles may be sent into spasm by the sudden onset. It is also possible that subjective hearing difficulties, such as understanding speech in background noise, may occur even without any recordable change to the audiogram. Insufficient is currently known about

acoustic shock. Hearing loss may or may not occur as a result of acoustic shock and other symptoms can arise. The following have been reported:

- High-pitched tinnitus
- Earache
- Recurrent stabbing pain
- Headaches
- Numbness
- Tingling in the face, neck, shoulder and arm on the affected side that fades over time
- Burning sensation around the affected ear that fades over time
- Ear fullness or blockage
- Light headedness
- Transient balance disorder
- Muffled hearing
- Hyperacusis
- Stress and anxiety.

Best practice is to issue headsets to specific individuals and to give them a choice between monaural and binaural headsets. No headsets should be unacceptably loud and they should be regularly cleaned and maintained. Call handlers should be trained to recognise faulty headsets and also to adjust the volume of their headset as appropriate, in particular to return the volume to its normal level after turning it up to hear a quiet call (Sprigg *et al.*, 2003). They should also be trained to recognise and report (Figure 1.6) incidents of possible acoustic shock. The employer has a duty, under the Reporting of Injuries, Diseases and Dangerous Occurrences (RIDDOR), to report incidents to the relevant enforcing authority (as detailed by the Health and Safety Executive. Under the 1995 regulations, these include any incident that results in an individual being unable to continue with their normal work for more than three consecutive days.

Leisure noise

Noise induced hearing loss is usually, but not exclusively, of occupational origin. The kind of noise to which the person is exposed has little bearing on the resultant hearing loss; if they are of the same intensity and duration, they will tend to produce a very similar hearing loss. Noisy leisure activities such as playing or listening to loud music, lawn mowing, do-it-yourself, hot air balloon trips, cinema visits, visits to trendy restaurants and motorcycling can all contribute to hearing loss.

Shooting is one of the most dangerous leisure activities as far as hearing is concerned. Men who are involved in target shooting, for example, are twice as likely to suffer hearing damage as those who do not shoot (Nondahl *et al.*, 2000). Time spent in noisy leisure activities raises the risk of exceeding the acceptable daily noise dose.

Reported Incident of Acoustic Shock			
Name of operator			
Date		Time	
Source of exposure			
Description of noise			
Details of headset			
Other relevant equipment			
Incident electronically recorded and copy kept	Yes/No Location of copy:		
Symptoms experienced (Tick those that apply and give details)	Hearing loss: Tinnitus: Numbness: Balance: Stress: Other:		
Reported to		Signed	
Signed (operator)			

Figure 1.6 An example of a form for reporting an incident of acoustic shock.

Possible causes of non-occupational hazardous noise, where it is sensible to use ear protection if the noise cannot be otherwise reduced, include:

- Shooting – peak pressure approximately 140 to 160 dBSPL, but may reach 165 dBSPL (Kryter and Garinther, 1996).
- Amplified music – levels of up to 120 dBA have been reported for in-car amplification (Axelsson, 1998); levels above 100 dBA are not unusual at pop concerts and discotheques (Laukli, 1998); a limit of 90 dBA for discotheques and of 100 dBA for concerts, which are generally attended less frequently, has been recommended by the World Health Organisation (1993).

- Personal stereos – levels of 90 to 100 dBA are common amongst young people and levels used may reach 105 dBA (Prasher and Patrick, 1998).
- Playing in orchestras – the fiddle produces levels of about 80 to 110 dBA; brass and woodwind produce levels over 90 dBA; percussion produces relatively low levels of ‘continuous’ noise but with peak exposures of up to 140 dBSPL (Wright Reid, 2001).
- Riding motor bikes – the noise limits for new motor bikes and for replacement exhaust systems are very stringent. However, above 40 mph, the noise of wind turbulence exceeds the noise of the bike. This noise can be in excess of 105 dBA at 70 mph and includes a high level of low frequency noise against which earplugs are less effective (Lower *et al.*, 1994; Jordan *et al.*, 2004).
- Using DIY power tools – many power tools emit sound pressure levels in excess of 85 dBA; for example a power jigsaw, 86.7 dBA, a hammer drill, 90.7 dBA; a rotary lawn mower, 96 dBA. Noise emission information is provided with DIY and gardening equipment but is not always easily accessible. Hearing protection should ideally be placed for sale with the equipment but often is nowhere near.
- Noisy bars and restaurants – levels of 90 dBA are common (Axelsson, 1998). The modern tendency to have wooden floors and bare surfaces increases sound levels by reverberation and levels between 85 and 100 dBA have been recorded.
- Arcade computer games – levels of about 90 dBA (Prasher and Patrick, 1998).
- Cinema attendance – levels of 100 to 110 dBA, and occasionally even higher. Trailers and commercials tend to be louder than the film itself although a maximum level of 85 dBA has been recommended for these by the British Standards Institute.
- Fireworks – Chinese firecrackers at two metres can produce 160 dBSPL (Prasher and Patrick, 1998).

Regulations for the reduction of noise exposure currently do not apply to those who attend noisy events but only to those who are working in the noise, although ear protection is sometimes made available at some of these events or venues, particularly where young children are involved. There is also no requirement to use ear protection for hobbies or home use, and public awareness and conformity is low. The experience of dulled hearing and tinnitus after exposure to noise should be treated as a warning of possible future hearing damage.

Tinnitus and noise induced hearing loss

Tinnitus is the subjective sensation of noise, without any external cause. It may appear to be in the ears or in the head and common descriptions include whistles, hissing, throbbing, pulsating and buzzing. Tinnitus can be intermittent or continuous. Throbbing or pulsating tinnitus is most usually linked to vascular problems, for example high blood pressure or a glomus tumour in the middle ear or the jugular vein, which pulsates with the heart beat. High tone tinnitus is very common, often with the pitch of the tinnitus being close to the area of greatest hearing loss. This may be because, when the hair cells in one area of the organ of Corti are damaged (Figure 1.7) the outer hair cells in nearby areas become over-active in an attempt to

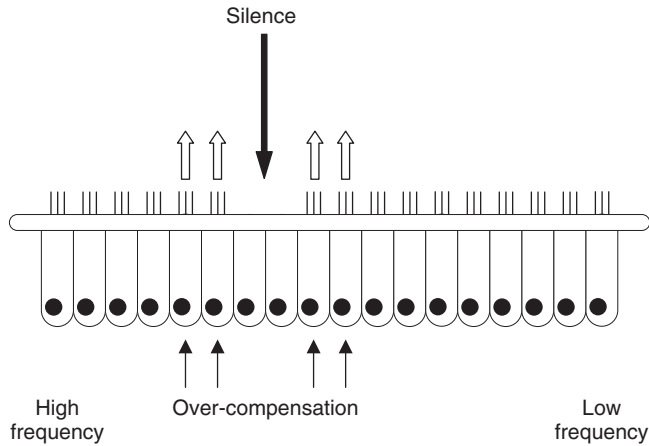


Figure 1.7 One possible cause of tinnitus may be lack of hair cell activity in one area of the cochlea, due to damage, being 'compensated' for by over-activity in nearby areas.

compensate. This over-activity could be the source of noise induced tinnitus. Wearing hearing protection is likely to make tinnitus appear worse. Hearing protection cuts out noise from the outside environment, which would normally help cover or 'mask' the tinnitus, and therefore the tinnitus becomes much more obvious and less tolerable.

Tinnitus is subjective and for some people it can be very troublesome. It may hinder concentration, prevent sleep, cause anxiety, irritability and other psychological problems. In severe cases, it has been known to lead to suicide. Tinnitus can be accepted as an additional handicap for which compensation is sometimes made.

Where the hearing loss is caused by impulse noise, that is relatively short duration noise of very high intensity, this is known as 'acoustic trauma'. Impulse noise may be caused by, for example, drop forges, presses, hammers, riveting, impact welding, nail guns, gunfire and explosives. Acoustic trauma can cause serious permanent destruction within the inner ear and, in some cases, there may also be ruptured eardrums and the bones of the ear may be dislocated or damaged. Acoustic trauma is often characterised by good low frequency hearing accompanied by a sharp high frequency drop in hearing (Figure 1.8). A flatter audiogram may be found when physical middle ear damage is also present. This is because conductive hearing loss tends to affect the low frequencies, whilst sensorineural loss tends to affect the high frequency region.

The effect of hearing loss on speech discrimination

Most hearing loss, including noise induced hearing loss, affects mainly the higher frequencies, see Figure 1.8. This is unfortunate since we rely on the high frequency sounds in speech to provide intelligibility. The low frequency sounds give volume and rhythm to speech, rather than clarity.

Speech sounds fall into two groups, vowels and consonants. Vowels (e.g. 'e', as in egg, or 'a', as in car) tend to be low frequency sounds and are relatively loud and

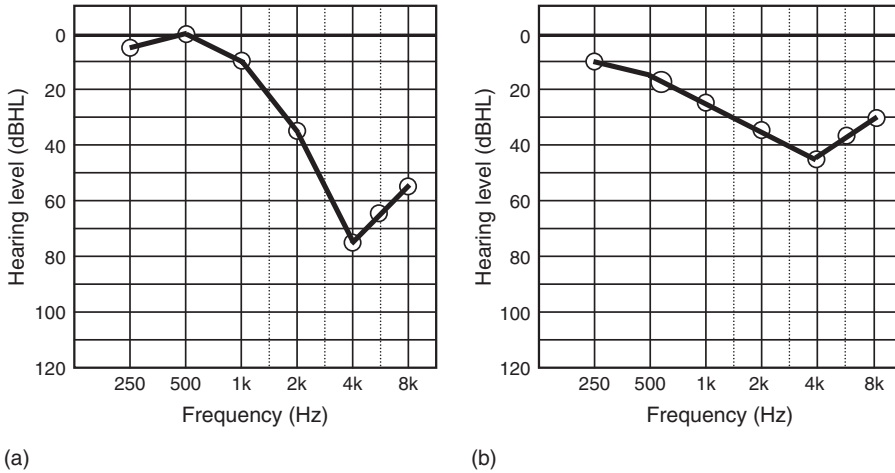


Figure 1.8 Example audiograms found with (a) noise trauma (b) noise induced hearing loss.

easy to hear. Consonants (e.g. ‘k’, as in ‘kick’, or ‘s’ as in sunshine) tend to be high frequency sounds and are relatively quiet and easily ‘lost’, especially when there is background noise. Hence it is common for individuals with hearing loss to hear the vowels well but to miss many or all of the consonants, which gives them the impression that other people are mumbling. It is possible to demonstrate the basic problem by looking at a sentence without consonants, for example Figure 1.9, and trying to guess its meaning.

Most people find this quite difficult to do this, although there is a wide variation in the level of individual skill to make use of the clues that are available. This is of course also true of people with hearing impairments and some will manage much better than others. In general, it is difficult to make sense of speech when the high frequency sounds are missing. It is much easier if the loss is a low frequency one (e.g. as in the case of Ménière’s disorder) as can be seen when the same sentence is presented with the consonants present and the vowels missing as in Figure 1.10.

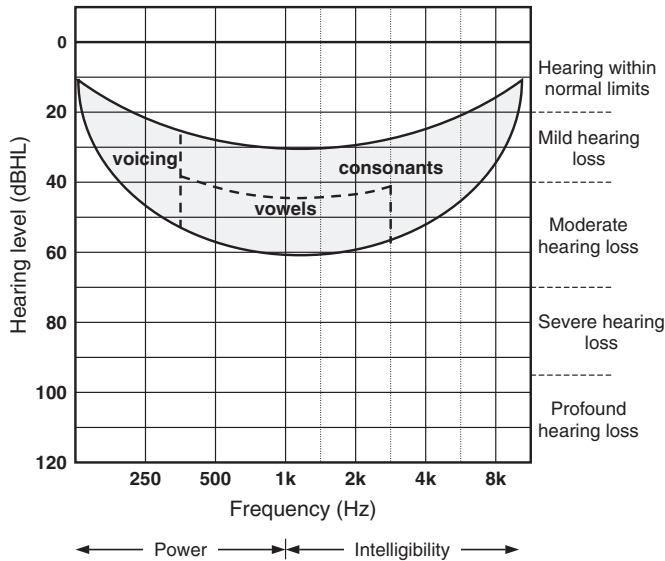
The effect of a hearing loss on the ability to hear speech sounds can be estimated from the audiogram. Figure 1.11(a) shows an audiogram with an area marked to indicate the approximate level and frequency of various sounds in

–a-- a-- –i-- –e-- u-- –e –i--.

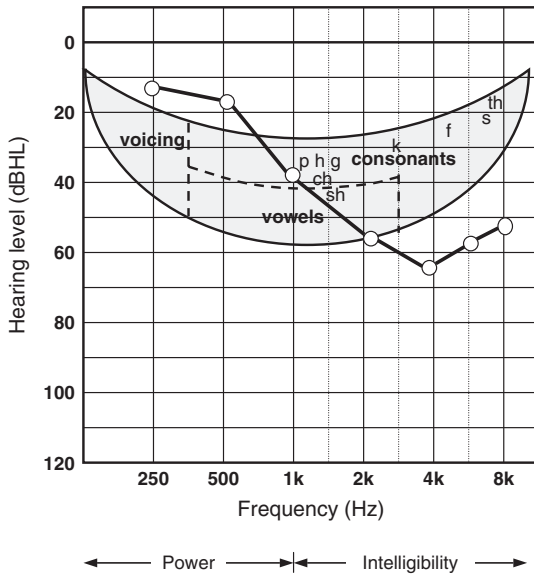
Figure 1.9 A well-known sentence presented without consonants.

J–ck –nd J–ll w–nt –p th– h–ll.

Figure 1.10 A well-known sentence presented without vowels.



(a)



(b)

Figure 1.11 (a) An audiogram form with the speech area shown. (b) A completed air conduction audiogram for the right ear indicating the speech sounds that are likely to be missed by this individual.

normal conversational speech. (This is based on the average long-term speech spectrum which is often known as the ‘speech banana’.) When someone’s hearing loss is plotted on the audiogram, it is possible to estimate what sounds they are likely to miss. Looking at Figure 1.11(b), it is possible to see that this individual will miss many high frequency sounds, such as s, f, th, k, p, h and g, unless they use a hearing aid.

Non-auditory effects of noise

The non-auditory effects of exposure to noise on health and well-being are less well defined than the effects of noise on hearing. These non-auditory effects may include:

- Annoyance and changes in social behaviour – There are individual differences in susceptibility to noise but noise may increase annoyance and aggression, reduce helping behaviour and influence judgement (Smith and Broadbent, 1991). Annoyance tends to increase if the noise is perceived as unnecessary, harmful or frightening and where managers are viewed as unconcerned about the noise (Borsky, 1969).
- Reduced efficiency – The effect of noise on performance is very real but is dependent on the interaction of many different factors, such as the nature of the noise, the personality of the individual and the nature of the task in hand. In general, momentary inefficiencies tend to be more likely to occur in conditions of loud noise (Broadbent, 1979). Performance may also be affected by the extra effort involved in listening and long exposure to noise causes fatigue (Smith and Broadbent, 1991). Memory tasks have been found to be impaired by the presence of speech but not the presence of other noise. Clerical tasks tend to be very little affected by noise. Introverts tend to prefer to work in silence and are less efficient in noise, whereas extroverts tend to prefer, and to work better in, varied auditory stimulation (Davies *et al.*, 1969). This also tends to be true of gender differences, females tending to work slower in noise, whilst it seems to have little effect on males (Gulian and Thomas, 1986).
- Reduced safety – There is some evidence that accidents are more frequent in areas of high noise. Jessel (1977), for example, found that accidents were three to four times more frequent in noisy situations than in quiet ones. Noise seems to affect safety and efficiency particularly at night (Smith, 1989).
- Physiological responses, for example increases in blood pressure and cholesterol – There is little change in physiological responses with noise below 70 dBA but changes become more pronounced as the noise level increases (Smith and Broadbent, 1991). There may be an increase in non-specific dizziness when noise and vibration are combined (Pykkö and Stark, 1985).
- Poor health – Communicating in noise increases the risk of such health problems as laryngitis and vocal cord polyps (Smith and Broadbent, 1991). It is also possible that noise lowers resistance to infection.

- Hormonal changes during pregnancy can affect cochlear function – A mild low frequency (500 Hz and below) hearing loss may occur throughout the pregnancy, together with an intolerance of loud noises during the third trimester and into the post-natal (postpartum) period (Sennaroglu and Belgin, 2001). Tinnitus may also become more noticeable. Recovery occurs during the post-natal period.
- Sleep disturbance – There is a 70 per cent probability of being awakened by noise of 70 dBA (Lukas, 1977). Performance may be affected by noise-disturbed sleep although this is not always the case. Day time noise may strain the central nervous system leading to a greater need for recovery during deep sleep.

The effects of hearing disorders on the ability to work in noisy environments

In general, it is not appropriate to prevent someone from working in noise because of a hearing disability, unless there are great health and safety risks and adaptations cannot be made that reduce the risks to an acceptable level.

Outer ear problems

Most types of ear disease, allergy or skin disorder will influence the selection of hearing protection at work. Earmuffs will usually be the preferred option but in some cases, it may be appropriate to use earplugs of different material. Hygiene will also be important.

Tinnitus

Tinnitus may affect the choice of hearing protection because its use can appear to increase the tinnitus. Earplugs tend to be worse than earmuffs and specialist suggestions for minimising the problems should also be considered. For example, it could be appropriate to find some way of introducing quiet sound (perhaps white noise or soft music) directly into the muff to mask the tinnitus. However, the noise attenuation of the hearing protection must not be affected.

Deafness

It is important to consider the noise levels in which someone with a hearing loss has to work and to ensure that their ‘residual’, or remaining, hearing is adequately protected. If the hearing loss is a conductive one, for example that caused by impacted wax, the hearing loss will provide natural added protection against noise damage. This is *not* so where the hearing loss is sensorineural, for example

in the case of noise induced hearing loss. A mild hearing loss is unlikely to have any direct effect on the ability to work. A more severe loss may be troublesome if it impairs communication and the hearing of warning signals. In most cases, it will be possible for the individual to continue in their job although it may be necessary to offer greater ear protection, extra education and counselling, and sometimes vibrating or flashing warning signals rather than auditory warning signals. People have a right to work and there are only a very few jobs where the health and safety issues cannot be resolved and the only option is to remove the individual from working on that job or in that area. Although the health and safety of the individual and their co-workers is paramount, this action should only be taken where there is a very real safety issue and only as a last possible resort. Such decisions should be taken by a medical practitioner.

Balance problems and/or visual disturbance

Some hearing disorders affect balance as well as hearing, for example Ménière's disorder. More rarely, visual disturbance may also occur, for example in some cases of an acoustic neuroma. Medical advice, regarding the health and safety aspects of the work undertaken, should be sought, especially where loss of balance could be a hazard. It may not be appropriate for the individual to climb ladders or to use a cherry picker, for example. Ménière's episodes may be brought on by exposure to loud noise and, if this is the case and the worker wishes to continue in the same job, extra hearing protection may be advisable. A medical opinion should be sought. Poor eyesight, in conjunction with hearing loss, can negatively affect communication ability. Regular vision screening may therefore be important in conjunction with monitoring hearing for certain individuals.

Summary

Any sound (occupational or leisure noise) above a certain level is likely to cause hearing loss. It is generally accepted that 70 dBA is a safe level of sound that should not cause hearing damage but the louder a sound is, and the longer that someone is exposed to it, the more likely is permanent hearing damage. Some workers may be especially vulnerable to noise damage and require special consideration, for example pregnant women, individuals with a pre-existing hearing problem and those who are working with solvents.

A '4 kHz notch' on the audiogram is a common characteristic of noise induced hearing loss and the hearing loss is often accompanied by tinnitus.

The hearing loss may impair communication and the hearing of warning signals. However, an individual should not be prevented from working in a noisy area because of a hearing disability, unless there are great health and safety risks and adaptations cannot be made that reduce the risks to an acceptable level.

Noise in the workplace

Changes in legislation

The Control of Noise at Work Regulations (2005) implement the Physical Agents (Noise) Directive 2003/10/EC and replace the Noise at Work Regulations of 1986. The Physical Agents (Noise) Directive specifies minimum requirements for the protection of workers across the European Union against health and safety risks arising from exposure to excessive noise. The Directive introduces new and more stringent requirements. Many industries that previously were not covered will now come under the regulations and many that previously were at or above the lower (first) action level will now come under the requirements of the upper (second) action level, examples of this within the food industry can be seen in Table 2.1.

Noise at work legislation

The Control of Noise at Work Regulations 2005

The main provisions of the Directive, which are contained in the regulations, are:

- Assessment of noise levels where workers are likely to be exposed to risks.
- Elimination of risks at source or reduction to a minimum.
- Appropriate health surveillance where the risk assessment indicates a risk to health.
- Averaging of exposure over 8 hours or a week in appropriate circumstances.
- The following actions to be taken where personal noise exposure exceeds 80 dBA (continuous noise) and 112 Pa (impulse noise):
 - Availability of hearing protectors
 - Provision of information and training about risks to hearing and the use of hearing protection
 - Availability of audiometric testing where there is a risk to health.

Table 2.1 Examples of noise levels found in the food industry and the effect of the 2005 noise regulations

Process	Approximate level (dBA)	Action level
Dough mixing	85	Upper
Packaging	85+	Upper
Milling	85+	Upper
Bottling	85+	Upper
Bread slicing	85+	Upper
Meat chopping	90+	Upper
Homogenising	90+	Upper
Powered meat sawing	90+	Upper
Hopper feeding	95	Upper
Hammer milling	95+	Upper
High speed bottling	100	Upper

- The following actions to be taken where personal noise exposure exceeds 85 dBA (continuous noise) and 140 Pa (impulse noise):
 - Establishment and implementation of a programme of technical and/or organisational measures intended to reduce exposure to noise
 - Marking, delimiting and restriction of access to areas
 - Mandatory use of hearing protectors
 - A right to hearing checks.
- A limit on personal noise exposure, taking account of any hearing protection worn, of 87 dBA (continuous noise) and 200 Pa (impulse noise).
- Derogation power if using hearing protection causes risks to health and safety. Non-application where there is a conflict with public service activities.

The action levels

Under the Control of Noise at Work Regulations 2005, there are two action levels and a limit value. The action levels are levels of noise exposure at which employers have to take certain actions to reduce noise and/or its effect on hearing. The exposure limit value considers the level of noise at the ear, taking into account the reduction provided by the hearing protection in use. Each action level has a separate specified limit for continuous noise and for impulse noise. The action level for continuous noise is based on an 8 hour average noise exposure level known as the ‘daily personal noise exposure level’ or $L_{EX,8h}$. Alternatively, the 8 hours can be calculated as an average over a week. The action level for impulse noise is a peak value.

- The first or lower action level for continuous noise is set at a daily or weekly exposure of 80 dBA and for impulse noise is set at a peak sound pressure of 112 Pa or 135 dBC. At or above the first action level, the employer must make hearing protection available and provide appropriate information and training

to ensure, as far as possible, that workers will understand why they should use it and know how to wear and use it properly.

- The second or upper action level for continuous noise is set at a daily or weekly exposure of 85 dBA and for impulse noise is set at a peak sound pressure of 140 Pa or 137 dBC. At or above the second action level, the employer must take all reasonably practicable measures to reduce noise exposure in ways other than by providing hearing protection using, for example, engineering controls. Reduction of noise at source is the best way of ensuring hearing conservation because hearing protection is only efficient if it is in good condition, fitted correctly and worn all the time of noise exposure. The use of hearing protection is mandatory while noise control measures are being implemented and also where it is not possible or practicable to reduce the noise to below the second action level. Where noise is likely to be at or above the second action level, the area must be demarcated and signed (Figure 2.1) as a 'Hearing Protection Zone'. Access should be restricted as far as possible and all people entering one of these areas, even if only passing through, must wear hearing protection.
- There is also a limit value, which is a daily or weekly exposure level of 87 dBA for continuous noise and a peak sound pressure of 200 Pa or 140 dBC. This is a limit at the ear (taking the attenuation provided by hearing protection into account) which must not be exceeded. If the exposure limit is reached, immediate action must be taken to reduce exposure to below these values.

Hearing protection must, as well as reducing or attenuating noise by the required amount, be suitable for the working environment and compatible with other safety equipment that is being used. If possible, a choice of suitable hearing protection should be provided as some workers will have an individual preference or they may not be able to use certain types of hearing protection

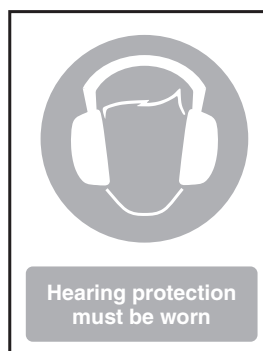


Figure 2.1 A sign (coloured blue and white) for the purpose of indicating that ear protection must be worn, as specified by The Health and Safety (Safety Signs and Signals) Regulations (1996).

because of ear infections or other health problems. Wearing ear protection is mandatory at or above the second action level. This means that workers must use it at all times when they are required so to do. There should be procedures in place to ensure that replacement hearing protection is available and that faulty protection is disposed of. It is a duty of the management to ensure that hearing protection is worn and being used correctly, and it is helpful to carry out spot checks to ensure that this is the case. Disciplinary procedures should take effect where any worker persistently fails to use their hearing protection correctly.

Noise can be a safety hazard, prevent hearing warning signals, interfere with communication and create stress. The Control of Noise at Work Regulations 2005 require excessive noise to be reduced at source wherever practicable. A noise reduction of only 3 dB, which may seem very little, equates to halving the intensity of the noise (because noise is measured on a logarithmic scale). In effect this means that, when the level is reduced by 3 dB, someone can work for twice as long in the noise yet have the same daily personal noise exposure.

Noise measurements must be taken by a competent person who will also determine the $L_{EX,8h}$. The sound pressure levels (SPLs) are usually measured for the different tasks carried out and at the different places in which the individual works. The $L_{EX,8h}$ can be calculated from these values and the time spent in each place or at each task, or it can be measured directly using a dosimeter. Where there is intermittent noise exposure, calculations can be averaged over a week rather than 8 hours if this is more representative.

The employer's obligations

Under the Control of Noise at Work Regulations 2005, the employer has legal duties to control the risks to health and safety that may occur through noise exposure, including when workers are working away from the main site. Although the Control of Noise at Work Regulations only apply to people at work, employers also have duties under the Health and Safety at Work Act 1974 to do what is reasonably practicable to safeguard the health and safety of other people who are exposed to noise risk through the company's activities. Action taken should be similar to that taken for exposed employees. Self-employed people are included as both employers and employees within the regulations and must therefore protect themselves from noise in the same way as other employers must protect their employees, except in as far as that, although it is advisable to have regular hearing checks, there is no requirement for the self-employed person to provide themselves with health surveillance.

There is a general obligation to assess health and safety risks under the Management of Health and Safety at Work Regulations 1999. The Control of Noise at Work Regulations 2005 extends this and, where noise is identified as a potential risk, the employer must now make a 'suitable and sufficient' risk

assessment to enable them to decide if action is required to control employees' exposure to noise. The risk assessment must be based on competent advice and:

- Identify the workers who are exposed above the lower action level.
- Contain measured noise levels, together with the type and duration of exposure, for all employees exposed above the upper action level.
- Identify measures (excluding hearing protection) needed to prevent risk from noise exposure or to reduce the risks to a minimum, for example by using alternative equipment or processes. If anyone is exposed above the exposure limit despite the control measures, immediate action must be taken to reduce exposure and to ensure it does not happen again.
- Consider the adequacy of hearing protection. Hearing protection should be used only as a last resort where it is not possible or practicable to reduce the level below the exposure action levels, or as an interim measure whilst engineering controls are put in place.
- Consider the effects of noise exposure on especially vulnerable individuals or groups. This includes individuals with pre-existing hearing problems, pregnant women, young people and those workers who are exposed to vibration or to certain chemicals, such as solvents.
- Consider the effects of noise exposure beyond normal working hours.
- Consider the information (from group data) available from health surveillance. This should provide guidance on the effectiveness of noise controls.
- Include information to permit compliance with other duties under the Regulations.

The employer is responsible for the recording of the major findings of the assessment. These should be retained together with the action plan (including justification of decisions) and a record of the measures actually taken. The risk assessment should be reviewed regularly and whenever there is a reason to suspect that the noise levels or exposure risks may have altered. The employer should also keep abreast of developments in legal requirements.

Employers are expected to provide health surveillance where the risk assessment indicates a risk to hearing. Health surveillance is obligatory where employees are exposed at or above the upper action level and for vulnerable employees exposed at or above the lower action level. Employers should make it clear to employees, preferably in writing, how the occupational health information is to be used and who could have access to it and the reasons for this access. Employers and managers should only have access to the amount of information necessary for them to carry out their management responsibilities.

The correct use of hearing protection should be encouraged and enforced and the company's Safety Policy should include a strong commitment to noise reduction and the use of hearing protection. Someone in authority should be given responsibility for the provision and maintenance of personal protection, they should also record details of the issue of protection, problems in use and arrangements for training workers on where and how to wear them. The protection should be inspected periodically and repaired or replaced as necessary. Suitable

storage and facilities for cleaning should be available. Training and education should be on-going and a system put in place for reporting faults or loss. Spot checks are needed to ensure compliance. If a worker is not using hearing protection correctly due to problems, these should be resolved. If a worker is not using hearing protection correctly without reason, the employee should be given a verbal warning. If any worker persistently continues not to wear the protection correctly, normal disciplinary procedures should be followed.

The employer should ensure as far as is practicable that the equipment provided in order to comply with the Regulations is properly used and well maintained. Regular checks should be carried out and any defects or problems that are noted or brought to the management's attention should be promptly remedied.

Understanding the Health and Safety Executive's categories

The Health and Safety Executive (HSE) suggest that the employee's hearing test results should be categorised according to their hearing loss. There are four categories and, in addition, cases of unilateral loss must be noted:

1. **Acceptable hearing** – Individuals in this category should be given education and training on the effects of noise and the correct use of hearing protection. They should to be monitored continuously under the hearing surveillance programme but no special action is required.
2. **Mild hearing impairment** – Individuals in this category must be given formal notification of their hearing damage and its implications. They must be retrained in the use of hearing protection and the importance of complying with hearing conservation measures.
3. **Poor hearing** – Individuals in this category must be referred to the Occupational Health Physician or their general practitioner (GP) if there is no occupational health physician.
4. **Rapid hearing loss** – Individuals in this category must be referred to the Occupational Health Physician or their GP if there is no occupational health physician. Future tests may need to be carried out at more frequent intervals.

Unilateral loss – Where there is a significant difference in hearing between the ears, the individual must be referred to the Occupational Health Physician or their GP if there is no occupational health physician.

The employee's obligations

Every employee has an individual responsibility for their own safety at work and that of their colleagues and members of the public (Health and Safety Act, 1974). The worker must not intentionally or recklessly interfere with or misuse anything

that has been provided for health and safety purposes. They must report any defect in safety equipment or procedures of which they are aware. They should be aware of the system for reporting defects and problems to the management. They must also co-operate with their employer in order that they can implement health and safety measures to comply with current legislation. They have a duty to comply with the measures introduced to meet the requirements of the Control of Noise at Work Regulations 2005. They must use any noise control measures and hearing protection in accordance with the instructions they are given, take care of their hearing protection and report any faults or problems.

Summary

The Control of Noise at Work Regulations 2005 replaced the Noise at Work Regulations 1989. They specify minimum requirements for protecting workers against the health and safety risks associated with excessive noise. This places duties on the employer to eliminate risks or reduce them to a minimum by means other than hearing protection. A major change in the regulations is the reduction of the action levels by 5 dB. The lower action level is 80 dBA for continuous noise and 112 Pa for impulse noise. The upper action level is 85 dBA and 140 Pa, respectively. In addition there is a maximum limit to exposure, taking account of ear protection. This is 87 dBA and 200 Pa respectively. The employer has a duty to provide information and training to the workers exposed to noise and to provide health surveillance where workers are regularly exposed at or above the upper action level. Hearing protection must be available from the lower action level but its wearing must be enforced at or above the upper action level. Employees must co-operate with the management with regard to the measures introduced to comply with the noise regulations.

Further reading

Health and Safety Executive, Guidance on the Control of Noise at Work Regulations, 2005.

Hearing conservation

Introduction

The link between exposure to excessive noise and the development of hearing loss is well established. The British government has issued guidance on noise at work since 1963. Exposure to high levels of noise may permanently damage the hair cells in the cochlea, although the degree of hearing loss acquired is dependent on the noise level and the duration of exposure. It is probable that much damage to the outer hair cells in the cochlea, causing distortion of sounds and difficulty in hearing in background noise, occurs even before any hearing loss can be measured by audiometry (Graham and Martin, 2001). Although the degree of hazard from noise increases rapidly with exposure above 90 dBA, there is evidence of residual risk to hearing down to at least 82 dBA (Health and Safety Commission, 2004). Individuals can vary widely in their susceptibility to noise damage and some employees are particularly vulnerable. Exact risk assessment for individuals is therefore very difficult to establish.

Noise induced hearing loss is preventable. The aim of a hearing conservation programme is to minimise damage due to excessive noise. Hearing conservation should be implemented as soon as a noise problem is suspected and all measures should be part of an integrated conservation programme if they are to be fully effective. The programme (Figures 3.1 and 3.2) will involve adequate record keeping and include several important steps:

1. Noise assessment and evaluation of risk
2. Action plan
3. Review and reassessment.

A new occupational hearing programme will bring problems to the surface and may bring forward cases for compensation but these cannot be 'buried under the carpet' and if a conservation programme is not put in place, the situation will be far worse in a few years' time.

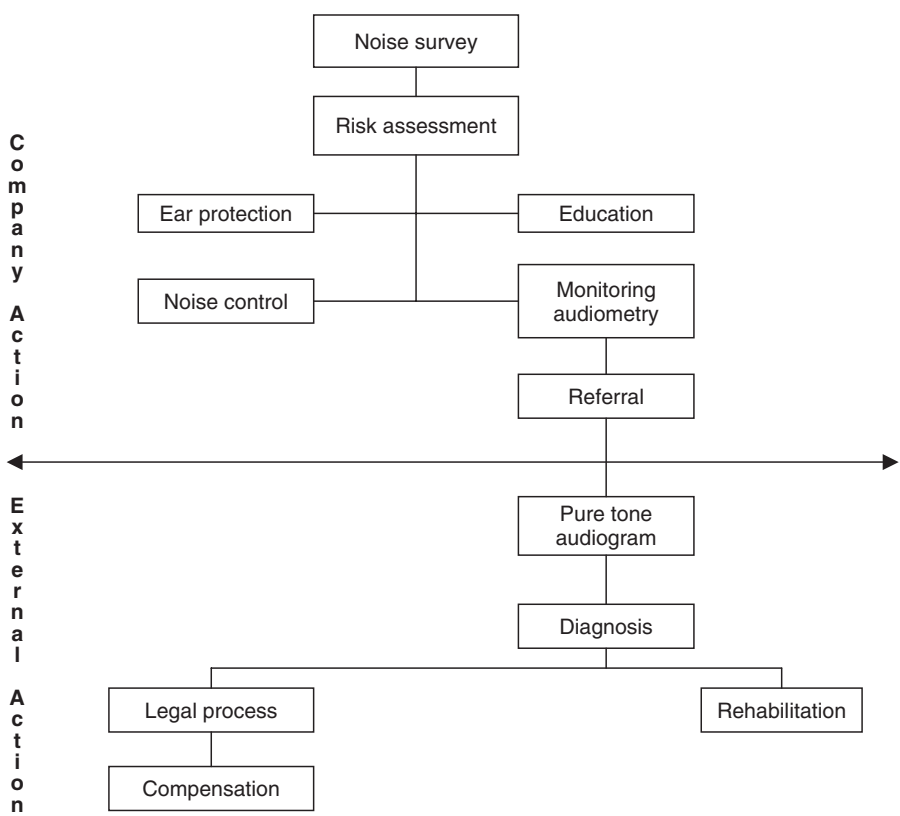


Figure 3.1 Components of a typical hearing conservation programme and some possible outcomes of referral.

Exposure action levels

The Control of Noise at Work Regulations 2005 specifies a lower exposure action level, an upper exposure action level and a fixed maximum exposure limit for any worker, which takes into account the effect of any hearing protection worn. The exposure level is set according to whether the noise is of an instantaneous nature or is a daily noise exposure level. The daily noise exposure includes all noise present including impulsive noise. A separate peak level is stated for instantaneous noise because very high levels can cause hearing damage, however short the length of exposure. The fixed maximum exposure limit represents an over-riding limit on noise level no matter how infrequently the worker is exposed.

The lower exposure level is 80 dBA for daily noise exposure and 112 Pa for instantaneous peak exposure. At the lower exposure level, ear protection must be made available. The higher exposure level is 85 dBA for daily noise exposure and 140 Pa for instantaneous peak exposure. At the higher exposure level, ear

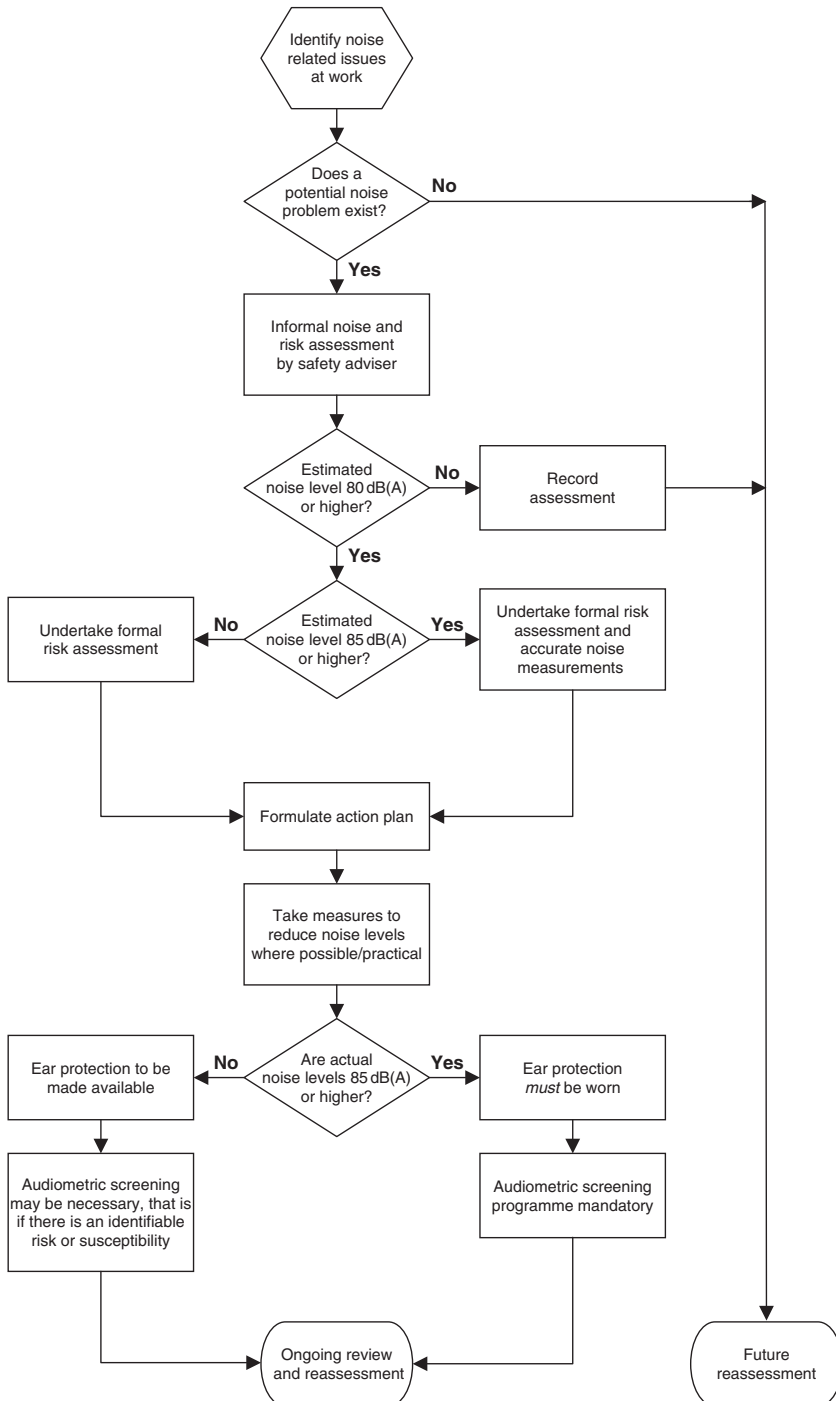


Figure 3.2 A company hearing conservation programme flow chart.

protection must be worn. The maximum exposure limit is 87 dBA for daily noise exposure and 200 Pa for instantaneous peak exposure.

The European parliament (2003) has also stated that:

Current scientific knowledge of the effects which exposure to noise may have on health and safety is not sufficient to enable precise exposure levels covering all risks to health and safety, especially as regards the effects of noise other than those of an auditory nature, to be set . . . Employers should make adjustments in the light of technical progress and scientific knowledge regarding risks related to exposure to noise, with a view to improving the health and safety protection of workers.

Noise exposure and evaluation of risk

Evaluation of risk

It is the employer’s duty to carry out a ‘suitable and sufficient assessment of risk’ (Health and Safety Commission, 2004). It usually falls to Safety Advisors to identify areas or equipment where conditions of noise may be hazardous. Initially a rule of thumb may be used to suggest areas of possible concern, based on the difficulty of being heard (Table 3.1) and the manufacturer’s information about the noise emission levels of the machinery may also help to provide a guideline. Many occupations involve potentially hazardous noise exposure and therefore threaten the hearing of workers. Examples of some of the types of tools and equipment that may produce a noise problem are given in Table 3.2. However, account must be taken of all noise, not just that produced by machinery.

If it appears that there may be a noise problem, it is necessary to identify those workers who may be affected, not only with regard to hearing damage but also, for example, interference with the ability to communicate or to hear warning signals. A risk assessment will estimate the level of risk by estimating

Table 3.1 A rule of thumb used to estimate probable noise levels

Rule of thumb	Listening check	Approximate noise level
One metre rule	Difficulty in being heard clearly, or having to shout to be heard, by someone 1 metre or 3 feet away	90 dBA
Two metre rule	Difficulty in being heard clearly, or having to shout to be heard, by someone 2 metres or 6 feet away	85 dBA
Normal conversation rule	The noise level is approximately the same as voice level when talking at a normal conversational distance	80 dBA
Risk doubles if	An increase in sound level can be noticed	+ ≥5 dB

Table 3.2 Examples of some sources of potentially hazardous noise

Continuous noise sources	Impact noise sources
Aircraft noise	Cartridge operated tools
Bottling plant	Detonators
Chainsaws	Drop forge
Compressors	Explosives
Drills	Guns
Diesel motors	Hammers
Fire alarms	Jolt-squeeze
Gardening and sports ground machinery	Punch presses
Gas liquefaction	Pneumatic tools
Grain dryers	Riveting
Grinding	
Helicopters	
Jet engines	
Metal working machines	
Milling machines	
Musical instruments	
Pig feeding	
Powered hand tools	
Printing and copying machines	
Tanks	
Tractors	
Traffic noise	
Turbo jet engine	
Weaving machines	
Welding	
Wood working machines	

the noise exposure. The noise level may be considered, at this stage, using as appropriate:

- Rule of thumb estimation
- Manufacturers' or suppliers' information
- Information on typical noise levels in certain industries
- A sound level meter to measure the noise level.

It is also important to be able to recognise certain factors, such as the use of ototoxic chemicals in certain processes or the vibration emitted by some equipment, which may interact with the effect of the noise to produce more severe hearing problems.

Action levels are specified by law. If the worker's daily exposure is below the lower exposure action values (the first action level), the risk of noise induced hearing loss is very low. Noise can still cause a nuisance below 80 dBA and, if practical, noise levels should be reduced further. A record should be kept of the current noise levels and it is important to ensure they are maintained at a level that will minimise

risk. At or above the lower exposure action value, a suitable risk assessment must be carried out. If the worker's daily exposure is likely to be at or above the upper exposure action value (the second action level), the risk assessment must include accurate sound level measurements. A competent person, who can supervise the collection of information and its use in the final assessment, must carry out all noise assessments.

Noise measurement

It is the employer's duty to assess and, where it appears necessary, to measure the noise levels at work, paying particular attention to the level, type and duration of exposure and any exposure to impulsive noise. The methods and equipment to use for noise measurement must be adequate to determine whether the exposure levels have been exceeded.

The amount of noise to which a person is exposed is called their 'noise dose' or daily noise exposure and is calculated from the noise level and the length of time in the noise. This is relatively straightforward if the noise is constant or regular and steady. Where the sound level varies throughout the day, the calculation is more complex and it is sensible to work on a worst case scenario. The L_{eq} is the 'equivalent continuous noise exposure', which means that, if it is not a steady noise, the noise exposure will be averaged throughout the working day. It is important to realise that decibels are logarithmic units and that they are not added or averaged in the same way as 'normal' numbers. The term 'daily personal exposure to noise' (shortened to $L_{EP,d}$ or $L_{EX,8h}$) is usually the equivalent continuous level (L_{eq}) over an 8-hour day. Where the noise exposure varies markedly from day to day, it may be measured over a week, which is taken to be five 8-hour days (International Standards Organisation, 1990). Where shifts are longer than 8 hours, noise exposure limits can still be calculated using the 'equal energy rule', which takes account of the noise levels and time of exposure. This will give lower limits than those applying to an 8-hour day (Table 3.3). However, consideration must also be given to the:

- Problems related to the use of hearing protection over such a prolonged period.
- The effect of fatigue and stress factors, related to long shifts, on noise risk.
- Decreased hours of recovery.

Table 3.3 Noise exposure limits for shifts in excess of 8 hours

Duration (hours)	Noise limit equivalent to (dBA)		
8	90	85	80
9	89.2	84.5	79.5
10	88.4	84.0	79.0
11	87.7	83.6	78.6
12	87.1	83.2	78.2
13	86.5	82.9	77.9
14	86.0	82.6	77.6
15	85.5	82.3	77.3

Before making the noise measurements, thought needs to go into deciding the type of information required. Various instruments and techniques may be used but, in order that sound level measurements are accurate, it is important that:

- The person undertaking noise measurements is suitably trained and competent.
- The equipment used is traceably calibrated.
- An appropriate standard method of measurement is used.

The person in charge of the programme does not necessarily have to take the noise measurements and it is common for specialists to be brought in on a consultancy basis. Instruments used for measuring noise include:

- A basic sound level meter.
- A sound level meter providing octave band analysis.
- An integrating sound level meter.
- A dosimeter.

All these are designed to provide objective measurements of the noise level and must be calibrated to ensure their accuracy. This is especially important when the equipment is being used to establish whether a sound level is above or below a pre-set action value. Where very high levels of noise are involved, extra precision is required. A difference of only 0.1 dB could, for example, prevent an aircraft from being allowed to operate from a large international airport. Routine calibration of sound level measuring devices may be carried out using a sound calibrator, which is a small instrument that can be coupled to the microphone of the sound level meter and which delivers a known sound signal. The sound calibrator must itself be calibrated at regular intervals.

A basic sound level meter consists (Figure 3.3) of a microphone that converts the sound into an equivalent electrical signal so that it can be processed electronically or digitally. Several different processes may be performed on the signal, including varying the signal level according to the frequency so that the sound level meter responds in approximately the same way as the human ear. The response of

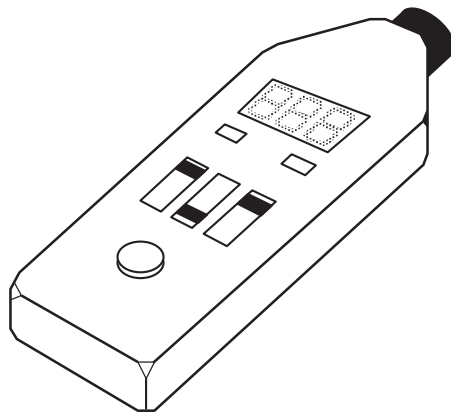


Figure 3.3 A basic sound level meter.

Table 3.4 Some scales used in measuring sound scales

Scale	Corresponding measure
dB (linear) or dBSPL	There is no weighting on the signal so it passes through the sound level meter unmodified. The reference level (0 dBSPL) corresponds to 0.00002 Pa.
Pascal (Pa)	Pascal (or newtons per square metre) is the SI unit for pressure measurement. ● 0.00002 Pa corresponds to the faintest sound a human can just hear. ● 20 Pa corresponds to a very loud signal (120 dBSPL), for example pop concert.
dBA	Corresponds to the sensitivity of the human ear at low sound levels. The dBA scale is used for most noise measurements.
dBb	Corresponds to the sensitivity of the human ear at medium sound levels.
dBc	Corresponds to the sensitivity of the human ear at high sound levels.
dBd	Standardised measurement of aircraft noise.

the human ear varies at different sound levels and this has resulted in the use of a number of different weighting networks (Table 3.4). The dBA scale is generally used for most noise measurements in the workplace. The ‘D’ network is used for aircraft noise. Where the noise is not modified, a linear or ‘flat’ scale is used or the measurement may be made in pascal (Pa) which is a measurement of pressure.

Octave band analysis is used where it is deemed necessary to provide more detailed information about the frequency content of the noise. Filters are used to divide the sound signal (20 Hz to 20 kHz) into frequency bands. Each filter will reject all frequencies outside its selected band (Figure 3.4). The bandwidth is generally one octave or one-third octave. One-third octave bands provide more detailed analysis than octave bands. An octave filter with a centre frequency of 1 kHz, for example, will admit only frequencies between 707 Hz and 1414 Hz, whilst a one-third octave filter with a centre frequency of 1 kHz will admit only frequencies between 891 Hz and 1122 Hz. In many cases, it is not necessary to use frequency

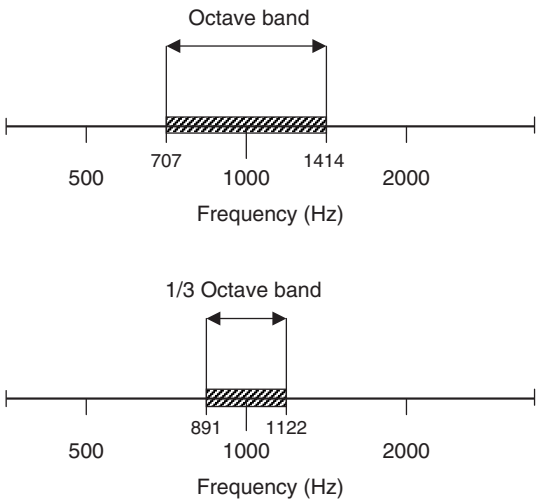


Figure 3.4 Octave and third octave band filters.

analysis and a measurement of the noise level both with and without the A-weighting may allow an adequate acoustic evaluation of the noise hazard (Graham and Martin, 2001).

A sound level meter may measure continuous sounds or specific events. Information from a basic sound level meter, set to A-weighting and 'slow' response, can be used to quantify noise exposure where the average noise level over a short period is typical of that over the whole working day. When an employee changes noise environments during the day, there are a number of practical methods that are widely used to establish the noise exposure:

1. Sound levels are measured in various areas throughout the work premises and these are combined with shift patterns or data from time and motion analysis. This system can be easily updated if shift patterns change. A noise map (Figure 3.5) may be produced for the workplace or certain areas

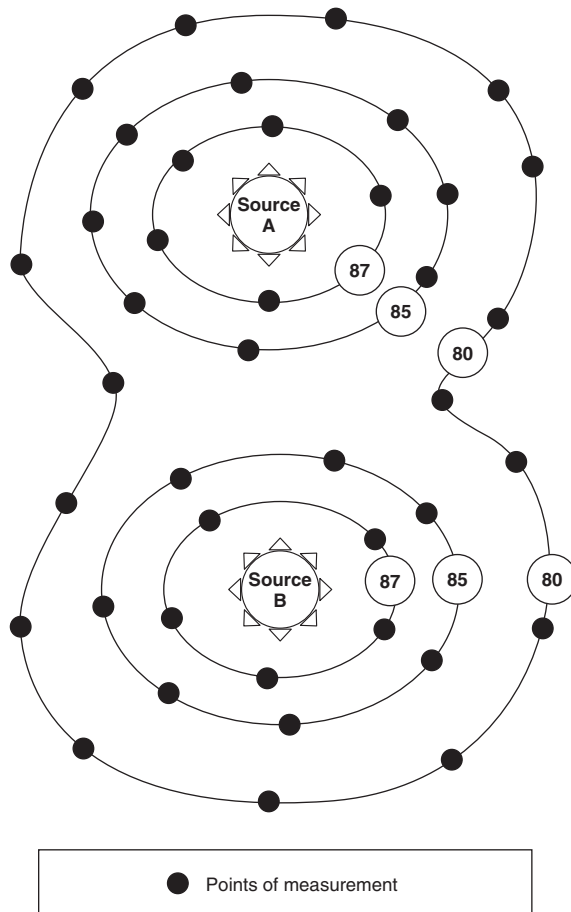


Figure 3.5 An example of a noise survey map.

within it. In order to produce a noise map, noise is measured at selected worker locations. These noise measurements are plotted on a plan of the workplace and joined by lines to identify areas where there are noise hazards. The more measurement points used, the more accurate will be the map. The employees need not be present when the measurements are made but these should be taken as close as possible to where the workers' ears will be positioned.

2. An integrating sound level meter is used, which measures sounds from specific events and integrates them to give an 'equivalent continuous sound pressure level' (L_{eq}). If the noise fluctuates over a wide range of levels or is irregular, intermittent or impulsive, an integrating sound level meter must be used (British Standards Institution, 1976).
3. Dosimetry is used to establish the noise exposure of an individual. A dosimeter is a small integrating sound level meter, which is worn with the microphone at or near the employee's ear. Measurement is usually taken throughout a typical working day.
4. For high level impulse noises, a sound level meter with an impulse noise facility is needed.

The action plan

Introduction

The action plan is the link between risk assessment and the control of the problems. It should include noise control, hearing protection, monitoring audiometry and education, with further attention given to:

- Particularly vulnerable individuals or sensitive risk groups and the adaptation of measures as necessary.
- The effect of noise on the warning signals needed to reduce risk of accidents.
- Consulting appropriate up-to-date published information (including advice provided by the HSE).
- Provision of adequate hearing protection, where risks cannot be reduced sufficiently by other means.
- Appropriate signage of areas at or above 85 dBA and 140 Pa.
- The keeping of adequate records.
- Checking the effectiveness of all measures taken to comply with the regulations.

The action plan should set out a prioritised list of actions to minimise noise exposure. The first priority should always be given to immediate risk, and urgent action is likely to involve introducing ear protection, whilst other measures are being investigated. In making a list of actions, good practice and industry standards should be considered, in addition to the requirements of the relevant regulations. Each action should be given a realistic timescale within which the work should be carried out and the whole plan should be under

the responsibility of a named person who has sufficient authority to be able to carry it out effectively.

Noise control

Noise control involves reducing noise at source to a minimum by the use of preventative methods including the use of alternative equipment, appropriate maintenance, adequate training and information for workers, design and layout of the workplace, isolating noise sources, and methods and organisation of work. Noise control tasks should be prioritised according to the level of noise contributed to the total noise exposure, as those making the greatest contribution to the noise hazard will also have the greatest effect on reducing personal noise exposure. Consulting specialist noise control engineers or giving adequate training in noise control to appropriate workers is often advisable and may save expensive misdirected efforts.

An effective noise reduction programme will have a positive policy to use low noise equipment wherever possible (whether hired or purchased) and will consider whether a change of the processes or the machines can reduce the noise exposure. The Provision and Use of Work Equipment Regulations (1998) states that all tools and machinery provided for employees' use must be suitable, including taking account of possible effects on the health and safety of the user. When considering new equipment, the manufacturers' noise emission data should be consulted. There is a legal requirement (British Standards, 1997) for equipment manufacturers to provide this information where the noise level, at operator position, is likely to exceed 70 dBA or 130 dBC for peak noises. They must also state the 'sound power level' or total noise emission of the equipment, if this is likely to exceed 85 dBA at operator position. Standard data is obtained under laboratory conditions and may under-estimate 'real life' exposure. The manufacturer should provide additional information to the purchaser in these cases but it is advisable to ask for realistic noise levels under all operating conditions. It may also be advisable to obtain a guarantee from the manufacturers that the noise will not exceed an agreed level when the machine is installed. Installation may include particular requirements that have to be followed to reduce noise and vibration. When the equipment is brought into use, noise emissions should be checked to ensure that they are not above the agreed levels. Good maintenance can help to keep equipment as quiet as possible. All equipment should be regularly checked to ensure that the noise level has not increased over time and there should be a system in place for the operator to report any problems. Wherever possible, purchases should be made from suppliers who design for low noise. Where the purchasing of noisy equipment cannot be avoided, a record should be kept of the reasons for the purchase and of the shortcomings of the equipment. This will help to guide future action and show how the employer's legal duties have been met. Noisy equipment should be used for as short a period of time as feasible and noisy work should be scheduled to take place when as few workers as possible will be present.

Careful design of the workplace, together with the use of sound absorbent building materials, can reduce the effects of noise emissions. The layout of machines can be such that noisy machines are placed away from otherwise quiet areas or they can be enclosed within a sound proof cover or placed behind a noise reducing barrier or screen. Such barriers should be placed close to the noise source or to the workers being protected and should be as high and as wide as space will allow. The barrier may not reduce reflected noise but using sound absorbent materials on the ceiling can help. Where it is difficult to control the noise, it may be practical to provide a noise shelter for the operator to give remote control of the work processes in hand. Such a shelter must be of a suitable design to be acceptable to employees, with regard to such aspects as size and ventilation. The shelter should be in use for the maximum time possible for each worker as any reduction in time spent within the shelter dramatically increases the noise risk. In other situations, increasing the distance between the workers and the noise source may be sufficient to reduce excessive noise exposure. Where appropriate, mufflers and silencers may be able to be fitted to reduce medium and high frequency noise. Low frequency noise is more difficult to reduce but, where control by other means has not been possible, active noise reduction can be used. This is a relatively costly specialist method of control that uses phase cancellation. The technique can also be employed within personal hearing protection or in noise reducing helmets, which may be a more cost-effective alternative.

The length of exposure to noise may be reduced by job rotation (alternating noisy tasks with quiet ones) or giving adequate rest periods away from the noise. It is important that rest and break areas are placed well away from noisy areas and that noise exposure (including music and speech, as well as machine noise) is minimised during these periods. If there is any noise exposure during these additional times, this must be taken into account when making a risk assessment.

Hearing protection

The first principle of hearing conservation is to prevent excessive noise from occurring and, where this is not possible, to take steps to remove the hazard. Adequate personal hearing protection must be made available and *should* be worn correctly at all times where it is not possible to reduce noise exposure below 80 dBA; it *must* be worn correctly at all times where it is not possible to reduce noise exposure below 85 dBA. It is always of primary importance to reduce noise at source but this may be prohibitively expensive, inconvenient or simply not possible. Even where adequate noise reduction is possible, personal hearing protection may be required whilst measures are being put into place. Hearing protection should be 'so selected as to eliminate the risk to hearing or to reduce the risk to a minimum' (European Parliament, 2003). Two main types of hearing protection are available, earmuffs and earplugs. Earmuffs generally provide more noise reduction than earplugs but there is a wide degree of variation

between individual types, and the manufacturer's data should be consulted when deciding on the protection to be supplied. Hearing protection must be fitted correctly and used all the time of noise exposure, otherwise its effectiveness will be greatly reduced. Earmuffs are easier to fit correctly than earplugs and their use can be readily monitored, however they have to be fitted tightly and so can be hot and uncomfortable to wear for long periods. Rest periods and job rotation may help to reduce the length of time for which hearing protection has to be worn.

The main problems encountered with personal hearing protection are that the employer may place undue reliance on it rather than taking adequate steps to reduce noise at source and that the ear protection may be unsuitable, poorly fitted or maintained and not worn as constantly as it should be. Hearing protection may also interfere with speech communication and warning signals and in this case hearing protection with a flat frequency response ('musicians' earplugs') may be preferable. The employer has a duty to ensure that workers are wearing their hearing protection and should:

- Have a safety policy that includes the need to use hearing protection.
- Place an appropriate person in charge of issuing hearing protection.
- Ensure replacement hearing protection is readily available.
- Carry out spot checks to ensure hearing protection is being used properly.
- Discipline any employee who persistently fails to use hearing protection properly.
- Ensure all managers set a good example by wearing ear protection at all times in noisy areas.

Exemptions ('derogations') to the use of hearing protection can be granted, but only where the use of hearing protection is likely to increase the risk to health and safety rather than decrease it, or for emergency services. The HSE issue exemption certificates. They are not given lightly and are regularly reviewed. The resulting risks must be reduced to a minimum and increased health surveillance must be put into place.

Monitoring audiometry

An on-going hearing surveillance programme should be introduced to monitor the hearing of workers exposed to noise. This involves hearing tests to detect early signs of noise damage. The aims of the programme are usually to safeguard the employees' hearing, to identify and protect employees who are at increased risk and to check the long-term effectiveness of noise control measures. Audiometry should identify anyone who is developing or has developed significant noise damage. A programme of monitoring audiometry must be put into place for:

- All employees who are exposed at or above the upper action level.
- All vulnerable or susceptible employees, who are exposed at or above the lower action level.

If a hearing loss is found to be developing, the worker should be warned and measures should be introduced to prevent further noise damage. It is important to check that the hearing protection being worn is of the type that was issued to that individual and that it has not been tampered with, that it is in good condition and is being worn in the correct manner. It may be necessary to retrain the worker in the correct use of their hearing protection. Exposure factors should be investigated and steps must be taken to preserve the employee's remaining or 'residual' hearing.

Audiometry should also alert management to those employees who are highly susceptible to noise damage, where additional measures may be needed. These measures may include issuing personally moulded earplugs or earmuffs with greater attenuation, more frequent hearing tests and extra education on why and how to avoid noise risk, or, usually as a last resort, removal from noise exposure.

Under normal circumstances, it is good practice to re-test an employee who is exposed to hazardous noise:

- every year for the first two years
- at three-year intervals thereafter, if there is no cause for concern.

Where hearing damage is known or thought to be occurring, the next hearing re-test should be repeated at a shorter than normal interval, for example in three months, six months or a year as appropriate. Employees who will be exposed to noise at work should be tested pre-employment or as early as possible in their employment. This first test forms the baseline for future comparisons and it is extremely important that it is accurate and can be shown to be so. A pre-employment audiogram will also indicate any pre-existing hearing loss, where there is an extra duty of care and the employer is required to consider extra precautions to prevent further hearing loss. It is advisable, wherever possible, to undertake a final hearing test before a worker leaves employment, as this can be used to help prove the limit of liability for any noise damage to hearing.

Careful records must be made and retained, in addition to the audiogram itself, which should include:

- The name of the adequately trained, competent person conducting the test.
- The serial number of the calibrated audiometer being used (and there must be a traceable calibration certificate available).
- The date of the test.
- The daily validation of the equipment.
- The background noise level.
- Any recent noise exposure of the person being tested (there should have been no exposure to excessive noise within at least the previous 16 hours; this is particularly important in the case of a baseline audiogram).
- A sufficiently full medical and work history.

Audiometric results should be explained to the individual concerned and, where there is any hearing damage, this should include:

- The significance of any hearing loss
- What will happen next

- The importance of complying properly with noise control and hearing protection measures
- Encouraging the employee to seek further medical advice, where appropriate.

Group data from audiometric testing can and should also be used to monitor the effectiveness of the whole conservation programme. For this purpose, it is often most helpful to look at specific problem areas. Trends may be indicated by statistical information, where there are substantial groups of people involved. Alternatively, where there are smaller numbers of workers involved, it may help to consider if there are any particular groups of workers that are beginning to develop noise-related hearing problems. One way of achieving this is to look at the number of individuals falling into the different health and safety categories, especially category two, which is a warning level. The results of anonymous data analysis should be made available to the employees or their safety representatives, as well as to the employers, and should be used to target noise reduction, education, compliance with hearing protection and noise control measures.

Seeking further medical advice

Where an employee's hearing is found to be within a referral category, they will be referred to a doctor. This may be the company's occupational health physician or, where there is no company occupational physician, to the employee's general practitioner (GP). An example of a letter of referral to a GP is given in Figure 3.6. The employee's consent should be obtained to contact the GP and to send a copy of the audiogram. It is advisable to obtain such consent from all employees at the beginning of their employment. Where this has not been done and an employee withholds their consent, the employee should be advised of the reasons for approaching the doctor and, if they continue to withhold consent, should be asked to sign a disclaimer. A full record should be kept.

Education and training to conserve hearing

A hearing conservation programme should (as well as reducing noise hazards) include increasing awareness through education. Education is a requirement for those responsible for the programme as well as for those affected by the measures. Employee representatives, such as safety representatives or trade union representatives, should, where possible and appropriate, be involved in the development of a hearing conservation programme, as this will assist in gaining the employees' acceptance. Most conservation measures rely to some extent on the co-operation of the employees to implement their policies, for example to ensure they do not exceed time limits in noise, to keep the doors to noisy areas closed and to use hearing protection correctly. Employee acceptance is therefore very important. Specific training will be required on, for example, the correct use of hearing protection but employees are most

Name and address of General Practitioner

Date

Dear Dr_____

Re: Name_____

DoB_____

Address_____

Following a routine occupational audiometric screening test on (date), Mr/Ms _____'s hearing was found to be bilaterally deficient in the high frequency region. In addition, previous test results indicate that this is a somewhat rapid deterioration.

Mr/Ms _____ has been an employee of _____ company for the last _____ years and has been exposed to metal cutting machine noise on a fairly regular basis. However, s/he assures me that s/he has worn suitable hearing protection when necessary. Generally, s/he experiences no difficulty in hearing but this may be due to his/her adapting to communicating with others at work whilst wearing hearing protection. S/he does not report any other health problems and describes him/herself as fully fit.

Owing to the level of hearing and its apparent rapid decrease, I would be grateful if you could see this patient with a view to ENT referral to establish cause.

I enclose copies of the relevant documents.

Yours sincerely

Dr _____

Occupational Health Physician

Figure 3.6 An example referral letter to a GP.

likely to protect themselves adequately if they understand the risks and the protection available. Education should be on-going and may take many forms, including talks, films, posters and leaflets. Some industries have special problems, for example some engineers and fitters are known to listen ‘diagnostically’ for machine noise, sometimes using a screwdriver from the

machine to their mastoid to listen to grinding and other noises. This practice may lead to hearing damage, usually over the frequencies 3, 4 and 6 kHz. Advice to all employees should explain the effects of noise on hearing, the systems in place to reduce harmful noise and their duty to comply with requirements, for instance by:

- Not entering noisy work areas unnecessarily and keeping doors to noisy areas closed.
- Wearing their hearing protection correctly at all times when working in or passing through areas where there is high noise exposure.
- Using correctly any equipment provided by the employer for noise control, for example not removing silencers, shields and barriers that have been fitted.
- Looking after all hearing protection provided to them.
- Reporting any equipment defects.

Noise induced hearing loss usually goes unnoticed in the early stages and people often become used to loud noise so they are no longer bothered by it. Consequently, workers are more likely to be aware of the discomfort and isolation of using hearing protection than its benefits. Explanations of why hearing protection is required, together with supervision to ensure it is worn correctly, is usually necessary and, without this, many workers will not comply. Monitoring audiometry itself can serve to educate, as routine hearing checks can help to convince employees that a real risk does exist. The tests also provide a regular opportunity to remind employees individually of the need to continue to protect their hearing.

The personnel involved in the audiometric programme

The audiometric testing programme should be under the responsibility of someone who is fully conversant with the technical and ethical aspects of audiometry. The employer should make clear the role and responsibilities of the person in charge and ensure that there is a protocol for reporting results back to individuals, unions and management. The designated person in charge will refer employees on when further medical advice is needed. The person in charge may or may not be involved in performing the tests but will be responsible for the quality of the service provided, maintaining the appropriate standards of testing and record keeping and the referral of individuals for further advice.

The person who carries out the tests should have undertaken an appropriate training course and have at least the following knowledge and competencies (Health and Safety Commission, 2004):

- A good understanding of the aims, objectives and techniques of industrial audiometry and how these relate to hearing.
- The ability to carry out proper otoscopic examination of the ear.
- The ability to ensure an appropriate test environment and to operate and maintain the audiometer and associated equipment.

- The ability to carry out the test procedure accurately and repeatedly.
- Understanding of the procedures that must be put into practice to ensure the confidentiality of personal health information (which includes audiometric results).
- Knowledge of how to assess and present audiometric results according to a defined system.
- When and how to seek further medical advice.
- Familiarity with the hearing protection in use by employees and the ability to teach employees to correctly fit, clean and maintain it.

Review and reassessment of noise risk

The noise risk assessment should be reviewed regularly and repeated as necessary. The Health and Safety Commission (2004) suggest that reassessments may occur on average every five years but that review and reassessments should be integrated into an on-going conservation programme rather than being carried out at set intervals. Reassessment may be needed when health surveillance indicates that employees' hearing is being damaged (which suggests that noise controls are not effective) or whenever changes occur that might impact upon the noise levels, for example changes in:

- the patterns of work
- the processes used
- the machinery in use
- technological knowledge.

It is always important to keep up to date, using such resources as HSE publications, trade journals and industry group meetings and publications. Reassessment is usually less work than the initial risk assessment.

Summary

Hearing conservation is needed when workers are exposed to loud noise. The Control of Noise at Work Regulations 2005 specifies lower action levels of 80 dBA for daily noise exposure, or 112 Pa for instantaneous peak exposures; upper action levels of 85 dBA for daily noise exposure, or 140 Pa for instantaneous peak exposures; maximum exposure limits of 87 dBA for daily noise exposure, or 200 Pa for instantaneous peak exposures.

If there appears to be a noise hazard that is at or above the lower action level, a suitable and sufficient risk assessment must be carried out. This will include accurate sound level measurements if the noise is likely to be at or above the upper action level. Where the employee's noise exposure varies during the day, calculations may involve time and motion study, or an integrating sound level

meter can be used. A small integrating sound level meter, known as a dosimeter, may be used to calculate an individual's exposure throughout a typical working day. The noise risk assessment must be reviewed regularly, and whenever there is any change that could impact on the noise levels.

An action plan forms the link between the risk assessment and the control of the problems. It sets out a list of prioritised actions, which will include such things as noise control, hearing protection, monitoring audiometry and education. Adequate records must be kept. Noise reduction is always of greatest importance. Hearing protection will be required whilst noise levels are being reduced or where it is not practical to reduce noise levels below the lower action level. At the lower action level, hearing protection must be provided; at the upper action level, hearing protection must be worn. Monitoring audiometry is mandatory where employees are exposed to noise levels at or above the upper action level and at the lower action level for those employees who are susceptible. Audiometric tests are usually carried out every year for the first two years and then, if there is no cause for concern, every three years. The personal audiometric records are confidential but an individual health record, which will be available to the enforcing authorities on request, should also be maintained.

Further reading

Health and Safety Executive information sheets, HSE Books.

South, T. (2004) *Managing Noise and Vibration at Work*, Elsevier Butterworth-Heinemann.

Personal hearing protection

Noise reduction and ear protection

Hearing protection should be considered as a last resort, to be used only after all possible steps have been taken to reduce noise at source, through engineering methods, for example by:

- using quieter processes
- using machines designed for low noise
- maintaining machines to ensure noise levels remain low
- siting machines away from people
- enclosing machines using sound absorbent materials.

If it is not possible to reduce noise sufficiently, ear protection is needed whenever people are working in hazardous noise. Ear protection protects the ears from new hearing damage and it is a legal requirement that whenever workers are subjected to noise at or above the first action level, ear protection must be freely available and *should* be worn. Although it is not a legal requirement at the first action level for ear protection to be worn, workers must be made aware that if they do not use it at all times when they are in noisy areas, their hearing is at risk. At the second action level, ear protection *must* be worn.

If ear protection is removed for even a short time the level of overall protection will be drastically reduced. For example, if the ear protection is worn for half the time of exposure, 30 dB of protection will drop to only 3 dB (Table 4.1). Similarly, an increase of only 3 dB in the noise level represents a doubling in the energy level (because the decibel scale is logarithmic) and can therefore result in the same hearing damage in half the time (Figure 4.1).

Ear protection zones should be clearly identified by signs. Within these areas ear protection should be worn at *all* times and by *everyone* who enters the area. Ear protection that carries a CE marking meets the essential safety requirements

Table 4.1 The reduction in attenuation when ear protection is not worn all the time

Percentage time used	Maximum protection (dB)
100	30
99	20
95	13
90	10
80	7
70	5
60	4
50	3

as set out in the British Standard, BS EN 352 (1-3): 2002, such as size, weight, durability and attenuation. Where appropriate, ear protection must also be compatible with other safety equipment including safety helmets.

There are three basic types of ear protection available:

- 1. Earmuffs or ear defenders
- 2. Earplugs
- 3. Semi-inserts.

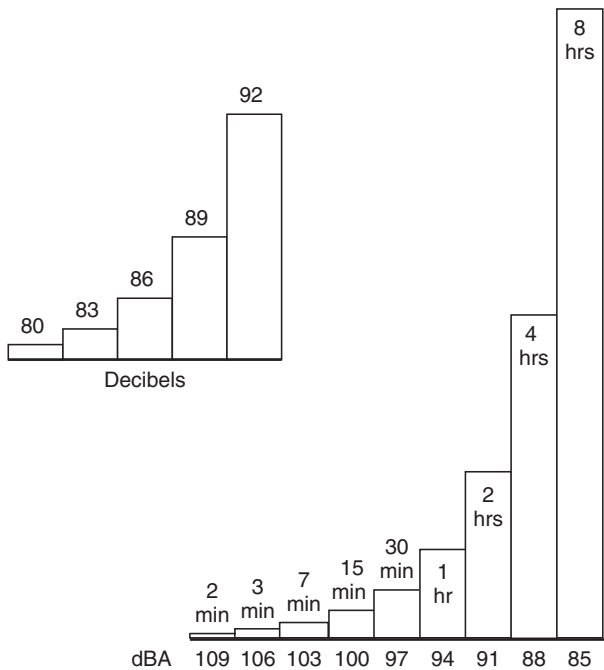


Figure 4.1 The time taken to reach an L_{eq} equal to 85 dBA as noise level increases, with decibel scale inset.

The correct ear protection must be chosen for the purpose required. The choice will depend on a number of factors including:

- The degree of attenuation required
- Compatibility with other safety equipment
- The need for communication
- Cost including cost of maintenance and replacement
- Comfort and personal preference.

Earplugs

Earplugs fit into the ear canal itself. There are several different types of earplugs available; these may be disposable, reusable or 'permanent' (long-lasting). Earplugs can be obtained with a cord or trace. The cord keeps the pair together and makes it less easy for them to be lost. They can also be hung around the neck when they are not in use – ideal for short periods in and out of noisy areas. Earplugs must be kept clean and they must be inserted using clean hands. If this is not the case, it may lead to cases of otitis externa. Ear plugs should not be used if the worker has an ear infection (although it may be possible to use earmuffs). The degree of attenuation provided generally varies markedly across the frequency range and is poorest in the low frequency region.

Disposable plugs

Disposable plugs (Figure 4.2) are most commonly made from plastic foam but may be of other materials such as glass down or wax. Foam plugs are probably the easiest to use. They can be rolled and compressed to fit into the ear and held in place while they expand to the shape of the individual ear canal. If fitted

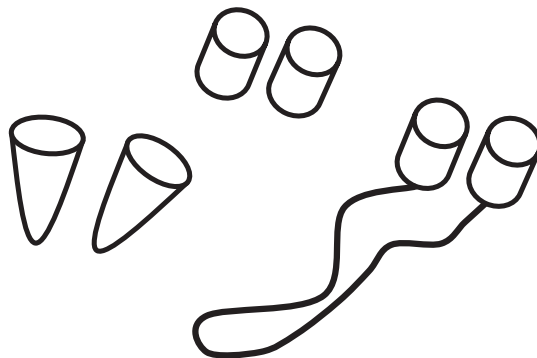


Figure 4.2 Disposable earplugs.

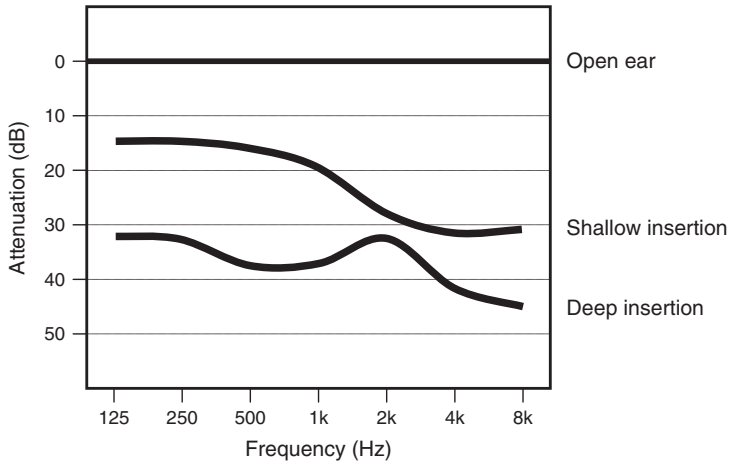


Figure 4.3 The effect of incorrect insertion on attenuation.

correctly, earplugs can be very effective. The degree of sound attenuation will vary but the maximum will be about 25–30 dB. If not fitted correctly, the attenuation may be markedly reduced. A sufficient supply of disposable plugs should be kept readily available, adjacent to hand washing facilities, just outside the ear protection zone.

Plugs must be fitted snugly (Figures 4.3 and 4.4) to afford maximum protection. This involves pulling the ear outwards and upwards with the opposite hand to open the ear canal whilst inserting the plug. The manufacturer's instructions

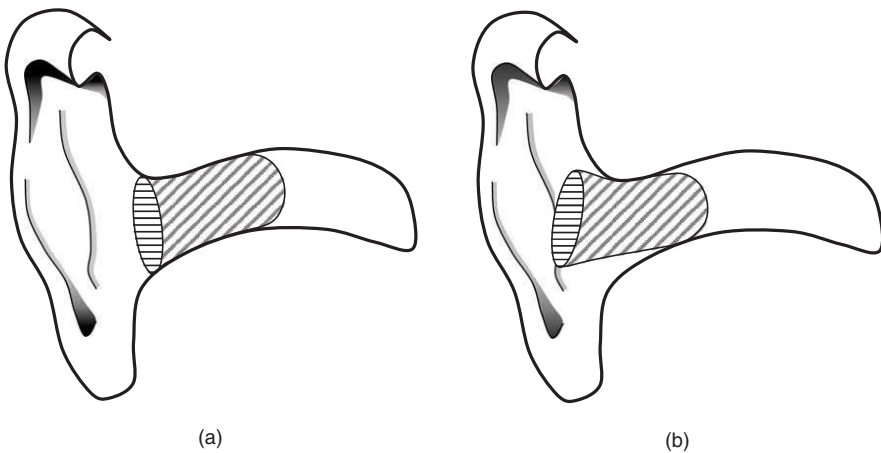


Figure 4.4 Correct and incorrect insertion of disposable earplugs: (a) correct insertion (b) incorrect insertion.

for insertion should be followed carefully. Hair must be kept out of the way. Ideally disposable plugs should be thrown away after one use but, if kept clean, some foam earplugs can be re-used for up to about a week.

Reusable earplugs

Reusable pre-moulded earplugs of soft flexible plastic (Figure 4.5) will fit the shape of most ear canals. These are sometimes available in a number of different sizes but they must fit snugly and it may be difficult to fit them successfully in ear canals of unusual size or shape. This type of plug can be washed with soap and water but in time will begin to harden and need replacing. The degree of sound attenuation will vary but the maximum will generally be in the region of 20–25 dB.

Personally moulded 'permanent' earplugs

Personally moulded or custom made earplugs (Figure 4.6) are made to fit the individual worker's ears and are most commonly made of silicone. They are produced in the same way as an ear mould for a hearing aid and therefore should fit comfortably but tightly in the ear. A good fit is somewhat dependent on the skill of the person who takes the impression of the ear from which the plug is made. The degree of sound attenuation will vary but the maximum will generally be 25–30 dB. These plugs are easier to fit correctly and the protection achieved is likely to be nearer to their assumed protection value.

Some earplugs are manufactured specifically for certain noise uses. These may include a noise filter to allow speech to be heard whilst filtering other frequencies. These are only suitable for the situations for which they are

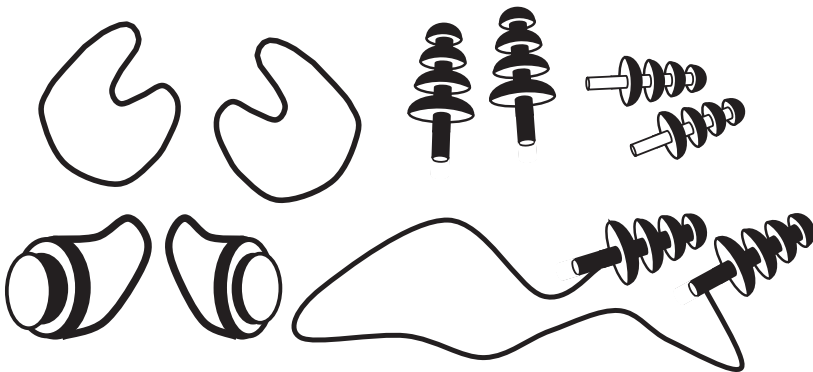


Figure 4.5 Reusable earplugs.

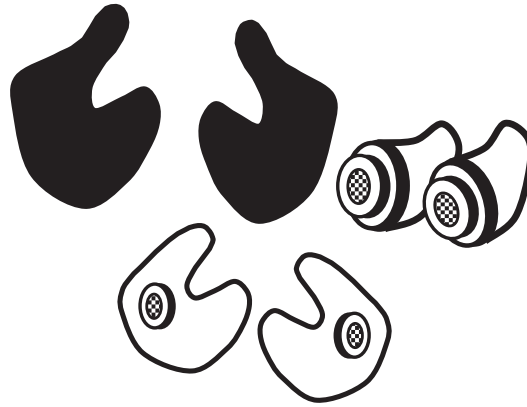


Figure 4.6 Personally moulded 'permanent' earplugs.

intended; most industrial noise is low frequency and low pass filters will render the plugs ineffective in many industrial situations. A variation on personally moulded plugs that offers 'flat' or 'even' sound attenuation across a broad spectrum of frequencies is the filtered musician's earplug (Figure 4.7). The frequency response of this plug follows the shape of the ear's natural response, so that music and speech can still be heard clearly but at a reduced level. They were called musicians' earplugs because they were developed for musicians who are exposed to high levels of sound for long periods of time (Wright Reid, 2001) but for whom normal ear protection was unsuitable

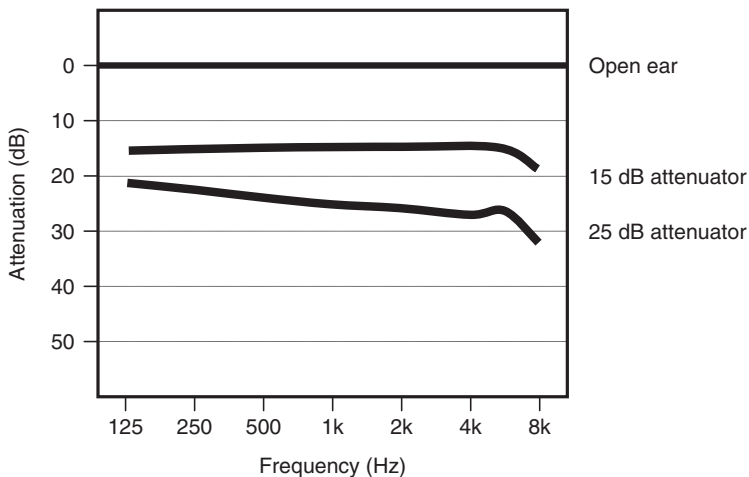


Figure 4.7 The flat attenuation response of musicians' earplugs.

because it altered the balance of sound. The plugs are available with a range of attenuation levels, between 9 and 25 dB. The musician's earplug is suitable not only for musicians but also for a wide range of industrial and leisure occupations to reduce dangerous noise levels evenly and thus preserve natural sound quality.

In the food industry, there is understandable concern that earplugs could fall into the food during the manufacturing process. It is therefore common practice to use plugs that are linked by a blue cord, which is easy to see. Blue is the only colour that is classified as a non-food colour. Earplugs for the food industry usually also contain sufficient metal content to be detected by a metal detector incorporated into the food production line machinery.

Semi-inserts

Semi-inserts are earplugs or caps positioned against or into the ear canal and attached to a small handle or a rigid headband (Figure 4.8) which is usually worn under the chin but may be worn around the back of the neck. The attenuation may vary depending upon the way that the equipment is worn and this should be checked on the manufacturer's information. Semi-inserts are easy to insert and remove and are therefore advantageous for workers who go into noise for short periods. Removal using the band or handle can help to keep the earplugs clean and they may be easier to see than earplugs, making it relatively easy to check that they are being worn. The degree of sound attenuation provided will vary but the maximum is likely to be no more than about 20–25 dB. It is very important that, when semi-inserts are fitted, they are pressed firmly against the opening of the ear canal in order to achieve the correct degree of attenuation. This pressure can make them uncomfortable to wear for long periods.

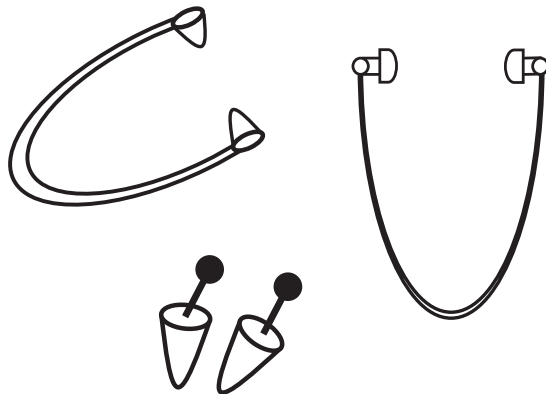


Figure 4.8 Semi-inserts.

Earmuffs

Earmuffs or ear defenders have the appearance of headphones (Figure 4.9); they consist of soft ear cushions that create a seal around the ears and hard outer cups that are joined by a headband. The cushions must be large enough to fit right over the ears. The seal may be disturbed if the earmuffs are worn over long hair or spectacles, or if the headband is bent.

Earmuffs or ear defenders are the most effective type of individual hearing protection and some may give up to about 50 dB of attenuation in certain frequencies. Generally the tighter and heavier the ear muffs, the more attenuation they provide. Earmuffs can be uncomfortably hot and heavy so they should be chosen to be as light as possible, provided that they will give sufficient attenuation. Disposable covers can be used to absorb sweat but these may reduce the attenuation provided. Where insert earplugs cannot be used, for example with minor diseases of the external ear, it may be possible to use earmuffs.

Passive (ordinary) ear defenders will provide protection against high levels of noise but can make hearing instructions, warnings and general communication difficult. Removing the defenders to listen to instructions is not an option because of an immediate loss of attenuation performance. Electronic ear defenders can enable the wearer to communicate, hear warning signals and/or be entertained at work. The use of level-dependent or peak-limiting ear defenders enables the wearer to hear instructions and warning signals in quiet situations, whilst electronically blocking out dangerous levels of noise. These are most suited to situations in which there is intermittent noise with a need to communicate in the quieter periods. Another problem with normal ear protection is that it is less effective in the low frequency region. The use of active noise reduction (ANR) ear defenders, which use an electronic noise cancelling system to provide additional noise reduction, may be particularly helpful when it is necessary to reduce low frequency noise, in the 50–500 Hz region. Some ear defenders are



Figure 4.9 Earmuffs.

manufactured specifically for certain noise uses. Some acoustic level-dependent earmuffs, for example, are effective against very high single impulse noise, such as firearms. Such defenders are only suitable for the situations for which they are intended.

There are hearing protectors that provide one-way communication facilities or entertainment programmes, for example radio reception; these may include a sound limiting system, for example to 70 or 75 dBA. It is important to check the level of the radio provided (or other sound levels, such as signals or messages, reproduced within the muffs) to ensure that this is not too loud. As 80 dBA is the first action level, this level of continuous sound should be considered too loud. Ear defenders with communication devices should still allow the hearing of external warning signals and checks need to be made to ensure this is the case. Ear defenders can be obtained with integrated receivers or two-way radios or even mobile phones. Figure 4.10 gives a résumé of the advantages and disadvantages of earplugs and earmuffs.

The use of plugs and muffs together is controversial. Maximal noise reduction can be achieved by wearing earplugs in combination with muffs but the effective attenuation is *not* found by adding the individual attenuation values together. The maximum possible protection is in the region of 50 dB and plugs used with a high performance muff will not increase the level of protection much but will be

Earplugs	Earmuffs
Advantages: ● Small and easily portable ● Able to be worn with glasses ● Able to be worn without compromising other safety equipment, for example helmets ● Lightweight ● More comfortable in hot and humid conditions	Advantages: ● Easier to fit correctly ● Can be readily seen therefore easy to monitor use ● Maximum attenuation possible for very noisy conditions ● Can be obtained in different colours for example for different attenuation levels to be used in different noise zones ● Possible to wear with minor ear infections
Disadvantages: ● Takes more time to fit ● Needs careful fitting ● More difficult to remove ● Clean hands important in fitting and removal ● Can irritate the ear canal ● Easy to lose ● Not easy to see and therefore difficult to monitor use	Disadvantages: ● Larger and heavier ● More uncomfortable in hot and humid conditions ● Difficult to use with glasses ● Difficult to use with other safety equipment, for example safety helmets ● Appropriate muffs must be obtained for use with helmets, visors, and so on

Figure 4.10 The advantages and disadvantages of earplugs and earmuffs.

heavier and less comfortable. If plugs and muffs are to be used together, the best combination is probably a high performance earplug used with a moderate performance earmuff. The protection achieved should be about 6 dB more than the better of the two individual ear protectors. Ideally test data should be obtained for the combination being used.

Ear protection must provide enough attenuation to reduce noise to below the action levels but significant over-protection is not helpful. Workers will be more likely to remove hot and heavy equipment, they may find it difficult to hear warning signals and other communication and they may be tempted to remove it to facilitate communication. Removal for even very short periods must be avoided. Protection should reduce the level at the ear to between 65 and 75 dBA. The Health and Safety Commission (2004) suggest that less than 70 dBA at the ear could cause communication difficulties and also cause the wearer to feel isolated.

Importance of fit and maintenance

Ear protection will provide the correct attenuation only if the fit is snug or tight. Where the seal provided by the device is not air-tight, which is difficult to achieve in practice, noise will leak through. Air leaks around or through the device can drastically reduce attenuation. It is therefore very important to achieve as good a fit as possible.

Ear protection must be compatible with other devices as necessary. For example, where ear defenders are worn with safety helmets, they must not restrict the movement of the shell of the helmet in case of impact. If the helmet is modified or if the clearance between the head cradle and the helmet outer shell is insufficient, the safety of the equipment is compromised. Earmuffs can be obtained which are specifically for use with safety helmets (Figure 4.11). These may be attached to the helmet or they may have the headband running behind the neck or under the chin. Where the muffs are worn with safety equipment, it is important to read the manufacturer's data, as the

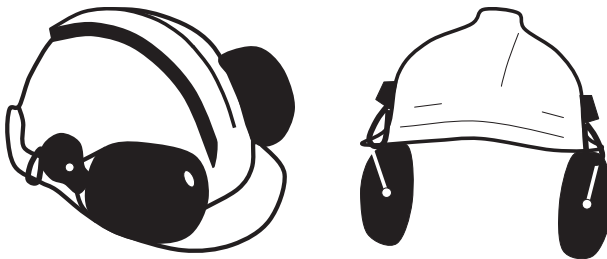


Figure 4.11 Safety helmets with earmuffs.

attenuation values may be different from the data for similar but ‘standard’ muffs.

Exemptions from using ear protection are only possible if the compulsory use of ear protectors may increase danger to a point where this outweighs the danger from hearing damage or where using ear protection is impractical, for example for the army on active service.

Ear protection must not only be provided but also maintained in good working order. The Personal Protective Equipment at Work Regulations 1992 state that ‘every employer shall ensure that any personal protective equipment provided to his employees is maintained (including replaced or cleaned as appropriate) in an efficient state, in efficient working order and in good repair’. The employee has a responsibility for their own safety and that of their colleagues and members of the public. They must not interfere with or intentionally misuse their hearing protection. It is important to check that there have been no unofficial modifications made, for example drilling through the muffs for ‘ventilation’. Employees also have a duty to report faulty or worn equipment that come to their attention. Hearing protection should be kept clean and checked regularly for wear and tear. Ear cushions and earplugs that are no longer pliable, and headbands that are stretched so that the ear cushions do not fit snugly to the head, should be replaced.

Earplugs should be cleaned in accordance with the manufacturer’s instructions and should never be passed on to another person for their use. Earmuffs should ideally be personally issued and used only by that individual but if they have to be used by different people, for example by visitors, they should be hygienically cleaned between usage or disposable covers used.

Workers are most likely to wear their ear protection if it is as light and comfortable as possible and if they have had some choice in the type. They are most likely to wear it correctly and all the time if they have had training in its fitting and understand the need to wear it. It is important that a competent person has responsibility for training workers in the use of ear protection. Workers need to know how to wear it correctly, how to care for it and store it, how to check it for damage and where to get a replacement when necessary (BS EN 458: 2001) and the importance of wearing it all the time in noise.

Attenuation

Noise may be measured in decibels (dB) at different frequencies, usually octave bands, or as a single overall dB level. The overall level is weighted to reflect the way we hear in the different frequency bands.

Attenuation is the term used for noise reduction, and is given as a number of decibels. In order to select hearing protection the attenuation value must be known. British Standards (BS EN 24869-2:1995) describe a number of

methods of estimating the sound pressure level (SPL) at the ear when wearing ear protection. The main methods are:

- Octave band analysis
- High Medium Low (HML)
- Single Number Rating (SNR).

Octave band analysis is the most accurate prediction method. Where octave band analysis is not available, a reasonable approximation of the protection can be gained using the HML figures or the SNR. The figures provided by the manufacturers give a guide to the potential hearing protection in decibels' attenuation and relate to the difference in hearing levels with and without hearing protection. The HML figures give an approximation of attenuation in each of three frequency bands: high, mid and low, whilst the SNR is a single figure of attenuation and involves the simplest calculation.

Manufacturers are required to give standard information (BS EN 352) that can be used in calculating the degree of hearing protection (Figure 4.12). Full information includes the mean attenuation, standard deviation and assumed protection values at each octave band centre frequency from 63 (optional) or 125 Hz through to 8 kHz. It is normal to work to the assumed protection value, which is the mean value minus the standard deviation. The information may be given in a simplified form, as High (H), Medium (M) and Low (L) attenuation values or as a SNR value.

Noise should be reduced to below the action levels. At the same time, over-protection is not advisable as it interferes with the ability to hear warning signals and to communicate. The type of protection used should be chosen according to the attenuation required. Although all ear protection should come supplied with information about the degree of attenuation, the assumed protection value will only provide a guide if the ear protection is well maintained, correctly fitted and worn for the entire time of exposure. Indeed, BS EN 24869-1: 1992 says that the method and procedures used in testing are not normally achieved under field conditions but this approach is used because it facilitates reproduction of the same results each time. In real life, the protection afforded is usually 5–10 dB less than under laboratory conditions but can be

Attenuation table – Tested to EN 352-2, CE marked								
Frequency (Hz)	63	125	250	500	1k	2k	4k	8k
Mean attenuation	13.7	11.2	19.1	25.7	29.2	32.0	36.8	39.0
Standard deviation	3.9	3.2	2.2	2.7	3.1	2.3	2.7	3.7
Assumed Protection (APV)	9.8	8.0	16.9	23.0	26.1	29.7	34.0	35.3
SNR = 27 dB	H = 31 dB			M = 24 dB			L = 16 dB	

Figure 4.12 An example of an attenuation table for a pair of earmuffs.

much less, due, for example, to the way it is fitted and deterioration of the ear protection in use. If the protection is not worn for the entire time of exposure, the situation is far worse.

Octave band analysis

An octave corresponds to a doubling or halving of frequency. Each frequency given in a table of attenuation, see Table 4.1, is the centre frequency of the octave band. Ideally the noise attenuation required will be calculated in each octave band as an unweighted L_{eq} , based on frequency analysis of the noise in the workplace. This will involve detailed noise measurement and a complex calculation using correction factors to convert to dBA. All protection provides less attenuation at low frequencies than at high frequencies and octave band analysis must be used if the noise has a significant low frequency content or if it is dominated by single frequencies.

HML (High Medium Low)

The HML figures are the preferred method where octave band analysis is not available. The A-weighted (L_A) and C-weighted (L_C) SPLs are required for this method, which is as follows:

The predicted noise level reduction (PNR) is calculated from the difference between the L_C and the L_A . If the difference is 2 or less, the PNR =

$$\frac{M - (H - M)}{4} \times (L_C - L_A - 2)$$

If the difference is 2 or more, the PNR =

$$\frac{M - (M - L)}{8} \times (L_C - L_A - 2)$$

The effective A-weighted SPL at the ear (L'_A) is calculated by subtracting the PNR from the A-weighted noise level (L_A):

$$L'_A = L_A - (\text{PNR})$$

If the noise is an impulsive noise such as gunfire or a drop-hammer, the ear protection must reduce the peak sound pressure to below the peak action level. If the noise is not dominated by low frequency components, the effectiveness of the protection can be estimated as follows:

1. Find the difference between the A-weighted and the C-weighted maximum sound pressure levels using the 'Fast' time weighting on a sound level meter.
2. Where this value is less than 5 dB the predicted reduction in the peak sound level is equal to the M value (of the HML values).

If the noise is dominated by low frequencies specialised measurements will be required.

Single Number Rating (SNR)

Ratings are used to provide a guide to potential hearing protection in dB attenuation. The rating system used in Europe is SNR, which relates to the difference in hearing levels without and with hearing protection.

To find the effective level of noise exposure in dBA, the noise level is measured (as a C-weighted L_{eq}) and the attenuation (SNR) provided by the ear protection is subtracted from this. For example:

$$L_C \text{ (C-weighted } L_{eq}) = 100 \text{ dBC}$$

If the SNR = 20 dB

$$L_C - \text{SNR} = 100 - 20$$

The effective SPL at the ear = 80 dBA

The ratings are obtained under laboratory (ideal) conditions and assigned by the manufacturers. They do not accurately reflect the protection afforded in real life, which is *usually* for earplugs about 6–10 dB less and for earmuffs about 2 dB less than under laboratory conditions but could be more. This is mainly due to the condition of the protectors and the way they are fitted. A safety margin should be used to allow for this and for earplugs; this should be a reduction based on two standard deviations rather than one. (Standard deviation is a measure of variability. It is a statistical term that tells how spread out numbers are from the average.) The ear protection must be used continuously during noise exposure to justify accepting the nominal attenuation value as an assessment of the protection efficiency. If the usage is not full time, the attenuation value falls so much that the choice of hearing protection is largely irrelevant.

Summary

Ear protection is available as plugs or muffs. They must be kept clean and fitted and worn correctly. There are different types of muffs and plugs, and these must be chosen according to the purpose for which they are required. Earmuffs can provide the greatest degree of sound attenuation, up to about 50 dB. Earmuffs should be as light and comfortable as possible so that workers are more likely to wear them all the time. If ear protection is not worn all the time, or if it is not worn correctly, the amount of attenuation it provides drops markedly.

Further reading

Prasher, D., Luxon, L. and Pyykko, I. (1998) *Protection Against Noise Vol. 2*, Whurr.
South, T. (2004) *Managing Noise and Vibration at Work*, Elsevier.

Organisation of an audiometric health surveillance programme

Producing an audiometric health surveillance programme

A health surveillance programme involves screening to detect early signs of hearing loss and introducing procedures to ensure that, where signs of hearing loss are found, appropriate action is taken. Employees who will be affected by the introduction of an audiometric health surveillance programme should be made aware of the implications of the programme, in particular its aims and objectives and the methods and procedures to be followed, including how results will be kept confidential and when and how employees will be referred on for medical advice.

Occupational audiometry is not a diagnostic procedure, although it may alert the tester to changes in hearing due to a variety of different causes. It is a monitoring procedure with the primary purpose of detecting early damage to hearing caused by noise at work. It should also facilitate the identification of individuals who are at increased risk, where the employer has an extra duty of care to protect them. It has the added benefit that it can be used to check the long-term effectiveness of hearing conservation measures.

Audiometric health surveillance involves regular testing of the hearing levels of all employees exposed at or above the upper or second action level of 85 dBA. Audiometric health surveillance may not be necessary if exposure is just above 85 dBA for only a very short time during the working week. Monitoring audiometry is not mandatory if the level of exposure is below 85 dBA unless there are

practical or health reasons to carry it out. For example, an increased risk of damage from noise may be indicated by:

- An existing hearing loss
- The existence of tinnitus (especially where this is troublesome)
- Marked temporary threshold shift (TTS)
- The medical history
- A family history of early deafness
- A history of previous noise exposure
- The results of previous hearing assessments
- Exemption from the use of hearing protection (where the use of protection would cause a greater risk to health and safety than that caused by not using it).

In these cases or where there is other cause for concern, audiometric health surveillance should be provided. Some firms choose to provide audiometric testing for all employees exposed at or above the first action level of 80 dBA, this may be best practice but is not a legal requirement.

An employee who is particularly susceptible to noise damage may also require a greater level of hearing protection and the existence of certain medical disorders can affect the choice of hearing protection. This could include conditions such as:

- Earache
- Otitis externa
- Discharge from the ear
- Recent ear operations
- Ear infections.

Audiometric health surveillance may also be introduced to ensure good hearing where this is an essential to fitness for work. Some elements of the programme may differ when this, rather than hearing conservation, is the overall aim. The hearing assessment could, for example, assess the hearing ability over the important speech frequencies or those particular frequencies required to hear warning signals. Alternatively a simple pass/fail screening procedure at a set level, for example 30 dBHL, could be used. Where good hearing is a requirement for the job, this should be made clear at the recruitment stage.

The responsibilities of personnel involved in the audiometric programme

The person in charge

The ultimate responsibility for the introduction and correct running of the audiometric health surveillance programme rests with the employer but they will normally appoint or designate a suitable person to be in charge of the programme.

The employer should ensure that the responsibilities of the person in charge are clear and that suitable procedures and protocols are set up. The person in charge will normally be responsible for the quality of the service provided and they should:

- Appreciate the aims and objectives of screening audiometry and how it fits into the overall hearing conservation programme.
- Understand the technical and ethical aspects of occupational audiometry.
- Ensure that appropriate standards are set and maintained during the testing procedure.
- Keep adequate records.
- Ensure confidentiality.
- Feed back the results in a suitable form to the individual employees and to the unions and the employers.
- Refer on for further advice as appropriate.
- Obtain baseline audiograms of all new employees, before noise exposure.
- Organise a regular schedule of hearing tests for employees exposed to noise (usually this involves annual tests for the first two years and then a test every three years).
- Organise more frequent testing where employees may be at higher risk.

The person in charge may or may not be involved in carrying out the actual testing. Examples of likely suitable persons might include:

- An occupational physician
- A nurse with special interest, appropriate training and experience
- An audiological scientist
- A trained audiologist.

Some companies contract out the provision of hearing tests (and, in some cases, all health surveillance) to external providers. This may be very helpful but it should be remembered that the employer remains responsible for the programme and for ensuring that further advice is obtained when this is indicated by the test results. The employer or the person within the company designated to be in charge must ensure that the external provider is competent and that there are procedures in place for the safe keeping of confidential records and for the results to be fed back to the company in an appropriate format.

The testers

Audiometric tests must be accurate and repeatable and carried out according to standard procedures. The tester must therefore be adequately trained and competent to carry out the tests. This will often mean attending a course for Occupational or Industrial Audiometry. The syllabus for such courses should include at least the following:

- The aims and objectives of occupational audiometry
- The relationship with hearing conservation

- Otoscope examination
- Audiometric screening test procedures
- The test environment
- Operating and maintaining the audiometer
- Calibration and validation of the audiometer
- Confidentiality of personal health records
- Categorisation of audiometric screening results
- Medical referral
- Hearing protection and its proper fitting and maintenance.

The screening test itself should take about 15 minutes but additional time will be needed, for example, for taking the case history. It is usually part of the duties of the tester to explain the test results to the individual employee, to provide informal education with regard to the use of hearing protection and the need for hearing conservation and to explain the importance of referral if an abnormality is found.

The physician

The results of the screening programme must be acted upon, including referring on for medical advice where hearing damage has been found. An occupational physician may be directly involved in the surveillance programme or, where this is not the case, referrals will usually be made to the individual's GP. The directive has been interpreted by the HSE for national practice, such that, where health surveillance is in the charge of a competent person, who is appropriately trained and experienced and who takes account of the HSE guidance, it is normal and acceptable practice that referral may be made to the GP on an 'as needed' basis, rather than all surveillance necessarily having to be directly under the supervision of a doctor.

Employees who fall into categories 3 or 4, or where unilateral hearing loss has been noted should be referred to the doctor. Any medical symptoms, such as dizziness, severe or persistent tinnitus, earache, fluctuating hearing loss, discomfort or fullness in the ear, or any hearing loss that is causing difficulty for the individual should also be referred. Where hearing loss is progressing at a faster rate than would normally be expected, even though the referral level has not been reached, the individual should be given advice on hearing conservation by an occupational physician or other suitably trained and experienced professional.

The role of the doctor usually includes:

- Assessing whether the hearing problem is likely to be due to noise alone or to other cause or causes requiring further medical investigation.
- Making further referrals, where this is indicated, to an ear nose and throat surgeon for diagnosis and treatment.
- Making arrangements for more frequent hearing checks.

- Considering whether a hearing aid, tinnitus masker or other device could be of benefit.
- Making further referrals, where this is indicated, for hearing aids or other appropriate devices (referral may be to the National Health Service or to a private hearing aid audiologist or dispenser).
- Providing advice on the effects of noise on hearing.
- Checking the use of hearing protection and giving guidance on its correct use.

The physician will often be required to give advice on fitness for work. If NIHL (or other hearing loss) is stable, continuing exposure to noise may be acceptable provided that adequate hearing protection is worn. However, if the hearing levels are so poor that any further hearing loss would be unacceptable, or if the hearing loss could affect the safety of the individual or of others, it is possible that the employee can no longer be considered fit for that particular job. Care must be taken when making this decision and detailed records should be made. Disability itself cannot be the reason for refusing to give work but neither can health and safety be put at risk. A profoundly deaf individual might wish, for example, to drive for a firm. This is not unreasonable and should be refused only on the grounds of deafness if it can be shown that there is a real safety issue, for example it might be reasonable to prevent this individual from driving a forklift truck where it is very important to be able to hear safety warnings. In addition, if special provision can be made which will make it possible for a hearing impaired worker to do a certain job or work within a particular environment, this provision should be made. There is considerable government help available to assist companies to finance special equipment (which may include hearing aids) that will allow a disabled worker to succeed in their employment.

The audiometric equipment

Introduction

An audiometer is a piece of equipment for establishing the presence and degree of hearing loss by comparing the individual's hearing levels with average normal hearing levels. Whatever audiometer is used it should comply with current British Standards (BS EN 60645-1: 2001). All industrial audiometers will have:

- A pair of earphones of the type TDH39 with MX4I/AR cushions. The right earphone is coloured red, the left is coloured blue. Some audiometers will also have an additional noise excluding outer headphone (Figure 5.8).
- A frequency range of at least 500 Hz, 1 kHz, 2 kHz, 3 kHz, 4 kHz and 6 kHz. The range may also include 250 Hz and 8 kHz.

- An intensity range of at least 0–70 dBHL. Most industrial audiometers have a wider range, between –10 and 90 dBHL or more. The volume or ‘attenuator’ control adjusts the intensity of the output in steps of 5 dB.
- A patient signal button, which operates a light on the audiometer control panel to indicate to the tester when the test tone has been heard.

Manual audiometers

Manual audiometers may be clinical (diagnostic) or industrial. The industrial audiometer (Figure 5.1) is much simpler than the clinical audiometer, and is required only to be able to provide a measure of hearing threshold by air conduction, without masking. In manual audiometry, the tester selects the frequency and the level of the tones and presents them according to an agreed procedure. The tones are switched on and off by pressing and releasing a tone presentation button.

Manual audiometry is suitable for most purposes and in the hands of a skilled operator may be faster to use than an automatic Békèsy machine. In industry, it is mainly used where there are small numbers of people to be tested but it should always be available, as there are some individuals who are unable to cope with self-recording audiometry, particularly of the Békèsy type.

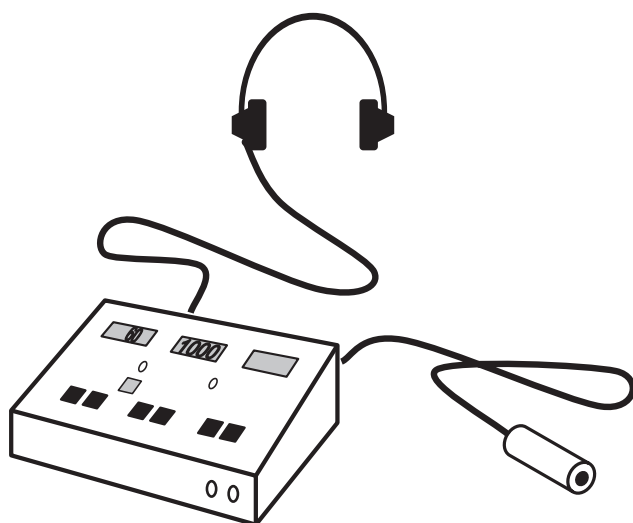


Figure 5.1 An industrial audiometer of the manual type.

Self-recording audiometers

Self-recording audiometers are advantageous where there are a large number of employees to test. Manual testing of large numbers can be very boring and accuracy may suffer, whereas self-recording audiometers are not dependent in the same way upon the tester's skill or concentration. However, it is important to maintain alertness, particularly in the subject but also in the tester. If the employee under test loses concentration, the test will be unreliable and will probably be rejected by the machine. The tester needs to keep an eye on the test to ensure that the employee is concentrating and that there are no problems. Usually, after watching the initial familiarisation, it will be possible to undertake some other activity in the same room. The tester should not leave the room during the test. Analysis will be carried out automatically by the machine according to the HSE categories but the tester should be aware of the procedure and should review the results before dismissing the worker.

Self-recording audiometers (Figure 5.2) will provide at least one of the following:

- An automated version of the manual test, where the presentation is automatic but the procedure is exactly the same as if the tester were testing manually.
- A Békèsy test, using pulsed tones that automatically change the frequency and ear of presentation according to a pre-set programme. The frequency and intensity range requirements are the same but the lowest frequencies are presented first, moving through to the highest. The employee under test controls the signal level by pressing a button. As long as the employee holds the button down, the signal will be automatically reduced in small steps. When the employee no longer presses the button (because the signal has become inaudible), the audiometer will automatically reverse the procedure and increase the signal level until the employee again presses the button (because the signal has

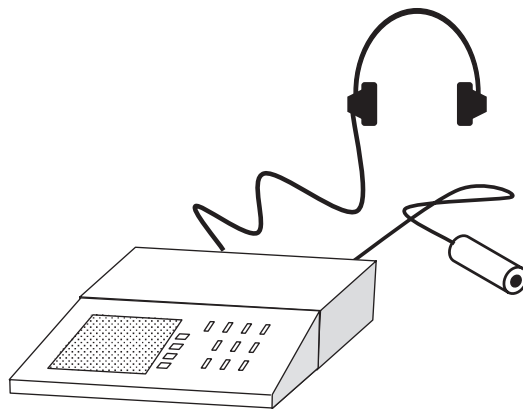


Figure 5.2 A self-recording audiometer.

become audible). In this way, the audiometer produces a zigzag trace of troughs and valleys, which are known as 'excursions'. The mid-point of these gives the threshold. It is important that the employee remains alert throughout the test. Some audiometers will automatically redo any unreliable parts of the test.

Pulsed tones may be easier for employees who suffer from tinnitus, as continuous tones are more easily confused with tinnitus than are pulsed tones. However, it is also possible to carry out a manual test using a pulsed signal, pulsed tones should only be used when necessary.

Computerised audiometers

Some audiometers are designed to interface with a personal computer or PC (Figure 5.3) via a standard RS232 interface. The advantage of integrating with a PC is that it can provide a comprehensive audiometric system, with such facilities as a paperless questionnaire, a choice of test methods and electronic recording of data which provides for data analysis and recall.

Comparing thresholds on manual and automatic Békèsy audiometers

A difference of 10 dB or more between audiograms of the same type is likely to indicate a threshold shift. However, hearing thresholds obtained using Békèsy audiometers are usually more acute (though not necessarily more accurate), by about 3 dB, than those produced using manual audiometers. This must be taken into

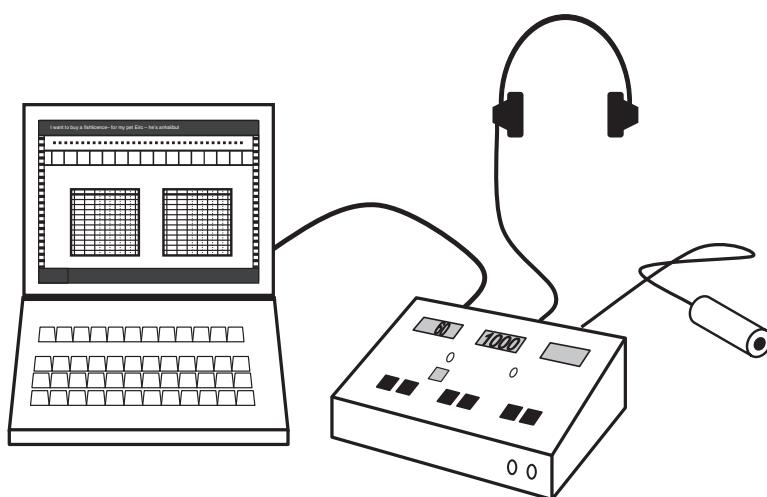


Figure 5.3 A PC-based audiometer.

account and, where a ‘manual’ audiogram is being compared to an earlier Békèsy audiogram, a 15 dB difference may be more appropriate as an indication of genuine threshold shift. It is not possible to use Békèsy audiometry with all employees and when manual audiometry has been used with a certain individual, it is often preferable to continue to use the same type of audiometry with that person.

The importance of manual handling procedures

If an individual has to move heavy audiometric equipment, it is important that they do so in such a way as to avoid personal injury and in line with the Manual Handling Operations Regulations. These regulations do not state specific weight limits that are considered safe to lift as it is almost impossible to establish universally accepted weight limits. Even a relatively light piece of equipment, for example 3 kg, lifted off a high shelf may create a manual handling risk. The weight of the load is only one factor as, for example, someone may take more care when moving a heavy item than a light one. The Health and Safety Executive (1992) provide guidance including the approximate weight recommendations shown in Figure 5.4.

When lifting and carrying items of equipment, these should be kept close to the body in order to reduce the stress on the lower back and make it easier to control the load. Lifting equipment from above the head, or from floor level, is most stressful and should be avoided if possible. If a load can be lifted and lowered safely, it

Lifting and lowering			
Position held		Men (kg)	Women (kg)
Straight arm	Close to the body	10	7
	Away from the body	5	3
Bent arm	Close to the body	20	13
	Away from the body	10	7
From floor level	Close to the body	10	7
	Away from the body	5	3

Figure 5.4 Manual handling – approximate guidelines for lifting and lowering assuming good working conditions.

can generally be carried safely for a short distance. Longer distances, for example over 10 metres, may produce fatigue and an increased risk of injury. The working environment should be safe for moving equipment, for example there should be no slip or trip hazards, such as slippery or uneven floors or poor lighting.

Individuals vary markedly in their ability to lift and carry equipment. There is no threshold below which manual handling operations are regarded as 'safe' but the risks involved are increased for certain individuals and particular care should be taken by pregnant women or anyone with relevant health problems. Age, fitness, gender and physical ability will affect the individual's capacity to lift and carry.

A substantial number of recurrent back problems are due to poor posture and everyday movement. Good posture during lifting is important. Gradual deterioration of weight bearing joints is normal with increasing age but can be accelerated by repeated stress and/or poor posture.

Audiometer maintenance

Audiometer calibration

An audiometer has to be accurate to be of any value and best practice is that all audiometers should be calibrated to British Standards at least annually. The earphones are an integral part of the audiometer and headphones and should be sent with the audiometer when it is calibrated. If the headphones are exchanged, re-calibration of the audiometer is required.

Calibration ensures that the audiometer conforms to BS EN 60645-1: 2001, which defines aspects of audiometer accuracy, including:

- Frequency accuracy must be ± 3 per cent.
- Purity must be such that the total harmonic distortion does not exceed 5 per cent.
- The attenuator 5 dB steps must be correct between ± 1 dB.
- Unwanted sound from the audiometer should be inaudible up to and including the dial setting 50 dBHL.
- The hearing level must be accurate to within ± 3 dB from 500 Hz to 4 kHz and to within ± 5 dB at 6 and 8 kHz.

Audiometers should be calibrated annually. A calibration certificate (Figure 5.5) will be issued after calibration and these must be retained with the audiometer.

Daily audiometer validation

Validation involves simple checks of functioning based on ISO 8253-1. Most of these checks should be carried out before the audiometer is used each day. Checks do not need to be carried out on those days when the audiometer is not being used, however they may have to be carried out more than once a day if the

Acoustronics Ltd. 104 Alexander St. Belham BM2 3GA											
											Certificate no. 042958
		CERTIFICATE OF CALIBRATION									
Equipment type		Diagnostic Audiometer									
Make	Acoustronics	Model no.	AT240	Serial no.	A11825						
Headphone	TDH39	Right S/N	60786	Left S/N	60778						
Bone vib.	B71	Serial no.	—	Date	31-03-2005						
Frequency (Hz)											
Frequency	125.0	250.0	500.0	750.0	1000	1500	2000	3000	4000	6000	8000
Actual	124.4	252.3	502.5	753.9	1004	1497	1992	2984	4049	6032	7982
Error %	-0.5	+0.9	+0.5	+0.5	+0.4	-0.2	-0.4	-0.5	+1.2	+0.5	-0.2
Air Conduction Sound Pressure Level											
Left	+0.6	0	-0.3	+0.2	+0.4	+0.3	-0.4	-0.6	-0.1	+0.4	0
After Adjust	+0.1	0	+0.1	+0.2	-0.1	-0.2	+0.2	-0.1	-0.1	-0.2	0
Narrow Band	-0.1	-0.1	+0.1	-0.1	+0.1	+0.1	0	0	0	0	-0.2
Right	+1.6	+0.5	-0.1	+0.2	0	-0.5	-0.8	-0.4	-0.9	-0.1	-0.4
After Adjust	+0.1	0	-0.1	+0.2	0	0	+0.2	+0.1	+0.1	-0.1	+0.1
Narrow Band	+0.1	0	0	+0.1	0	0	0	+0.1	0	0	0
Bone Conduction Force Level											
Bone Vibrator	—	+0.8	+1.1	+0.9	+0.8	+1.6	+1.9	+1.3	-2.7	—	—
After Adjust	—	+0.8	+1.1	+0.9	+0.8	+1.6	+1.9	+1.3	-2.7	—	—
Calibrated to BS EN ISO 389				Signed _____				Date _____			

Figure 5.5 An audiometer calibration certificate.

audiometer is moved, for example to another site. There are certain checks that need only to be carried out once a week rather than daily.

Validation checks should include:

Daily

- Ensure the headphones are the correct ones for the audiometer, that is ear-phone serial numbers tally with the instrument serial number.
- Straighten any tangled leads. Ensure all connections are firm and giving good contact.
- Check the battery state if applicable.

- Switch on the equipment and allow adequate warm-up time.
- Check all knobs and switches function in a silent, click-free manner and that the operation is such that no noise radiated from the audiometer is audible at the client's position.
- Check that the client's signal system operates correctly.
- Check the output levels for all tones at a just audible level (i.e. about 10 dB above your threshold). Repeat for each earphone.
- Check at approximately 60 dBHL for unwanted sounds, noise, hum or cross-talk, or for any other noticeable distortion or other problems. Check both earphones and across all frequencies. Flex the leads to check for intermittency due to broken wires.
- On automatic recording audiometers, check the marker pens and the mechanical operation.
- Clean and examine the audiometer and its attachments. Check the earphone cushions and the plugs and leads for signs of wear or damage. Replace as necessary.

Weekly

- Check the tension and the swivel joints of the earphone headband.
- Make an approximate calibration check by performing your own audiogram or that of a known subject. Variation of no more than 5 dB is acceptable at any frequency.
- Listen at low levels for signs of unwanted sounds, noise, hum and crosstalk.
- Check 'talk through' (the client communication speech circuit).

Periodically, the required date of the next laboratory calibration should also be checked, so that re-calibration of the equipment can be arranged prior to the expiry of the current calibration certificate.

To save time, an acoustic ear (Figure 5.6) may be used to speed up the validation process by automatically checking thresholds at a pre-set level, usually

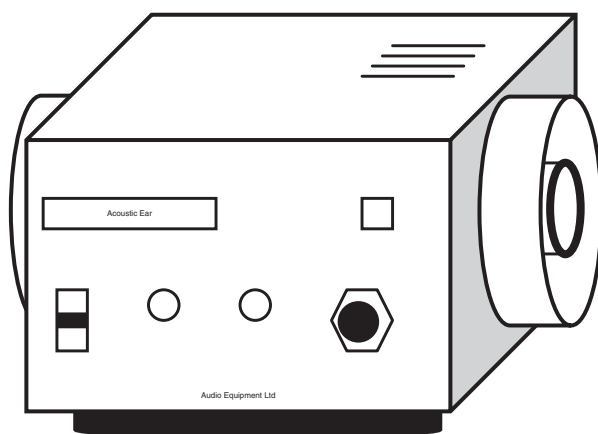


Figure 5.6 An acoustic ear.

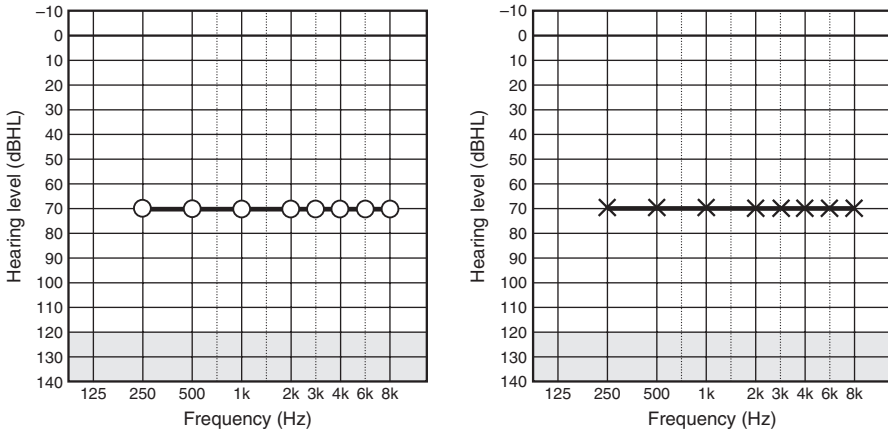


Figure 5.7 A totally flat audiogram that could have resulted from faulty equipment (or been produced by an acoustic ear).

60 or 70 dBHL (Figure 5.7). The acoustic ear also has the advantage of providing an objective calibration check which is more reliable than using one's own audiogram.

Factors affecting the accuracy of audiometry

Introduction

Audiometry will only be of value if it is carried out under appropriate conditions and to high technical standards. A small degree of variation may occur between audiograms when taken by different testers at different times or even by the same tester on the same day. This could be due to all kinds of factors unrelated to hearing threshold, for example concentration. British Standards (BS 6655) suggests that variability on retest as a result of standard deviation is approximately 3 dB at frequencies up to 3 kHz and rising to 6 dB at 8 kHz. Variation of 5 dB is therefore ignored when comparing audiograms.

Accuracy can be affected by:

- The tester
- The equipment
- The background
- The person under test
- Temporary threshold shift (TTS).

The tester

The tester may improve results by inadvertently cueing the employee to the signal presentation by creating light, movement or noise at the same time as the tones are presented. They may respond to this cue even though they cannot hear the tone, making their results seem good when in reality they could be very poor. For example, the light on the audiometer panel comes on when the sound is being presented and care should be taken that the panel light or a reflection of it cannot be seen. If the tester wears glasses these may reflect the light, therefore seating position is very important even within the booth. It is also possible for the tester to give audible clues, which are usually because they are too heavy handed and press hard on switches when a light pressure is all that is required. The audiometer rocking on the table might cause noise, for example, which is likely to be at a low frequency that might be heard. Placing the audiometer on a soft surface such as a cloth or a telephone directory will usually cure this problem.

The equipment

Faulty equipment will produce inaccurate results. The audiometer should be checked daily before use but faults can occur suddenly. Any audiogram which appears extremely unusual, including if it is a flat line (Figure 5.7) should alert the tester to possible equipment or other error. If faulty equipment is suspected, the headphones should be taken off the client, and the tester should check the functioning of the machine. Replacement of faulty parts, such as headphone leads, should be possible when spare parts are available or obtained but if headsets need to be exchanged re-calibration will be needed. Although specialist equipment is needed for calibration, it can be carried out on-site if the audiometer is to be in constant use and no spare is available. Temporary 'repair' if only one earphone is faulty is comparatively easy as the left and right plugs at the back of the audiometer can be swapped around, or the earphones can be reversed, to obtain results from each ear using the one working earphone. This is not however a long-term solution! Faulty internal components will almost always necessitate return to the manufacturers for repair.

The background

The employee's hearing levels may be elevated (worsened) in a number of ways, for example by background noise or other background distractions. Visual distractions may cause the client to lose concentration. Hence there should be no interruptions nor should it be possible to see other activities going on nearby. Noise is a particular problem because it can interfere with hearing the signal tones. The test room should therefore be sited away from obvious noise sources.

The person under test

The employee's results may be affected by tiredness, stress, tinnitus, illness, lack of understanding or by lack of concentration, giving results worse than would otherwise be the case.

Results may be falsified intentionally either by making them better, usually in order to obtain work, or by making them worse, usually in order to obtain maximum compensation.

Temporary threshold shift

Temporary threshold shift (TTS) is the change in hearing thresholds, that is deafness, brought about by exposure to loud noise. If a hearing test is carried out whilst the employee is suffering from temporary threshold shift, the results will be worse than their 'real' hearing threshold levels. After a period of rest the hearing should fully recover. The length of the rest period will depend upon the degree of temporary threshold shift. Ideally the test would be carried out after 48 hours away from noise, for example on Monday morning before work starts. This is often not possible to achieve and 16 hours away from noise is accepted as a reasonable rest period before testing. It is extremely important that the first, baseline audiogram is carried out under ideal conditions so that it is known to be accurate and that the conditions under which it was taken are recorded so that this can be proven if necessary. Tests after the baseline should ideally be carried out after a suitable rest period. Where this cannot be arranged, it may be acceptable to reduce noise exposure by ensuring that adequate (possibly extra) hearing protection has been worn prior to the test and at least a short period of rest taken immediately prior to the test. It is less of an issue after the baseline audiogram, because temporary threshold shift is not affecting the audiogram significantly if the hearing levels have not shifted by more than 5 dB. If the thresholds have shifted significantly (by 10 dB or more), the test should be repeated after a suitable period without noise exposure, prior to any action being taken.

The effect of leisure noise

It is important to remember that leisure noise may be a cause of noise induced hearing loss as well as occupational noise. Current regulations apply only to those who are working in noise and there is no requirement to use hearing protection for hobbies or home use. If someone has engaged in a noisy hobby or if they have been exposed to noise whilst travelling to work, they may have temporary threshold shift which will affect the hearing test results. Employees should be instructed to stay away from noise for at least 16 hours prior to their test or to use adequate hearing protection where this is not possible.

Possible causes of non-occupational hazardous noise that could cause TTS and that should be avoided wherever possible prior to the hearing test, include:

- Shooting
- Amplified music (including in-car entertainment)
- Personal stereos
- Playing in orchestras
- Riding motor bicycles
- Driving with the windows open
- Using DIY tools
- Hot air balloon rides
- Noisy bars and restaurants
- Arcade computer games
- Cinema attendance
- Fireworks.

The hearing test is also usually a good opportunity to discuss noise exposure outside work and its potential to increase hearing damage.

Non-organic hearing loss

Non-organic hearing loss generally refers to malingering, that is inventing or enhancing a hearing loss for financial gain (compensation). Malingerers often present with a (false) unilateral loss, presumably because they believe this will be easier to feign. Also commonly, they may present with a true hearing loss that has been exaggerated, for example they might have a moderate hearing loss but pretend to have a profound loss. The tester should always be alert for the possibility of malingering when anyone is intending to apply for compensation. It is sometimes possible to pick up clues that the person is malingering from the way they act. For example, many people do not know what it is like to be deaf and over-exaggerate lip-reading or they forget their degree of deafness and answer an interesting question when the circumstances are such that they should not have heard. If non-organic loss is suspected, the test should be repeated prior to referral. Often the two test results will differ by more than 5 dB because it is difficult to remember the precise levels previously given. This information will be useful to the physician on referral.

The test environment

Background noise can have a significant effect on audiometric results. Ambient noise levels must therefore be very low to ensure the noise does not elevate threshold results. Acceptable environmental noise levels, to ensure accurate results down to zero can be obtained, are given in Table 5.1. These levels are

Table 5.1 Example noise reduction within a booth

Frequency (Hz)	Noise reduction (dB)
125	18
250	32
500	38
1 k	44
2 k	51
4 k	52
8 k	50

usually only met in industry by using a sound-attenuating booth. Ideally, octave band analysis of the noise levels within the booth should be carried out but, if this is not possible, it is generally acceptable to carry out hearing tests where the ambient noise level is 30 dBA or less. Where levels are a little above this, using noise excluding earphones may help the situation (Figure 5.8).

The test booth

Audiometric booths are sound-treated enclosures intended to reduce or attenuate sound levels by a given amount (Table 5.1) and they cannot *guarantee* an adequately quiet noise environment. Siting the booth away from noise sources, such as machinery, lifts, traffic, office typing, toilets and so on is therefore very important. British Standards suggest the highest permissible external noise level is 59 dB at 500 Hz (Table 5.2). The booth should also be sited away from any other distractions.

An acoustic booth for occupational audiometry (Figure 5.9) must satisfy the requirements of ISO 6189/ BS 6655. It must have a window and the door should

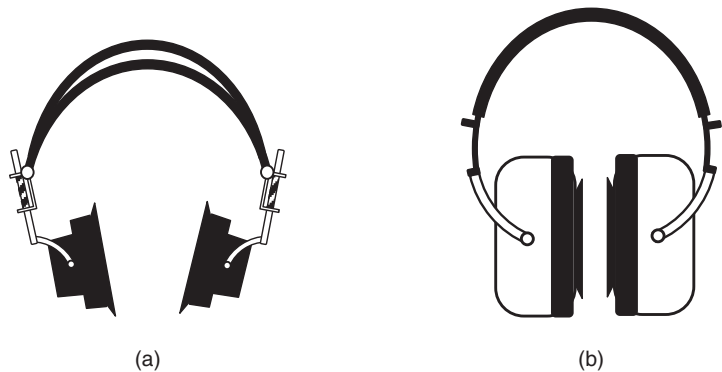


Figure 5.8 Audiometer earphones: (a) normal headphones (b) noise excluding headphones.

Table 5.2 Permissible ambient noise levels appropriate to achieve measurements down to 0 dBHL

Frequency (Hz)	Permissible noise levels for audiometric testing (dBSPL)	Noise reduction provided by standard audiometric earphones	Noise reduction achieved using a typical noise excluding headset	Noise reduction achieved using a typical sound booth
125	43 (47)	4	9	18
250	28 (33)	5	13	32
500	9 (18)	9	24	38
1 k	7 (20)	13	30	44
2 k	6 (27)	21	39	51
4 k	7 (38)	31	44	52
8 k	10 (36)	26	35	50

Note: The numbers in brackets in the second column relate to the ambient noise levels permissible using standard earphones (based on ISO 6189: 1983). Audiometry for conservation purposes is not recommended if the ambient noise exceeds these levels. The degree of noise reduction provided by a typical noise excluding headset and by an acoustic booth is given. Noise excluding cups enclose the standard earphones and the figures given in the fourth column are therefore for the cups and earphones together.

be fitted with a handle inside or a magnetic seal so that the person being tested does not feel cut off and can easily get out. If the audiometer has a ‘talk through’ facility, its use can help relieve the fear of isolation. The booth should be of sufficient size to avoid a feeling of claustrophobia, which the Department of Health and Social Security recommend should be a minimum of 1.2 m long by 1 m wide with a height of 2 m. There must be adequate ventilation, which must be silent.

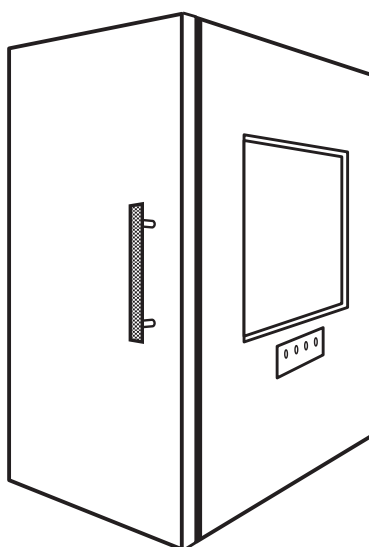


Figure 5.9 An audiometric booth.

The person under test should be seated comfortably so that they feel in contact but are unable to see the hand movements of the tester, which could lead to inaccurate hearing thresholds.

Summary

A health surveillance programme involves monitoring to detect early signs of hearing loss and involves procedures to ensure that appropriate action is taken where signs of hearing loss are found. It involves regular audiometric testing of all employees exposed at or above the upper or second action level of 85 dBA and of vulnerable employees at or above the first action level of 80 dBA. The ultimate responsibility for the introduction and correct running of the audiometric health surveillance programme rests with the employer who will normally designate a suitable person to be in charge of the programme. Audiometric tests must be accurate and standard procedures have to be followed. A small degree of variation may occur between tests and variation of 5 dB is therefore ignored when comparing audiograms. Hearing thresholds obtained using Békèsy audiometers vary from those produced using manual audiometers by an additional 3 dB. Test results should be explained to the individual employee, education should be provided with regard to hearing conservation and the use of hearing protection. The importance of medical referral should be explained when any abnormality is found.

The audiometer used for testing may be manual or automatic and should comply with current British Standards. A subjective check on the audiometer's accuracy should be carried out daily in use. Accuracy can be compromised if audiometry is not carried out under appropriate conditions and to high technical standards. Background noise levels must be very low to ensure the noise does not affect the test results. These levels can usually only be met in industry by using a sound-attenuating booth sited away from obvious noise sources.

Further reading

Health and Safety Executive (1992) *Manual Handling. Manual Handling Operations Regulations 1992*.

Health and Safety Executive Guidance on Noise Regulations. www.hse.gov.uk

Auditing and record keeping

Keeping adequate records

Records of the risk assessment

The main findings of the risk assessment and any action plan, together with the measurement data, should be recorded and preserved, in a suitable form that is readily retrievable and easily understood. The exact format of the record will depend upon the situation for which it is required. The Health and Safety Commission (2004) suggest that a minimum record will include details of:

- Workplaces, areas, jobs or people included in the assessment
- Locations and duration of the measurements taken, together with details of any noise controls being used at the time
- Work patterns and estimations of daily noise exposure
- Daily personal noise exposures, where these have been calculated
- Peak noise exposure levels, where these have been measured
- Sources of noise
- Any further information necessary to help comply with the legal duty to reduce noise exposure (e.g. details of instruments used and their calibration; a noise map or plan showing noise levels in various areas with a record of who works there and typically for how long; recommended actions for noise control)
- Dates of measurements
- Details of the competent person or persons who undertook the measurements and who took responsibility for the calculations and the conclusions drawn.

The risk assessment should lead to the identification of individuals requiring hearing monitoring. A list of these employees should be sent to the Occupational

HEALTH SURVEILLANCE REQUEST FORM

FROM _____

DEPARTMENT _____

LOCATION _____

TO THE OCCUPATIONAL HEALTH DEPARTMENT

_____ Department risk assessment has identified that the following individuals require health surveillance in accordance with Health and Safety legislation. A copy of the risk assessment is attached.

NAME	LOCATION	EXT.

Please organise appointments and undertake appropriate hearing monitoring for the individuals listed above.

Following this please notify me of their fitness to work or their failure to attend.

Signed _____

Date _____ Extn. No. _____

Figure 6.1 An example of a health surveillance request form.

Health Department (Figure 6.1). A copy of the risk assessment should be attached. It is also useful to maintain a training record, signed by the individual concerned, to show that training in the use of hearing protection etc. has been given and who carried it out.

Personal audiometric records and their use

Personal medical records remain confidential and should not be shown to non-medical staff, including the employer, without the worker's written permission. The individual's questionnaire should be used to obtain further relevant information, and is particularly useful when assessing the likely cause of any hearing loss. The individual's audiogram should be categorised using the current and previous test results. Looking back to the base-line audiogram may also add information. Notes should be made of all findings, including those during otoscopic examination. Important features should be recorded in such a way that they are readily seen when the next hearing test is carried out. It is sensible to have checks in place to ensure that adequate steps have been taken when a worker has been found to need warning or referral for medical advice. A record should be kept of any advice, investigation or treatment has been suggested and given or obtained.

Separate individual health records should be kept up to date and must be made available to the enforcing authorities on request, as part of their checks on compliance with the regulations.

Access to audiometric records

Personal medical information

The personal audiometric records are regarded as medically confidential and will normally be held by the occupational health professional in charge of the testing programme. Where the occupational health professional is not medically qualified, the holding of information should have been agreed between the employer and the employees or their representatives, when the programme was developed. The same rules of confidentiality will apply as if it was a doctor holding the records. Medical information must not normally be placed in the personnel records nor made available to the employer, unless the worker involved has given their informed written consent. Consent must be explicit, so that the worker knows what data is involved and what use is to be made of it. The worker should normally sign to indicate their positive agreement (Figure 6.2). The consent must have been freely given, refusal to give, or withdrawal of, consent must not have any employment repercussions for the worker.

Medical data is deemed 'sensitive data' and it should be treated as requiring high security. Health records should therefore be given special treatment. The Information Commission (2004) suggest that health records be kept in a separate data base or provided with separate access controls or that they might be kept in a sealed envelope in the worker's personal file. Managers and human resources personnel are not usually qualified to interpret the medical records and this should be left to doctors, nurses or other appropriate health professionals. Access should be on a strictly 'need to know' basis. In general, managers only need to know about the worker's fitness to work, although it may sometimes be necessary to know about a worker's health in order to protect him or her or others.

MEDICAL CONSENT FORM

Name

DOB

Address

Med. ID

Postcode

Telephone

GP Name

GP Address

Postcode

Telephone

I hereby grant _____ (*company*) access to my medical records. I understand that my details will be held in strict medical confidence and are subject to the Data Protection Act 1998.

I **request/do not request*** to see medical information supplied to _____ (*company*) under the terms of the Access to Medical Reports Act 1988.

Signed

Dated

* Please delete as appropriate

Figure 6.2 An example of a consent form.

Anyone having access to medical details should be contractually bound to equivalent conditions of confidentiality as a medical practitioner.

A non-medically qualified person in charge of the programme is ‘obliged to forward the information (where problems have been identified) to the worker’s GP or consultant’ (Health and Safety Commission, 2004). The employee’s consent should be obtained. This is often obtained at the stage of the pre-employment medical or at the beginning of employment. Workers should not be asked to give permission for disclosure of more than is necessary. Only the specific information needed should be elicited and workers should be aware of why health information is being collected and be given clear written details of who will have access to the information and under what circumstances.

Questionnaires used should also only elicit information that is relevant and necessary and they should be designed by the occupational health professionals so that they conform to this. They should be checked to ensure that they do not lead to discrimination under the Disability Discrimination Act 1995.

Individual health records

A further individual health record that does not contain any personal medical information should also be kept, for as long as the employee is under health surveillance. This health record will include:

- The name and identification details of the worker
- Their history of noise exposure
- Their fitness for work
- Any restrictions imposed upon them
- Any other outcome of the audiometric health surveillance programme.

Health records should be kept up to date and made available to the enforcing authorities on request.

Anonymous group data

Anonymous group data can be maintained, without consent, in such a way that it does not reveal any individual's hearing thresholds or compromise confidentiality. This is useful for identifying areas or tasks where noise control measures are not working or to allow comparison of success between different control measures.

Retaining records

The original records must be kept 'for at least as long as the individual remains under health surveillance' (Health and Safety Executive, 2004). Enquiries regarding an individual's hearing may arise many years after they have left the workplace or after noise exposure has ceased and it is therefore sensible to retain the records for a number of years. The precise number of years for which these records should be kept is a matter of some debate, as different sources suggest records should be retained for a varying number of years, generally 5 to 50 years after the last date of entry, for example regulation 11 of The Control of Substances Hazardous to Health suggests it should be at least 40 years from the date of the last entry made in them. Some degree of common sense should prevail and the age of the employee may affect the decision reached. The records of an employee who leaves at an early age may need to be kept for a considerable number of years, whereas someone who retires from work at 65 would be 105 (or more probably deceased) if their records had been kept for 40 years.

If a business ceases trading, the health records should be offered to the HSE, who may suggest they are returned to the individuals concerned.

The Data Protection Act

The requirements of the Data Protection Act (1998) and the Freedom of Information Act (2000) will apply to the records being retained as part of an audiometric programme. There are legal obligations to protect personal data and ensure its accuracy and to allow individuals access to their personal records. The Data Protection Act relates to information concerning a living person who can or could be identified by the information. There are eight principles of compliance, which are that data must be:

1. Fairly and lawfully processed
2. Processed for limited purposes
3. Adequate, relevant and not excessive
4. Accurate
5. Not kept for longer than is necessary
6. Processed in line with the individual's rights
7. Secure
8. Not transferred to countries without adequate protection.

The individual, company or organisation who decides why data is being held and the way in which it is processed is known as the 'data controller' and it is their responsibility to comply with the requirements of the Act. Processing personal data has a wide meaning and includes:

- Obtaining, holding and maintaining data
- Organising, retrieving, consulting and amending data
- Disclosing, erasing or destroying data.

Any individual can require the data holder to confirm if they are holding any personal data about them and, if so, to say what it is and to whom it could be passed. Someone who handles data only in accordance with instructions from a third person is not a data controller. Personal data held electronically, for example on a computer, and in certain paper-based systems are covered by the provisions of the Act. A data controller must notify the Information Commissioner, giving certain details, including the type of information they hold and the purposes to which it will be put. The Information Commissioner maintains a public register. Failure to notify (including annual renewal) or to register any changes is an offence, as is unauthorised access to or disclosure of personal information. The Information Commissioner's address is:

The Information Commissioner
Wycliffe House
Water Lane
Wilmslow
Cheshire.

Assessment of the effectiveness of the hearing conservation programme

It is essential that the results of audiometric testing are available in a form that will allow the employer to check that hearing conservation measures are effective. Anonymous group information should be provided for this purpose. This can be analysed and presented in such a way that it can be used to check if there are any specific groups of workers whose hearing has deteriorated, for example in one particular area or in certain shifts. Computer programs are particularly helpful in presenting this kind of statistical analysis and some computerised audiometers can be programmed to provide this information.

In some cases, it may be convenient to present the analysed data as simple charts or tables to show the percentage of workers falling into each category. This is particularly useful if it is broken down into, for example, locations or specific jobs. If the work force has remained reasonably stable, the information can also be compared with the data from previous tests. The type of analysis that is undertaken will often depend on the number of workers involved. Where the analysis indicates that there is a potential problem, further investigation and action is required, which is likely to include:

- Reassessment of noise exposure levels, taking particular account of any changes in the working environment
- Re-education of workers in the use of hearing protection and other conservation measures
- Reassessment of hearing levels at a shorter interval than normal.

Auditing

Introduction

The occupational health staff carrying out the hearing tests will carry out certain on-going checks but a regular internal audit will be needed to examine procedures in a systematic manner. The audit process will generally involve a number of stages from planning through to follow-up (Table 6.1). The planning stage involves deciding who should be audited on what, by whom, when, where and how. The programme organiser will normally make these decisions but discussion with the staff involved is helpful in deciding what to look at, and what to look for, as well as increasing the staff's feeling of involvement. A schedule and checklist are important aids that will help to establish and maintain focus on essential features. The auditing schedule should be agreed with the staff. The organiser may be the auditor if they have sufficient expertise in the areas they will be auditing. Alternatively, the auditing may be shared and one option is to use staff to audit one another, which may serve to encourage the adoption of good practices and possibly lessen staff stress. The audit itself is usually in two parts. The documentation, records and equipment are

Table 6.1 Stages of an audit process

Stage	Process	Actions
1	Planning	● Discuss with staff (what/who/how/when to audit?) ● Draw up checklist
2	Conducting	● Documentation and equipment audit ● Staff interviews and assessments
3	Outcomes	● Consider positive and negative outcomes ● Agree corrective actions
4	Report	● Oral or written ● Individual or group
5	Document	● Main findings ● Corrective suggestions with time line
6	Follow-up	● Confirm corrective actions have been implemented ● Confirm corrective actions are effective

normally reviewed first to ensure these are complete and adequate to comply with the areas and issues identified in the checklist. The second part involves checking and assessing the processes used by members of staff. This is usually by interview and observation. The outcomes of the audit must be summarised and recorded. They should also be reported orally or in writing, to each individual or to the group, as appropriate. Corrective action should be agreed, together with a timescale for this action. A further follow-up, on a formal or informal basis, should also be organised.

An internal audit is an effective management tool to ensure that a pro-active ‘best practice’ approach to hearing conservation is adopted. An on-going schedule of internal audits will help the person in charge of the programme to ensure that staff are maintaining their knowledge and skills and using the correct methods and procedures. The auditor needs to be aware that observation places stress on the member of staff being audited and the exercise should be undertaken in a positive and helpful way. Credit should always be given for good practice; where practice is poor, problems are often due to inadequate training and direct criticism is usually unhelpful. Reasons for carrying out auditing are likely to include:

- To assess the level of compliance with legislative requirements, for example health and safety and data protection regulations
- To assess the level of compliance with the company’s policies and procedures
- To provide information for assessing the success of the conservation programme
- To identify gaps and weaknesses in the system.

Audiological checking and auditing

Individual testers should carry out certain checks every time a hearing test is carried out. These checks will include:

- A visual check to ensure the audiometer is within its calibration date and that there are no obvious faults.

- An auditory check to validate the audiometer's calibration and that the background noise is within acceptable limits.
- Examination of the individual health records to check that any restrictions or other outcomes have been acted upon.
- Questioning of the individual to highlight any significant changes in noise exposure and/or medical history.
- Inspection of individual hearing protection to check its condition and adequacy for the noise levels in which the employee is working. Any changes in working practices or noise levels may indicate a requirement to recalculate exposure and the adequacy of hearing protection.

In addition to these routine checks, auditing will involve the person in charge of the programme in making regular checks on documentation, personnel and equipment to ensure that all procedures are being followed. An audit will usually check that all the necessary systems exist and that they are adequate. Such checks will usually include:

- Checking documentation including all questionnaires, audiograms, reports, forms and other paperwork to ensure it:
 - conforms to guidelines, policies and procedures
 - has been correctly completed
 - is being accessed only by the appropriate staff.
- Observation of testers (at least annually) to ensure (Figure 6.3) that they are adhering to accepted:
 - audiometry procedures and standards
 - hygiene procedures, including correct cleaning procedures for equipment and correct disposal of consumables.
- Ensuring that a record is maintained of testers' names and qualifications. It is also a good idea to keep a record of details of all relevant training courses attended.
- Ensuring all audiometers have been calibrated at least annually and checked daily using personal baseline validation procedures whenever the audiometer has been in use. This will involve examination of calibration files and subjective validation records. The calibration records (Figures 6.4 and 6.5) should be retained with the calibration certificates, which must be to the recognised British Standard and will be supplied by the firm carrying out the calibration. It is good practice to affix a label to the audiometer giving the date of its last calibration and the date when it is next due for calibration. The audiometer should not be used if it is outside its calibration date.

Validation records (Figure 6.6) involve the tester checking their own audiogram, taken on the day of use, against their personal baseline audiogram. There should be no more than 5 dB of difference between these to accept that the audiometer is still in calibration. An alternative method of validation is to use a piece of equipment known as an acoustic ear. This provides an objective response at a pre-set sound level, for example 70 dBHL, with a printout of the result.

AUDIOLOGICAL AUDIT

Site/Department _____

Name _____ Position _____

Qualifications _____ Start date _____

Date	Activity observed	Details of observation	Action taken/required

Training requirements/comments

Auditor _____ Signed _____ Date _____

Figure 6.3 An example form to be used to record observation of staff.

- Noise levels recorded in the sound booth (at least annually). The personal baseline audiogram will act as a daily subjective check that noise levels have not increased as any significant increase in background noise will result in a change in the thresholds recorded on the personal audiogram.
- Observation of the work environment and its noise levels (at least annually). The person in charge of the programme should be aware of the noise levels in which the employees are working. The findings of the noise survey and risk assessment should be referred to but it is also good practice to visit the noisy areas personally and ask engineers or line managers about any changes.

Calibration Record

Re.: ISO 8253-1 (Stage A)

Equipment Make & Model:						Serial No.:				
Week Commencing	Baseline Tests & Calibration Checks	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Faults Found/Actions Performed	Signed

Figure 6.4 A calibration record form.

Calibration Record

Re.: ISO 8253-1 (Stage A)

Equipment Make & Model: AMPLIVOX 620					Serial No.: 4216					
Week Commencing	Baseline Tests & Calibration Checks	Mon	Tues	Wed	Thur	Fri	Sat	Sun	Faults Found/Actions Performed	Signed
17/10/06	PTA recorded/Stage A checks	✓	✓	✓	✓	✗	✗	✗	BC signal low – replaced lead	J. Smith
24/10/06	Audiogram recorded/Stage A	✓	✗	✗	✗	✓	✗	✗	None	H. Brown
01/11/06	Full Audiogram & Stage A checks	✗	✓	✗	✗	✗	✗	✗	None	A. Jones
08/11/06	Not in use									
15/11/06	Not in use									
22/11/06	Not in use								Despatched for annual calibration	J. Smith
29/11/06	Not in use									
05/12/06	PTA recorded/Stage A checks	✓	✓	✗	✗	✗	✗	✗	None	M. Black

Figure 6.5 An example of a completed calibration record.

VALIDATION (PERSONAL BASELINE) RECORDS

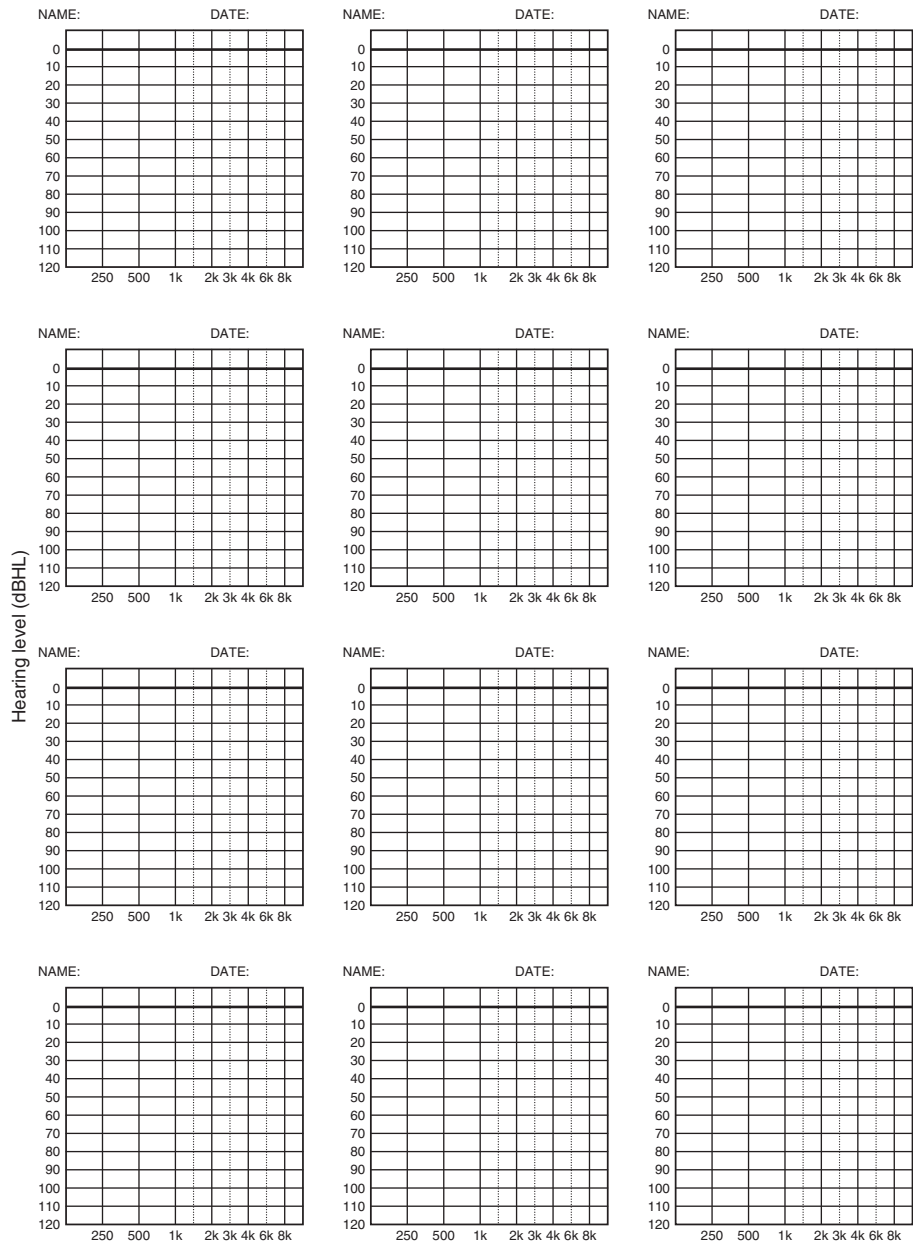


Figure 6.6 Validation (personal baseline) records.

Summary

Accurate records need to be kept, maintained and used. The personal medical records, including audiograms and case histories, are confidential and will be used by the occupational health department to ensure the health of the individual. The individual health (audiometric) records are work-related hearing records that have to be made available to the enforcing authorities on request. In addition, anonymous group data will be kept that can be used to monitor the effectiveness of the hearing conservation programme. Auditing involves formal checks to ensure adherence to all the required standards and procedures.

II

Occupational Audiometry

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Case history and otoscopic examination

The condition of the employee for testing

On the day of the test it is important that the employee is in a suitable condition for testing. In particular they should be:

- Alert and ready to co-operate
- Generally well and free from colds
- Free from noise exposure.

It is important for the employee to be tested before any exposure to noise as recent noise exposure can cause temporary threshold shift, that is temporary deafness. Ideally there should have been no noise exposure for 48 hours before the test but, as this is often impossible to achieve, in practice 16 hours without noise exposure is generally considered acceptable. Temporary threshold shift is a temporary elevation in hearing threshold after exposure to noise. The degree of deafness depends on the noise level and its duration and the time away from the noise before testing. The greater the threshold shift, the longer the rest time is required before testing. Particular care must be taken when obtaining the first baseline audiogram as it is imperative that this is accurate. For tests after the first baseline, if it is not possible to test before

noise exposure on the day of test, the test results may still be acceptable if suitable ear protection has been worn and a rest period taken before testing. Further hearing assessment without noise exposure will have to be undertaken if the audiogram shows an identifiable threshold shift, in comparison to the previous audiogram.

The case history

The need for a case history

A case history must be taken to investigate possible causes of hearing loss whether by previous noise exposure or other cause. This information will assist in decision-making following audiogram analysis and classification. The case history is usually obtained through a questionnaire, which should cover relevant areas but not include a great deal of unnecessary information. Two questionnaires are needed. The first questionnaire will, of necessity, be longer than those given with subsequent tests as it should help to identify possible causes of pre-employment hearing loss. It will include:

- Previous noise exposure at work and leisure
- Any history of injury to the head or ears
- Ototoxic drugs
- Relevant illnesses
- Previous ear disease
- Family hearing problems.

Subsequent questionnaires need only to record changes in the history since the last test. These will include any changes in noise exposure, medication, illnesses and so on but do not need to repeat information which will not have changed, for example childhood information, previous work history and the family history of deafness.

Administration of the questionnaires

The case history may be obtained by asking the employee the questions and recording their answers or it may be provided in a form that the employee can complete in advance and bring to the test with him or her. Each method has its advantages and a combination of both may be useful for the initial questionnaire. It is often helpful for the employee to have time to recall and check childhood illnesses, family history and previous work history for example. Many people do need to look these things up or ask relatives in order to ensure the information is acceptably accurate. When a case history has been

AUDIOMETRIC RESULTS CONSENT FORM	
I _____	
(Name)	
of _____	
(Company/Employer)	
Consent to hearing tests being carried out by the _____	
	(Company)
Occupational Health Department and for a report on the results to be made available to my employer for the purpose of hearing conservation.	
Signed: _____	Date: _____

Figure 7.1 An example of an audiometric results report consent form.

answered in advance, it is helpful to go through the answers with the employee to ensure everything has been answered fully and to obtain further details where appropriate.

The case history is generally signed as a true and accurate record by the employee and countersigned by the person taking the history. Consent may also be obtained at this point for the test results to be made available to the employer in order to protect against the risk of hearing damage from noise exposure (Figure 7.1).

It is often helpful to ask the employee to bring their ear protection with them to the test. This allows the tester to check its condition, whether it is the correct protection for the work undertaken and whether it is being correctly worn.

Constructing the first questionnaire

The questions in the case history will include many standard ones but to some extent questions will be individual to the industry and situation. Therefore there is not one suggested questionnaire but examples that may help when deciding the questions to be included.

A questionnaire to be completed by the employee will need to be preceded by an introduction explaining the need for the case history. It may also include a paragraph reminding the employee to avoid noise exposure before the test and to bring their hearing protection with them. All the questionnaires included here present guidelines only and should be modified to the circumstances and purposes of their use.

General information required is likely to include some or all of the following:

- Name
- Date of birth (or age)
- Job title
- Department
- Manager
- Shift
- Noise level if known
- Start date of this employment (or length of time in this employment)
- National insurance number or clock number.

There will need to be questions to investigate the medical history. These may include, for example, some or all of the following questions:

- Do you think your hearing is normal?
- Have you ever seen a doctor or been to hospital with regard to your hearing/ears?
- Do you have a hearing aid?
- Have you ever received any compensation for hearing problems?
- Have you ever had ear disease/trauma to the ears/tinnitus?
- Do you have deafness in the family?
- Have you had any childhood illnesses, for example mumps, measles, chickenpox, tuberculosis, meningitis?
- Are you taking any medication (ototoxic drugs)?

There will need to be questions to investigate the social and work history. These may include, for example, some or all of the following questions:

- Have you ever been exposed to gunfire or explosion?
- Have you been exposed to loud music on a regular basis?
- Have you ever worked with noisy tools or equipment?
- Have you ever previously had a hearing test or worn ear protection?
- Do you wear a headset regularly?
- Have you been exposed to loud noise in the last 16 hours? If so were you wearing ear protection?
- Have you had a cold, flu or sinus problem in the past 3 days?

Figures 7.2–7.4 show examples of both examiner-completed and employee-completed initial questionnaires, both of the kind that can be given to the employee and those that are filled in by the tester.

Review questionnaires

The questionnaires that follow the first one do not need to repeat history that will not change. Figure 7.5 shows an example of a employee-completed review questionnaire. Review questionnaires will obviously have to provide sufficient

OCCUPATIONAL HEALTH AUDIOMETRIC QUESTIONNAIRE		
Surname _____	Job Title _____	
Forename _____	Shift _____	
Date of birth _____	Manager _____	
Audiometer calibration date _____	Length of time in the company _____	
Audiometer verification date _____	Ear protection worn today Y/N	
<u>WORK HISTORY:</u>		<u>Details:</u>
Have you previously worked in noise?	Y/N	_____
Have you previously worn ear protection?	Y/N	Always/occasionally/seldom Type _____
Have you previously had an audiogram?	Y/N	
Have you ever been exposed to shots or blasts?	Y/N	_____
Do you wear a headset regularly?	Y/N	Single sided (L/R)/double/in ear
<u>SOCIAL HISTORY:</u>		<u>Details:</u>
Do you regularly use DIY power tools?	Y/N	_____
Do you or have you ever been shooting?	Y/N	_____
Do you attend loud music venues regularly?	Y/N	_____
Do you ride a motorbike?	Y/N	_____
Do you play or sing in an orchestra/group?	Y/N	_____
Do you have any other noisy hobbies?	Y/N	_____
Do you use ear protection for your hobbies?	Y/N	Always/occasionally/seldom Type _____

Figure 7.3 An example of an examiner-completed initial questionnaire (2).

(continued)

general information to indicate to which employee they relate, and the following are useful to include:

- Name/clock number
- Age
- Has your job changed since your last hearing test? If so has the noise level changed?
- Do you consider your hearing has changed since your last hearing test?

MEDICAL HISTORY: Have you ever seen a doctor about your hearing? Do you think your hearing is good? Do you have noises in the ears (tinnitus)? Have you ever had: • ear disease or discharging ears? • ear surgery? • head trauma? • wax removed? • any childhood illnesses? • ototoxic drugs? • high blood pressure? Are you currently taking any medication? Do you have any deafness in your family? Do you use a hearing aid?	 Y/N Y/N Y/N Y/N Y/N Y/N Y/N Y/N Y/N Y/N Y/N Y/N	Details: _____ _____ Persistent/annoying/buzzing/ringing _____ _____ _____ _____ _____ _____ _____ _____ _____ _____
Do you today have any of the following? Pain in the ear Left Right No Discharging ear(s) Left Right No Ringing in the ear Left Right No Ear blockage Left Right No Have you been exposed to loud noise without ear protection within the last 16 hours? Yes/No Have had a cold, flu or sinus problem within the last 3 days? Yes/No		
Employee's signature _____ Date _____		
RESULTS: Otoscopic results _____ Hearing test results: Category _____ Action _____ Examiner's signature _____ Date _____		

Figure 7.3 (continued)

- Do you have to shout to make yourself heard at 10 feet away or less?
- Do you usually wear ear protection?
- Have you had any ear problems since your last hearing test?
- Have you experienced any tinnitus since your last hearing test?
- Have you had any head injury since your last hearing test?
- Have you taken up any noisy hobbies since your last hearing test?
- Are you currently taking any medication?

Noise and Health Questionnaire		
Please answer these questions as accurately as you can. We need this information to help us to interpret your hearing test. It is very important that you are not exposed to loud noise for 16 hours prior to your test. This includes listening to loud music on your way to work and riding a motor bike. Avoidance of loud noise will help to ensure your results are as accurate as possible. If it is impossible to avoid noise exposure, you should use additional hearing protection during the pre-test period. Please bring any ear protection that you usually use to the test with you.		
Surname _____ Forenames _____ Date of birth _____ Age _____ National Insurance No. _____ Date started employment here _____ Job Title _____		
Please tick Yes (Y) or No (N):	Y	N
Does your job expose you to high levels of noise?		
Do you have to shout to make yourself heard at a distance of 10 feet away?		
Do you usually wear ear protection?		
If so, do you wear it all the time?		
Have you ever had an ear or mastoid operation?		
Have you ever had a perforated eardrum?		
Did you suffer from frequent ear infections as a child?		
Do you currently suffer from frequent ear infections?		
Have you had measles, mumps, chicken pox, tuberculosis, scarlet fever, meningitis or diphtheria?		
Have you ever had a head injury or concussion?		
Do you ever suffer from noises in your head or ears?		
Do you suffer from dizziness/giddiness?		
Do you suffer from high blood pressure?		
Have you ever had any medication that the doctor has said could affect your hearing?		
Have you been exposed to any solvents?		
Is there any deafness in your family?		
If so, give details:		
Who? _____		
Cause? _____		
Do you hear better or worse in noise?		
Do you have wax removed from your ears?		
If yes, when?		

(continued)

Figure 7.4 An example of a employee-completed type of initial questionnaire.

Have you had any previous employment where you have had a noisy job? If so, give details: 1. Job title: _____ Employer: _____ Dates of employment: From _____ to _____ What type of ear protection did you use? ear plugs/inserts/muffs/other Did you wear ear protection all the time? 2. Job title: _____ Employer: _____ Dates of employment: From _____ to _____ What type of ear protection did you use? ear plugs/inserts/muffs/other Did you wear ear protection all the time? 3. Job title: _____ Employer: _____ Dates of employment: From _____ to _____ What type of ear protection did you use? ear plugs/inserts/muffs/other Did you wear ear protection all the time?		
Have you ever: Been exposed to gunfire or an explosion? Served in the armed forces? Been a member of an aeroplane cabin crew?		
Do your hobbies include: Smoking? Shooting? Motorbikes? Attending discotheques or pop concerts? Playing in an orchestra or band? DIY?		
The information given above is correct and complete to the best of my knowledge. I consent to my employer being informed of the overall results of my hearing test i.e. whether my hearing is normal or there is evidence of a hearing loss. I understand this information will be used only in order to protect the hearing of any employee at risk from noise exposure. Signed: _____ Date: _____		

(continued)

Figure 7.4 (continued)

FOR MEDICAL USE ONLY:

Day of the working week _____ Hours worked before test _____

Hearing protection used today Yes/No

Otoscopic Examination:

Result of this audiometric test: _____

Category: _____

Comments and follow-up action: _____

Examiner signature: _____ Date: _____

Review date: _____

Figure 7.4 (continued)

Otoscopy

Otoscopic examination

Otoscopy is the visual inspection of the outer ear. It involves looking at the pinna, ear canal and eardrum. An otoscope is used for inspecting the canal and the drum. An otoscope, also called an auriscope, consists of a magnifying lens, a funnel or speculum and a case containing batteries (Figure 7.6). There should be at least three sizes of specula available, small, medium and large. ‘Size’ refers to the diameter of the ear tip inserted into the ear canal, not the length of the speculum; extra length specula are available but are not normally required and are not recommended for use by inexperienced practitioners.

Before testing an employee’s hearing, an inspection of the outer ear should always be undertaken to ensure there are no conditions that would prevent an accurate test from being carried out or that should be referred on for medical treatment. The examination should include observation of the pinna, the canal and the eardrum to ascertain:

- Is the condition of the skin healthy?
- Are there any scars in front of or behind the pinna?
- Is the ear canal normal?
- Is the amount of wax in the ear excessive?

<u>Review Audiometry Questionnaire</u>																					
Please answer these questions as accurately as you can. We need this information to help us to interpret your hearing test. It is very important that you are not exposed to loud noise for at least 16 hours prior to your test. This includes listening to loud music on your way to work and riding a motor bike. Avoidance of loud noise will help to ensure your results are as accurate as possible. If it is not possible to avoid loud noise, please ensure you wear adequate hearing protection throughout the whole time of exposure. This will help to ensure your test is as accurate as possible. Please bring with you to the hearing test: a) this form completed, and b) any hearing protection that you usually use. Surname _____ Forenames _____ Age _____ Job Title _____ Please circle Yes (Y) or No (N): <table style="width: 100%; border: none;"> <tr> <td style="width: 80%;">1. Has your job or the noise level changed since your last hearing test?</td> <td style="width: 20%; text-align: right;">Y/N</td> </tr> <tr> <td>2. If so, does your job expose you to high levels of noise?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>3. Do you have to shout to make yourself heard at a distance of 10 feet away?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>4. Have you been given ear protection?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>5. If so, do you wear it all the time/occasionally/seldom/never?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>6. Have you had any difficulty with your ears or hearing since your last test?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>7. Have you had your ears syringed?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>8. Have you experienced noises in your head or ears since your last test?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>9. Have you taken any medication since your last test?</td> <td style="text-align: right;">Y/N</td> </tr> <tr> <td>10. Have you taken up any noisy hobbies e.g. shooting or motorcycling?</td> <td style="text-align: right;">Y/N</td> </tr> </table> The information given above is correct and complete to the best of my knowledge. I consent to my employer being informed of the overall results of my hearing test i.e. whether my hearing is normal or there is evidence of a hearing loss. I understand this information will be used only in order to protect the hearing of any employee at risk from noise exposure. Signed: _____ Date: _____		1. Has your job or the noise level changed since your last hearing test?	Y/N	2. If so, does your job expose you to high levels of noise?	Y/N	3. Do you have to shout to make yourself heard at a distance of 10 feet away?	Y/N	4. Have you been given ear protection?	Y/N	5. If so, do you wear it all the time/occasionally/seldom/never?	Y/N	6. Have you had any difficulty with your ears or hearing since your last test?	Y/N	7. Have you had your ears syringed?	Y/N	8. Have you experienced noises in your head or ears since your last test?	Y/N	9. Have you taken any medication since your last test?	Y/N	10. Have you taken up any noisy hobbies e.g. shooting or motorcycling?	Y/N
1. Has your job or the noise level changed since your last hearing test?	Y/N																				
2. If so, does your job expose you to high levels of noise?	Y/N																				
3. Do you have to shout to make yourself heard at a distance of 10 feet away?	Y/N																				
4. Have you been given ear protection?	Y/N																				
5. If so, do you wear it all the time/occasionally/seldom/never?	Y/N																				
6. Have you had any difficulty with your ears or hearing since your last test?	Y/N																				
7. Have you had your ears syringed?	Y/N																				
8. Have you experienced noises in your head or ears since your last test?	Y/N																				
9. Have you taken any medication since your last test?	Y/N																				
10. Have you taken up any noisy hobbies e.g. shooting or motorcycling?	Y/N																				
<u>FOR MEDICAL USE ONLY</u> Day of the week: _____ Hours worked before test: _____ Hearing protection worn: Usually: Y/N Today: Y/N Type: ()Plugs ()Muffs ()Other Ear examination results: R. _____ L. _____ HSE classification category: 1 2 3 4 Action required: ()None ()Warning ()Referral																					

Figure 7.5 An example of a employee-completed review questionnaire.

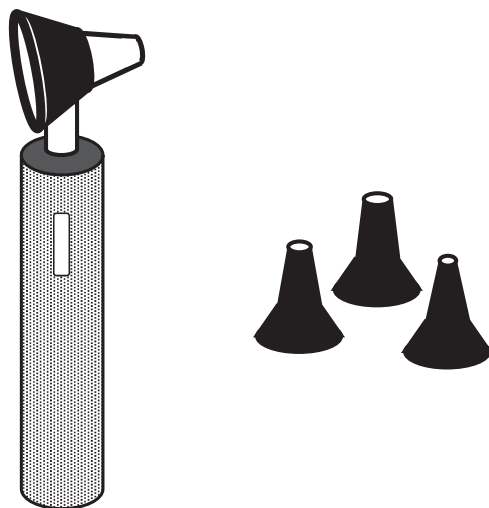


Figure 7.6 An otoscope and specula.

- Is the eardrum normal?
- Does the employee require medical referral?
- Can the hearing test proceed?

1. Is the condition of the skin healthy?

The tightly adhering skin lining of the outer ear is well supplied by nerve endings and inflammation may lead to discomfort or pain. Skin conditions will usually need to be referred for medical treatment if this is not already in hand. Clinical judgement is needed to decide if the test can go ahead. Testing should not be carried out where there is infection or discharge or where there is any risk to the patient or to the equipment. If the test is to be carried out, extra care will need to be taken during otoscopy itself and in placement of the headphones. Extra care will also be needed in cleaning the headphones after use.

2. Are there any scars in front of or behind the pinna?

Scars may indicate ear surgery, which should have been noted in the questionnaire. If not, further questioning is needed to ascertain the history.

3. Is the ear canal normal?

Any growths or foreign bodies should be referred for removal or treatment where appropriate. If they are not blocking the ear, it will sometimes be possible to continue with audiometric testing. Clinical judgement is needed to make this decision.

4. Is the amount of wax in the ear excessive?

As a rule of thumb, the amount of wax in the ear is not excessive for acceptable hearing thresholds to be obtained if at least 10 per cent of the eardrum can be viewed. However, if the eardrum cannot be seen clearly, it is possible that conditions needing medical referral may be missed. For this reason, it is

generally considered that not more than 50 per cent of the drum should be obscured by wax. Hearing tests should never be carried out when wax is completely blocking the ear, as the results are likely to be considerably worse than they would be otherwise.

5. Is the eardrum normal?

Some conditions of the eardrum necessitate medical referral. Training and experience is needed to recognise abnormalities and clinical judgement is needed to decide when to refer and whether or not the test should proceed.

The otoscopic procedure

Hygiene is important throughout otoscopy. Hands should be washed before and after ear inspection. Use of an otoscope with disposable specula is recommended. A new speculum should always be used for each patient. If re-usable specula are used these should be cleaned and disinfected after use and stored dry in a closed container. The speculum should be wiped with an alcohol wipe before use. The same speculum can be used on each ear of the employee if there is no sign of infection, blood or discharge. Where there are signs of any of these, the speculum used should not be used for the other ear and should be disposed of safely as clinical waste.

The procedure for otoscopic examination is as follows:

1. Explain what you are about to do and obtain the patient's permission for the examination.
2. Check the pinna for scars and signs of inflammation.
3. Check the canal entrance to choose the correct size of speculum. This should be as large as is compatible with the size of the ear canal. Inexperienced practitioners often use a speculum that is too small. There are very few adult ears that are really small. A medium/large speculum will give more light and facilitate a better view.
4. Attach the appropriate speculum to the otoscope without handling the tip of the speculum. Wipe the tip with an alcohol wipe.
5. You should be at, or slightly below, the level of the employee's ear in order to get a clear view. The ear canal runs upwards in adults. Sitting or kneeling is therefore recommended and also provides the safest position for carrying out the procedure.
6. Hold the otoscope with the barrel pointing to the side and brace your hand against the side of the patient's face (Figure 7.7) and the possibility of any discomfort can be minimised by co-ordinating head and otoscope movements.
7. Place the speculum a little way into the entrance to the canal. Gently lift the pinna upwards and slightly backwards to straighten the canal. Carefully insert the speculum as far as necessary to obtain a good view of the canal and drum but not so as to cause discomfort, remembering that the canal is both sensitive and delicate, particularly in the bony portion.
8. Repeat on the other ear.



Figure 7.7 Holding the otoscope.

OTOSCOPIC EXAMINATION					
Normal	Left ()	Right ()	Wax blockage (full)	Left ()	Right ()
Perforation	Left ()	Right ()	Obstruction (partial)	Left ()	Right ()
Scarring	Left ()	Right ()	50% eardrum visible	Left ()	Right ()
Discharge	Left ()	Right ()	—% eardrum visible	Left ()	Right ()
Fluid	Left ()	Right ()	Abnormal pressure	Left ()	Right ()
<u>Details and comments:</u>					

Figure 7.8 An example of a method of recording the otoscopic results.

- 9. Dispose of (or sterilise) the speculum.
- 10. Keep a record of the condition of the ear (Figures 7.8 and 7.9) and refer any patient whose ears are blocked with wax or where there is any cause for concern. The record is usually kept at the end of the case history questionnaire.
- 11. If the ears are clear of wax and in a suitable condition for testing, proceed.

Wax

Wax is a normal secretion from the glands near the entrance to the ear canal. New wax is a colourless liquid but with time it hardens and changes colour. Relatively new wax is golden and moist but old wax is dark brown and hard. The

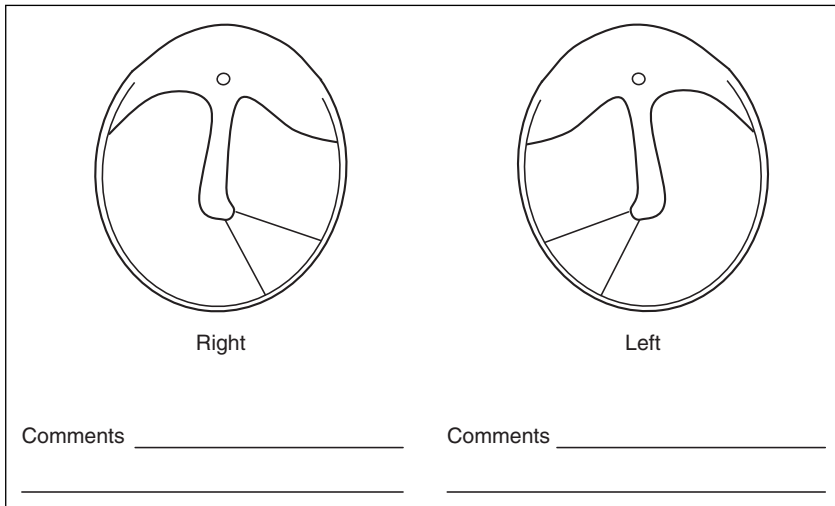


Figure 7.9 An alternative method of recording otoscopic results.

amount of wax produced varies from person to person. In addition, older people tend to have drier wax and their skin growth is slower. This means the wax does not drop out of the ear as quickly and wax blockages are more common with increasing age. There are also some racial differences in the wax produced, for example Chinese wax is drier and more like paper than European wax. Keratin may also accumulate and cause a blockage in the ear canal. Keratin is usually much whiter than wax.

Wax (or cerumen) may accumulate in the ear canal and, if it blocks the ear canal, will impair hearing. Such impairment may be very slight or as great as about 30 dB. In some cases it may also cause a low-pitched tinnitus. Wax may tend to accumulate more than normal when earplugs (or hearing aids) are worn as the wax cannot fall out of the ear. Wax that builds up to the point that it blocks the ear canal will need to be removed before the hearing is tested. Preparatory solutions are available for treatment of wax but some of these are rather harsh, particularly those containing hydrogen peroxide. Wax may be softened using warm olive oil for a few days; sometimes this treatment is sufficient in itself but removal by a trained medical practitioner (usually a nurse, doctor or ENT surgeon) may still be necessary. Medical removal may be by syringing with warm water or dry removal using suction or a wax tool (Figure 7.10). Professional judgement is needed to choose the method to use and this is usually based on the condition of patient, skin, eardrum (if known) and cerumen. The ear should ideally be allowed recovery time of up to three weeks before the hearing test is carried out. Where necessary, the test can be undertaken sooner as long as the patient shows no discomfort and extra care is taken.

Excessive wax may also affect hearing aid use, causing poor hearing and/or a whistling noise (feedback) from the aid as the sound is reflected back from the

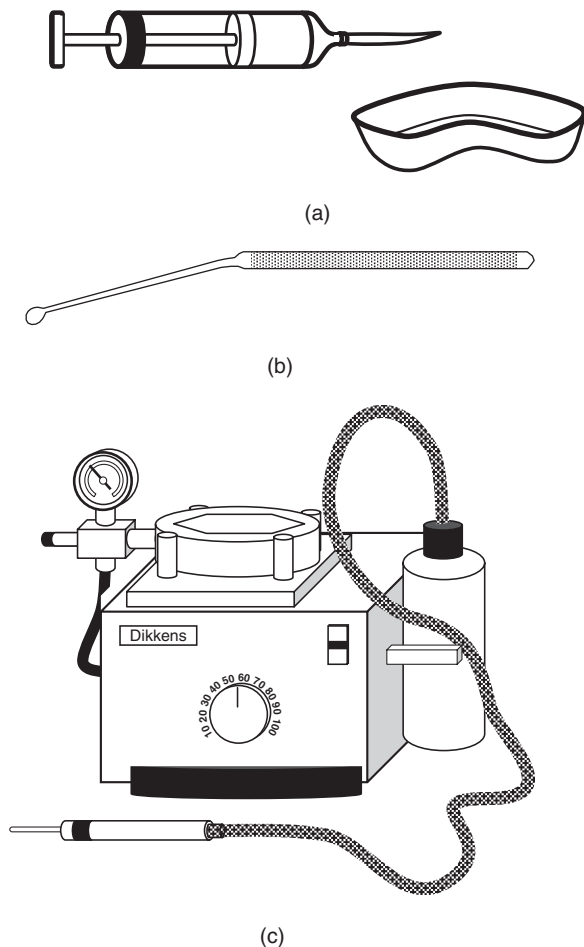


Figure 7.10 Wax removal equipment: (a) Ear syringe; (b) Wax removal tool and (c) Vacuum pump.

wax blockage and re-enters the microphone. Hearing aid users who make a lot of wax may find regular removal, for example every six months, may be helpful.

The eardrum

The normal eardrum

The eardrum (tympanic membrane) is oval in shape and concave. It is positioned at an angle of about 55° to the floor of the ear canal. Its normal shape and position are such that when viewed through an otoscope, the light emitted by the otoscope

is reflected back, not as a pool but as a small cone that follows the line of the jaw. This reflected light is known as ‘the light reflex’.

The eardrum is divided into two sections: a lower elastic fibrous section (the pars tensa) and a smaller upper section (the pars flaccida) which has fewer and less well organised fibres.

Around the edge of the pars tensa, the membrane is thickened to create a rim of fibre and cartilage known as the annular ligament. This fits in a groove in the bony ear canal and acts to hold the eardrum in place. There is no ligament towards the top of the pars tensa because here it turns to form the malleolar folds that run to the handle of the malleus. These folds separate the eardrum into its two sections.

The handle of the malleus can be seen centrally. The central point at the base of the handle of the malleus is called the umbo. The malleus is the first bone in the middle ear and its handle is attached in the fibrous layer of the pars tensa. Small blood vessels can often be seen, particularly around the handle of malleus. The short process of malleus can be seen as a white bump towards the top of the eardrum.

There is some variation in the colour of a normal eardrum but it is usually pale grey and semi-transparent. The eardrum separates the outer ear from the middle ear. If the eardrum is particularly translucent, it may be possible to see the incus (the second bone of the middle ear) and even the entrances to the Eustachian tube and to the round window.

Abnormalities of the eardrum

It is important to know what a normal healthy eardrum looks like in order to recognise abnormalities, when these are seen.

Colour

The eardrum is normally semi-transparent and pearly grey. Abnormal colour may suggest certain disorders, Table 7.1. Some of these will need to be referred. In the case of sclerosis, the hearing test will usually continue.

Table 7.1 Abnormal colours of the eardrum

Colour	May indicate
Bright red	Infection
Dark and dull	Fluid
Chocolate brown	Dried blood
White	Chalky deposits/sclerosis/scarring
Yellow	Pus

Perforations

A perforation is a hole in the eardrum. The patient may be aware or unaware of it. Reference to the cause, if known, should be made in the case history. Often the cause will be a history of recurrent infection. A perforation may be large or small, positioned centrally or towards the top or sides, dry or discharging, long-standing or recent. Medical referral may be appropriate. Generally, perforations are termed 'safe' if they are small and situated centrally in the pars tensa, as such perforations do not usually cause any problems and will spontaneously heal. Large perforations or repeated perforations do not always heal. In some cases, eardrum repair will be appropriate, usually to prevent infection. Perforations are termed 'unsafe' if situated in the pars flaccida or towards the annular ligament. Hearing tests can usually be continued if the perforation is dry but hygiene is particularly important to prevent infection. Previous perforations may leave the eardrum weakened and more prone to re-perforation. Water should not be allowed to enter the middle ear and therefore syringing should never be carried out if a perforation is suspected or if there is a history of previous perforations. The use of eardrops is also to be avoided.

Scarring

Sclerosis due to scarring (or fibrosis) occurs after inflammation and involves hardening of the tissues. Scars may also show as patches of white on the eardrum, due to deposits of calcium on the drum. These sometimes occur in a horseshoe shape where there has been a grommet in the drum at some earlier time. Sometimes a perforation may heal but the fibrous layer does not grow back. In this case, the area will be weak and may look like a thin layer of skin only or even appear as if it has not healed at all. Scarring may have a slight (but permanent) effect on the hearing levels but this is generally nothing to worry about and does not prevent the hearing test from being carried out.

The cone of light

The reflected light from the otoscope may be displaced, misshapen, dull or missing. This is usually related to current or previous Eustachian tube dysfunction. It could be due the employee having, for example, hayfever, sinusitis, a cold, influenza or otitis media (less common in adults) or to a past history of otitis media. If the problem is current, the hearing test should be re-appointed for a time when the employee will have recovered.

Summary

The hearing test should be preceded by undertaking a case history and otoscopic examination. The case history is usually in the form of a questionnaire, which is used to assist in decision-making following audiogram analysis and categorisation.

The initial questionnaire will be quite lengthy as it must investigate possible causes of pre-employment hearing loss including previous noise exposure, illness, injury, medication and any relevant family history. Subsequent questionnaires can be shorter as they need only to record any changes.

Otoscopic examination involves inspection of the outer ear to ensure its condition is suitable for testing. Certain conditions may be referred on for medical treatment. The otoscopic procedure should be carried out in such a way that it provides a good view of the ear, whilst ensuring the safety and comfort of the patient.

Further reading

Hawke, M., Keene, M. and Alberti, P.W. (1990) *Clinical Otoscopy. An Introduction to Ear Diseases*, Churchill Livingstone.

Audiometric techniques for occupational health monitoring

Introduction

Industrial audiometry involves monitoring workers for hearing problems due to excessive noise exposure at work. However, monitoring will identify all hearing problems, whether due to noise exposure or other causes. Some information regarding the possible cause may be obtained from the case history, otoscopic examination, the audiogram configuration and tuning fork tests but diagnosis will be obtained only by medical referral for further investigation.

Occupational hearing monitoring involves simple air conduction tests, using an audiometer and its attached headphones. The results of air conduction tests will show whether or not there is a hearing loss and (in most cases) the degree of that loss. They will not show whether the loss is temporary or permanent, conductive or sensorineural. Also, the results should be interpreted with caution if there is a marked difference in hearing ability between the left and the right ears, or where either ear has a threshold of 40 dB or more at any frequency. In these cases, it is possible that the hearing levels shown on the audiogram may not be true, due to a phenomenon known as 'cross-hearing'. This is where a loud sound applied to one ear will pass through the bones of the skull and may be heard in the opposite ear. The true threshold of the test ear may be worse than that shown on the audiogram but this will only be known after referral for diagnostic audiometry.

Monitoring audiometry may be undertaken manually or automatically. Automatic audiometry is also known as 'self-recording'. The audiograms are presented somewhat differently in manual and automatic audiometry. The audiogram forms used in

each case can be seen in Figure 8.1. The manual audiogram form shown here has been adapted for industrial use. In many cases, a diagnostic audiogram form is used. The graph is identical but the symbols and information presented here have been tailored to the particular use to which it will be put. For occupational monitoring purposes, only air conduction symbols are required and if no response occurs at the maximum output level of the audiometer, an arrow is drawn from the corner of the appropriate symbol (EN 26189:1991) as shown in Figure 8.1(a). Symbols meaning 'no response' should not be connected with the line to symbols representing measured thresholds.

There are three methods widely used in industrial audiometry (Figure 8.2) these are:

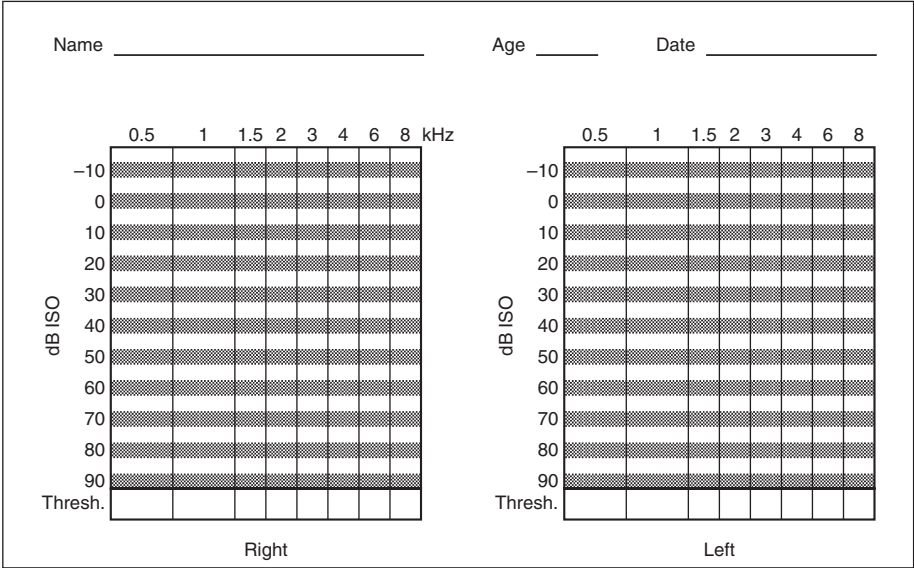
1. Manual – BSA Recommended Procedure or Hughson-Westlake Procedure.
2. Automatic or Self-recording – Automated Hughson-Westlake or 'auto-threshold'.
3. Automatic or Self-recording – Békèsy.

All of the accepted methods and techniques are equally valid. There will be a slight difference between the results obtained when using manual or auto-threshold methods and Békèsy methods because it is a little easier to hear pulsed tones rather than continuous tones (Békèsy tests use pulsed tones). Usually the same method will be used each time for each individual. However, if a manual or an automated Hughson-Westlake audiogram is to be compared with one taken using a Békèsy audiometer, it has been suggested (Robinson and Whittle, 1973) that 3 dB should be added to the hearing levels found using the Békèsy audiometer. This is described in the British and European standards (EN 26189: 1991).

The decision as to whether to use a manual or an automatic technique is generally made through the personal choice of the operator where both types are available. Automatic audiometry is usually easier to use where there are many people to test. Békèsy is the most widely used automatic test mainly because it tends to be the quickest option. There are, however, a small number of people who are unable to perform Békèsy audiometry. This is generally because their reactions are not fast enough or they do not fully understand what they have to do. Manual audiometry is used in these cases, and some audiologists use it as a matter of choice. Manual audiometry can be a little quicker than Békèsy in the hands of a skilled tester but the audiogram also has to be categorised manually and for most people this adds to the time involved and to the risk of error. Continuously carrying out manual audiograms can become very tedious and this may lead to errors if the tester's attention to detail wanes.

Some audiometers will perform both manual and automatic tests whilst others will only carry out one method. Whatever the type of audiometer used, the results can be accepted only if the audiometer is calibrated annually and validated daily in use to ensure its accuracy. The test must also be carried out in suitably quiet conditions, which will generally mean that the ambient or background noise level should not be more than 35 dBA (British Society of Audiology, 2004).

Normally, a case history and otoscopic examination will precede the audiometric test. Where there are no contra-indications to testing, the employee will



(b)

Figure 8.1 (continued) (b) An example of an audiogram form for an automatic audiometer.

	Advantages	Disadvantages
Manual	● Can test those unable to perform Békèsy ● Can accept alternative responses ● Can be quicker in the hands of a skilled professional	● Incorrect responses may be accepted ● Require experience to test accurately
Automated	● Automatic 'manual' method ● Tester error ruled out ● Can be used with minimal training	● Will accept a response of any duration, including if the button is touched in error ● The computer can accept incorrect responses ● Can be lengthy
Békèsy	● Quickest test for inexperienced testers ● Least possibility of error for inexperienced testers ● Permits tester to carry on other work nearby ● Pulsed tones less easily confused with tinnitus	● Monotonous test ● Slower reactions will lengthen the time taken as frequencies have to be repeated ● A small number of people unable to perform Békèsy

Figure 8.2 A comparison of audiometric methods widely used in occupational health.

much as 30 dB, and therefore wax should be removed where necessary, prior to testing. This is usually carried out at the worker's GP's surgery and ideally there should be up to three weeks, or as specified by a medically qualified person, between wax removal and the hearing test. Where this is not practicable, a wait of several days will usually suffice if extra care is taken.

Wax blockage (total or partial) may also prevent an adequate inspection of the ear canal and eardrum during otoscopy. As a rule of thumb, wax should be reported if there is more than a 50 per cent obstruction but the test results may be accepted if at least 10 per cent of the eardrum is visible. Testing may be postponed in cases of inflammation of the outer ear, eczema of the pinna or suspected otitis externa, in order to avoid hygiene and comfort issues.

General illness or fatigue can affect the worker's ability to concentrate, so testing should not be carried out where these are suspected. Workers should also be required to arrive at least 5 minutes before the test begins in order to prevent any errors due to physical exertion. Ear infections may directly affect hearing levels, as may conditions such as colds, catarrh and hayfever. In these cases, the hearing may well be worse than usual for that person and it is best to postpone the hearing test until the condition has cleared. If the test is carried out, the results may be accepted if the results are no worse than the previous audiogram. In the case of a baseline or first audiogram, testing conditions need to be such that the accuracy of the audiogram is assured.

The worker should avoid exposure to loud noise for at least 16 hours (Health and Safety Executive, 1995), but ideally 48 hours, prior to testing. If this is not possible, hearing protection with high attenuation should be worn at least on the day of the test and, ideally, also on the day before the test. In addition, the worker should be kept out of noise for at least 15 minutes prior to the test (EN 26189:1991).

Approximately 15 minutes should be allowed for screening audiometry, whichever technique is used; more than 20 minutes could fatigue the worker and affect the test results. The whole procedure including otoscopic examination and a pre-completed case history will usually take about 20 minutes. When the test is completed the audiogram should be signed by the tester (and optionally by the employee), it should also be dated and a record kept of the type and the serial number of the audiometer used.

Preparation for the test

Instructions to the employee under test

The instructions given to the employee are very important and they must be simple and clear. The employee has only to respond when the sound is heard, no matter how loud or how quiet it is and no matter in which ear it is heard. The response should be maintained for the entire length of the signal. The employee should also be told how the test can be stopped if the sound is uncomfortably loud or there are any other disturbing events. Self-recording audiometry follows a set pattern and starts with the same ear each time (this may be left or right depending on the

equipment). Manual audiometry usually starts with the better ear, if there is one, as this makes the task easier for the person under test, particularly when they have not had a hearing test before.

The British Society of Audiology (2004) suggests wording that can be used in instructing the person under test:

I am going to test your hearing by measuring the quietest sounds that you can hear. As *soon* as you hear a sound (tone), press the button (or raise your finger). Keep it pressed (or raised) for as long as you hear the sound (tone), no matter which ear you hear it in. Release the button (or lower your finger) as soon as you think you no longer hear the sound (tone). Whatever the sound and no matter how faint the sound, press the button (or raise your finger) as soon as you think you hear it, and release it (or lower it) as soon as you think it stops.

Alternative wording is quite acceptable as long as it covers the same points. The same wording can be used for both manual and self-recording audiometry but, when using Békèsy audiometry, it may be preferable (EN 26189: 1991) to use a slight variation on this wording, for example:

I am now going to test your hearing. As soon as you hear a bleeping sound, I want you to press the button. Keep the button pressed until you no longer hear it. Then release the button. Keep repeating this procedure. Try to react quickly; *as soon as* you think you hear the bleeps, no matter how faint they are, press the button and release it *as soon as* the sound goes away. I am going to test each ear separately, starting with the left (or right) ear.

A simple written version of the instructions can also be provided if required; this may be particularly helpful for employees who have not had a previous hearing test. The employee should be asked if they understand the instructions and they should be informed that they can interrupt the test if they experience any disturbing events (or discomfort). Tinnitus is common in cases of noise induced hearing loss and, if the employee has reported tinnitus, they should be advised to ignore the tinnitus as far as possible (unless there is discomfort or the tinnitus is exacerbated). If the employee appears to be or reports that they are confusing the tinnitus and the test sounds, this observation including a note of the frequency or frequencies and ear(s) involved in any possible confusion should be written on the audiogram form.

If, at any time during manual audiometry, the switch button is not working, the employee can be asked to raise a finger in response to the sounds. This will still allow them to indicate silently for the entire length of the sound presented and is quite acceptable as a response to the tones. This option does not exist in self-recording audiometry.

Instructions should be given before fitting the headphones, as the headphones will make it more difficult to hear the instructions, especially if they are of the noise excluding type. Some audiometers provide a 'talk through' button to allow speech, via a microphone, to be heard clearly inside the headphones. This is particularly helpful where a worker has to be re-instructed during the test.

The person being tested should be seated comfortably and should not be disturbed or distracted during the test by anyone or anything unrelated to the test. It is important that the employee is seated such that they cannot see the audiometer panel or the tester's movements when delivering the signal. The tester should not leave the room during the test.

Fitting the headphones

Any spectacles, earrings or head ornaments should be removed and hair swept away from the ears before fitting the headphones, as these articles may affect the correct placement of the earphones. Hearing aids, if these are worn, must also be removed before fitting the headphones. To leave hearing aids in place would produce incorrect results and could cause discomfort or damage to the person being tested. Aided hearing cannot be tested in this way. When it is necessary to know someone's hearing levels whilst using hearing aids, an aided hearing test is performed, which involves using loudspeakers in a calibrated 'sound field'. This is a specialist test. An approximation of the hearing levels achieved can be calculated if the gain provided by the hearing aid is known but this is also somewhat of a specialist area. Alternatively, the individual can be observed or tested in the real working conditions to ascertain if he or she can hear sufficiently well to meet the requirements of the situation.

Headphones should be fitted, or at least checked and adjusted, by the tester. The worker should be instructed not to touch the headphones thereafter. The red earphone should be placed on the right ear and the blue one on the left ear. The headband should be tight so that the earphones fit snugly and the earphones should be placed centrally over the ear canal. This latter point is particularly important to check carefully when using noise excluding headphones (Figure 8.3) as the outer cup may look perfect whilst the inner earphone is not positioned correctly. The incorrect positioning of the earphone over the ear canal may produce a 'notch', or worsening of the hearing loss, of about 10 dB or more on the audiogram usually at 6 kHz but sometimes at 8 kHz (Flottorp, 1995).

Familiarisation and monitoring

Prior to the start of familiarisation, it is suggested (EN 26189: 1991) that the employee should have a rest period of at least half a minute.

The familiarisation procedure is to allow the employee to get used to the sounds before the full test begins. All the testing procedures allow for familiarisation as this is very important to ensure the results are acceptable. In Békésy audiometry, some observation by the tester of the early excursions (the zigzag lines that indicate the results) is necessary. In manual audiometry and when using the automated Hughson-Westlake procedure, there is an extra check at 1 kHz, in the first ear only, which ensures that the results from the first ear have not changed due to further learning occurring after the familiarisation period.

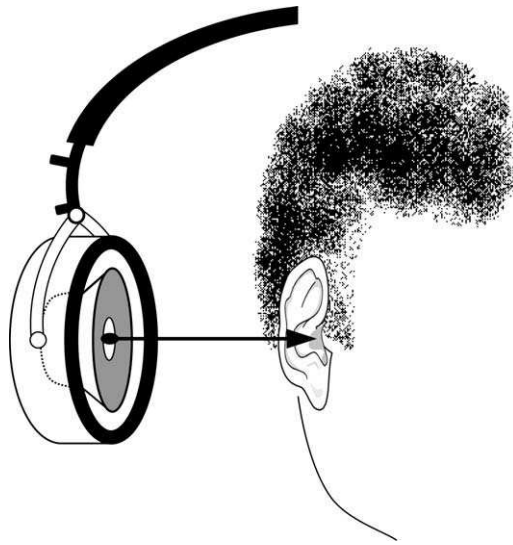


Figure 8.3 Careful positioning of noise excluding headphones.

Manual audiometry

The manual test method

The threshold of hearing can be thought of as the quietest sound that someone can just hear and, in many cases, they may be a little unsure if they really heard it or not. The threshold is defined as: ‘the lowest level at which responses occur in at least half of a series of ascending trials with a minimum of two responses required at that level’ (British Society of Audiology, 2004).

At each frequency, the hearing threshold is the lowest level at which the person responds to the ascending signals, that is coming out of silence. At least two responses (out of three or four) are required at the same level. Responses are only counted in the ascending mode, which is following a 5 dB increase in the sound. Responses to descending signals, that is following a 10 dB decrease in the sound, are not counted. This is because it is easier to hear the sound when it is anticipated and less easy to hear it coming out of silence.

Many people use the term ‘Hughson-Westlake’ to mean auto threshold. However, the Hughson-Westlake method of finding hearing threshold can also be used as a manual method. The accepted Hughson-Westlake method (which is a modified version of the original) is virtually identical to the British Society of Audiology (BSA) recommended procedure for manual audiometry. The difference between the two methods rests on the number of responses presented at each level when finding threshold. The BSA method requires two correct responses out of four presentations, whilst the Hughson-Westlake method requires two correct responses out of three presentations. Either is acceptable and should make no significant difference to the results.

The frequencies tested in industrial audiometry differ somewhat from those tested for other audiometric purposes. In industrial audiometry, it is not a requirement to test 250 Hz or 8 kHz, although 8 kHz is recommended (EN 26189: 1991) as it is useful when trying to ascertain the possible cause of a hearing loss. The frequencies 3 kHz and 6 kHz are always tested because these frequencies are particularly important for indicating probable noise damage. For other audiometric purposes, 3 kHz and 6 kHz are generally optional (British Society of Audiology, 2004) and, for this reason, these frequencies are marked on the standard audiogram form only as unlabelled broken lines.

Whilst self-recording audiometry always starts with the same ear, in manual audiometry the tester will always start with the better ear. This is because the person being tested will find the task easier if they are able to get used to the test and its requirements whilst they are using the ear which hears relatively well. If there is no noticeable difference between the ears, either ear may be tested first.

Presentation of the frequencies also differs between the tests. Self-recording audiometry starts with the lowest frequency and increases up the frequency range. Manual audiometry starts with 1 kHz because this is a sound that is easily recognisable and which most people hear relatively well even with a hearing loss.

The test signals should be pure tones of 1 to 3 seconds duration with varying gaps of 1 to 3 seconds between signals. The tester should be careful to avoid a rhythmic presentation by using a wide variety of signal and gap lengths. If the signals are presented in a predictable rhythmic way, the employee may guess when the signal has been presented even if it has not been heard and therefore respond to signals actually below their threshold. If the tester thinks they may be too rhythmic in their presentation, it is a good idea to ensure they give an occasional long gap (of about 3 seconds) before presenting the next tone. Signals must be presented with no visual, auditory or other cues that might help in guessing when an 'inaudible' signal has been presented. It should be realised that, as well as responding to cues subconsciously, employees may consciously try to alter their thresholds by attempting to:

- worsen their threshold to improve their chances of receiving compensation, that is 'malingering'
- improve their threshold in order to be accepted for a new job or to retain their present job, which requires good hearing.

In manual audiometry, each frequency is tested in turn, in the following order: 1 kHz, 2 kHz, 3 kHz, 4 kHz, 6 kHz, (8 kHz), 500 Hz. The first frequency tested, 1 kHz, is tested again before testing starts on the second ear. This is because some people's results improve as testing progresses and they understand better what is required of them. If the 1-kHz result is the same or only 5 dB different on re-testing, the results can all be accepted as correct. If there has been a change of 10 dB or more the results are all suspect and the full test of the first ear should be repeated. If there is a 5 dB difference in the results, the best result is accepted as the correct one and therefore joined with the other results by a solid line. The re-test at 1 kHz is carried out only on the first ear as, if there is no error, it is assumed that the results from the second ear will also be correct.

The first tone presented at each frequency should be well above threshold, without being uncomfortably loud. This is so that the employee can hear clearly the type of sound to listen for and so that the tester can ensure that the employee is responding correctly and for the full duration of the tone. The first tone presentation is therefore held for a relatively long time (about 3 seconds) and the tester should ensure that the employee is holding the button down for the entire length of the signal. If not, the instruction should be repeated and the response checked before continuing with the test. With experience, the tester will be able to predict an appropriate starting point, which should be about 30 dB above the employee's known or estimated threshold. It is usually sufficient to start with a 40 dBHL signal for, in most cases, a 40 dBHL tone will be effective and it will not be too loud. However, if there is no response to a 40 dBHL tone, the signal should be raised in 20 dB steps until there is a response, up to a maximum level of 80 dBHL. If the level reaches 80 dBHL without a response, the tone should be increased thereafter in 5 dB steps, watching to ensure that no discomfort is experienced (many industrial audiometers do not reach above this level).

Each time the worker being tested hears a signal, the next signal should be presented at a level 10 dB lower, continuing until the level is below threshold (i.e. not heard) and therefore there is no response to the signal. After a null response the next signal should be 5 dB higher. If necessary, continue to increase in 5 dB steps until a response is given. After the response, the level should be decreased by 10 dB and another series of ascending 5 dB steps begun again. This procedure of decreases and increases is continued until there have been two responses at the same level in the previous two, three or four responses in the ascending mode. This level is the threshold of hearing. Threshold is the lowest level at which responses occur in at least 50% of a series of ascending trials with a minimum of two responses required at that level. The procedure (which must be strictly adhered to, always going down in 10 dB steps and up in 5 dB steps) is repeated at each frequency, always starting at a clearly audible level. The basic method is given in Figure 8.4, see also Figure 8.5.

At each frequency	Manual method
	i) The tone is presented first at 40 dBHL (or 30 dB above the assumed threshold) for about 3 seconds and the tester should check that the employee responds correctly and for the entire duration of the signal. If the employee does not respond, the tone should be raised by 20 dB with re-instruction if necessary.
	ii) The level of the tone is reduced in steps of 10 dB until the employee cannot hear the tone (i.e. fails to respond).
	iii) When the employee fails to respond to the tone, the level of the tone is increased in steps of 5 dB until they respond again.
	Steps (ii) and (iii) are repeated until the threshold is found.
	<i>Threshold</i> is taken as the lowest level at which the employee responds to two out of three (or four) ascending signals, that is coming out of silence.

Figure 8.4 The method for manual audiometry.

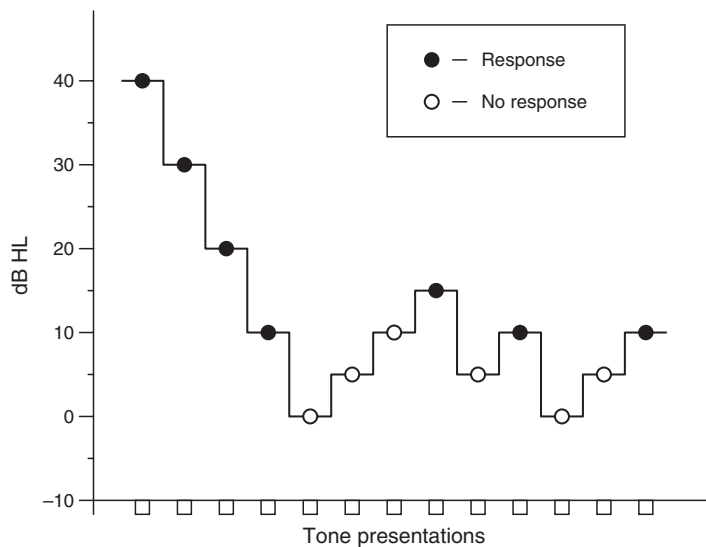


Figure 8.5 The basic manual method used in occupational audiometry.

Results are recorded on an audiogram form using, or based on, the BSA standard format (see Figure 8.1). The symbols used for air conduction results are a cross for the left ear and a circle for the right ear. Appropriate colours, red for right and blue for left, may be used if desired. Each threshold value should be marked by the appropriate symbol and the accepted thresholds should then be joined using a solid line, as shown in the example in Figure 8.6.

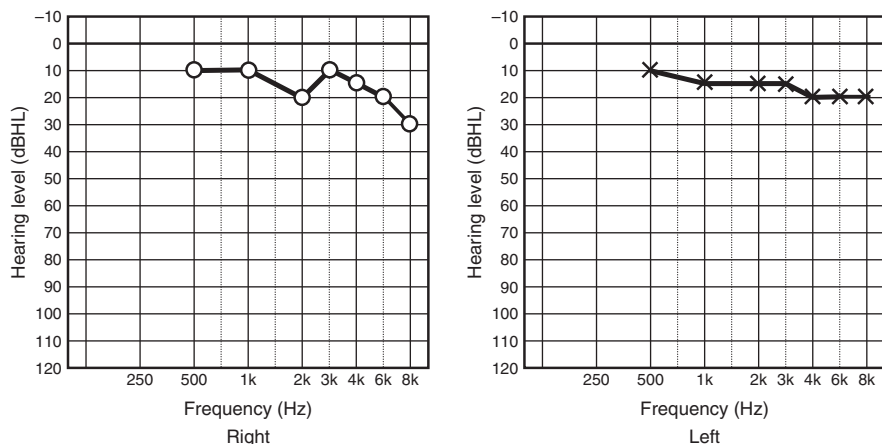


Figure 8.6 An example of an audiogram completed for occupational health purposes.

Self-recording audiometry

Automated Hughson-Westlake audiometry

Many industrial audiometers offer an automated version of the Hughson-Westlake test. This is basically the same as the manual method except that the audiometer automatically carries out the procedure, plots the results and categorises the audiogram. The results are plotted on an audiogram form as shown in Figure 8.7.

Békèsy audiometry

The Békèsy test is carried out using an industrial Békèsy audiometer. This audiometer delivers a series of short pulses of tone, which automatically reduce or increase in level. The frequencies tested are normally in the order of 500 Hz, 1 kHz, 2 kHz, 3 kHz, 4 kHz, 6 kHz and 8 kHz.

Békèsy audiometry requires the audiometer to sweep automatically through the test frequencies, which are presented on the horizontal axis of the graph (Figure 8.8). Intensity in decibels is shown on the vertical axis. Tones are presented in 1 dB steps, at a set rate of attenuation per second, and the audiometer plots up and down lines, or traces, of decreased and increased intensity (known as excursions). The attenuation rate is usually 5 dB per second, which is the preferred rate (EN 26189: 1991).

In order to record excursions, the employee is required to hold the signal button down as soon as the pulsed tone is heard. Whilst the button is held down, the signal will gradually decrease (which is recorded on the graph as an upward line) until a level is reached at which it is no longer heard. At this point, the employee must release the button. On release, the signal will gradually increase (recorded as a downward line) until it is heard once more and the button is therefore pressed again. This reverses the procedure.

A pattern of zigzag lines is produced across each frequency in the range tested. These upward and downward lines indicate the increases and decreases in signal level and form peaks and troughs for each test frequency.

A period of familiarisation is built into the automatic schedule prior to the test itself. The task is usually practiced at 500 Hz. The familiarisation time is very short, with about 30 seconds usually being sufficient. The tester should observe the tracings during the early part of the test. Once the tester is satisfied that the excursions are reliable, the employee can continue with the test without further assistance.

Ideally, the number of excursions per frequency should be at least six. The excursion lines on the vertical axis should extend for 10 to 15 dB and there should be no significant variation between them. For this to happen, the employee must be alert and react quickly by pressing the button immediately the signal is heard and releasing it immediately the signal goes away.

The initial trace obtained during familiarisation is ignored, as is the first reversal after a change of frequency. Once the trace is stable, the threshold of hearing can be calculated by averaging the peaks and troughs produced at each frequency.

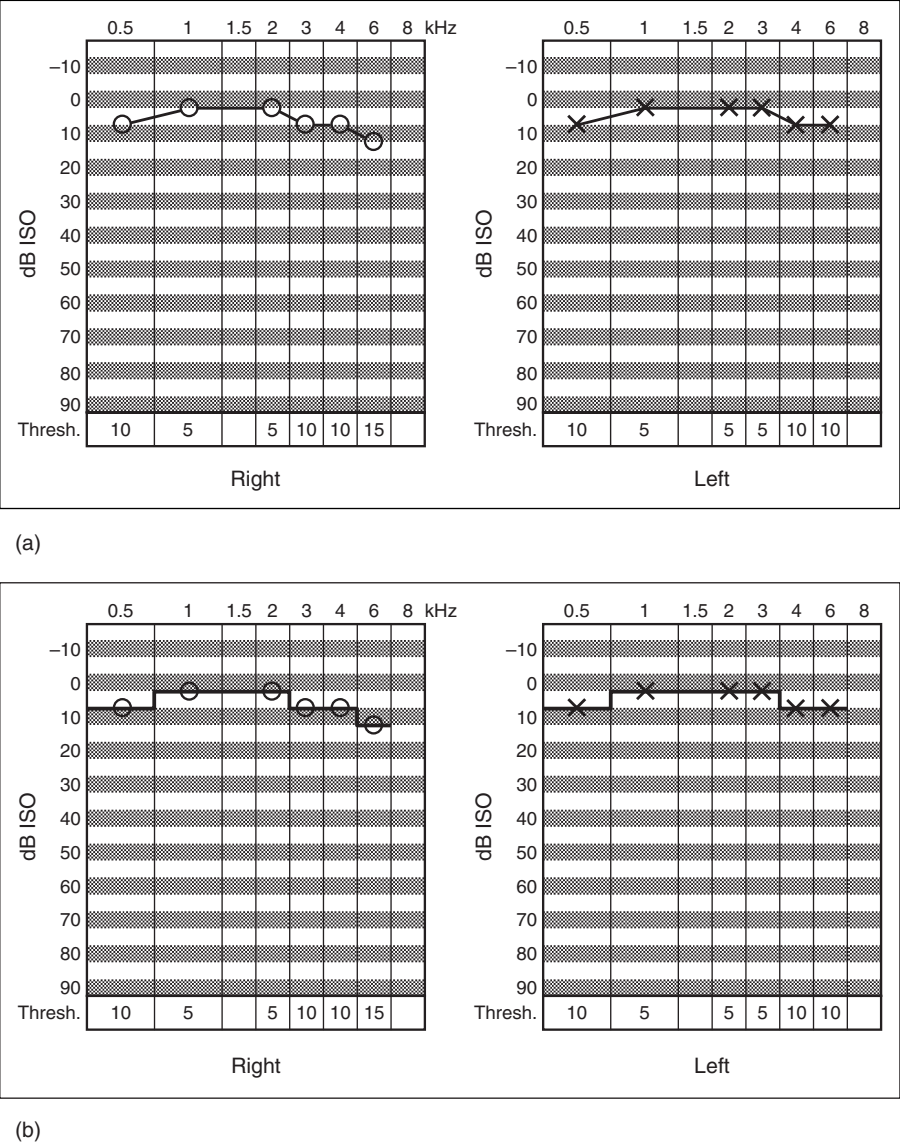


Figure 8.7 Automated Hughson-Westlake audiograms. (a) A completed automated Hughson-Westlake audiogram. (b) An alternative method of presenting an automated Hughson-Westlake audiogram.

The threshold is the mean of these two averages. The audiometer will identify all frequencies with unreliable tracings, which should then be repeated in order to obtain accurate results.

Problems do occur when an employee is slow to react. Delayed reaction time when pressing and releasing the button will result in larger excursions, of as

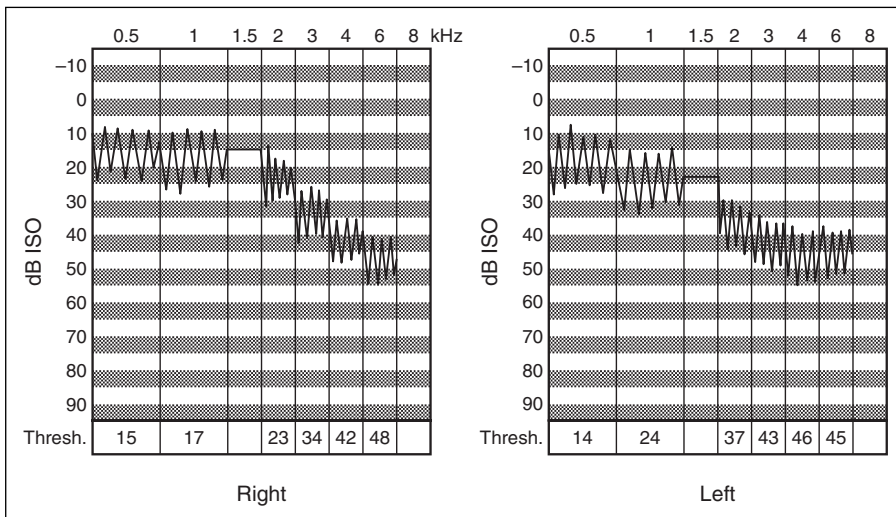


Figure 8.8 An example of a completed Békèsy audiogram.

much as 30 dB. Fewer excursions for each test frequency will be produced and the result will usually be rejected by the audiometer. On some machines, it is possible to slow down the attenuation rate to 2.5 dB per second and lengthen the seconds per frequency rate to accommodate sufficient time for an adequate number of excursions. This may make it easier for someone with slow reactions.

A visual estimation of the hearing threshold is usually possible but unnecessary, as the audiometer will present the average result for each frequency. Self-recording audiometers will also automatically categorise the threshold of hearing according to the Health and Safety Executive (HSE) categories. An example of a self-recording audiogram form is given in Figure 8.7(a) and an example of a completed Békèsy tracing is shown in Figure 8.8.

Summary

Industrial audiometry involves screening workers to find hearing problems due to excessive noise. The screening test will indicate all kinds of hearing loss, not just those due to noise exposure. Diagnostic testing will be required to establish the type and possible cause of the loss but some indication of the likely cause may be obtained from the case history, otoscopic examination, the audiogram shape and from tuning fork tests.

The hearing test can be automatic (self-recording) or manual. The self-recording test may be either a Békèsy test or an automated version of the manual test. Békèsy testing involves the use of short pulsed tones, which continuously increase and decrease in volume. The worker has to press a switch button if they hear the sound and release it if they cannot. This results in a series of zigzag traces, the average of

which is taken to be the hearing threshold. Automatic Békèsy audiometers are widely used in industry as Békèsy testing is particularly useful where large numbers of people need to be tested. However, manual audiometry should always be available for the small number of people who are unable to perform the Békèsy test. In manual audiometry, the tester presents a pure tone of one to three seconds duration. If the worker under test hears the sound, they press the button. The signal is reduced in steps of 10 dB until the sound disappears. The signal is then increased in small steps, of 5 dB, until it is heard again. This procedure is repeated until threshold is found. Hearing threshold is the level at which a person can just hear. It is taken as the lowest level at which the person being tested can hear at least two out of three, or two out of four, signals presented, but counting only those presented coming out of silence (i.e. in the ascending mode).

The audiogram obtained must be placed in one of the Health and Safety Executive (HSE)'s categories; this will be done automatically by a self-recording audiometer or must be worked out by the tester when using a manual machine.

Further reading

British Society of Audiology (2004). Recommended procedure. Pure tone air and bone conduction threshold audiometry with and without masking and determination of uncomfortable loudness levels. British Society of Audiology, Reading. European Standard EN 26189: 1991. Acoustics – Pure tone air conduction threshold audiometry for hearing conservation purposes.

The audiogram and its categorisation

The audiogram

An audiogram is a graph on which are plotted the results of the hearing test. In occupational health monitoring, only air conduction results are obtained so the graph is a relatively simple one to read. When diagnostic testing is undertaken, other results will also be plotted. The audiogram shows the frequencies tested along the horizontal axis and the hearing level in decibels along the vertical axis. The vertical axis runs the opposite way to most graphs, with the lowest numbers at the top.

Test results that are plotted by hand will be plotted on an audiogram using the BSA standard format. The symbols used are a cross for the left ear and a circle for the right ear. The use of colours is optional but red denotes the right ear and blue the left ear. Each threshold value is marked using the appropriate symbol and these are joined by a solid line. Audiograms are too small to include a line for every 5-dB increment and results that fall between the 10-dB increments (45 dB, 75 dB, etc.) have to be plotted between the lines. The frequencies 750 Hz, 1.5 kHz, 3 kHz and 6 kHz are represented only as dotted lines and are not labelled.

Test results that are plotted automatically by the audiometer will use an audiogram form that is basically the same as the British Society of Audiology (BSA) format. However, there will be slight differences, for example light and dark bands may be used rather than single lines to show the hearing level (Figure 9.1). The graph may also omit 250 Hz as this is not usually tested when screening for noise damage.

The completed automatic tracing may be shown as a standard audiogram with standard points connected by straight lines or as a series of zigzag lines (Békèsy).

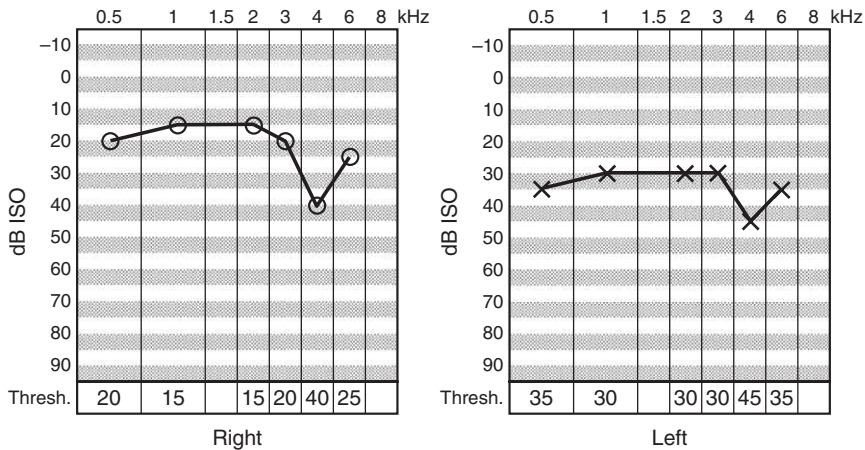


Figure 9.1 An audiogram plotted by an automatic audiometer.

The Health and Safety Executive method for evaluating audiograms

Introduction

The Health and Safety Executive (HSE) propose a method of categorising audiograms that provides occupational health personnel with a method of assessing the level of hearing damage that provides defined steps to follow in every case. They suggest (Health and Safety Executive, 2004) that an ideal categorisation scheme will:

- Identify all who are starting to develop noise induced hearing loss
- Identify all those in whom noise induced hearing loss is developing rapidly
- Identify other hearing disorders that would benefit from medical referral
- Identify where more frequent hearing assessments may be necessary
- Be easily understood by all involved (including employers and employees)
- Enable logical data analysis to facilitate comparison of particular jobs, locations and so on in order to assess the effectiveness of noise control measures.

After a noise and health questionnaire has been completed, a hearing test should be carried out in which all the appropriate frequencies from 500 Hz to 8 kHz are tested, so that the audiogram provides a complete picture of testing by air conduction. It is good practice to ask the worker to bring their hearing protection to each test so that it can be inspected and the way it is worn can be checked. The method of categorisation is intended to simplify results but is not intended to replace professional judgement. The following quality control issues should also be taken into consideration:

- Background noise
- Calibration and validation of equipment used

- Time away from noise before the test and the possibility of temporary threshold shift (TTS)
- Repetition of all tests showing a change of 10 dB or more since the last test
- Comparison with the baseline audiogram.

It is absolutely vital that all conditions for the baseline test meet the necessary requirements and can be shown to do so (through careful record-keeping) in case a later claim should be made for industrial deafness. Best practice is to repeat all baseline audiograms because they are such an extremely important point of reference. It is also best practice to offer the employee a copy of their audiogram after each test. Repeat audiograms are normally carried out every three years but where there is any concern about hearing levels, or where the noise exposure has altered, audiograms should be repeated at a shorter interval.

Calculating the summed hearing levels to decide the appropriate category

Although the frequencies 500 Hz and 8 kHz are tested and provide useful information, only the frequencies 1 kHz, 2 kHz, 3 kHz, 4 kHz and 6 kHz are used in the calculations. These are the frequencies considered to be most highly related to noise induced hearing loss (NIHL). The method (Figure 9.2) involves the following:

- Add the hearing levels obtained at 1 kHz, 2 kHz, 3 kHz, 4 kHz and 6 kHz to give one summed value for each ear. Compare these values with those given in Table 9.1, which takes account of age and gender. This will place the individual into category 1, 2 or 3, which relate to acceptable hearing ability (within normal limits), mild hearing impairment (warning level) and poor hearing (referral level), respectively.
- The hearing levels at 3 kHz, 4 kHz and 6 kHz should be added together to determine whether there has been a reduction in hearing levels of 30 dB or more since the previous test, which should have been within the last three years. This will place the individual in category 4, which relates to rapid hearing loss and necessitates referral. It is also desirable to assess audiograms that fall into categories 1 and 2 to provide early warning of any hearing loss that appears to be progressing at a faster rate than might normally be expected, taking account of the age and gender of the individual. Where there are concerns about changes in hearing thresholds or where noise exposure has altered, a repeat audiogram may be required earlier than the next routine test.
- The sum of the hearing levels at 1 kHz, 2 kHz, 3 kHz and 4 kHz should be added together, for each ear, to give two single-summed values. The difference between the value for the left and right ear should be determined. If the difference is more than 60 dB (i.e. 61 dB or more), the individual has a unilateral hearing loss that requires medical referral.

Where the results of the hearing test have not led to medical referral but the individual has reported other symptoms, such as pain, discharge, dizziness, or

Category 1, 2 (normal or mild hearing loss)	Category 3 (poor hearing)	Category 4 (rapid loss)	Unilateral
Add the hearing levels at: (1 + 2 + 3 + 4 + 6) kHz		Add the hearing levels at: (3 + 4 + 6) kHz	Add the hearing levels at: (1 + 2 + 3 + 4) kHz
Compare each ear with value for age and gender (Table 9.1)		Compare each ear with previous test <3 years ago	Compare left with right
Cat. 1 = L + R < given value Cat. 2 = L or R ≥ given value	Cat. 3 = L or R ≥ given value	Reduction in either ear ≥30 dB = category 4	Difference between ears >60 dB = unilateral
Assess audiograms and history for concerns. Where appropriate medical referral or repeat audiogram earlier than normal interval.	Medical referral		

Figure 9.2 The HSE method for categorising audiograms.

severe or persistent tinnitus, or where it is clear that the loss has become a handicap to the individual, medical referral is also appropriate. This should lead to diagnosis of the cause of the problem, which in some cases may be nothing more than the effects of normal ageing, when a hearing aid may be of benefit.

Table 9.1 The HSE’s classification of audiograms according to age and gender

Sum of the hearing levels at 1 kHz, 2 kHz, 3 kHz, 4 kHz and 6 kHz				
Age group	Males		Females	
	Warning level	Referral level	Warning level	Referral level
18–24	51	95	46	78
25–29	67	113	55	91
30–34	82	132	63	105
35–39	100	154	71	119
40–44	121	183	80	134
45–49	142	211	93	153
50–54	165	240	111	176
55–59	190	269	131	204
60–64	217	296	157	235
65	235	311	175	255

The Health and Safety Executive’s categories and the actions required

The results obtained from the hearing test will place the employee in one of the Health and Safety Executive (HSE)’s four categories according to hearing loss, and, in addition, cases of unilateral hearing loss will also be noted. A chart that may be helpful in recording results is given in Figure 9.3 and a résumé of the main points of the categorisation system can be found in Figure 9.6.

- 1. Acceptable hearing ability
Category 1 applies where the sum (for either ear) is below the warning level given in Table 9.1. In general, no special action is required. However, all individuals should be given advice about the effects of noise on hearing and the correct use of hearing protection. It may also be appropriate, in order to reinforce the importance of this advice, to provide an informal warning of the possibility of slight early damage, where the initial hearing level (from the baseline audiogram) was particularly good but there is a noticeable shift in hearing levels or where the audiogram configuration suggests very early signs of possible noise damage.
- 2. Mild hearing impairment
Category 2 applies where the sum (for either ear) is equal to or exceeds the warning level given in Table 9.1. Although the individual is not yet at a level requiring referral, the following actions should be taken:
 - Formal notification of the presence of hearing damage, see the example in Figure 9.4. This notification should include the degree of loss and the implications for further damage, in addition to ways of minimising further damage. The information should be given both verbally and in writing.
 - Retraining of the individual to ensure the correct use of hearing protection and to reinforce understanding of the effects of noise on hearing and the importance of complying with all hearing conservation measures.

kHz	1 + 2 + 3 + 4 + 6		3 + 4 + 6		1 + 2 + 3 + 4	
	L	R	L	R	L	R
Sum						
Category						
Action						

Figure 9.3 A simple chart to assist in calculating and recording an audiogram’s categorisation.

Name: _____Date: _____

Mild Hearing Impairment Warning Notification.

The results of your hearing test on (date) have indicated that you have a mild hearing impairment in comparison with other (men/women) of your age group. This may be due to exposure to noise at work and/or involvement in noisy hobbies in your spare time. This kind of hearing loss is irreversible. Hearing deteriorates as we get older but this hearing problem will add to any age-related hearing loss you develop. This means you may become deaf earlier than other people in your age group.

Finding this hearing loss developing now, before it becomes severe enough to require medical referral, allows you to prevent further damage, which will be likely if you continue to be exposed to high noise levels without adequate hearing protection. This is a warning to you to ensure that you prevent further damage.

You should ensure you take the following steps:

- Wear your hearing protection *at all times* in high noise levels.
- Check that you are wearing your hearing protection *correctly*.
- Comply with all hearing conservation measures that are in place.
- Report any increase in noise levels that you notice or any problems with your hearing protection. Do not work in noise without adequate well-maintained hearing protection.
- Wear hearing protection when you are involved in noisy hobbies or activities.

Signed: _____Position: _____Date: _____

Figure 9.4 An example of a formal notification of the presence of mild hearing damage.

3. Poor hearing

Category 3 applies where the sum (for either ear) is equal to or exceeds the referral level given in Table 9.1. The individual’s hearing has now reached a level requiring referral and the following action should be taken:

- The audiogram should be brought to the attention of a doctor. This will be either the occupational health physician or the employee’s GP.
- The individual should be advised of the findings (Figure 9.5).
- Future hearing tests may need to be carried out more frequently than at three yearly intervals. The actual interval required will be a matter for professional judgement based on the rate of deterioration, the noise levels and any other relevant factors.

4. Rapid hearing loss

Category 4 applies where the previous test was carried out within the last three years. The hearing thresholds at 3 kHz, 4 kHz and 6 kHz should be added together to give a sum (for each ear) for the current audiogram and for the previous audiogram if this is not already recorded. If a reduction of 30 dB or more is found between the current and the previous audiograms,

Name: _____	Date: _____
<u>Referral Notification</u>	
The results of your hearing test on (date) indicate that you have 1. poor hearing. 2. rapid hearing loss. 3. a unilateral hearing loss. <i>(Select relevant problem, 1, 2 or 3, and appropriate explanation, below).</i> This means that: 1. Your hearing is worse than normal for your age group and this could be due to exposure to high levels of noise at work and/or in your hobbies. 2. Your hearing has deteriorated rapidly since your last hearing test and this could be due to exposure to high levels of noise at work and/or in your hobbies. 3. Your hearing is much worse in one ear than the other. This is not usually due to noise and is more likely to be due to an infection or other disorder, which requires further investigation and advice or treatment as appropriate. You are therefore being referred to your GP who will investigate the extent and cause of the hearing damage and take appropriate action. It is extremely important that you conserve your remaining hearing and that you follow the advice you have been given in this regard. We will continue to monitor your hearing.	
Signed: _____	Position: _____ Date: _____

Figure 9.5 An example of a referral notification.

the individual's hearing has reached a level requiring referral and the following actions should be taken:

- The audiogram should be brought to the attention of a doctor. This will be either the occupational health physician or the employee's GP.
- The individual should be advised of the findings, see Figure 9.5.
- Future hearing tests are likely to be needed more frequently than at three year intervals. The actual interval required will be a matter for professional judgement based on the rate of deterioration, the noise levels and any other relevant factors.

Unilateral hearing loss

The hearing thresholds, from the current audiogram, at 1 kHz, 2 kHz, 3 kHz and 4 kHz should be added together to give a sum for each ear. If the difference between the left and the right ear sum is more than 60 dB (i.e. 61 dB or more), there is evidence of a significant difference between the ears and the following actions should be taken (unless of course appropriate action has already occurred):

- The audiogram should be brought to the attention of a doctor. This will be either the occupational health physician or the employee's GP.
- The individual should be advised of the findings, see Figure 9.5.

Category		Sum of thresholds at	Action
1.	Acceptable hearing ability	(1 + 2 + 3 + 4 + 6) kHz In both ears this is below the warning level for the appropriate age and gender given in Table 9.1	None
2.	Mild hearing impairment	(1 + 2 + 3 + 4 + 6) kHz In one or both ears this is ≥ the warning level for the appropriate age and gender given in Table 9.1	Warning
3.	Poor hearing	(1 + 2 + 3 + 4 + 6) kHz In one or both ears this is ≥ the referral level for the appropriate age and gender given in Table 9.1	Referral
4.	Rapid hearing loss	(3 + 4 + 6) kHz Where the previous test is within the last 3 years and there is a reduction in hearing level ≥30 dB in one or both ears.	Referral
Unilateral		(1 + 2 + 3 + 4) kHz The difference between this sum in each ear is >60 dB.	• Advise individual • Referral

Figure 9.6 A resumé of the HSE categorisation system.

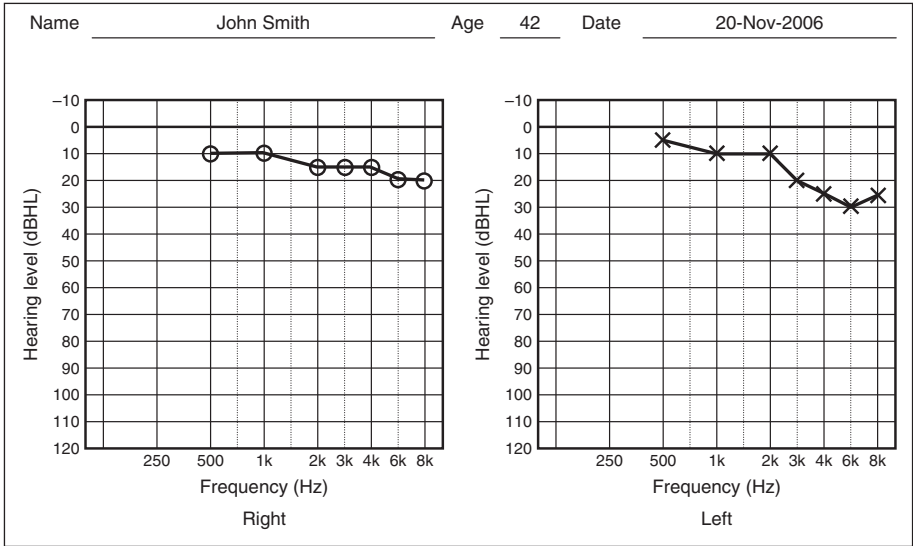
Worked examples

John Smith is a worker who was tested on entry to his present employment, his baseline audiogram is shown in Figure 9.7 and its categorisation is shown in Figure 9.8. The following year, his hearing is being checked again. Figure 9.7 also shows his current audiogram and its categorisation is shown in Figure 9.9.

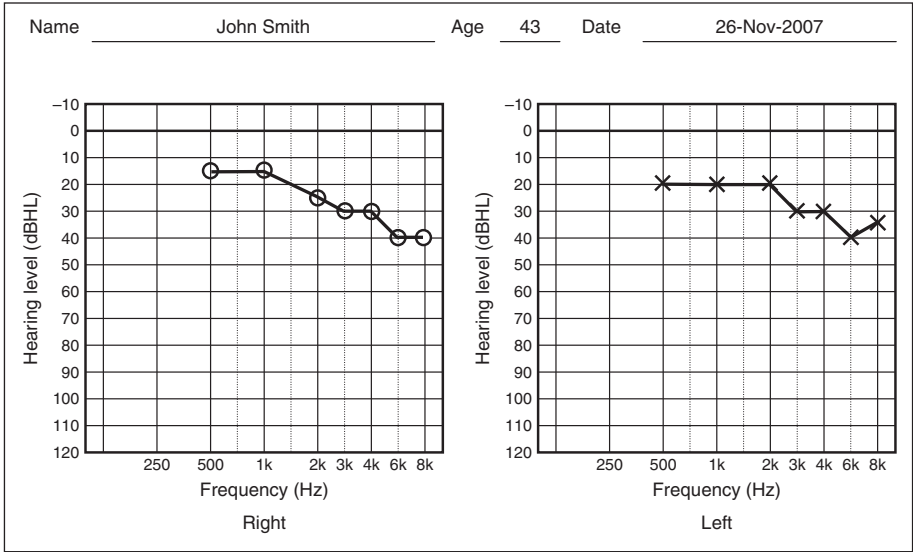
The Health and Safety Executive (HSE) former five categories

Introduction

Prior to the introduction of the 2005 changes in noise regulations, five categories were used to assess occupational audiograms. These categories will continue to be present in the workers’ records for some considerable number of years, and it



(a)



(b)

Figure 9.7 John Smith's (a) baseline and (b) current audiograms.

is therefore important to understand how they were reached and what each category means, although it will no longer be necessary to use this more complex system for current and future audiograms. The audiogram was assessed according to the method given below and the results obtained placed the worker into

kHz	1 + 2 + 3 + 4 + 6		3 + 4 + 6		1 + 2 + 3 + 4	
Ear	R	L	R	L	R	L
Sum	75	95	50	75	55	65
Category	1: Acceptable hearing ability					
Action	No Special action					

Figure 9.8 Categorisation of John Smith’s baseline audiogram (male, age 42).

kHz	1 + 2 + 3 + 4 + 6		3 + 4 + 6		1 + 2 + 3 + 4	
Ear	R	L	R	L	R	L
Sum	140	140	100	100	100	100
Category	2: Mild hearing impairment		4: Rapid hearing loss			
Action	Warning		Referral			
Overall result	Category 4: Rapid (mild) hearing loss: Refer for medical advice. Check hearing conservation measures. Retest in 6 months.					

Figure 9.9 Categorisation of John Smith’s current audiogram (male, age 43).

one or more of five categories, each of which required certain stated actions. This could be a confusing and unsatisfactory system to use, not least because (Health and Safety Executive, 2004):

- The numbering system was not logical and was not consistent with the severity of the problem.
- There were no meaningful labels attached to the categories.
- The employee often had to be placed in more than one category (Figure 9.10).

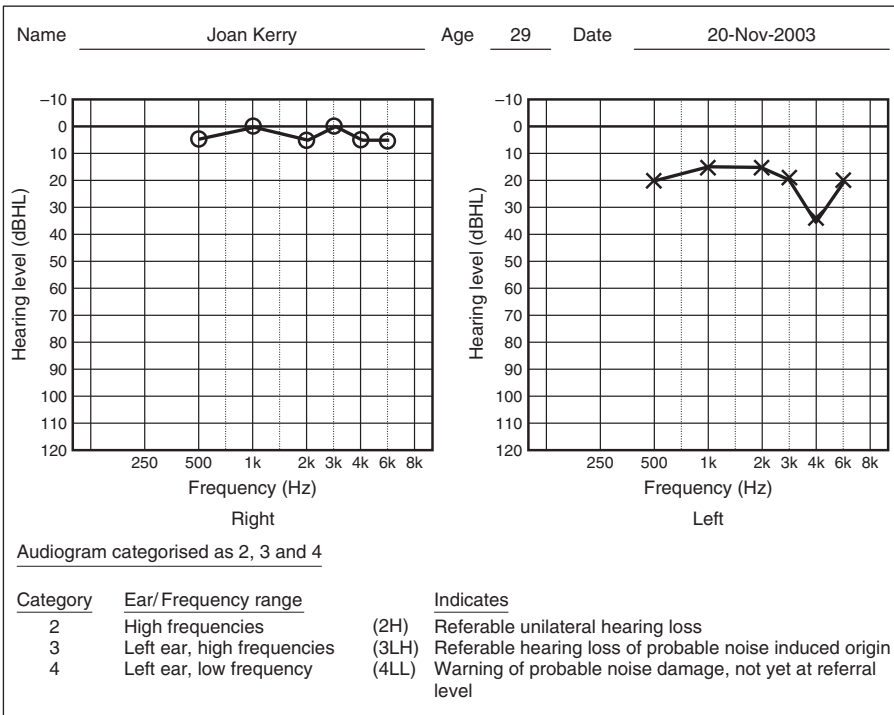


Figure 9.10 An example audiogram and its former categorisation.

- Comparisons were only made with the previous audiogram with no requirement to look back at the baseline at any point.
- Results from 500 Hz carried equal weight even though they are generally less important than results from other frequencies used to detect and categorise noise induced hearing loss.
- No gender difference was included, although women generally have better hearing than men.
- Little account was taken of the rapid decline in hearing due to presbycusis over the age of fifty, since all workers over this age were placed together in one band. This biased the results of the scheme, which was intended to identify noise induced hearing loss rather than presbycusis.
- Steady deterioration in hearing due to probable noise hearing induced loss was not always picked up for some considerable time, particularly where an individual's pre-employment hearing levels were good.
- There was insufficient quality control built into the system.
- The data used for comparison with age was insufficiently sensitive, had never been validated and was inconsistent with data from the Medical Research Council's National Study of Hearing (Davis, 1995).

Assessment of the Audiogram according to the former categorisation system

The method of assessment involved the following steps:

- 1. Note the age of the employee and the date when the audiogram was taken. If there is a previous audiogram, the interval between this audiogram and the previous one should also be found and noted.
- 2. The audiogram is evaluated by adding the hearing levels in two bands – the high frequencies and the low frequencies – in the following manner:
 - i) The low frequencies (500 Hz, 1 kHz and 2 kHz) are added together to give one number for each ear.
 - ii) The high frequencies (3 kHz, 4 kHz and 6 kHz) are added together to give one number for each ear.

This gives a set of four results for each audiogram. Care should be taken not to include hearing levels at any other frequencies (e.g. 250 Hz or 8 kHz) in the calculations.

Figure 9.11 suggests a simple chart that may be helpful in understanding the calculation and recording of these results. When the categories have been printed on an audiogram they will normally appear in an abbreviated form, with the category first, followed by the ear and then the frequency, for example 3LH refers to category 3 in the left ear, high frequencies. Sometimes the letter ‘B’ will be used to signify that both high and low frequencies have been affected in that ear, for

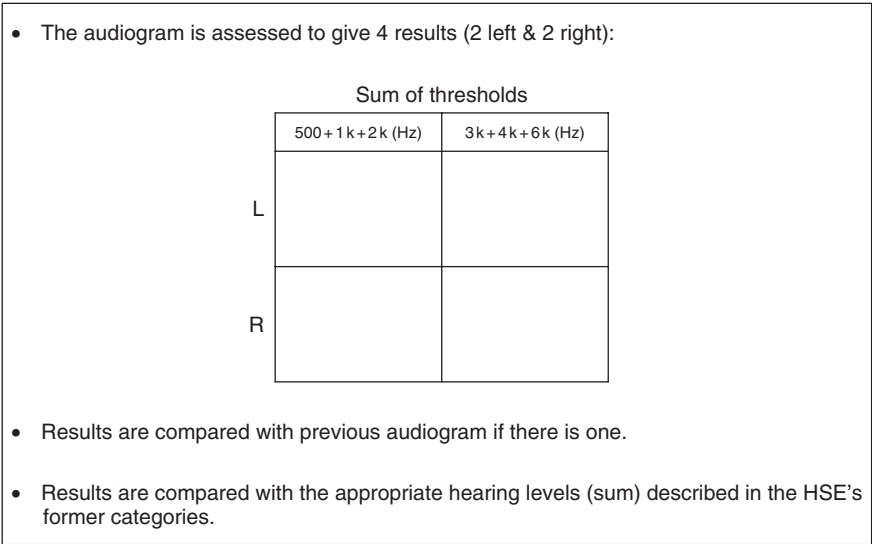


Figure 9.11 A simple chart to assist in calculating and recording the former audiogram categories.

example 3LB refers to category 3 in the left ear but both high and low frequencies are affected. The results obtained placed the employee in one or more of the HSE's former five categories according to hearing loss.

The former category 1

A previous audiogram was required to place a worker in this category, therefore, if the audiogram is a baseline, category 1 cannot be applied. This is the only category that required reference to a previous audiogram. Category 1 applied where there was a change in hearing loss since the last test. This was a change in any one or more of the four results (low frequency, high frequency, left and right ears). The interval since the last test was important as it determined the amount of change required to place the worker in this category. If the last test was three years ago or more, a change of at least 45 dB had to be observed but, if the last test was less than three years ago, the change had to be at least 30 dB. The former category 1 indicated a rapid change in hearing levels that could be due to noise or some other cause. The audiogram had to be brought to the attention of the designated medical practitioner who would decide on the action to be taken and the company was required to take steps to prevent further deterioration in hearing.

The former category 2

Former category 2 applied where there was a one-sided (unilateral) hearing loss or a significant difference between the ears. This category considered the difference between the left ear and the right ear. To place a worker in this category a difference of 46 dB or more was required in the low frequencies and/or a difference of 61 dB or more in the high frequencies. This former category was not usually due to industrial noise but could have been due to a disorder of the auditory nerve or some other cause. The audiogram had to be brought to the attention of the designated medical practitioner who would decide on the action to be taken.

The former category 3

Former category 3 applied where significant noise induced hearing loss was likely. A worker was placed in category 3 according to the results calculated from their audiogram compared with a table of values for the appropriate age group. Each of the four results was compared in turn. A worker was placed in category 3 if the hearing loss in the low frequencies and/or high frequencies, in either or both ears, exceeded the referral level given in Table 9.2. This former category 3 reflected changes in hearing levels that suggested probable noise damage. The audiogram had to be brought to the attention of the designated medical practitioner who would decide on the action to be taken. The worker had to be formally notified of the presence of hearing damage and the company was required to investigate the cause of the hearing loss and to take steps to prevent further deterioration.

Table 9.2 Values by age for the former categories 3 and 4

Age	CAT. 3: Referral		CAT. 4: Warning	
	Frequency range			
	Low	High	Low	High
20–24	60	75	45	45
25–29	66	87	45	45
30–34	72	99	45	45
35–39	78	111	48	54
40–44	84	123	51	60
45–49	90	135	54	66
50–54	90	144	57	75
55–59	90	144	60	87
60–64	90	144	65	100
65	90	144	70	115

The former category 4

The former category 4 applied where there was a suggestion of noise induced hearing loss. A worker was placed in category 4 according to the results of their current audiogram, which were compared with a table of values for the appropriate age group. Each of the four results was compared in turn. A worker was placed in category 4 if the hearing loss in the low frequencies and/or high frequencies, in either or both ears, exceeded the warning level given in Table 9.2, but had not yet reached the referral level. The employee had to be formally notified of the presence of hearing damage and counselled to ensure that they understood the significance of their hearing status and the need to comply with hearing conservation measures. The rate of progression of the hearing loss had to be carefully monitored and the audiogram would be repeated earlier than the normal interval if this was indicated. Exposure factors were to be investigated, which could highlight a particular noise problem.

The former category 5

The former category 5 applied where hearing levels were within ‘normal’ limits. Any worker who did not fit into categories 1 to 4 was placed in category 5. There was no specific action required but monitoring audiometry had to continue to be carried out at regular intervals.

Summary

An audiogram is a graph showing the results of a hearing test, on which the frequencies are indicated along the horizontal axis and the hearing threshold level in decibels along the vertical axis. The symbols used are a cross (blue) for

the left ear and a circle (red) for the right ear and these are joined by a solid line. The Health and Safety Executive (HSE) method of categorising audiograms, using the frequencies 1 kHz, 2 kHz, 3 kHz, 4 kHz and 6 kHz, provides occupational health personnel with a relatively simple method of assessing the level of hearing damage and defined steps to follow in every case. Each worker tested will be placed into one of the four categories, according to hearing loss, and note is also taken of cases of unilateral hearing loss.

Categories 1 to 3 are based on the sum of the hearing levels at 1 kHz, 2 kHz, 3 kHz, 4 kHz and 6 kHz. Category 1 refers to acceptable hearing ability and applies where the sum falls below the warning level; no special action is required. Category 2 refers to mild hearing impairment and applies where the sum is equal to or exceeds the warning level; formal notification of the presence of hearing damage is required, together with retraining of the individual in hearing conservation. Category 3 refers to poor hearing and applies where the sum is equal to or exceeds the referral level; the individual's audiogram should be brought to the attention of a medical practitioner.

Category 4 refers to rapid hearing loss and applies where the previous test was carried out within the last three years. The hearing thresholds at 3 kHz, 4 kHz and 6 kHz are summed and compared with the relevant sums from the previous audiogram. If a reduction of 30 dB or more is found between the current and the previous hearing tests, the individual's audiogram should be brought to the attention of a medical practitioner and future hearing tests may need to be carried out more frequently than normal. Unilateral hearing loss is determined by comparing the sum of the hearing thresholds at 1 kHz, 2 kHz, 3 kHz and 4 kHz to see if there is a difference between the left and the right ear that is more than 60 dB. Where this exists, the individual should be advised of the findings and the audiogram brought to the attention of a medical practitioner.

The HSE's former five category system will remain on the records of earlier audiograms but is no longer used to categorise audiograms.

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III

Action and Referral

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Causes of hearing loss and the role of the physician

The occupational physician's role

Introduction

When an employee has been found to have a significant hearing loss, a second audiogram should be performed, preferably within a month. It is very important that this audiogram should be taken in quiet conditions and when the employee has not been subjected to excessive noise levels at work or leisure. This will often need to be on a Monday morning but it is also important that there has been no leisure exposure to loud noise ideally in the 48 hours, but certainly for no more than 16 hours, prior to the test. The audiogram, together with the questionnaire and any other information including the previous audiogram, if one exists, should be obtained by the Occupational Health physician who will arrange to see the employee. It is helpful if the occupational physician is aware of the working conditions and an occasional walk through the plant, noting the noisy areas, will help them to appreciate where people work.

The medical assessment of hearing

The purpose of the medical assessment, as a part of the hearing conservation programme, is to ensure that nothing is missed and that the required actions are

taken. It is therefore necessary to check and explore the information obtained and to act upon it, for example:

- The work history and any noise exposure are explored. Questioning attempts to bring to light any noise exposure not yet disclosed.
- The employee’s medical questionnaire is reviewed to elaborate on any relevant aspects. Questions regarding hereditary hearing loss, trauma, disease and medication may be repeated and further enquiries made into social history and hobbies.
- The employee’s use of hearing protection is investigated. The employee may be counselled on the need for hearing conservation and the correct use of hearing protection.
- The audiogram is reviewed.
- Otoscopic examination is performed to rule out wax or any outer ear abnormality.
- Tuning fork tests are carried out to establish the likely site of the problem and thus whether the loss is sensorineural or conductive.
- The results of the case history, otoscopic examination, audiogram and tuning fork tests are brought together to establish a likely cause and to decide on the further action necessary, for example with regard to:
 - fitness for work
 - hearing conservation, the adequacy of hearing protection and the possible need for more frequent future hearing tests
 - referral to the GP for treatment and further investigation of medical symptoms.
- The audiogram and its significance should be discussed with the employee. Table 10.1 indicates degrees of hearing loss, which may be helpful in this discussion.

All information should be recorded and summarised with, if possible, an opinion as to the probable cause. Consent must be obtained for the results of the assessment to be sent to the individual’s GP for further investigation or for results to be made available to the employer. An example of a referral letter is given in Figure 10.1. If a further opinion or advice is required, this may be obtained from an audiologist or from an Ear, Nose and Throat (ENT) consultant, the latter would usually be via the worker’s GP. Figure 10.2 shows an example of a detailed report that might be obtained from an audiologist direct to the occupational physician or occupational health adviser, which can be particularly helpful in reaching decisions at work. Figure 10.3 shows an example of a medical report from an ENT consultant to the GP.

Table 10.1 Degrees of hearing loss

Average hearing threshold level (dBHL)	Degree of hearing loss
–10–20	None (normal)
21–40	Mild
41–70	Moderate
71–95	Severe
Over 95	Profound

Dear Dr _____,

Re: Mr _____.

I saw Mr _____ following routine occupational hearing tests. His audiogram shows a (mild/moderate/severe/profound) hearing loss and I enclose a copy.

Mr _____ has worked as a _____ for _____ years and his past occupational history includes:

- any noisy occupations
- any hobbies leading to exposure
- relevant medical history
- relevant family history
- past and present drugs.

The (degree/type of) hearing loss (does/does not) appear to be consistent with (the noise exposure/noise induced hearing loss) and I would be grateful if you would arrange to see Mr _____, with a view to possible ENT referral.

Yours sincerely,

Dr _____

Occupational Health Physician/Consultant.

Figure 10.1 An example of a referral letter from the occupational health practitioner to an employee's GP.

Audiological Report: _____ (date)

Re: _____ (name)
 _____ (address)

DoB _____

History

I saw _____ (name) for audiological and hearing aid assessment on _____ (date). She reported a family history of hearing problems, although she did not know of any other family members with a hearing loss as severe as her own. It seems that she may have suffered a blast injury in 1943, when a land mine exploded near her school, shattering the windows and causing _____ (name) to fall down the stairs. It is possible that this could have had an additive effect, worsening a genetic hearing loss of late onset. She first noticed the problem in her late thirties, since which time it has gradually deteriorated.

(continued)

Figure 10.2 An example of an audiological report.

Audiological Examination

On otoscopic examination, both eardrums and canals appeared normal (apart from a small spot in the left ear canal). Impedance measurements indicated normal middle ear pressure.

The pure tone audiogram shows a bilateral profound sensorineural hearing loss with evidence of loudness recruitment. The loss is reported to be accompanied by severe tinnitus.

The audiogram is attached [See Figure 10.5(a)].

Aided Hearing

Until about five years ago, _____ (name) was usefully aided bilaterally. At this time, the right ear lost its remaining useful hearing. She uses an NHS hearing aid in the left ear and also possesses a private in-the-ear hearing aid. This in-the-ear hearing aid is peak clipped at 110 dB, which is just below her uncomfortable loudness level (ULL). Since the hearing has continued to deteriorate, neither aid now provides sufficient benefit. This is borne out by the aided audiogram [see Figure 10.5(b)].

Prognosis

_____ (name)'s hearing has progressed until there is virtually no usable hearing in the right ear and a profound loss in the left. It seems likely that the left ear may continue to deteriorate. The current level of aided hearing provides very little hearing for speech. Without lipreading, _____ (name) is unable to follow even a simple sentence. Nevertheless, the hearing which remains in the left ear provides cues which she uses effectively to supplement the lipread pattern. Indeed, her lipreading skills are exceptional and she manages well in a one-to-one situation, although only with great concentration and effort. In groups, the situation is much more difficult and the loss restricts her social and her working life.

Discussion and opinion

_____ (name)'s hearing loss is profound. The following should be considered when planning suitable employment:

1. _____ (name) is an expert lipreader and uses her small amount of remaining hearing to support this. Any further loss of hearing would severely disadvantage her. It is therefore inappropriate for her to be working in a noisy situation.
2. _____ (name) is unable to hear warning signals and should not be working alone.
3. Communication is difficult and only successful on a one-to-one basis.
4. Due to the profound degree of hearing loss, there are no further hearing aid options available and tinnitus maskers would be inappropriate.
5. The hearing loss falls within the criteria for cochlear implantation and I have suggested that _____ (name) speaks to her own doctor to explore this as a possibility. In my opinion, she would be an excellent candidate for consideration for a cochlear implant. An implant would provide much improved hearing, which should allow her to cope well in quiet working conditions and greatly improve her overall quality of life.

Signed _____ Date _____
(Qualifications and title)

Figure 10.2 (continued)

Dear Dr _____,

Re: _____ (name, date of birth)
_____ (address).

Many thanks for your referral.

Problems

Significant bilateral hearing loss in the right ear, ranging from 30 to 85 dBHL, and in the left ear, ranging from 40 to 80 dBHL. Mrs _____ also has tinnitus but this is not a significant issue for her.

Risk factors

1. Noise at work: Mrs _____ stated that she has spent about sixteen years working in a weaving factory.
2. Raised cholesterol and high blood pressure but these are currently relatively stable.

Findings

The ear drums are both normal on examination. Hearing loss as detailed above. The audiogram is consistent with noise damage. This appears to be the most likely cause of the hearing loss although there are other possible causative factors.

Plan

I have suggested digital hearing aids and placed her on the waiting list for NHS digital hearing aids. I have also explained the need to wear adequate hearing protection at all times and the risk of increasing loss if she fails to do so. If she does use adequate hearing protection and her hearing is stable, continuing in her current job should be acceptable. However, I assume her hearing will also be monitored by the Occupational Health Department at work and if her hearing should deteriorate further, she will need to find a quieter occupation.

Yours sincerely,

(Consultant Otolaryngologist)

Figure 10.3 An example of an ENT report to the GP.

Fitness for work

The occupational health physician will have to make decisions regarding fitness for work. It is important that health and safety is not compromised and any hearing loss should be discussed openly and honestly. Work in many noisy environments can be safe if the hearing loss is moderate and stable and the worker understands the problem. However, the worker may need to hear specific warning signals, understand instructions and may be required to communicate in meetings, on the telephone, in background noise and in groups all of which may be difficult, stressful and sometimes unsafe. Adequate ear protection must be worn even though hearing is even

more difficult when wearing ear protection. It is also advisable to carry out a hearing test every six months for at least the first two years in the job, as any further drop in hearing is usually noticeable fairly quickly. An example of a hearing assessment (fitness for work) report is given in Figure 10.4. It may be inappropriate for someone with a hearing loss to continue working in noise if their loss may cause danger to them or to others, or if they do not wear adequate hearing protection or the work is extremely noisy. It may also be inappropriate if the hearing loss is unstable or appears to be increasing or is so severe that further loss would not be acceptable.

In general, it is necessary to decide if a hearing loss is likely to prevent the employee from doing a particular job and certain questions may need to be answered, including:

- What is the degree of the hearing loss?
- Is the hearing loss stable?
- Will the loss and any tinnitus be heightened by further noise exposure?

Hearing Assessment (Fitness for Work) Report	
Name: _____	
Home Address: _____ _____ _____	
Employer: _____	
Department: _____ Shift: _____	
Occupation: _____	
As a result of hearing assessment, the above named is:	
Apparently free from any hearing defect that would impair their capacity to undertake the duties specified for this post (delete if not applicable).	
Comments: _____ _____ _____	
Signed: _____ (Medical Officer/Nursing Officer in charge)	
Position/Qualifications: _____ Date: _____	
Signed _____ (Employee/Candidate)	
Date: _____	

Figure 10.4 An example of a hearing assessment (fitness for work) report.

- Will the employee accept that they may be susceptible to a worsening hearing loss?
- Is the degree of loss so severe that further loss is unacceptable?
- Can the employee function in the environment?
- Can they work safely in this environment?
- Can they hear warnings? (e.g. forklift truck drivers must be able to hear sirens).
- Do they pose risks to fellow workers?
- Do they pose risks to themselves?

In exceptional circumstances, a doctor may indicate that it may no longer be appropriate to keep a worker in their current job. Sometimes audiograms based on the speech frequencies (Figure 10.5) or on the frequencies of particular warning signals may be helpful in reaching a decision. Figure 10.6 presents a simple method of calculating the percentage of speech sounds that the individual will hear by counting the dots (Mueller and Killion, 1990) within the speech area shown on the audiogram. The hearing thresholds are plotted on the audiogram form and the number of dots falling within the residual speech area, that is that part of the speech area that can still be heard, are counted.

Disability should be accommodated as far as possible but in a noisy industry it may pose a risk to the individual and to others. The physician has a duty of care both to the worker and to the employer and management. Potential problems should be discussed honestly and as openly as possible with all concerned. In some industries, it may be possible to reduce noise exposure or to find a quieter job for the worker. The Joint Service System of Medical Classification, JSP 346, used by the armed services, gives a series of categories to establish fitness for work as shown in Table 10.2. In this system, which is known as the 'PULHHEEMS' system, physical capability (P), upper limbs (U), locomotion (L), hearing acuity right and left (HH), visual acuity right and left (EE), mental capacity (M) and stability (S) are all graded. Hearing acuity is always graded first in the right ear and then in the left, thus the first H refers to the right ear and the second H to the left ear. The system is used to obtain and record a standardised picture of health and functioning:

- On recruitment
- Every five years during service from the age of 30 (more often over the age of 50)
- On demobilisation (for called up reservists) and/or discharge (termination of service employment).

The hearing is assessed by adding the thresholds at 500 Hz, 1 kHz and 2 kHz to give a low frequency result. The thresholds at 3 kHz, 4 kHz and 6 kHz are added to give a high frequency result. Whichever result, low or high, is the greatest determines the overall H score for that ear. The H score therefore does not provide any information as to whether the hearing loss in that ear is high or low frequency and in some instances speech pattern recognition is accepted as being a better indicator of hearing function than the H grades. Hearing is seen as being within acceptably normal limits if it falls into H1 or H2. Any drop in hearing that alters the H grade (except H1 to H2) must be referred for an ENT opinion. Some jobs do not require perfect hearing, although it is likely that this will be required

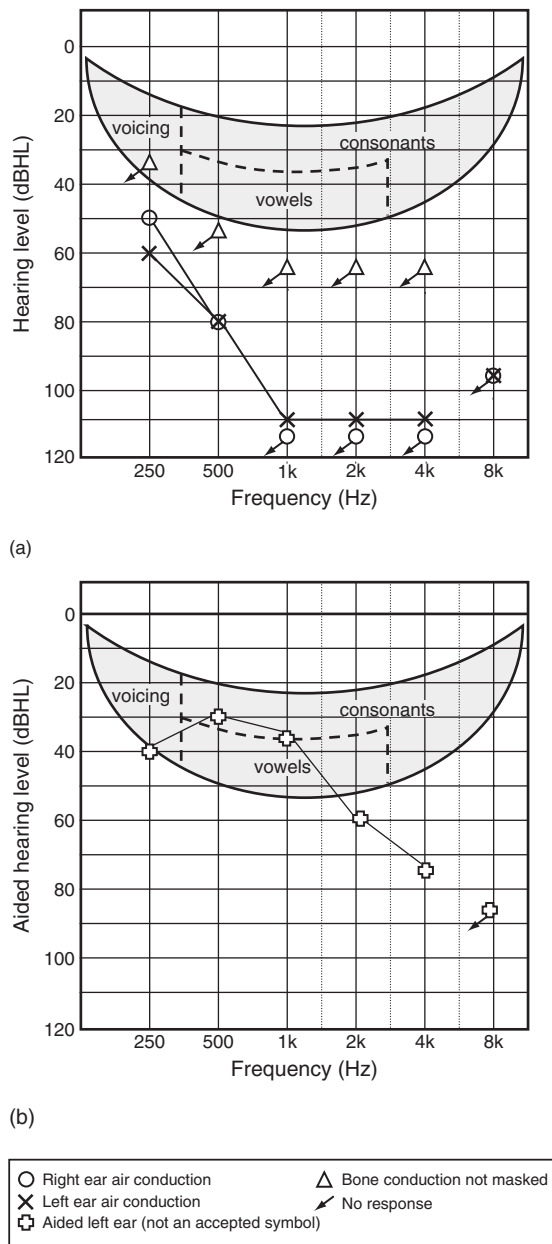


Figure 10.5 (a) A diagnostic audiogram with the speech area shown. The hearing loss is profound and the individual is unable to hear any speech whatsoever; (b) An aided audiogram for the left ear indicating that this individual is likely to hear vowel sounds only very faintly and to miss virtually all consonant sounds even with excellent hearing aids.

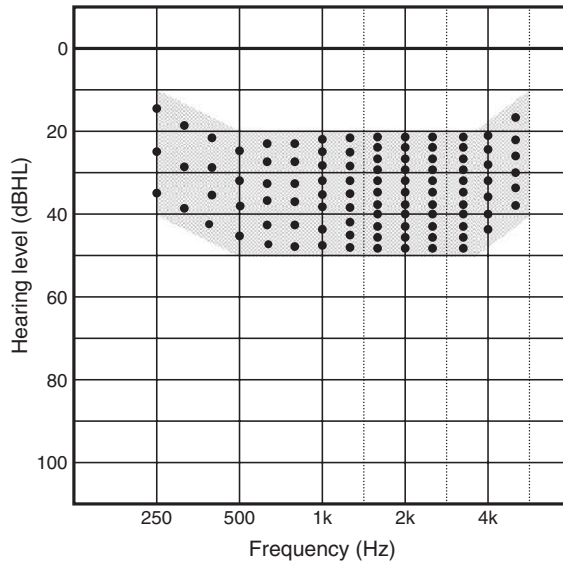


Figure 10.6 An audiogram form showing the speech area with dots to indicate speech sounds. By counting the dots that are within the given thresholds, an approximate percentage speech perception score is obtained.

for positions such as air crews, sonar operators, divers and where it is important to hear verbal instructions. Where service personnel are considered suitable to continue in employment despite a hearing loss, appropriate controls and education are to be put in place. The HH categories only refer to hearing acuity; ear diseases are additionally considered but as part of the P category. Consideration, including clinical assessment by an ENT consultant if appropriate, is given to whether ear disease is likely to lead to future incapacity.

Where hearing is a major safety issue, some employers carry out a daily 'safety check' to screen hearing at, for example, 30 dBHL to establish fitness for work. If the test is not passed, whether due to wax or any other cause, the worker is not certified fit for work on that day. If there is a specific requirement for good hearing for work, this should be made clear at the recruitment stage. Decisions as to whether to employ or not on the grounds of hearing loss should be reasonable and practicable and made on a case-by-case basis. All decisions should be well documented. The employee can be asked to sign a disclaimer after discussion at the recruitment stage with a copy given to management and the employee. If the recruit will not consent to this or will not agree to wear adequate hearing protection, employment may not be appropriate.

Other factors may also need to be considered, for example are hearing aids being used? If so, will they be used at work and is it safe to use hearing aids here? Does the individual use one aid or two? If they only use one aid, this may compromise their ability to hear well as, in addition to not being able to hear

Table 10.2 Hearing categories used by the armed services to guide fitness for work under the PULHHEEMS system

Category	Sum of low frequencies (dB)	Sum of high frequencies (dB)	Meaning	Outcome
1	Not more than 45 (i.e. 0–45) (Except royal navy only: No single level to be more than 20)	Not more than 45 (i.e. 0–45) (Except royal navy only: No single level to be more than 20, except 6 kHz which to be no more than 30)	Good hearing	H1 and H2 generally accepted as normal hearing
2	Not more than 84 (i.e. 46–84)	Not more than 123 (i.e. 46–123)	Acceptable practical hearing	H1 and H2 generally accepted as normal hearing
3	Not more than 150 (i.e. 85–150)	Not more than 210 (i.e. 124–210)	Impaired hearing. The bilateral hearing level generally considered unfit for entry	Will be referred to ENT
4	More than 150	More than 210	Very poor hearing. May be able to continue service in a particular trade, especially if unilateral	If category 4 is considered too bad, for example certain audiogram configurations, it automatically becomes category 8
8	More than 150	More than 210	Hearing so poor as to be unfit for service. Invaliding required	For compensation/pension purposes as a prescribed disease, hearing handicap must average 50 dB or more over the frequencies 1, 2 and 3 kHz

from the deaf side, hearing in noise is far worse. Noise can be upsetting to a hearing aid user, so that the aids may need to be turned down in noise, reducing the ability to hear well even further. Also a hearing aid user's aids may break down and it is necessary to know if the individual will then be able to function safely.

The type of hearing loss may have some bearing on employment, for example if the hearing loss is conductive, working in noise is unlikely to cause further loss. Conductive hearing loss gives much less disability and some natural protection against noise damage. However, where hearing is important for the occupation, the individual may still not be able to function adequately in the role. If hearing aids and/or other devices are appropriate to enable an individual to function at work, the employer has some responsibility to provide these. Access to work, the government scheme, may provide some financial assistance.

Type of hearing loss: tuning fork tests

Introduction

Industrial or occupational audiometry provides information about the degree of hearing loss but not the type of hearing loss. Information may therefore be sought from tuning fork tests to indicate the probable site of the problem before further diagnostic assessment is undertaken. Tuning forks are used in simple brief tests to determine the type of loss, that is whether the hearing loss is:

- Conductive (due to a problem in the outer or middle ear), or
- Sensorineural (due to a disorder in or beyond the cochlea).

A tuning fork is a simple device, made of steel, aluminium or magnesium, that vibrates when struck. Its prongs, or tines, move alternately away from and towards each other and produce a relatively pure tone. The fork should not be struck heavily or on a hard surface, as this would introduce harmonics and the tone would no longer be pure. The tuning fork should be held by the stem and struck, about two-thirds of the way along the tines, on a rubber pad or on the knee or elbow. Alternatively, the fork may be plucked at the top of the tines. Tuning forks for audiometric investigation require a flat base. Tuning forks are inexpensive, light, small and portable, and tuning fork tests can be performed on employees whose loss is not too severe to be able to hear the vibrations of the tines. The preferred frequency of the tuning fork to be used is 512 Hz (British Society of Audiology, 1987). Other frequencies (Figure 10.7) may also be used but very high tones fade too quickly to be of much use, whilst very low tones may produce vibrotactile results, that is they may be felt rather than heard. Whatever frequency is used, the results obtained apply only to that frequency.

The tuning fork tests most widely used are the Weber and Rinne tests, which together provide a reliable indication of the type of hearing loss.

The Weber test

The Weber test establishes in which ear the tone is perceived. The employee must first be asked if they have a poorer ear and, if so, which ear. The tuning fork is struck, and its base is placed on the forehead (Figure 10.8). A hand should be gently placed to support the back of the head. The employee is asked where they hear the tone.

- With normal hearing or an equal hearing loss, the tone will be heard in the midline.
- With a unilateral or an unequal sensorineural loss, the tone will be heard in the better ear.
- With a unilateral or unequal conductive loss, the tone will be heard in the poorer ear! This is likely to occur because the better ear is able to hear background noise, which masks the tone to some extent. The ear with the conductive loss has no such interference and hears the tone clearly by bone conduction (BC).

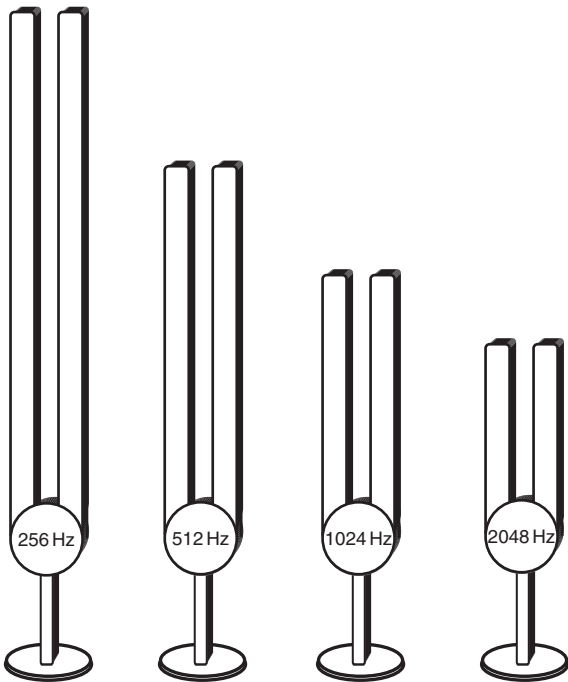


Figure 10.7 Tuning forks.

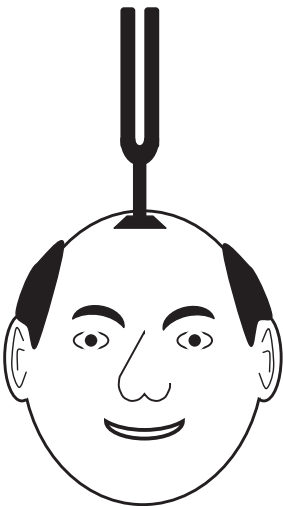


Figure 10.8 The Weber test.

The Rinne test

The Rinne test compares sensitivity by air conduction and bone conduction in one ear at a time (Figure 10.9). The tuning fork is struck and held with the tines in line with, and about 2.5 cm from, the canal's entrance for 2 seconds. The tuning fork is then moved quickly so that the base is pressed firmly against the mastoid, again for 2 seconds. A hand should be held against the opposite side of the head to provide counter-pressure. The employee is asked if they hear the tone louder at or behind the ear.

- With normal hearing, air conduction is more efficient than bone conduction (BC); the tone is therefore heard loudest at the ear. The same result is obtained with most sensorineural losses. This is known as a Rinne positive.
- With a conductive loss, the tone appears louder by bone conduction. This is a Rinne negative.
- Where there is a 'dead' ear or a severe to profound sensorineural loss on the test side the tone may also appear louder by bone conduction. This is known as a false Rinne negative. This result is due to cross-hearing, that is hearing the sound in the opposite ear.

When a Rinne negative result appears to contradict the Weber test result, a false Rinne negative can be suspected. Rubbing the tragus of the ear (masking) may help to prevent cross-hearing but cannot be carried out accurately.

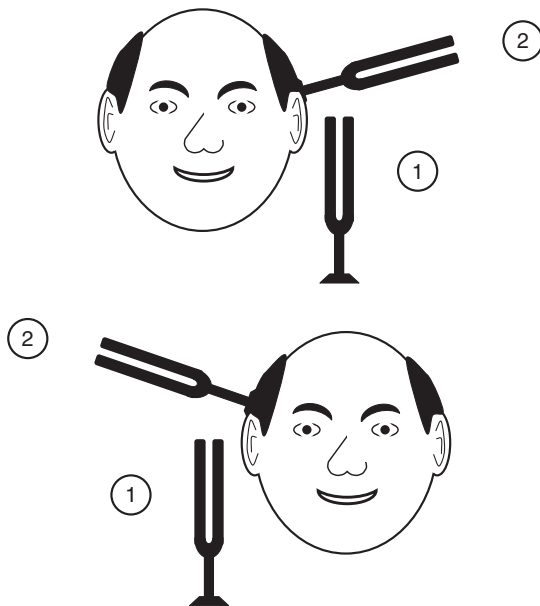


Figure 10.9 The Rinne test.

Common causes of hearing loss

Types of hearing loss

Hearing loss can be due to a wide range of causes, both conductive and sensorineural; example audiograms are shown in Figure 10.10. Conductive hearing loss is caused by some abnormality of the outer and/or middle ear. The inner ear is capable of functioning normally but the sound reaching it is reduced in level. If the conductive pathway is completely blocked, the reduction in sound (the conductive hearing loss) will be in the region of 60 to 70 dB. A pure conductive hearing loss cannot be total. Many conductive causes can be remedied medically or surgically, and alternatively hearing aids can be used usually with excellent results. Conductive hearing losses muffle sound and tend to be worse in the low frequencies than in the higher frequencies. Common causes of conductive hearing loss include impacted wax, otitis externa, otitis media and otosclerosis.

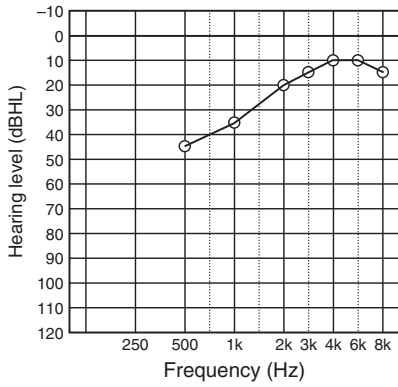
Sensorineural hearing loss describes the type of hearing loss caused by some abnormality in the cochlea, auditory nerve, or in the brain (also known as central hearing loss). Damage most commonly occurs in the cochlea and the higher frequencies are usually most affected. The perception of abnormal loudness growth may also occur, which is where a person cannot hear low levels of sound but when sounds increase they rapidly become too loud. The most common causes of sensorineural hearing loss are presbycusis and noise induced hearing loss. Other causes include vascular disorders, ototoxic drugs, genetic cause, Ménière's syndrome, certain diseases (e.g. mumps, measles, meningitis, flu, shingles, maternal rubella), head injury and acoustic neuroma.

Mixed hearing loss is the term used where elements of both sensorineural and conductive hearing loss are present. The type of hearing loss cannot be established on the basis of an air conduction audiogram alone. Tuning fork tests will provide some indication of the type of loss but diagnostic audiometry is required for an accurate assessment.

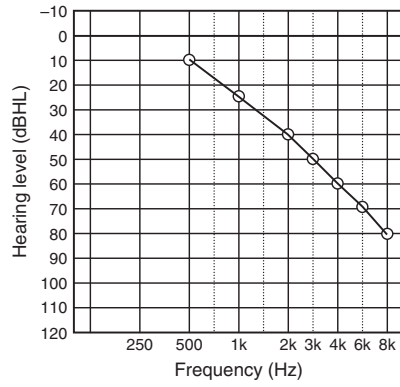
Common causes of conductive hearing loss

Otitis externa

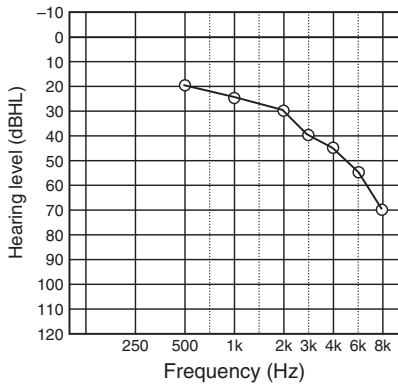
Otitis externa is inflammation of the outer ear. Inflammation is the reaction of the tissues to infection. Swimmers often suffer from otitis externa, because the earwax has been dissolved by water leaving the sensitive skin of the ear canal susceptible to bacteria or fungi. The symptoms of otitis externa include pain and swelling. The ear canal feels blocked and there may be a discharge. The pain may be worse on swallowing or when moving the ear. A hearing test is not appropriate until the condition has been alleviated. The individual should be advised to seek treatment from their GP, which will usually involve the use of eardrops. To avoid the condition, moisture should not be left in the ears after bathing.



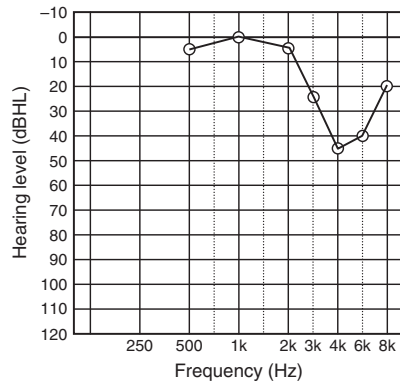
(a) Ménière's disorder or conductive loss



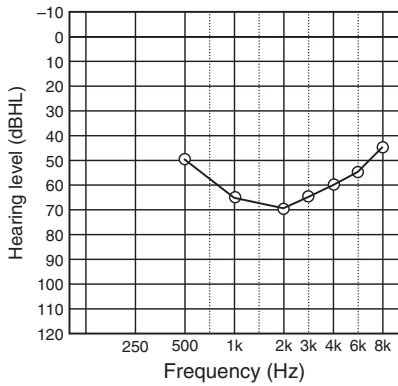
(b) Ototoxicity



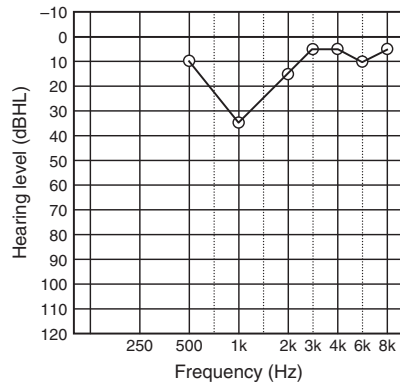
(c) Presbycusis



(d) Noise induced hearing loss



(e) Viral cause



(f) Hereditary or genetic

Figure 10.10 Possible example audiograms for specific causes of hearing loss (right ear only shown)

Stenosis

Stenosis or atresia is a blockage of the ear canal. Atresia is usually used to mean a complete closure or absence of the canal that has been present since birth. Stenosis is usually used to mean an acquired partial closure or extreme narrowing of the ear canal. This may be due to an overgrowth of the bone, polyps or even a collapse of the cartilage of the ear. Hearing loss will only be present if the ear canal is completely blocked.

Impacted wax and foreign bodies

New wax is soft, moist and light in colour and should fall naturally out of the ear. Wax serves to help keep the canal clear of dust, dirt and foreign bodies, for example hairs, metal filings and insects. Some foreign bodies that manage to enter the ear are safe and will be carried out of the ear canal naturally with the wax migration, other foreign bodies may need medical removal. Foreign bodies do not usually cause a noticeable hearing loss unless they totally block the ear canal. Sometimes, however, wax may accumulate in the ear canal and eventually block it. Old wax gradually becomes dark and hardens and if it blocks the ear canal it will cause a hearing loss. The actual degree of hearing loss is very variable but a loss of 10 to 30 dB is not unusual. If the ear is blocked with wax, hearing tests are inappropriate.

Wax is usually softened for a few days before removal; warmed olive oil is good for this purpose although there are also many proprietary solutions. Wax removal may be carried out by syringing, dry removal (using wax removal tools) or suction. Syringing is not appropriate if there is a history of ear perforation.

Otitis media

Otitis media is inflammation of the middle ear due to dysfunction of the Eustachian tube. The normal functions of the Eustachian tube are drainage and the maintenance of middle ear pressure. If the air pressure in the middle ear is not the same as ambient atmospheric pressure, the middle ear function may be adversely affected. In the early stages of 'negative pressure', that is lower pressure in the middle ear, a very slight hearing loss may result, which is often unnoticeable (although it can be a great problem in someone who already has a severe or profound hearing loss). The majority of cases of otitis media are sub-clinical, that is they do not show any symptoms although the hearing is reduced. However, on-going negative pressure can cause a retraction pocket or a cholesteatoma (a skin cyst in the middle ear). If not treated this cyst can erode the bone, cause facial paralysis, dizziness and eventually erode into the mastoid cells and into the brain cavity.

Continued negative pressure causes fluid to exude from the walls of the middle ear and will result in a temporary hearing loss. If the Eustachian tube blockage is due to 'mechanical' dysfunction, the fluid will be sterile (serous).

Usually the condition resolves itself and decongestants may help to drain and dry up the fluids. If the problem is long-standing or recurs frequently, 'glue ear' may result. This is where the fluid in the middle ear becomes sticky and glue-like and will not drain away. It may be necessary to puncture the eardrum and suck out the fluid. A grommet (a small plastic tube) will then be inserted in the eardrum to facilitate drainage and pressure equalisation until the condition clears. Tonsils and adenoids may be removed and sinuses may be washed out, if necessary. Grommet insertion will have an immediate effect on hearing although it may continue to improve further over the following few weeks. Grommets usually drop out of the drum into the ear canal after about six weeks to six months and the drum will heal. If the problem is thought to be long term, T-tubes which are long-term grommets may be inserted. Grommets cause no discomfort but will not normally be inserted more than a maximum of about three times as they may weaken or scar the eardrum, which could cause re-perforations and/or some hearing loss in adult life. The perforation left by a T-tube sometimes fails to heal. Alternative treatments for otitis media with effusion (fluid) include the use of hearing aids, homeopathic remedies, dietary adjustments (e.g. reduction of intake of dairy products and of sugar), acupuncture and cranial osteopathy.

If the blockage is due to the spread of an upper respiratory tract infection, the fluid will be infected pus that will cause painful pressure on the eardrum and may cause the drum to burst, as the infection tends to weaken the eardrum. A perforated eardrum will relieve the pressure and usually also the pain and the Eustachian tube will eventually clear and open again. A perforation in the lower central part of the eardrum is considered safe and will usually heal uneventfully after a few weeks, although it may leave an area of scar tissue. Upper respiratory tract infections are common in children because the child's immune system is less efficient than the adult's. Middle ear infection may be treated with antibiotics and analgesics.

Otitis media occurs in a small number of adults but is much less common than in young children. It is prevalent amongst babies and young children and tends to be most problematic prior to the age of about eight to ten years. This is because the Eustachian tube is more horizontal, smaller, wider and less rigid in young children and therefore more liable to collapse and to the spread of infection.

Otosclerosis

Otosclerosis is a condition in which new spongy bone grows around the stapes footplate. The bone gradually hardens and fixes the stapes in the oval window. This results in a gradually worsening conductive hearing loss, reaching a maximum loss of about 70 dB when the stapes no longer moves. The condition tends to run in families and to affect women more than men as it is accelerated by hormonal activity, for example in pregnancy. Treatment is by hearing aids and/or surgery. The stapes is removed in an operation, known as a stapedectomy, and replaced by prosthesis.

Common causes of sensorineural hearing loss

Ototoxic drugs

The use of certain drugs and exposure to certain chemicals, for example the solvent styrene, are known to cause or increase the risk of sensorineural hearing loss. Ototoxic drugs (Table 10.3) may produce deafness, vertigo and/or tinnitus, either temporarily or permanently. Potentially any drugs could be ototoxic if taken in large quantities or by someone who is particularly susceptible. Many medications cause reversible deafness and the hearing will recover when the drug is no longer taken, unless large doses are used, in which case the loss may become permanent. The likelihood of permanent ototoxic damage may also be increased if excretion is impaired due to renal disease. Some common drugs and substances are ototoxic including aspirin, ibuprofen, alcohol and tobacco. Neomycin, which is found in some over-the-counter medications is highly toxic to the ear and even given topically can cause damage and should never be used if the ear is perforated.

Vascular causes of hearing loss

Circulatory problems, such as high blood pressure, raised blood glucose levels in diabetes, arteriosclerosis and auto-immune disease can cause progressive hearing loss because the cochlea depends upon a sufficient supply of oxygen and nutrients to maintain its function. Some vascular or blood vessel disorders can cause sudden hearing loss including, for example, arterial occlusion and haemorrhage. Even a brief disruption to the blood supply can result in permanent damage.

Table 10.3 Examples of drugs that may be ototoxic

Group	Generic name	Notes
Aminoglycoside Antibiotics	● Gentamycin *V ● Streptomycin *V ● Amikacin *D ● Neomycin *D ● Kanamycin *D ● Tobramycin *V D	Incidence of toxicity may be as much as 25%. Risk increases with increasing dose. May cause mainly vestibular damage (*V) or mainly cochlear deafness (*D). Hearing loss may progress even after the drug has been stopped. High-pitched tinnitus may be the first symptom of a hearing problem. If the drug is not discontinued, hearing loss may develop after a few days. Hearing loss may be permanent and in some cases total. Aminoglycosides are usually reserved for serious infections or where other antibiotics are ineffective.
Other antibiotics	Erythromycin	Hearing loss is usually reversible.
	Vancomycin	Hearing loss is often irreversible.
	Capreomycin	A treatment for TB. Usually reversible.
	Minocycline	Used for sexually transmitted diseases e.g. syphilis. Can cause vestibular problems after only one or two doses. Reversible.

Table 10.3 Examples of drugs that may be ototoxic—Continued

Anti-malarial	● Chloroquine ● Hydroxychloroquine ● Quinine	Causes deafness. Reversible if in low doses. Quinine, used for cramps, and quinidine, used for cardiac rhythm disorders. Can cause hearing loss and tinnitus.
Anti-cancer (anti-neoplastic or chemotherapeutic)	● Carboplatin ● Cisplatin ● Vinblastin ● Bromocryptine ● Nitrogen mustard ● Bleomycine ● Carboplatinum ● Methotrexate	Drugs that are used to treat cancer. Cause irreversible high frequency hearing loss, tinnitus and sometimes vestibular damage. Hearing loss may develop even after the drugs has been discontinued.
Loop diuretics	● Ethacrynic acid ● Sodium ethacrynate ● Furosemide ● Bumetanide	Loop diuretics are the only diuretics that seem to be ototoxic. Used to treat fluid retention and occasionally for high blood pressure. Usually reversible but sometimes permanent, especially if combined with aminoglycosides.
Glucocorticosteroids	● Prednisone ● Adrenocorticotrophic hormone	Can cause irreversible hearing loss.
Non-steroidal anti-inflammatory (NSAIDs)	● Aspirin ● Naproxen ● Diclofenac ● Diflunisal ● Fenoprofen ● Phenylbutazone ● Piroxicam ● Flurazepam ● Tolmetin ● Sulindac ● Ibuprofen	Can cause tinnitus and hearing loss with prescribed high dosage, for example taken for rheumatoid arthritis. All reversible. As a rule of thumb, the hearing loss in decibels is approximately equal to the serum salicylate concentration in decilitres, for example a 50 dB hearing loss is produced by a concentration of 50 mg/dl.
Beta-blockers/ Cardiac medications	● Practolol ● Metoprolol ● Flecainide ● Procainamide ● Lidocaine	Can cause irreversible hearing loss
Anti-convulsants	● Phenytoin ● Ethosuximide	Can cause vestibular damage
Mood-altering (psychopharmacologic) agents	● Alprazolam ● Oxazepam ● Prozac ● Fluoxetine ● Doxepin	May cause sensorineural loss
Other medications	● Thalidomide	Now withdrawn. May cause permanent sensorineural loss.
Chemicals	● Cyclohexane ● Dichloromethane ● Hexane ● Lindane ● Methyl-chloride ● Methyl-n-butyl-ketone ● Perchlor-ethylene ● Styrene ● Tetrachlor-ethane ● Toluol ● Trichloroethylene	May cause permanent sensorineural loss.
Other substances	● Alcohol ● Tobacco	May cause sensorineural loss

Presbycusis and the effects of ageing

Presbycusis is defined as a hearing loss in a person over the age of 60, where the cause is not known. It is likely that the hearing loss is the result of ageing, including a lifetime's exposure to, for example, *normal* levels of noise, medication, stress, alcohol and so on possibly with some cell degeneration. The hearing loss due to presbycusis is usually sensorineural, bilateral, high frequency and progressive. The loss tends to progress slowly in the early stages but the speed of progress increases with increasing age.

Age is the greatest risk factor for hearing loss (Davis, 1996) but true 'presbycusis' is not the commonest cause. Lim and Stephens (1991) found presbycusis in less than 20% of a group of patients over 60 years old who were referred for hearing aids. If the hearing loss is due to a known cause, even though related to age (such as noise or diabetes), it is not classed as presbycusis.

Noise induced hearing loss

Noise induced hearing loss refers to a hearing loss that is the direct result of exposure to excessive noise. The effect that the noise can have depends upon a number of factors, including individual susceptibility. Noise induced hearing loss is usually but not exclusively of industrial origin. The kind of noise to which the person is exposed has little bearing on the resultant hearing loss, for example exposure to machine noise or to loud music, if they are of the same intensity and duration, may produce a similar hearing loss. The hearing loss is generally sensorineural in nature, gradual, and affects the high frequencies first but as it progresses it will also affect mid and lower frequencies although to a lesser extent. There is often a 'notch' in the audiogram at 4 kHz. Tinnitus frequently accompanies the hearing loss.

Acoustic trauma is a noise induced hearing loss caused by brief exposure(s) to high level impulse noise, often in the region of 130 to 140 dB SPL. The hearing loss will be sudden and may be accompanied by pain and sudden tinnitus. The audiogram usually shows a high frequency (sensorineural) hearing loss but if damage has also occurred in the outer or middle ear, causing perforation of the eardrum or ossicular discontinuity, the loss will have a low frequency conductive element. Overall the audiogram will then be likely to show a flat (mixed) hearing loss. In some cases, there may be some improvement in hearing levels in the days immediately following the acoustic trauma.

Head trauma

A head injury can cause damage to the ear. A blow to the head creates a pressure wave in the skull which travels through the skull to the cochlea and can cause temporary or permanent hearing loss, usually mainly in the high frequency region. A skull fracture extending into the occipital or squamous portion of the temporal bone may include fracture of the cochlea with irreversible hearing damage. Meningitis can occur as a complication of a temporal bone fracture.

Non-syndromic hereditary hearing loss

Some adults develop a sensorineural hearing loss, often high frequency, in middle age from about the age of 40. Often there is a history of early hearing loss in close family members. Children are sometimes born with a permanent sensorineural hearing loss of genetic origin. This may be a recessive condition where it is not immediately obvious that it is genetic in origin.

Syndromic hearing loss

A syndrome is a group of symptoms that tend to occur together. Examples include:

- Paget's disease – This is a recessive skeletal disorder in which the head and long bones progressively enlarge. Mixed hearing loss may occur due to new bone formation.
- Down's syndrome – This is a chromosomal abnormality with a high incidence of occurrence. Ear symptoms may include small pinnae, narrow ear canals, narrow Eustachian tubes, frequent otitis media and ossicular abnormalities. Hearing loss is common. This may be conductive, sensorineural or mixed. Presbycusis appears very early, often from around the age of twenty.
- Waardenburg's syndrome – This is a dominant genetic condition. Features may include a white forelock, different coloured eyes and hearing loss. The hearing loss is congenital, bilateral and sensorineural and may be progressive. It may be mild to profound and is often worst in the low and mid frequencies. Some individuals with this syndrome have no organ of Corti.
- Usher's syndrome – This is a recessive genetic condition. The baby is born with bilateral congenital sensorineural hearing loss which may be moderate to profound. Vision progressively degenerates from early teens. Initially this is noticed as night blindness, leading to tunnel vision and eventual blindness. The syndrome is relatively common amongst the congenitally profoundly deaf (possibly as much as 10 per cent of this population).

Acoustic neuroma

An acoustic neuroma is a non-cancerous tumour growing from the sheath of the eighth nerve. The tumour usually grows slowly but may eventually cause death. An acoustic neuroma may be difficult to diagnose as it may be completely asymptomatic and the individual may be unaware of it until it is too late. However, it often causes symptoms such as headaches, nystagmus, visual disturbance, balance problems, unilateral hearing loss and occasionally seizures. The symptoms may be such that an acoustic neuroma can affect safety at work. A hearing loss due to an acoustic neuroma is almost always unilateral but it can be of any degree or configuration. Treatment usually involves surgical removal and, in some cases, there may be some hearing left when the tumour has been removed.

Ménière's disorder

Ménière's disorder or syndrome is due to excess endolymphatic fluid (hydrops) in the cochlea. The symptoms may be difficult to distinguish from an acoustic neuroma. Ménière's disorder is characterised by episodes of:

- Fullness in the ear (increased fluid pressure)
- Fluctuating low frequency hearing loss
- Fluctuating low frequency 'roaring' tinnitus
- Vertigo.

Ménière's syndrome may be idiopathic (of unknown cause) or secondary to a problem such as neurosyphilis, viral infections, head trauma, multiple sclerosis, immune disease or otosclerosis. Other risk factors include hereditary predisposition, migraine, stress, fatigue, use of medication, drinking excessive alcohol, a history of food allergies, smoking and recent viral illness. (It is thought possible that the condition could be caused by a viral infection of the endolymphatic sac, but this is unproven.) The disorder most commonly begins between the ages of thirty and fifty and, in around 80 per cent of cases, it is unilateral. A physical distention of the membrane by excessive fluid pressure may cause mechanical disturbance of the cochlear and vestibular function. It is thought that bad attacks may be due to the increased fluid leading to a break in the membrane that separates the perilymph from the endolymph. Loud sounds may induce an attack.

Treatments may be dietary, stress management, medical or surgical. Salt should be restricted in the diet to help reduce fluid retention. Aspirins and non-steroidal anti-inflammatory drugs such as ibuprofen, caffeine and chocolate should be avoided as these can cause tinnitus. Chocolate can also trigger migraine. Smoking should be cut out as nicotine constricts blood vessels and therefore can restrict the blood supply to the inner ear, which increases the symptoms. Whilst stress does not cause Ménière's disorder, it can be a factor in failing to prevent, or to cope well with, attacks. The disorder can be managed medically in most cases and drug treatments include the use of steroids, anti-depressants, antihistamines (e.g. Betahistine or 'Serc'), diuretics, labyrinthine sedatives, anti-vertigo and vasoactive drugs. Valium and benzodiazepines can prevent an attack, by acting directly on the nerve controlling balance and its central connections to the brain, but should not be taken regularly as they can be habit forming. Gentamycin or streptomycin can be used to reduce vestibular function and thus reduce or eliminate attacks of dizziness but side effects can be marked, for example loss of hearing on the treated side occurs in about 30 per cent of those treated with intratympanic gentamycin. Another possible line of treatment is local over-pressure therapy, where low pressure pulses are transmitted to the round window in an effort to stimulate the flow of endolymph. This method is controversial and unsuitable for some patients.

Surgical treatment is used when medical treatment fails to relieve the vertigo. The type of operation depends upon the degree of hearing loss as one objective is to retain as much hearing as possible. Surgical treatments include:

- Endolymphatic sac decompression – This conservative procedure drains excess endolymph through a shunt inserted into the endolymphatic sac. This operation usually preserves hearing and the results can sometimes last for a number of years. However, the shunt can easily become clogged and vertigo is only controlled in just over half of the patients. The procedure is therefore not widely used.
- Vestibular nerve section – The vestibular nerve is cut which permanently cures vertigo in almost every case and usually leaves the hearing intact, although there is a risk of fluid leakage and possible meningitis.
- Labyrinthectomy – The inner ear labyrinth on the affected side is removed or destroyed and the vestibular nerve is cut. This results in permanent total hearing loss. Vertigo is eliminated in almost every case although there will be temporary loss of balance whilst the patient relearns to balance.

Infections

Examples of bacterial and viral infections that may cause hearing impairment include:

- Cytomegalovirus (CMV) – This virus causes an interuterine infection, known as cytomegalic inclusion disease, caught from the mother possibly during birth. In the mother, symptoms are mild and may pass unnoticed. For the child, the consequences may include growth retardation, mental retardation, hyperactivity, convulsions, facial weakness and bilateral sensorineural hearing loss. In the most severe cases of cytomegalic inclusion disease, the baby may die.
- Maternal rubella – This is a virus that may cause, in the child, heart disease, hearing loss, sight problems, dental abnormalities, psychomotor problems, behavioural problems and mental retardation. The hearing loss is sensorineural and bilateral, and tends to affect the low and mid frequencies more than the high frequencies.
- Meningitis – This is inflammation of the meninges surrounding the brain that can occur as a complication of otitis media and may cause sudden severe to profound bilateral sensorineural hearing loss. Unilateral hearing loss occasionally occurs and additional handicaps are common.
- Mumps – This is a contagious childhood disease. When the child recovers from mumps, permanent total unilateral deafness may have occurred. Bilateral deafness is rare.
- Measles – This is a contagious childhood disease that involves the respiratory tract and is often complicated by otitis media. It may cause deafness due to viral invasion of the inner ear via the bloodstream or through purulent labyrinthitis that has developed from otitis media. The hearing loss may be complicated by a conductive element.
- Syphilis – Congenital syphilis symptoms may include dental abnormalities, vestibular dysfunction, sensorineural hearing loss and mental retardation. The

hearing loss may occur suddenly in early childhood and is usually bilateral and severe to profound. Congenital syphilis may cause death in severe cases.

- Shingles – This infection is caused by the herpes zoster virus (as is chickenpox). If it affects the eighth cranial nerve, it may cause a sensorineural hearing loss with pain.
- Influenza and common cold viruses – Sensorineural hearing loss may occur due to direct infection of the inner ear via the bloodstream or through purulent labyrinthitis that has developed from otitis media. The virus may cause degeneration of the organ of Corti, the vestibular system and the eighth nerve. A mild to profound hearing loss and vestibular symptoms may result.

Causes of tinnitus

Tinnitus is common with hearing loss. Causes of tinnitus include:

- Noise
- Ear obstructions
- Hearing loss
- Ototoxic drugs
- Vascular disorders, for example pulsating carotid artery, jugular bulb tumour, and artery or vein malformations
- Ménière's disorder
- Acoustic neuroma.

Low frequency tinnitus may occur with conductive hearing loss and with Ménière's disorder. High frequency tinnitus may occur with many other causes of sensorineural hearing loss, including noise induced hearing loss. Vascular disorders may cause pulsating tinnitus. Treatment for tinnitus depends in part on the cause but in general treatments include surgery, drugs, masking devices (similar to hearing aids but presenting a relatively quiet masking noise), biofeedback, reassurance and behavioural modification. Most individuals with tinnitus also have some hearing loss and in many cases hearing aids, together with an explanation of the problem and reassurance, will help with both the hearing loss and the tinnitus.

Summary

The physician has an important role to play in the conservation programme. Any significant hearing loss must be investigated to ascertain the cause and treatment required. Causes of hearing loss may be conductive (e.g. impacted wax, otitis media, otosclerosis) or sensorineural (e.g. presbycusis, noise induced, ototoxicity, Ménière's disorder). The occupational physician will normally explore the case history, carry out otoscopy, review the audiogram and carry out tuning fork tests. They will form an opinion and refer on, via the GP, to an ENT consultant where

appropriate. They may counsel the employee, usually discussing the audiogram and its significance, the conservation programme and the use of hearing protection. They may also have to make decisions about fitness for work and future work in noisy conditions.

Further reading

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Diagnostic audiometry

Introduction

Diagnostic audiometry involves further testing under controlled conditions in order to obtain accurate information regarding both the degree and the type of hearing loss. These test results will be used, together with other information, to reach a diagnosis as to the cause of hearing loss.

Diagnostic pure tone audiometric tests

Threshold testing by air conduction (AC)

The pure tone air conduction test will be repeated and should give results within 5 dB of the occupational test results (if the original test was taken under suitable conditions). A diagnostic audiometer will have a wider range of testing levels and the ability to use 'masking' (Figure 11.1). Masking is the raising of the threshold of hearing for one sound by the presence of another sound. In everyday life we are often aware of the effect of masking, for example, when trying to hear on the telephone against a background of noise from the television.

In this initial test, the employee's left and right ears are tested separately by using headphones. However, where there is a significant difference between the ears, there is a danger that the sound may cross the skull and be perceived by the better ear on the opposite side to that being tested. This is known as 'cross-hearing'. For example, if an employee has one normal ear and one dead or profoundly deaf ear, when a tone is played into the deaf ear the employee will not respond and the tone will be

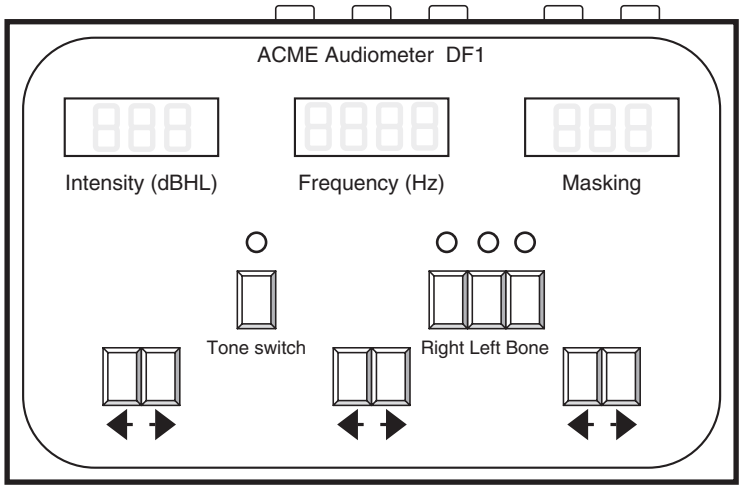


Figure 11.1 The audiometer panel showing the controls.

raised. However, when the tone reaches a sufficient level it will cross the skull (by bone conduction). The normal ear will now hear the tone as a very quiet sound and the employee will respond but, to the tester, it looks as if the poorer ear is responding.

Cross-hearing only causes a problem in testing when the ears have very different thresholds. In fact, it will not cause a problem unless the difference between the ears is at least 40 dB (the amount of sound lost in crossing the head can be 40–80 dB). On the audiogram, the threshold of the poorer ear often appears to follow the shape of the better threshold and it is therefore called a shadow curve (Figure 11.2).

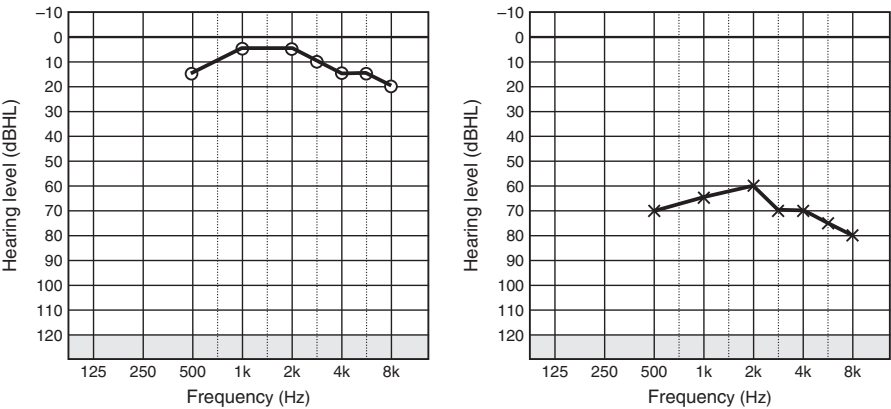


Figure 11.2 An audiogram in which the poorer threshold could be incorrect due to cross-hearing.

The problem of cross-hearing can be overcome by using masking. This involves introducing noise into the good ear so that it cannot hear the tone and the true threshold of the poorer ear can be determined. The threshold with masking may remain unchanged (in which case the original threshold was correct) or it may worsen (Figure 11.3). Where the threshold changes, the new thresholds are recorded on the audiogram and the incorrect thresholds (i.e. the shadow curve) are shaded in. If the thresholds have been masked but remained unchanged, no alteration has to be made to the audiogram, although many audiologists like to write under the audiogram that masking has been used or to half-shade the appropriate threshold symbols (Figure 11.4). The maximum level of masking noise used at each frequency masked should also be noted on the audiogram form.

Masking will be required to ensure the accuracy of air conduction test results where there is a difference of 40 dB or more between the air conduction thresholds of the two ears, or between the air conduction and not masked bone conduction thresholds. The better hearing ear receives the masking noise and the worse ear is re-tested. The assumption is that the better hearing ear is hearing the air conduction test signal by cross-hearing via bone conduction.

Threshold testing by bone conduction (BC)

In order to establish if any hearing loss is conductive (due to a problem in the outer or middle ears) or sensorineural (due to a problem in or beyond the cochlea) testing by bone conduction is also carried out. Figure 11.5 shows a clinical audiometer with the bone conduction transducer. In air conduction tests, sound is sent via headphones, through the normal route of the outer and middle ears. When testing by bone conduction, a bone conduction transducer is placed on the mastoid process behind the ear and the sound is sent directly to the cochlea via the bones of the skull, thus avoiding the outer and middle ears. If there is no problem in the

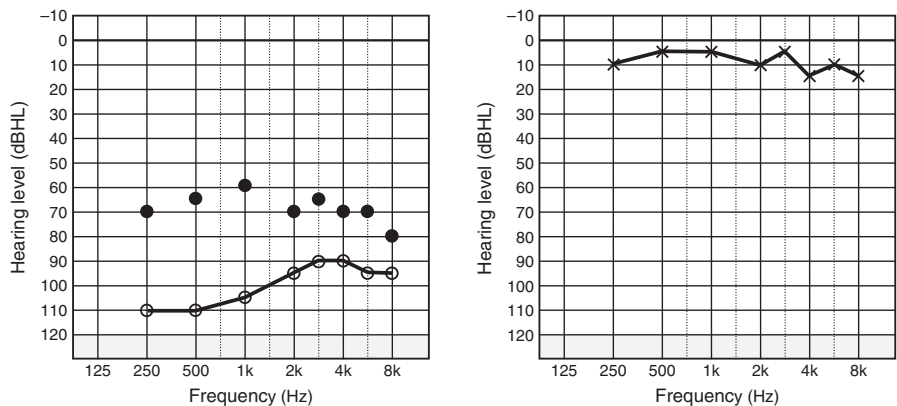


Figure 11.3 An audiogram where masking has been applied and the threshold of the worse ear has changed.

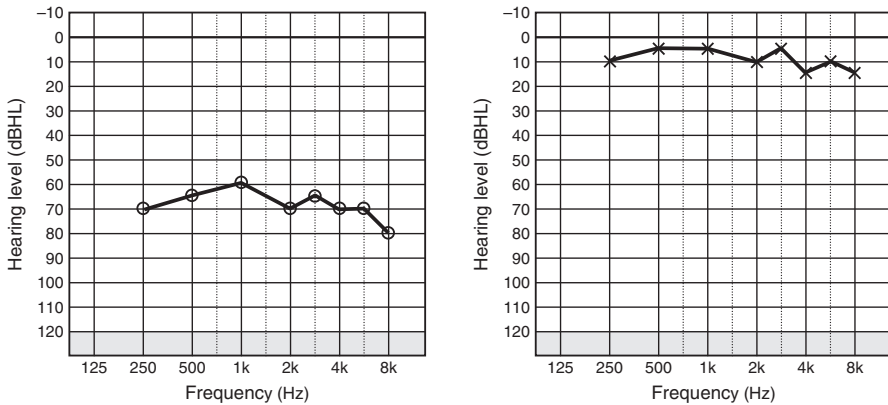


Figure 11.4 An audiogram where masking has been applied and the threshold of the worse ear has not changed.

outer or middle ears, the results by air conduction and by bone conduction should be the same (5 dB difference is acceptable). If the air conduction results are below the bone conduction results by 10 dB or more, that is there is an 'air-bone gap', this indicates a conductive hearing problem.

The bone conduction transducer is placed behind the ear on the mastoid. The ear with the poorer air conduction threshold is used and the frequencies tested are normally 500 Hz, 1 kHz, 2 kHz and 4 kHz (The British Society of Audiology, 2004). The frequency 250 Hz is not usually tested as responses may be due to feeling vibration, rather than hearing, even at low sound levels.

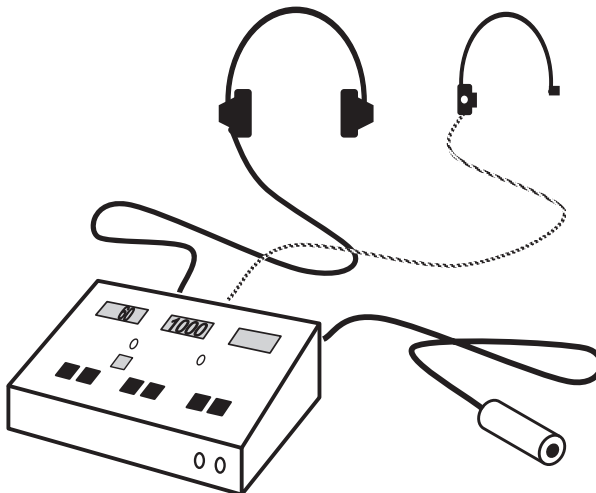


Figure 11.5 A clinical or diagnostic audiometer with earphones and bone conduction vibrator.

Frequencies above 4 kHz are also not tested, at these high frequencies the sound radiating through the air from the vibrator is greater than the vibration through the bone. This can also be a problem at 3 kHz and 4 kHz but an ear plug may be inserted in the ear canal of the test ear to attenuate the airborne sound when testing these frequencies. Frequencies of 2 kHz and below must be tested without an ear plug as thresholds at low frequencies will be improved through occlusion, which increases the sound level.

Thresholds for air conduction reflect the total hearing loss, whereas thresholds by bone conduction reflect the degree of sensorineural problem. The difference between the two, the air-bone gap, indicates the degree of any conductive element (Figure 11.6).

Bone conduction signals will be conducted through the entire skull and will therefore reach both cochleae regardless of where the bone vibrator is placed. If an unequal sensorineural loss exists, bone conduction testing will normally only reflect the thresholds of the better cochlea. Bone conduction thresholds are plotted with a triangular symbol, usually drawn in black and placed on the left or right graph according to the side on which the bone conduction vibrator was placed. This indicates only the position of the vibrator, not which ear received the signal, since both ears will receive a bone conducted signal.

Bone conduction with masking must be carried out if it is necessary to determine the precise degree of sensorineural hearing loss in each ear. Cross-hearing is a major problem with bone conduction. Sound applied to one side of the skull will be heard at the other side at almost the same level. This means that even if the difference between the two ears is as little as 10 dB, the bone conduction signal is likely to be heard more prominently in the better ear, wherever the vibrator is placed. In

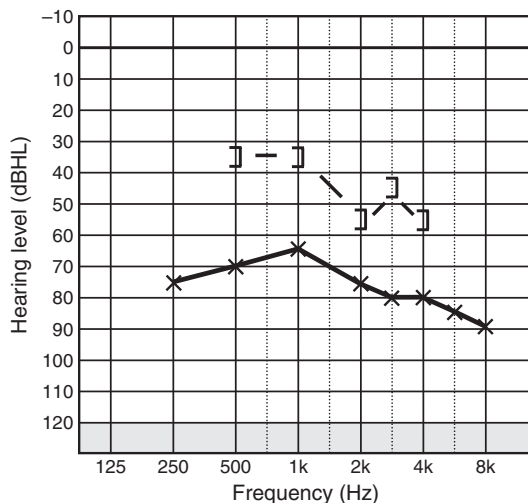


Figure 11.6 An audiogram for the left ear showing elements of conductive and sensorineural (i.e. a 'mixed') hearing loss.

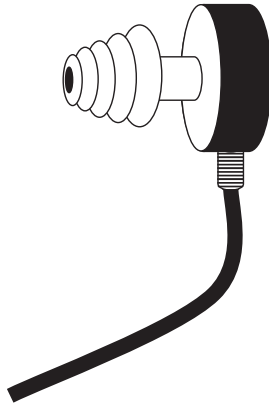


Figure 11.7 An insert receiver for applying masking noise.

practice, if the bone conduction threshold is better than the worst air conduction threshold by 10 dB or more, at any frequency, masking is used to obtain results for each ear separately. Where masking has been used to obtain results from each ear separately, the symbol drawn on the audiogram will be a square bracket, opening towards the test side. The bone conduction results are joined by a dotted line.

The method for masking bone conduction is basically the same as for air conduction. However, masking noise may be applied to the non-test ear either via a headphone or an insert receiver (Figure 11.7). An insert receiver has the advantage of being physically easier to use and the masking levels required may be lower. Where headphones are used, it is very important to ensure that the test ear is not covered, as covering the test ear will appear to improve the bone conduction thresholds of that ear, which is known as the ‘occlusion effect’.

Occlusion occurs because a sound that is transmitted as a vibration through the skull not only travels to the cochlea directly but will also reach the external ear canal. Reflections from the walls of the canal enhance the sound, which is then passed via the eardrum as additional sound. If there is a middle ear impairment, the occlusion effect may not be seen, as the passage of sound through the middle ear is reduced.

Other diagnostic tests

There are many other diagnostic tests which may be carried out to provide further information. Conductive hearing loss can be separated from sensorineural loss relatively easily using pure-tone audiometry and the possible cause of any conductive hearing loss can usually be suggested from the results of tympanometry, which is a widely used quick, easy and non-invasive test. Differentiating the various types of sensorineural hearing loss is more difficult and a number of specialized audiometric tests may be used to distinguish sensory (or cochlear) from neural (or

retrocochlear) hearing disorders. The results of a number of different audiometric tests are normally used in conjunction with information revealed from the case history and physical examination. Information that may help to distinguish cochlear disorders from those that occur beyond the cochlea (retrocochlear) may be obtained, for example, from acoustic reflex tests, speech audiometry, Békèsy audiometry, tests of abnormal loudness growth (recruitment) and tests of auditory adaptation or tone decay. Otoacoustic emissions (OAEs) are used to test the activity of the outer hair cells in response to sound stimulation. There is now also a test to find dead regions in the cochlea, where there are no surviving inner hair cells. Further specialised audiometric tests may be required in cases of tinnitus and where malingering is suspected. All these tests can provide powerful clues to the site and probable cause of sensorineural hearing loss.

Tympanometry

Tympanometry is a test used to detect disorders of the middle ear. It involves objective measurement of middle ear mobility (movement) and middle ear pressure. The results from tympanometry are recorded on a tympanogram, which is a graph on which the compliance of the eardrum is plotted on the vertical axis, against the air pressure of the ear on the horizontal axis.

To understand the test procedure, consider the effect of hitting a tennis ball against a brick wall and then against a sheet on a washing line. Clearly the brick wall, being stiffer and less compliant or yielding than the sheet, will produce greater reflection of the ball. The response to stiffness is applied in tympanometry, where the amount of reflection of a low frequency pure tone determines the mobility or movement of the middle ear system.

The test proceeds once the ear canal is sealed off with a soft plastic tip, not dissimilar to one of the semi-inserts used for hearing protection. This plastic tip holds the end of a probe consisting of three rubber tubes that are connected to:

1. A miniature loudspeaker that emits a tone of fixed frequency and intensity.
2. An air pump that varies the air pressure within the ear canal by automatically sweeping across from +200 to -400 decapascal (daPa).
3. A tiny microphone that picks up the varying sound level in the ear canal as the pressure changes.

Tympanometry begins with a positive increase of +200 daPa air pressure in the ear canal, which displaces the eardrum from its resting place and causes it to stiffen. The stiffened drum will reflect much of the low frequency tone. As the pump gradually decreases the air pressure, the ear drum becomes more flaccid. Maximum compliance is reached when the air pressure in the ear canal is the same as the air pressure in the middle ear, allowing sound energy to pass readily through the ear drum with little reflection. The air pump continues to reduce the air pressure in the ear canal, which then becomes negative when compared to the pressure in the middle ear. Once again the ear drum stiffens up and the reflection

of the tone increases in the ear canal. The microphone picks up the changes in sound energy, which are recorded and appear as a 'mountain peak' on the tympanogram at the position of maximum compliance. A peak at or near 0 daPa is seen in a normally functioning middle ear.

Compliance is measured in cubic centimeters (cc) or millilitres (ml). There is a relationship between pressure levels and volume. The magnitude of pressure per cc depends on the extent that pressure is able to disperse. If the pressure is confined in a small space, it will increase per cc but if it is given the freedom to disperse into a bigger cavity, the pressure will reduce. This can be seen when someone walks on fresh powdery snow and the penetration from pressure on the snow is considerable. With snow skis the pressure is dispersed across a larger surface resulting in less penetration of the snow.

When the ear drum has a perforation, the probe tone is allowed to disperse into the middle ear, giving a large cc reading. If the movement of the middle ear system is impeded by fluid (serous otitis media), or fixation along the ossicular chain (i.e. otosclerosis), then poor compliance will confine the probe tone to a greater extent giving a smaller reading in cc. Two characteristics of the tympanogram are of interest:

1. The shape
2. The air pressure at the point of maximum mobility or compliance.

Common shapes related to certain ear conditions are reasonably easy to recognise by comparison with the shape of an average normal tympanogram. For example, fluid in the middle ear prevents the eardrum from moving freely at any point and the resultant shape is a flat line. Where low pressure without fluid is present in the middle ear, the normal curve is displaced and such a condition may be responsible for slight 'unexplained' conductive hearing loss. Some common shapes are shown in Figure 11.8.

Otoacoustic emissions

Otoacoustic emissions (OAEs) are sounds created by the movement of the outer hair cells in response to low levels of sound stimulation. OAEs will only be present in healthy ears, that is as a rule of thumb, where hearing levels are better than 35 dBHL. A probe (containing a miniature loudspeaker that generates brief sound stimuli known as clicks and a microphone) is introduced into the ear canal. Clicks are generated and the resulting emissions are separated from the background noise and measured to give information on the frequency range from 1 to 4 kHz. OAEs are widely used for screening hearing in neonates. In early noise induced hearing loss, there is usually significant loss of outer hair cells, although this may not yet be reflected in the pure tone audiogram. OAEs may show the loss of cochlear function even when the audiogram is normal or only slightly affected. Outer hair cell damage predisposes the ear to further noise damage so early warning of an impending

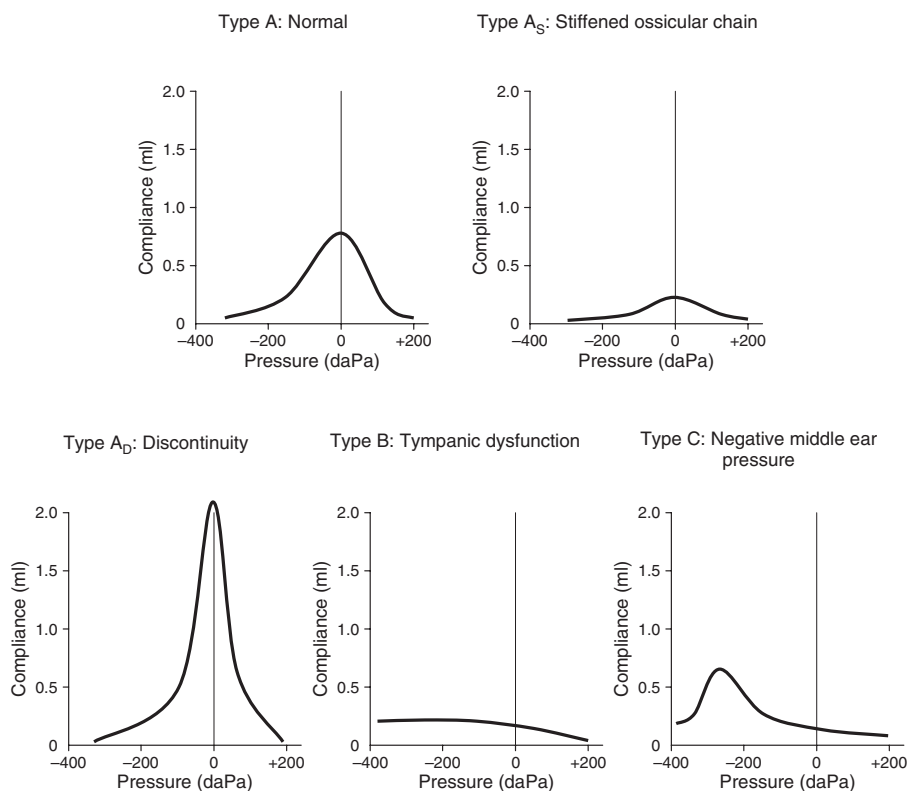


Figure 11.8 Common tympanogram configurations.

problem is important if further damage is to be avoided. OAEs may also be used (Figure 11.9) to show that hearing is within acceptable limits as:

- A screening tool for fitness for work
- A test for feigned deafness
- A test to indicate early signs of auditory disorders (including noise induced loss)
- An objective test for difficult to test individuals.

The TEN (HL) test for dead regions in the cochlea

Noise induced hearing loss can progress from outer hair cell damage only, to include progressive deterioration of the inner hair cells and supporting structures and eventually the total destruction of the cells in the cochlea. Areas with no functioning cells are known as 'dead regions'. On the pure tone audiogram these regions show as areas of worse hearing. It is not possible to tell whether the hearing is poor or non-existent in this area because loud sounds will produce sufficient movement of the basilar membrane to stimulate nerve cells in adjacent areas. The TEN (HL) test involves using masking noise to prevent the adjacent

	Normal hearing thresholds	Reduced hearing thresholds
Normal OAEs present	Normal hearing (better than 35 dBHL)	• Non-organic hearing loss • Autism • Attention deficit • Central hearing disorders
No OAEs	• Tinnitus (OAEs may be abnormal) • Excessive noise exposure • Ototoxicity • Vestibular disorders	• Presbycusis • NIHL • Genetic hearing loss • Conductive hearing loss

Figure 11.9 Auditory conditions and the presence of otoacoustic emissions.

live areas from responding to the specific frequency signal. If there are dead regions normal hearing aids cannot provide assistance and may increase distortion and the likelihood of feedback (whistling) from the hearing aid as high levels of amplification have to be used. However, it may be possible to use special hearing aids, known as frequency transposition aids, that shift the frequency to stimulate nearby live areas without causing these additional problems.

Interpretation of the diagnostic audiogram

Introduction

An audiogram will show thresholds by air and bone conduction. The results should be interpreted in terms of the:

- total amount of hearing loss, that is the loss by air conduction
- sensorineural element, that is the loss by bone conduction
- conductive element, that is the gap between the air conduction and the bone conduction readings.

An audiogram that illustrates normal hearing will show a line of both air and bone conduction symbols along, or close to, the 0 dBHL line near the top of the audiogram.

Sometimes in an audiometric test, the responses obtained may be due to feeling vibrations, rather than true hearing. This is a particular problem when testing by bone conduction, since the levels that produce ‘vibrotactile’ results are much lower than by air conduction (Figure 11.10). The audiologist will be aware when there is a need to question the validity of test results so that a sensorineural loss is not mistaken for a mixed loss.

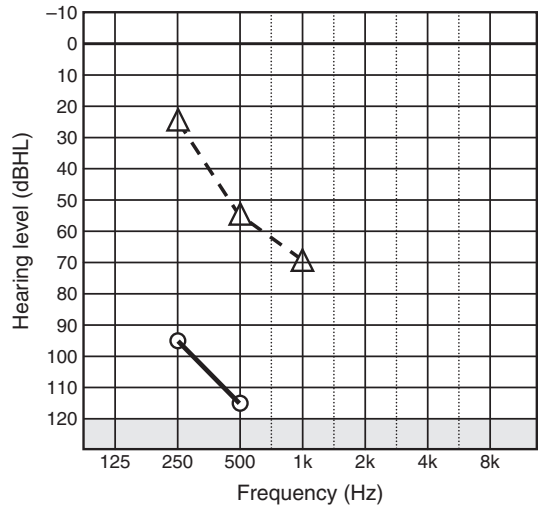


Figure 11.10 An audiogram indicating the minimum levels at which results could be due to vibrotactile responses.

It is often helpful to be able to describe the degree of hearing loss and the British Society of Audiology (2004) use the classification given in Table 11.1, which relates well to the effect on the hearing of speech. The single figure of hearing loss is obtained by taking an average over the five frequencies: 250 Hz + 500 Hz + 1 kHz + 2 kHz + 4 kHz. To give a full description, an indication of the audiometric configuration should be included. Although the descriptions are not standardised, the terms given in Table 11.2 are often used.

The audiogram form should record the serial number of the audiometer used and the type of earphones and bone vibrator used, the tester’s name, the date of the test and the date of the last objective calibration. Unless the test is recorded electronically, the audiogram should also be signed by the tester.

Conductive hearing loss

A purely conductive hearing loss produces thresholds by air conduction that are poorest in the low frequencies. Figure 11.11 shows examples of audiogram configurations for certain conductive conditions. Theoretically, the bone conduction thresholds should be 0 dBHL. In practice, this is not always true as the condition of the middle ear can have a slight effect on bone conduction sensitivity. For example, with:

- Otitis media, there are sometimes reduced high frequency thresholds by bone conduction.
- Otosclerosis, a dip in the bone conduction thresholds can often be seen at 2 kHz; this is known as Carhart’s notch.

Table 11.1 Description of the degree of hearing loss based on an average of thresholds at 250 Hz, 500 Hz, 1 kHz, 2 kHz and 4 kHz

Average hearing level	Hearing loss description
0–40 dBHL (Any value better than 0 dBHL is given the value 0 dBHL)	Acceptable hearing
20–40 dBHL	Mild loss
41–70 dBHL	Moderate loss
71–95 dBHL	Severe loss
Greater than 95 dBHL ('No response' is given a value of 130 dBHL)	Profound loss

Table 11.2 Terms used in the description of audiogram configuration

Term	Description
Flat	A loss that does not rise or fall more than 5 dB per octave
Gradually sloping	A loss that falls by 5–10 dB per octave
Precipitously or sharply falling (often also referred to as a 'ski-slope')	A loss that falls 15 dB or more per octave
Abruptly falling	A loss that is flat or gradual in the low frequency region but then falls sharply
Rising or reverse audiogram	A loss that increases by 5 dB or more per octave
Trough (sometimes also referred to as a 'cookie bite')	A loss that falls in the mid-frequency region (1–2 kHz) by 20 dB or more in comparison with the loss at 500 Hz and 4 kHz

Sensorineural hearing loss

Where a hearing loss is indicated but all of the air and bone conduction symbols appear within 10 dB of each other, this indicates a purely sensorineural loss equally in both ears (i.e. a bilateral sensorineural hearing loss of equal degree). Figure 11.12 presents examples of audiogram configurations for various sensorineural conditions.

Dead regions in the cochlea may be suspected where the hearing loss on the audiogram is 80 dB or greater. Dead regions can only be confirmed by specialised testing and cannot be assumed from the audiogram alone, although they are thought to be relatively common with hearing loss caused by:

- Sudden noise exposure, such as shooting
- Ménière's disorder
- Genetic disorders, especially low-frequency hearing losses and those with a 'cookie bite' configuration.

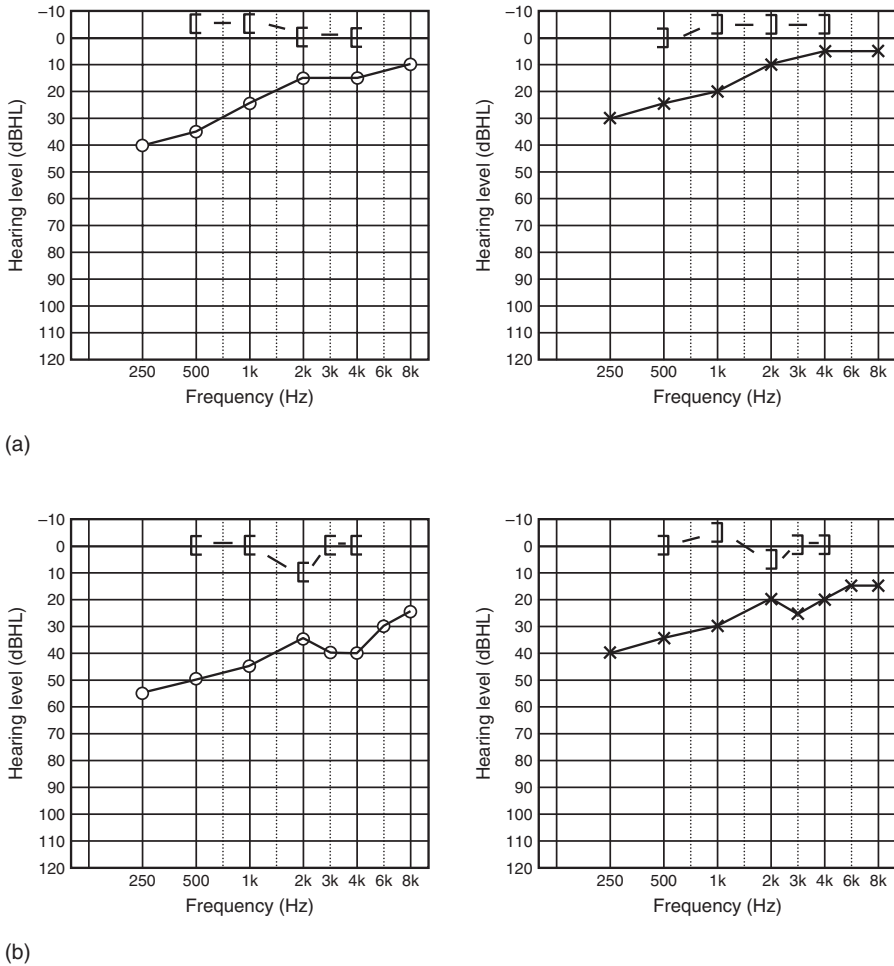
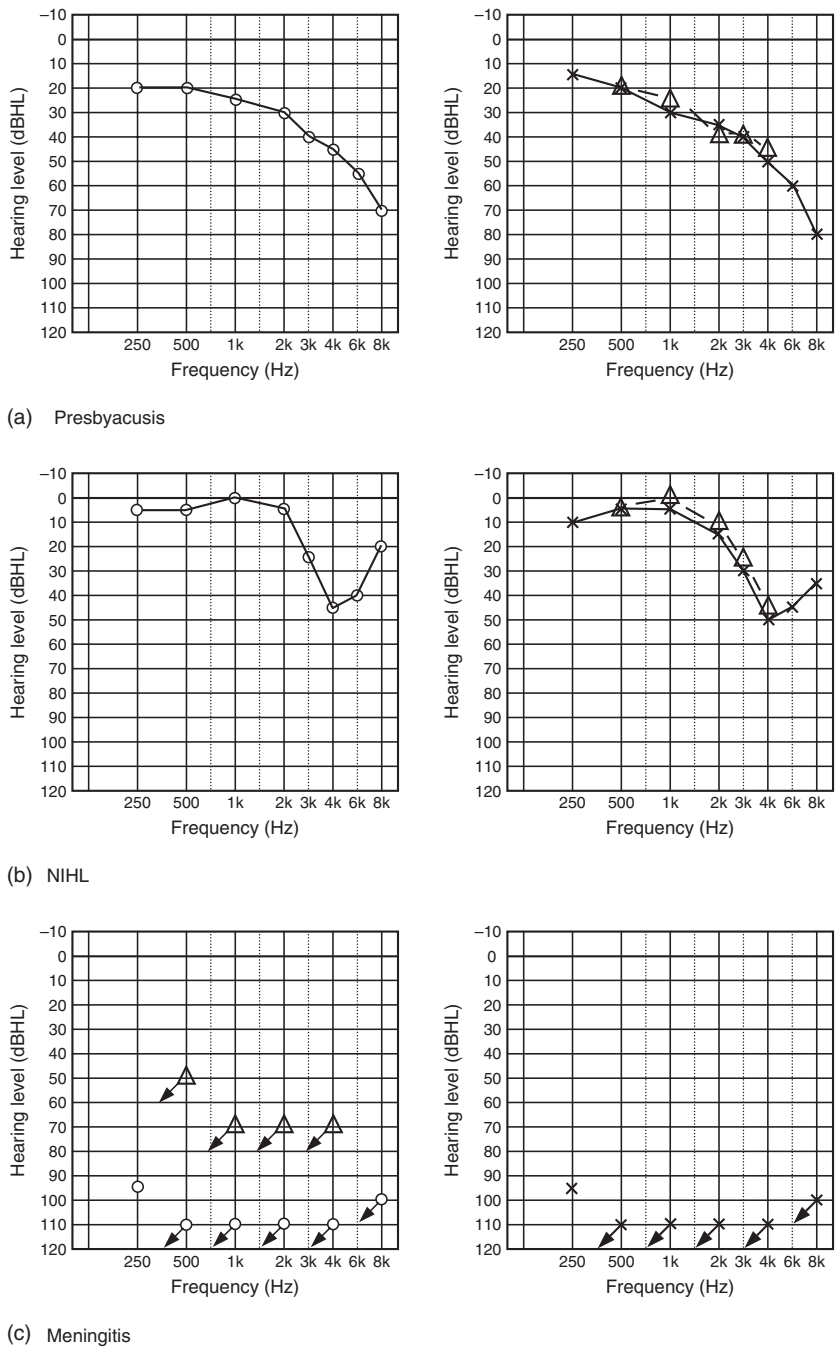


Figure 11.11 Examples of audiogram configurations for various conductive conditions: (a) Otitis media; (b) Otosclerosis.

Tinnitus assessment

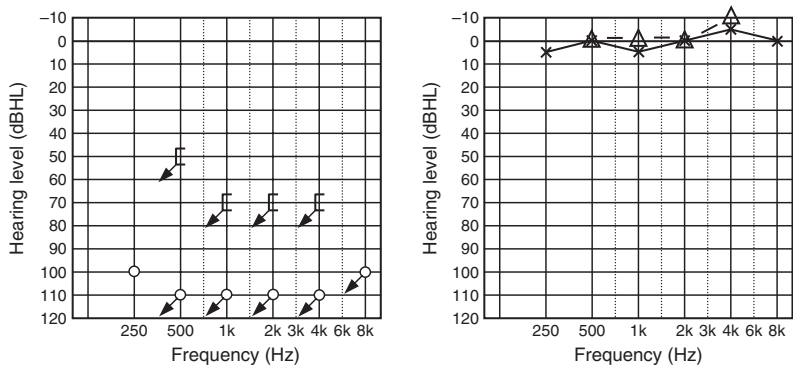
Tinnitus assessment is usually undertaken by an audiologist. There are many ways of assessing tinnitus, but the two most common involve:

1. Recording the employee's subjective account, often as part of the questionnaire.
2. Matching the pitch and loudness using an audiometer. A number of methods are available for pitch matching using the audiometer but the one most commonly used is the adaptive (bracketing) method.

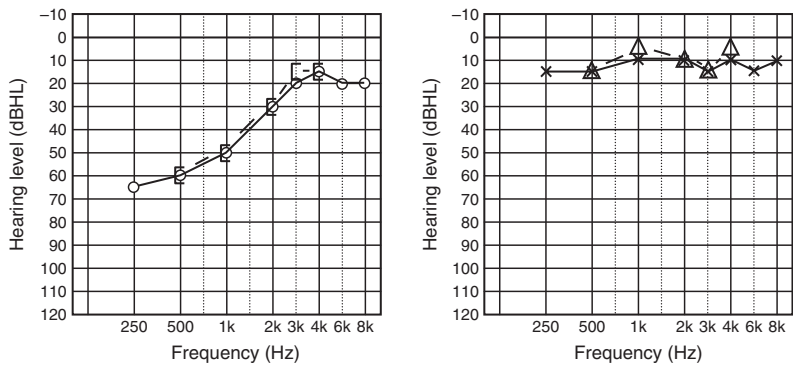


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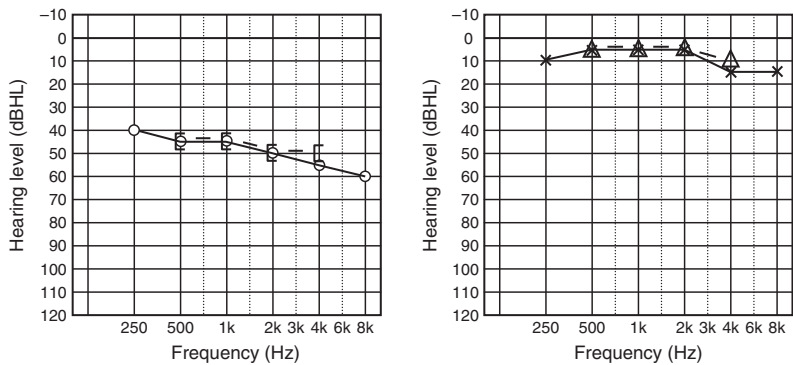
Figure 11.12 Examples of audiogram configurations for various sensorineural conditions.



(d) Mumps



(e) Ménière's disorder



(f) Acoustic neuroma

Figure 11.12 (continued)

- Frequency matching involves presenting tones in turn, starting at one extreme of the audiometer range (e.g. 8 kHz) and at a level estimated to be comfortable. The selection of 8 kHz or 125 Hz is made on the basis of the patient's description of their tinnitus. The tone is presented at 20 dB above the pure tone hearing threshold level for about 2 seconds and repeated as necessary allowing time between presentations for the patient to make the necessary comparison. If the patient says that their tinnitus is lower, the audiologist presents the next tone at the lowest frequency available on the audiometer (e.g. 125 Hz). The gap between the frequencies presented is narrowed, a step at a time, until the pitch nearest to the tinnitus has been located.
- Loudness matching is normally performed at the frequency nearest that of the tinnitus. One method for loudness matching involves presenting the tone initially at threshold (based on the audiogram). The tone is then held for 2 seconds and increased gradually in 5 dB steps until the patient judges the tone to be of equal loudness to the tinnitus.

It is important to realize that severe tinnitus may appear very quiet when matched to external sounds. Levels as low in intensity as 10 dB above threshold can be perceived as being very severe, perhaps because of the inescapability, or possibly due to the effects of recruitment (abnormal loudness growth). Although the effect of tinnitus is subjective, its severity can be judged on a graded scale, such as that put forward by McCombe *et al.* (1999) (Table 11.3).

Tinnitus frequently causes stress, anxiety and poor concentration. Patients may feel their tinnitus impairs their ability to hear clearly. This may be due to an associated hearing loss, rather than the tinnitus itself. Careful history-taking and assessment are important. The case history may include such questions as:

- How long have you had tinnitus?
- What does it sound like? (e.g. ringing/whistling/whining/humming/buzzing/roaring/rushing/ticking/clicking)
- Is it pulsating/continuous/occasional?
- Is it in one or both ears or centrally in the head? (commonly the left ear is reported to be the most affected (González and Fernández, 2004))

Table 11.3 A scale of severity for tinnitus

Scale	Psychological reaction to tinnitus
1. Slight	Tinnitus heard only in quiet, very easily masked, not troublesome.
2. Mild	Forgotten during activities, masked by environmental sounds, may sometimes interfere with sleep.
3. Moderate	Noticed even with background environmental noise, less noticeable when concentrating, can carry on with daily tasks, sometimes interferes with sleep and quiet activities.
4. Severe	Present almost all of the time, can interfere with quiet activities and ability to carry out daily tasks, complaint recorded by GP, hearing loss likely (but not essential), disturbs sleep pattern.
5. Catastrophic	Tinnitus symptoms severe, documented medical consultation, hearing loss likely (but not essential), associated psychological problems recorded in medical notes.

- How severe is it and how does this affect you? (e.g. does it stop you from sleeping/prevent you from concentrating/cause stress?)
- When is it most troublesome?
- What if anything makes the tinnitus worse? (e.g. silence/loud noise/stress/certain foods).
- Have you received any medical advice or help with regard to your tinnitus?
- Have you been exposed to any loud noise or taken any relevant medication?

In general, tinnitus is worst when there is no external noise to mask or cover it. Some relief may therefore be achieved through the use of masking instruments. These usually have the appearance of hearing aids but emit masking noise (often a rushing sound, known as 'white noise') at a low level for the patient. Masking noise can be presented alone for someone with normal hearing or combined into a hearing aid where there is a hearing loss. Relaxation training therapy may also be helpful for tinnitus sufferers. Relaxation does not act directly on the tinnitus and can be most successful if relaxation techniques are used generally, not only when the tinnitus is bad. If the tinnitus seems intrusive during the quiet of relaxation sessions, a background of quiet music will usually help to overcome this problem.

Testing for malingers

Non-organic hearing loss is one that is not of organic origin. It may be feigned or exaggerated, often for financial gain, but it may also be a true hearing loss (psychogenic). Psychogenic hearing loss is a rare genuine condition in which the hearing loss is of psychological origin. It is usually due to hysteria and at one time was termed 'hysterical deafness'. This kind of hearing loss is usually bilateral and profound. Generally, non-organic hearing loss relates to malingering.

Malingering is the deliberate faking of a hearing loss for personal gain, usually for compensation purposes. Many malingers do have a genuine hearing loss but exaggerate the loss to increase their claim, others sometimes pretend to have a one-sided or unilateral loss.

Non-organic hearing loss should be suspected whenever results of the different parts of the hearing assessment are at variance, for example:

- Repeat audiograms vary by more than 10 dB
- Excellent speech discrimination despite a hearing loss that seems severe
- No evidence of cross-hearing in a unilateral hearing loss
- No response to bone conduction when the bone vibrator is placed on the 'deaf' ear
- A flat audiogram across the frequency range.

The audiologist has to determine whether a hearing loss exists and what are the true threshold levels. Many tests for non-organic hearing loss exist. Most tests set out to confuse the patient in order to provide evidence of non-organic hearing loss. Some tests also indicate the approximate hearing threshold. Tests include the following.

Attention raising techniques

Attention raising techniques may be used to obtain true thresholds. These are very simple and are most widely used for testing children. The patient is asked to say 'yes' when they hear a tone and 'no' when they do not. Many unpractised people will say 'no' to presented tones that are below their feigned threshold. As long as the audiologist is sure no visual or timing clues have been given, true thresholds may be obtained in this way. Unfortunately more complex tests are usually (although not always) required to ascertain true thresholds for adults.

The Lombard test

The Lombard test is based on the principle that a normally hearing person will raise their voice in the presence of background noise. The patient is asked to read aloud and, at some unannounced time, masking noise is introduced and gradually increased in intensity. If the hearing loss is genuine, the noise will have no effect until it at least exceeds the deafness. If the hearing loss is faked, the patient will raise their voice without realizing it.

The delayed speech feedback test

The delayed speech feedback test involves the use of a tape recorder that has separate record and playback heads. The patient's voice is recorded as he or she speaks and played back with a very slight delay of 0.1–0.2 seconds. The creation of delayed feedback disturbs the speaker's speech pattern, causing slowing, stuttering or other difficulty. If the hearing loss is genuine, delayed feedback at intensities below threshold will have no effect on the speech.

The Stenger test

The Stenger test is the most widely used to identify monaural non-organic hearing loss. It employs the principle that, where two tones of the same frequency are presented simultaneously, only the loudest one is heard.

A two-channel audiometer is used to introduce a tone 10 dB above the threshold of the 'better' ear. The patient should respond. A tone is then presented to the 'deaf' ear 10 dB below its given threshold. Tones are presented simultaneously to both ears, 10 dB above the threshold of the better ear and 10 dB below the threshold of the 'deaf' ear.

A patient with a genuine hearing loss will continue to respond to the tone that is 10 dB above threshold in the better ear. If the patient does not respond, this is because he or she can only hear the tone in the 'deaf' ear, which he or she refuses to admit. The patient can only hear the loudest tone and does not realize there is still a tone above threshold in the 'better' ear. The test can be continued to reveal approximate true thresholds.

Speech tests

There are many different speech tests available. In general, speech tests require the listener to repeat lists of words or sentences given at varying sound levels. The lowest sound level at which the listener achieves 100% (or their best score) should correlate with the hearing threshold. Where it does not, a likely reason may be malingering. One simple speech test which (though it would not be used in diagnostic clinics) could be used by an occupational health professional to provide some indication of possible malingering, when combined with other results and observations, is as follows: A list of spondaic words (words with equal stress on each syllable) is presented. For example blackboard, toothbrush, retail, football, mushroom, bankrupt, grandma, touchstone, gridlock, freefall, ballcock, mousetrap, dogleg, tailgate, footbridge, birthday, birdbath, wholesale, taxbreak, grapefruit.

The first item in the list is given at a level which is easily heard and with each succeeding word the level is decreased by 5 dB. The listener must repeat each word in turn. The threshold of understanding the words should be 5–10 dB above the average pure tone threshold.

Cortical evoked response audiometry (CERA)

Cortical evoked response audiometry is an objective hearing test which is often useful in the estimation of hearing thresholds in medicolegal cases. Electrical activity can be generated or 'evoked' from the central nervous system in response to sound stimulation. This evoked activity is recorded and separated from random brain activity to give a waveform (Figure 11.13) which can be evaluated

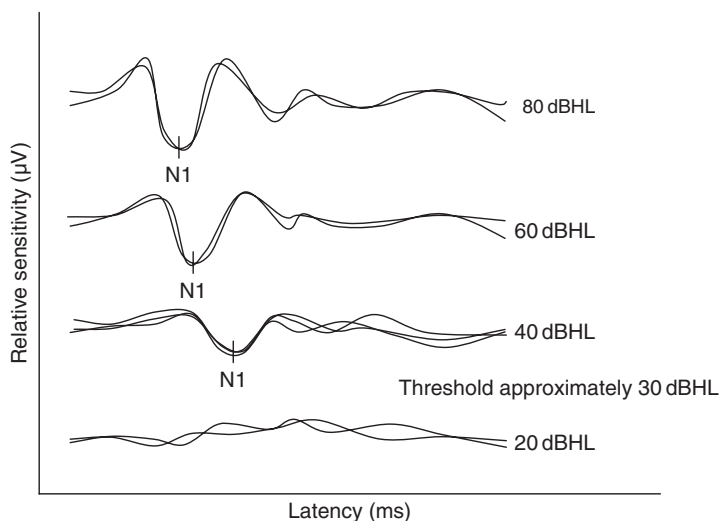


Figure 11.13 Waveforms obtained through cortical evoked response audiometry.

by a trained audiologist. In order to carry out CERA successfully, the patient must be passively co-operative but also awake; excessive movement or sleep can alter the electrical activity and interfere with obtaining the hearing test results.

Summary

Diagnostic audiometry seeks to obtain accurate information regarding both the degree and the type of hearing loss to assist in reaching a diagnosis as to the cause of hearing loss. Pure tone audiometry will include air conduction and bone conduction tests. Masking will be used where necessary to ensure the results have not been affected by cross-hearing and therefore mistakenly accepted as correct. Hearing losses may be described as being mild, moderate, severe or profound, depending on the level of hearing loss present. With regard to type of hearing loss, they may be due to sensorineural or conductive causes or a combination of the two, which is termed a 'mixed loss'. Several further diagnostic tests may be carried out to provide more information. One test which is usually undertaken as part of the diagnostic battery of tests is tympanometry, which is a quick, easy and non-invasive test that may suggest the possible cause of any conductive element in the hearing loss. A number of tests also exist for suspected cases of malingering, which is relatively common where claims are being made for compensation. These tests attempt to establish whether or not there is a hearing loss and ideally to establish the true hearing thresholds.

Further reading

- British Society of Audiology (2004) *Recommended Procedures* BSA, 80 Brighton Rd, Reading, RG6 1PS.
- Roeser, R.S., Valente, M. and Hosford-Dunn, H. (2000) *Audiology Diagnosis*, Thieme.
- Tate Maltby, M. (2002) *Principles of Hearing Aid Audiology*, 2nd ed., Whurr Publishers.

Rehabilitation and compensation

The meaning of hearing loss

The effects of noise induced hearing loss (NIHL)

Even a low level of noise, for example 85 dBA, will typically produce a hearing loss of around 10 dB after 10 years of exposure (Institute of Sound and Vibration Research, 1994). Higher levels of noise exposure will cause a greater degree of hearing loss (Table 12.1). Noise induced hearing loss generally affects mainly the higher frequencies. The greatest hearing loss is usually centred on 4 kHz, although the first changes in young people, exposed to noise for up to about two years, could be at 6 kHz with this moving to 4 kHz after about two to five years of exposure (McBride and Williams, 2001).

In some work environments, it is also possible that the greatest loss may be more common at other frequencies, especially 6 kHz. The Health and Safety Executive suggest that intense low frequency noise may cause maximum hearing loss in the 500 Hz region and intense high frequency noise loss at 6 or 8 kHz. The following examples of noise induced hearing loss with varying frequency notches have been noted:

- Musicians tend to have a loss centred on 6 kHz (Wright Reid, 2001).
- Fitters who use a screwdriver from the mastoid to the machine for diagnostic purposes (e.g. to hear grinding) may have a wider 'notch' with greatest damage across the range of 3 to 6 kHz.
- Gold miners may have a wider 'notch' with greatest damage across the range of 3 to 6 kHz (Soer *et al.*, 2002).
- Aircraft engineers may have a higher incidence of the greatest hearing loss centred on 6 kHz.

Table 12.1 The typical effect of noise on hearing loss over a 10-year period

Noise level (dBA)	75–79	80–84	85–89	90–94	95–99	100–109	110+
Median hearing loss (dB)	2.6	5.9	10.1	15.0	20.5	26.6	45.0

- Impulse noise, such as a drop forge hammer or an air blast circuit breaker may cause a 6 kHz dip (McBride and Williams, 2001).
- Soldiers exposed to light firearms may show a dip at 6 to 8 kHz (McBride and Williams, 2001).

Although a 6 kHz dip on the audiogram may be linked to exposure to certain types of noise, it must be treated with caution as a feature of NIHL because it has also been suggested that a 6 kHz notch could be a common incidental finding unrelated to exposure to noise. There are two possible reasons for this:

1. A 6 kHz notch may be apparent if the headphones are not correctly aligned with the ear canal (Flottorp, 1995).
2. Normal hearing was standardised with reference to the hearing of a group of otologically normal young adults (BS EN ISO 389–1: 2000). As human hearing is not equally sensitive across the frequency range, the average SPL was found at each frequency (Table 12.2) and called 0 dBHL. ‘Normal’ hearing should therefore show as a flat line graph on the audiogram but it may be that the reference standard at 6 kHz was set several dB too low, which could have the effect of producing a 6 kHz notch on a normal audiogram (Robinson, 1988).

Noise induced hearing loss may be accompanied by other problems, including tinnitus, recruitment, hyperacusis and diplacusis. Tinnitus is common with noise induced hearing loss and is reported by at least a quarter of those people who report noise induced hearing loss. Tinnitus may also occur with noise exposure in the absence of any hearing loss and it has been suggested that twice as many people may suffer from tinnitus as do from noise induced hearing loss (Health and Safety Commission, 2004).

Table 12.2 The average threshold sound pressure levels (SPLs) given by the British Standard (BS EN ISO 389-1: 2000) as equivalent to 0 dBHL

Frequency (Hz)	Equivalent SPL	dBHL level equivalent
125	45.0	0
250	25.5	0
500	11.5	0
1 k	7.0	0
2 k	9.0	0
3 k	10.0	0
4 k	9.5	0
6 k	15.5	0
8 k	13.0	0

The effects of hearing loss on speech discrimination

Someone with a high frequency hearing loss (i.e. most sensorineural hearing losses, including noise induced) will first notice a loss of hearing for high pitched sounds such as the door bell and the telephone. In time, speech will also become increasingly difficult to follow. The higher frequencies are very important for hearing speech clearly. There are also particular problems, with high frequency hearing loss, in hearing in background noise. This lack of clarity is due to reduced hearing for the consonant sounds, which carry most of the meaning in English. The frequencies from 500 Hz to 4 kHz are most important for understanding speech.

Spoken words consist of vowel and consonant sounds:

- Vowels are lower in frequency and louder than consonants. Vowels give speech its volume, rhythm and intonation.
- Consonants are higher in frequency and quieter than vowels and are therefore easily 'lost', or 'masked', in noisy situations. Consonants carry most of the meaning in speech. Those that are particularly difficult to hear include: t, p, h, f, k, s, th.

It is most difficult to hear in noisy situations. In quiet conditions and in one-to-one situations, many people with high frequency deafness, such as that experienced with noise induced hearing loss, can manage quite well. In group conversations or in conditions of background noise, speech and noise tend to merge together so that it can be much more difficult, or even impossible, for someone with a hearing loss to separate the sounds they want to hear from those they do not. This is partly due to the defective hearing mechanism, which is particularly noticeable where there is cochlear damage (as in noise induced hearing loss). It may also be due to hearing better in one ear than the other, as good binaural hearing is needed to hear well in noise.

The effects of hearing loss on the ability to work

Hearing loss may cause difficulties in the workplace. The degree of difficulty will obviously be related to the level of hearing loss. The greater the individual's deafness, the more severe will be the problems. There may be difficulties with:

- hearing warning signals
- misunderstandings, for example, of instructions
- ability to hear adequately when wearing ear protection
- hearing clearly on the telephone
- additional stress, strain and fatigue
- communicating in background noise
- hearing in meetings
- communicating with colleagues
- following groups conversations.

Compensation

Disability terminology

The original International Classification of Functioning, Disability and Health (ICF) (World Health Assembly, 1980) suggested the use of the three terms to classify functioning and disability:

1. Impairment – ‘Any loss or abnormality of a psychological or anatomical structure or function.’ With regard to hearing loss, this refers to the hearing loss in decibels as shown on an audiogram.
2. Disability – ‘Any restriction or inability (resulting from an impairment) to perform an activity in the manner or within the range considered normal for a human being.’ With regard to hearing loss, this refers to an inability to hear speech. Hearing disability is given as a percentage. Many schemes of assessment use an indirect scale of disability, derived from data based on the average speech perception ability of test groups compared with their degree of hearing loss (King *et al.* 1992).
3. Handicap – ‘Any disadvantage for a given individual, resulting from an impairment or a disability, that limits the fulfilment of a role that is normal . . . for that individual.’ With regard to hearing loss, this refers to limitations in the fulfilment of the life role of the hearing impaired individual. This means the degree of disadvantage a person suffers, which is highly individual and difficult to quantify. It depends on such things as gender, age, social and cultural factors, for example an elderly person who is unable to go out would be unlikely to suffer the same degree of handicap as a younger person in employment and with an active social life. Handicap is usually ‘measured’ using questionnaires that rate the degree of difficulty perceived by the individual in various situations.

Although the ICF (World Health Assembly, 2001) has now moved away from a ‘consequences of disease’ classification to a ‘components of health’ classification, the original classification remains helpful in understanding human functioning and its restrictions in relation to hearing loss. (An overview of the new classification is given in Figure 12.1.) With regard to compensation, it may be useful to consider that a hearing impairment at an early stage may not necessarily be noticed, does not usually affect functional progress and is unlikely to be compensated. A hearing impairment becomes a disability when it bothers the person, but it becomes a handicap when it affects the person’s ability to function in work or life generally. Compensation is awarded for the handicap or reduction in quality of life that the individual suffers. However, it is much easier to calculate disability than individual handicap. Out-of-court settlements and pensions are usually calculated according to the disability, whilst court settlements are more likely to consider the effect of the hearing loss on the individual, their work and their lifestyle.

A single figure of hearing level will provide a guide as to how easy or difficult it is to understand conversational speech, as shown in Table 12.3, and is often

	Components	Domains	Constructs	Positive aspect	Negative aspect
Part 1: Functioning and disability	Body functions and structures	Body functions and structures	Change in body functions or structures	Functioning Activities Participation	Impairment, limitations
	Activities and participation	Life areas (tasks and actions)	Capacity to execute tasks in standard environment and performance in current environment		Disability
Part 2: Contextual factors	Factors in environment	External influences on functioning and disability	Impact of features of world (physical, social, attitude)	Facilitators	Barriers, hindrances
	Personal factors	Internal influences on functioning and disability	Impact of personal attributes		

Figure 12.1 An overview of the current International Classification of Functioning, Disability and Health.

Table 12.3 A guide to hearing level and hearing disability

Hearing level (dBHL)	Speech understanding	Degree of disability
Less than 25	No difficulty	No significant disability
25–39	Difficulty with faint speech	Slight disability
40–54	Difficulty with normal speech	Mild disability
55–69	Difficulty with loud speech	Marked disability
70–89	Unable to understand speech unless amplified	Severe disability
90+	Unable to understand well even amplified speech	Extreme disability

used to assess hearing disability. Although there is no agreed standard way of calculating the single figure, it is usually taken as the mean or average of the hearing loss at 1 kHz, 2 kHz and 3 kHz.

A guide to hearing level and hearing disability

Claims and their calculation

Claims normally have to be brought within five years after discovering the problem in hearing, although there is no maximum limit on how long it may be before the hearing problem is discovered. There are several methods of calculating the

degree of hearing loss and the percentage disability for the purpose of compensation. The British Standard method of estimating hearing handicap (BS 5330:1976) calculates the degree of hearing loss as the average of the hearing thresholds at 1 kHz, 2 kHz and 3 kHz (King *et al.*, 1992) (Figure 12.2). If the hearing loss in both ears combined (binaural), averaged over these frequencies, is equal to or greater than 30 dBHL, this is deemed sufficient to cause a handicap. This assumes that the hearing thresholds in each ear are substantially similar. However, some schemes require an average of 50 dBHL to deem the loss compensable. Other frequency combinations, such as 500 Hz, 1 kHz and 2 kHz, are sometimes used to make the calculation but ‘without demonstrable superiority’ (BS 5330:1976).

The American Medical Association (AMA) and the American Academy of Otolaryngology (AAO, Canadian Centre for Occupational Health and Safety, 2002) use the four frequencies, 500 Hz, 1 kHz, 2 kHz and 3 kHz. Their calculation uses 25 dB as their ‘low fence’, that is the minimum hearing level assumed to cause disability, and 92 dB as equating to 100 per cent disability (Figure 12.3). In reality, there is probably no ‘low fence’, as there is no distinct value of hearing level below which there is zero disability, or following which there is a rapid increase in disability (King *et al.*, 1992). Instead there is a smooth curve of disability against hearing loss, starting from the zero and rising with increasing hearing threshold level. From this curve (Figure 12.2) it can be seen that 20 per cent disability corresponds to 30 dBHL. Many compensation schemes accept 20 per cent

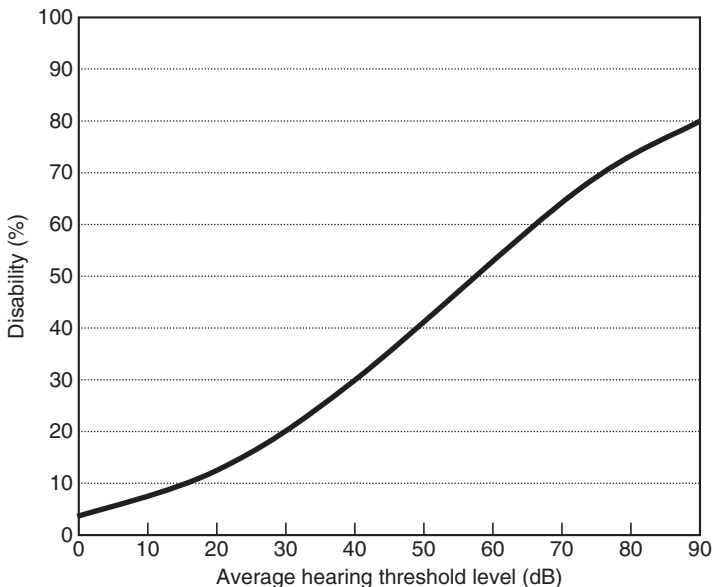


Figure 12.2 The relationship between hearing disability and hearing impairment based on the average hearing level at 1 kHz, 2 kHz and 3 kHz in the better ear.

		Left (dBHL)	Right (dBHL)
Frequency (Hz)	500	35	30
	1 k	55	45
	2 k	80	60
	3 k	90	85
Calculation	Sum of hearing thresholds	260	220
	Average hearing threshold	65	55
	'Low fence'	25	25
	Amount by which low fence exceeded	40	30
	Multiply by 1.5	($\times 1.5$)	($\times 1.5$)
	= Impairment (%)	60	45
Binaural Disability	(100% = 92 dBHL)		
	Multiply better ear by 5	45 $\times$ 5 = 225	
	Add to poorer ear	225 + 60 = 285	
	Divide total by 6	285/6 = 47.5%	
	= Disability (%)		

Figure 12.3 Hearing disability calculated using the AMA/AO formula.

disability as the entry level for compensation but the decibel level taken to equate to 20 per cent disability can vary (e.g. 20 per cent might be 30 dBHL, 40 dBHL or 50 dBHL in different schemes).

The value of compensation will be decided by the courts or, in the case of out-of-court settlements, usually by the pension funds (e.g. War Pensions) or the insurance companies. Figure 12.4 shows an example report from an ENT consultant commenting on the injuries sustained by a worker exposed to noise. The courts are not restricted to considering the frequencies 1 kHz, 2 kHz and 3 kHz and do not have a 'low fence'. They are likely to include higher frequencies when making their decision and may also consider compensation for lesser degrees of hearing loss. The actual amounts awarded vary widely from relatively small sums to many thousands of pounds. For example, damages of approximately £165 000 were awarded to a 56 year old driver of heavy plant vehicles in Wales in 1998 (Clement-Evans and McCombe, 1998). His average binaural hearing loss, calculated over the frequencies 1 kHz, 2 kHz and 3 kHz, was approximately 29 dB and the noise level in which he had been working was agreed as being 111 dB. The award was reduced to reflect hearing loss prior to 1963, which is the date after which it is accepted that an employer should have foreseen the risk of hearing damage. This reduction in award was based on the deduction of approximately 3 dB from the hearing loss, as having been caused pre-1963. The award included an amount for tinnitus, hyperacusis and depression and this was not reduced, as it was thought that, on the balance of probabilities, the man would not have suffered tinnitus or depression if he had not been exposed to noise after 1963. Hearing loss caused prior to the date of employment will also reduce any claim as long as it can be shown that it was a pre-existing condition. Pre-employment testing is therefore extremely important.

Medical Report on: _____

Date of birth: _____

Date of examination: _____

Occupation at time of examination: _____

I am in receipt of written instructions dated _____ from _____, Occupational Health Adviser, _____ Company Ltd, requesting medical examination of the aforementioned worker and to produce a written report stating the injuries sustained as a result of noise exposure, the current state of hearing and an assessment of prognosis with regards to hearing, taking account of any relevant medical history. A copy of the hearing test (pure tone audiogram) dated _____ is attached. The hearing test was performed by _____ (qualifications _____). The test was performed on a correctly calibrated audiometer using standard audiometric testing criteria. I have also reviewed the hospital records relating to attendance at the ENT clinic on _____ (dates) with the accompanying hearing tests.

_____ has been aware of deafness since _____. She feels that the deafness affects both ears and has been progressively worsening but that it is noticeably worse in the _____ ear. She has been aware of bilateral tinnitus for the same length of time. The tinnitus is much worse in the left ear where it is constant and causes significant problems in sleeping at night. She feels that they cannot do their current work as wearing ear defenders (a requirement for the current work) makes the tinnitus worse. She initially consulted the ENT department at _____ hospital in _____ with this complaint. The hearing test at this time confirmed the presence of bilateral high frequency deafness thought to be due to noise exposure. She was given advice with regard to coping with the tinnitus and discharged. She was referred again to the ENT department in _____ where she was reviewed by myself. A repeat hearing test at this time showed no significant change in hearing levels over the preceding five years. Once again, it was felt that she would benefit from a hearing aid. She was reassured to that effect and discharged back to her doctor's care.

_____ reports that she has worked for _____ Company Ltd for _____ years, most of this time in very noisy environments. She reports that she has always worn ear defenders when working near noisy machinery.

I examined _____. Her eardrums appeared healthy and the remainder of the ENT examination was unremarkable. Her hearing test performed on _____ showed bilateral asymmetrical high frequency sensorineural deafness. The hearing loss was indeed worse in the _____ ear. The pattern of hearing loss showed a significant notch at 4 kHz which is a characteristic feature of noise induced hearing loss. Her hearing thresholds show only very marginal deterioration compared with the earlier tests of _____ (date).

The pattern of hearing loss and the history of noise exposure would lead me to conclude that her hearing loss is due to cochlear damage. This is also the likely cause of the constant and sometimes disabling tinnitus. In the majority of people such tinnitus would be treated by way of masking. This is the situation when external noise from outside sources would mask the noise generated by the damaged cochlea. This is proving to be a problem due to the requirement for wearing ear defenders, which in effect cancel any noise from the outside environment and will obviously make her tinnitus much more obvious and less tolerable.

In answer to the questions posed in your letter:

1. The damage sustained by the cochlea is permanent.
2. The hearing will not improve and indeed she will experience gradual deterioration in her hearing due to age related changes.

(continued)

Figure 12.4 An example of a medical report on the injuries sustained by a worker exposed to noise.

3. With regards to treatment, the only option to _____ with regards to her hearing is the use of a hearing aid. With regards to the tinnitus, almost all treatment options revolve around the use of masking, whether by increasing awareness of environmental and background noise or by the introduction of a specific tinnitus masker (a device that looks identical to a hearing aid but is in effect a noise generator).

4. The use of ear defenders will produce a negative effect for this individual as it will mask the background noise necessary for the masking effect used by the ear to suppress tinnitus. In addition, wearing ear defenders whilst using tinnitus maskers is at best difficult and may be impossible.

I hope that the above information answers your questions and enables you to plan suitable employment for _____.

If you have any further questions regarding the above, please do not hesitate to contact me again.

Yours sincerely,

Consultant Otolaryngologist

Figure 12.4 (continued)

The Department of Social Security (DSS) has a limited scheme of compensation for certain specified industries. This compensation scheme averages across the frequencies 1 kHz, 2 kHz and 3 kHz, and uses 50 dBHL as the minimum loss eligible for compensation. Tinnitus may also be taken into account. Negligence does not have to be shown to obtain compensation but other conditions applied are very stringent.

The ‘Quality Adjusted Life Years’ (QALY)

The ‘QALY’ is used by the Health and Safety Commission (2004) to estimate a minimum monetary value of individual hearing loss. This approach is not intended for compensation purposes but considers the impact of hearing loss on quality of life, including the need for hospital treatment and restrictions on work and social activities, and attempts to reflect the actual value of the loss to the individual concerned. The ‘QALY’ index uses 0 to equate to death and 1 to equate to full health. The calculation is based on an annual value (Table 12.4) extended over the period during which the hearing loss is expected to continue. A hearing loss of 50 dB or more over a period of 40 years is said to represent a 10 per cent reduction in life quality and is given a present value of £96 000. A hearing loss between 30 and 49 dB is treated as having half this value and a sliding scale is used for hearing losses less than this. In these calculations, a life of 40 years is assumed after 10 years of noise exposure and life of 10 years is assumed after 40 years of noise exposure.

Table 12.4 Reduction in life quality due to hearing loss and its estimated monetary value

Hearing loss (dB)	Reduction in life quality (%)	Annual value (£)
50+	10	4200
30	5	2100
20–29	2.5	1050
15–19	1	420
10–14	0.25	105
<10	0	0

Auditory rehabilitation

A definition of auditory rehabilitation

Aural rehabilitation is ‘those professional efforts designed to help a person with a hearing loss. This includes services and procedures for lessening or compensating for a hearing impairment and specifically involves facilitating adequate receptive and expressive communication’ (American Speech and Hearing Association, 1984). It may include the identification and diagnosis of the hearing loss, counselling, hearing protection, fitting hearing aids and other related equipment, lipreading or ‘speech-reading’ and communication training, and tinnitus therapy.

Counselling

Most noise induced hearing loss occurs gradually over time and at first it is usually not noticed. The individual may well be unaware of a problem when hearing loss is discovered through audiometric monitoring. However, further investigation will sometimes produce comments such as:

- ‘The wife says I’m deaf’
- ‘People mumble’
- ‘I have to turn the television off to hear what my children say’
- ‘I don’t always hear the telephone.’

Such comments indicate that signs of the hearing problem were present but were ignored.

Counselling has to take account of individual differences, for example in age, stage of life, gender, psychological state, and work, social and life factors. These individual differences will all have an effect on the degree of handicap experienced by the person. This is in addition to variables related to the hearing loss itself. A hearing loss which is sudden, for example due to impact noise or explosion, will have a greater impact than a gradual

hearing loss where the individual has had time to adapt and develop coping strategies.

An audiologist, a hearing therapist and/or a speech and language therapist may undertake a counselling role. However, counselling for the employee will probably start in the Occupational Health Department at work with an explanation of hearing loss and the need for hearing conservation, including the proper use of hearing protection. Where a hearing loss falls into a warning or referral category, the significance of the hearing test results should be explained to the employee. A copy of their audiogram may be made available to all employees, in which case an explanatory sheet that can be handed out with the audiograms may be useful. An example of a written explanation of a Békèsy audiogram is given in Figure 12.5.

Name: _____Date: _____

Please find attached the results of your hearing test.

The information given below is provided to help you to understand these results.

The hearing test is intended to allow us to monitor changes in your hearing from one test to the next and, where necessary, to facilitate changes in your working practices to prevent worsening of any hearing loss. The results you have received consist of a graph, known as an audiogram, and a letter that notifies you of any further action that is needed.

On the audiogram, you will see a zigzag line. This indicates the quietest sounds you can just hear in each ear. The degree of hearing loss can be seen by looking at the left hand side of the graph where the axis shows the volume of the sound measured, in decibels (dB) from -10 to 90 dB. Average normal hearing at age eighteen to thirty will fall in the region of -10 to 20 dB. The further down the graph is the zigzag line, the worse your hearing. Hearing tends to deteriorate as we get older and your hearing may be expected to be poorer than 20 dB if you are over thirty years old. If your hearing is significantly below what is normal for your age, you will be referred for medical advice. This referral is to confirm the hearing test results and investigate the nature and cause of any hearing loss.

1. If the zigzag line falls between 21 and 45 dBHL, this is a slight hearing loss, which will usually cause no problem in quiet conditions but hearing may prove more difficult in background noise.
2. If the zigzag line falls between 46 and 70 dBHL, this is a moderate hearing loss. Usually this does not present too much of a problem when talking one-to-one in quiet conditions or on the telephone. However, the television will often be too loud for other people and hearing in groups and in conditions of background noise may be very difficult. At this level, hearing aids are usually very helpful.
3. If the zigzag line falls between 71 and 90 dBHL, this is a severe hearing loss, which is usually a communication problem even when talking one-to-one in quiet conditions or on the telephone. At this level, hearing aids are usually a necessity and other devices, such as an amplified telephone, may also be very helpful.

Figure 12.5 An example of a written explanation of a Békèsy audiogram.

Use of hearing aids

Hearing aid types

If there is a permanent hearing loss, the employee may need to use hearing aids. Many people find it difficult at first to accept the need for hearing aids and try to hide the fact that they have any hearing problem. It usually takes time (from a few days to several years) for people to come to accept their hearing loss and the need for hearing aids but talking about the problems may help in the adjustment process. To explore hearing-related problems at work, it is important to know about the individual's work environment, their job and the particular tasks in which they are involved, for example do they need to use the telephone or attend meetings. As well as the obvious need to ensure that ear protection is adequate and being worn correctly, there may be assistive listening devices that will help with specific problems. Tinnitus maskers, hearing aids, amplified telephones, inductive loop systems, special microphones (infra-red or radio hearing aids), flashing or vibrating alarms are all examples of devices which may be provided by the individual, the workplace or the government (often through 'Access to Work' at the local job centre).

Most modern hearing aids are digital and of the behind-the-ear, in-the-ear or in-the-canal type (Figure 12.6). Behind-the-ear (BTE, also called postaural) hearing aids consist of a plastic case, containing the electronics, that sits behind the pinna

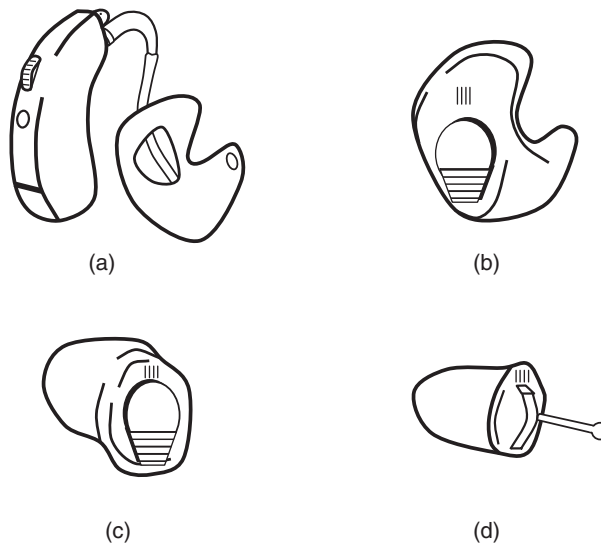


Figure 12.6 Examples of types of hearing aids available. (a) A behind-the-ear (BTE) hearing aid with earmould attached. (b) A full shell in-the-ear (ITE) hearing aid. (c) An in-the-canal (ITC) hearing aid. (d) A completely-in-the-canal (CIC) hearing aid.

and a plastic tube through which the sound is fed to an earmould, which fits in the concha and leads into the ear canal. In-the-ear hearing aids are placed inside the pinna, in the concha and/or canal. All in-the-ear hearing aids consist of only one part, into which the electronics are built. Many hearing aids are very small, the smallest of all being the completely-in-the-canal (CIC) type, which, as their name suggests, fit into the ear canal and are often almost 'invisible' in the ear. All hearing aids make sounds louder and also change the sound quality to some extent. Although they do not provide completely normal hearing they should give considerable benefit. Digital hearing aids encode the sound digitally and can be obtained with automatic functions such as reduction of background noise, automatic volume adjustment and protection from uncomfortably loud sounds. Wearing two hearing aids, especially where both ears have an approximately equal hearing loss, is usually very helpful. Binaural listening, that is using two ears, helps the brain to make sense of what it hears, in particular with regard to hearing in noise and the location of the sound. Hearing aids work less well in conditions of noise and at a distance, although background noise can be reduced to some extent, for example by:

- shutting the door to noisy areas
- turning off the TV or radio when conversing
- using hearing aids with sophisticated noise reduction programmes.

Sound field audiometry

The audiogram records the individual's hearing threshold when hearing aids are *not* worn. Hearing aids, if worn, must be removed before carrying out an occupational hearing test.

If it is necessary to know what an employee can hear wearing their hearing aids, an estimate of the level of hearing with hearing aids (aided thresholds) can be made by adding the amount of 'gain' or amplification provided by the hearing aid, where this is known, to their pure tone threshold levels. Where it is important to know more accurate aided thresholds, 'sound field' or 'free field' audiometry is used. This is a specialist area of testing, which is used most frequently with deaf children to verify hearing aid performance. Warble tones (tones that vary around a central frequency) or narrow bands of noise are presented through loudspeakers. Pure tones cannot be used because they are very prone to interference from reflections of the sound, which can cause significant variations of sound level even with a slight head movement. The sound field equipment must be calibrated and used within an adequately sound-treated room. Test results are normally given in dBA and can only be made directly in dBHL if the room is calibrated such that the dial reading is in dBHL. Alternatively, results can be converted from dBA to an approximate dBHL level by subtracting 5 dB at 4 kHz and 10 dB at 8 kHz. No adjustment need be made at 500 Hz, 1 kHz and 2 kHz.

In general, it may be more useful to observe the employee in the situation where they will be working to see if they can cope adequately in the real conditions.

Assistive listening devices

There are many devices that may be used with or instead of hearing aids to help overcome specific problems. Examples of those that may be relevant to the work situation include:

- The induction loop system is designed to overcome problems of background noise by transmitting the sound directly to the listener. The loop system involves a cable being fitted around the area required and attached to an amplifier. Sound is fed to the amplifier from a microphone or directly from a telephone, television or other device. Many, but not all, hearing aids are fitted with a loop receiver (known as a 'T' switch) that can pick up the signal. Anyone wearing a hearing aid switched to 'T' who walks into the loop area will receive the sound being fed into the system. When the 'T' switch is operative, the microphone is automatically switched off. Background noise is therefore greatly reduced and does not interfere with hearing the sound coming directly through the loop.

Some hearing aids have an 'MT' switch, where the microphone remains active. When using the 'MT' switch, the listener can hear around him or herself in addition to hearing the direct sound from the loop. For example, the listener could hear the church congregation singing, in addition to hearing the minister. The disadvantage of this is that any background noise is not cut out, which could interfere with hearing the direct sound.

If a loop system is to be fitted in a public place it must meet all the requirements of IEC 60118-4 and specialist installation is essential. A domestic loop system is not suitable for use in a public place. Most loop systems are not portable but where different venues are used for meetings, a portable loop system can be obtained. If the loop is only required for the telephone, special telephones can be obtained that include a small loop system within them.

- An amplified telephone may be used by an employee who has a slight to moderate hearing loss and who does not wear a hearing aid or by an employee who has a hearing aid without a 'T' switch.
- Radio hearing aids are designed to overcome the problems of background noise by transmitting the sound directly to the listener using radio (FM) waves. Sound is fed to a small transmitter from a microphone or directly from another device, such as a tape recorder. The system is very efficient, highly portable and usually individual but it tends to be expensive.
- Infra-red systems are similar to radio hearing aids but use infra-red light waves to transmit the sound. Infra-red signals cannot pass through walls and cannot be used outdoors as sunlight causes interference.
- Alarm systems may be obtained that use vibration or flashing lights to alert the hearing-impaired employee. These are generally used for people with severe or profound hearing loss but might sometimes be appropriate with lesser degrees of loss.

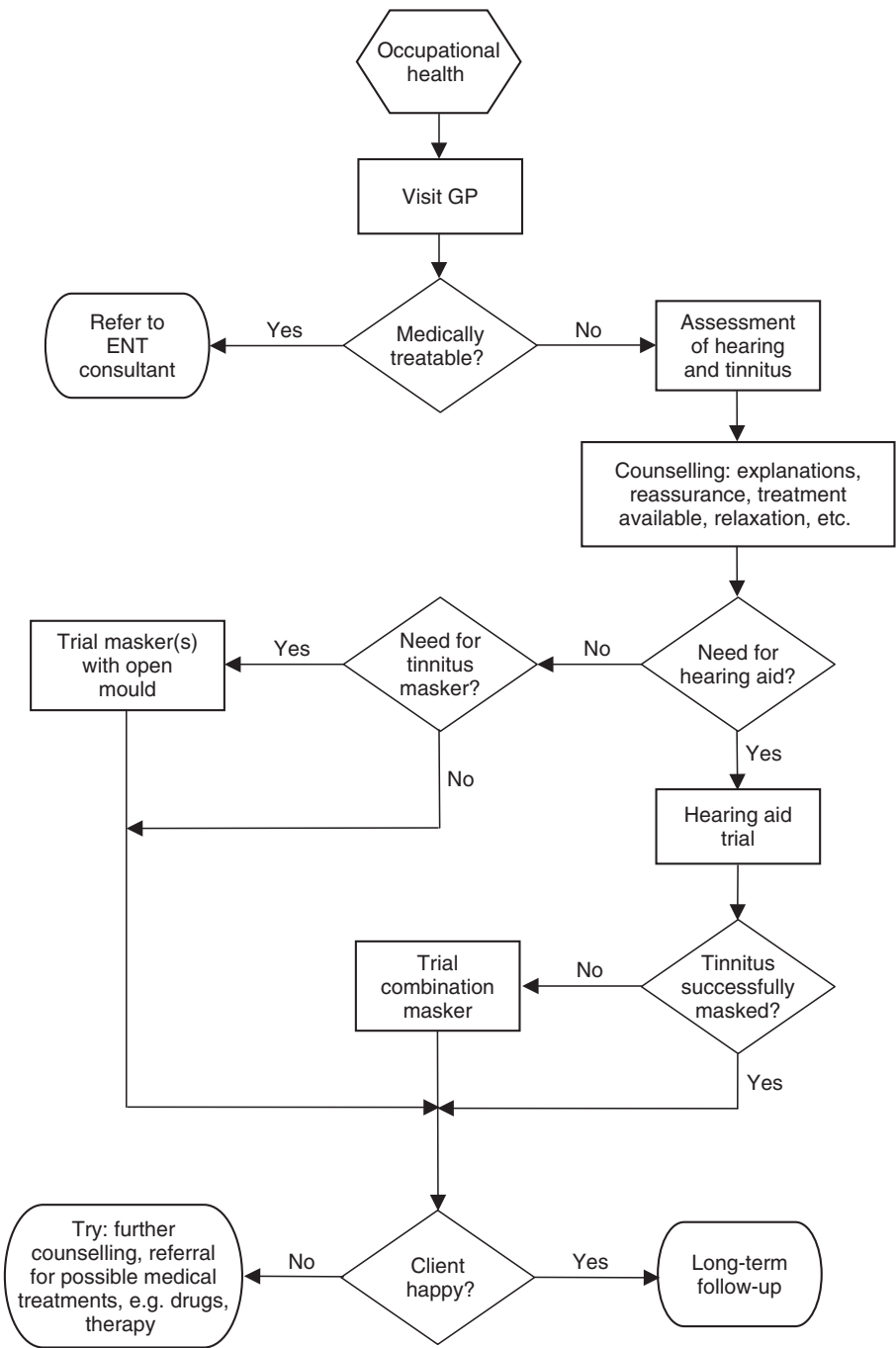


Figure 12.7 A possible route for tinnitus sufferers.

Tinnitus

There are many causes of tinnitus such as wax, high blood pressure and otosclerosis, but tinnitus is a very common complaint after exposure to noise and may occur with noise induced hearing loss or on its own. It is thought that tinnitus is often due to the response of abnormal hair cell activity in the cochlea, in or around the damaged areas. Thus in noise induced hearing loss, there is often tinnitus having a frequency close to 4 kHz, which is usually the area of greatest hearing loss. The level of tinnitus is usually about 5 to 20 dB above the hearing threshold but such levels can be perceived as being very severe to the sufferer.

Tinnitus management

A tinnitus programme can be followed after medical advice has been obtained. Such a programme may be offered through the National Health System or privately, often through a hearing aid dispenser. A possible route for tinnitus sufferers can be seen in Figure 12.7. A tinnitus programme will involve assessment of the tinnitus and its severity and often a trial period with tinnitus maskers. These are devices that look like hearing aids but emit a low-level noise that is intended to 'mask' or cover (wholly or partially) the tinnitus sound. A masker will usually be fitted to the ear using an open ear mould as the use of a solid mould can magnify the tinnitus, as well as reducing environmental sounds that could help to make the tinnitus less noticeable. In addition, a solid mould may reduce the ability to hear generally and thus also hamper communication. Often there is a hearing loss as well as tinnitus and, where this is the case, the use of a hearing aid may amplify background noise to a level that partially masks the tinnitus.

Masking should be used only as much as required and high levels of masking noise should not be used. In a few cases, maskers may cause tinnitus to stop for a length of time following the masking period but, in almost all cases, counselling and relaxation training therapy are an important part of the rehabilitation process.

Summary

Noise induced hearing loss (NIHL) is the result of exposure to high levels of noise. The hearing loss will be sensorineural, high frequency and may show a 'notch' on the audiogram, usually at 4 kHz. A notch found at 6 kHz may be indicative of noise induced hearing loss, but some caution should be evidenced when reaching this conclusion. The hearing loss is often accompanied by tinnitus. With a typical noise induced hearing loss, it is often possible to manage quite well in quiet conditions and when conversing on a one-to-one basis, but it may be difficult to follow group conversations and to separate words from background noise. Compensation for noise induced hearing loss is usually calculated as a percentage disability based

on the hearing levels averaged across the frequencies 1 kHz, 2 kHz and 3 kHz. Auditory rehabilitation may start in the occupational health department with advice on ear protection and explanations of hearing loss and will often go on to involve professional help and the use of hearing aids.

Further reading

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- Hazell, J. (2000) *Tinnitus*, Churchill Livingstone.
- Jastreboff, P.J. and Hazell, J.W.P. (2004) *Tinnitus Retraining Therapy*, Cambridge University Press.
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Background Science

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Basic anatomy and physiology of the ear

Introduction

The ear (Figure 13.1) can be thought of as being divided into three parts known as the outer ear, the middle ear and the inner ear:

- The outer ear consists of the pinna or auricle, the ear canal and the eardrum.
- The middle ear is an air-filled cavity within the skull. A chain of three bones, or ossicles, runs across the middle ear cavity and connects the outer ear to the inner ear.
- The inner ear contains the cochlea, or organ of hearing, and the balance mechanism. Signals from the nerve endings in the cochlea are transmitted along the auditory nerve to the hearing centres of the brain.

The purpose of the ear is to enhance and transmit sound to the coding mechanism of the brain, which will interpret the meaning of the sound.

The outer ear

The pinna

The part of the ear, commonly referred to by members of the public as ‘the ear’, is the pinna or auricle, which is a convoluted structure of pliable cartilage covered by a tight layer of skin. Anatomical names for features of the pinna are shown in Figure 13.2. The main function of the pinna is to collect sound waves, particularly from a forward direction, and to funnel them into the ear canal. The shape of

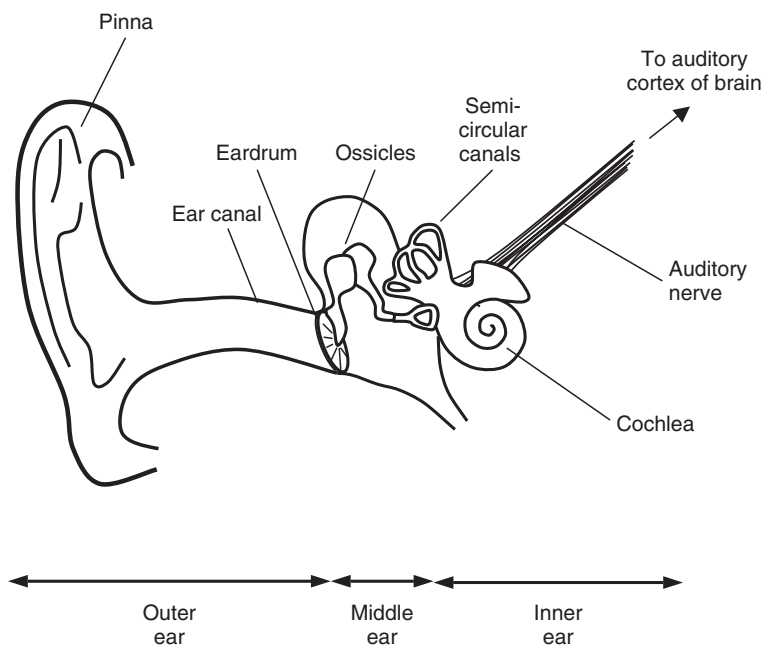


Figure 13.1 The divisions of the ear.

the pinna causes the sound to be amplified by about 5 dB, especially in the higher frequency region around 5.5 kHz. The pinna also plays an important part in the localisation of sound. If the pinna is absent, there is a loss of sound reception of about 5 dB and localisation is more difficult.

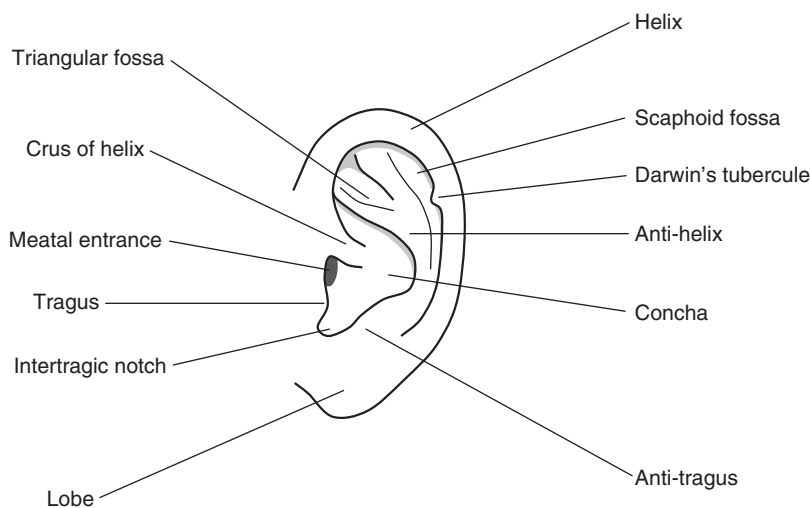


Figure 13.2 Anatomical features of the pinna.

The ear canal

The cartilage from the pinna extends to form the first one-third of the ear canal or external auditory meatus. The inner two-thirds of the canal is formed of bone. The whole canal (Figure 13.3) is covered by skin which becomes very thin in the deeper parts of the ear canal. Sound resonates within the canal, causing high frequencies around 3 kHz to be amplified by about 10 to 15 dB.

The canal is approximately 2.5 cm long and ends at the eardrum. It is not a flat tube, the outer one-third runs slightly upwards and backwards, whilst the inner two-thirds run downwards and forwards. Hence it is important to lift the pinna upwards and backwards when viewing the eardrum. Near to the eardrum the floor of the canal dips to form a small recess where debris sometimes collects. The external ear is well supplied with sensory nerves and the inner bony part of the ear canal is particularly sensitive.

The outer (cartilaginous) part of the ear canal contains hairs and three types of glands:

1. Ceruminous or wax glands
2. Apocrine or sweat glands
3. Sebaceous or oil glands.

These glands secrete wax, which consists of cerumen mixed with sweat and oil. The wax mixes with skin debris and should migrate naturally outwards, eventually falling out of the ear. Wax is colourless and moist when first secreted but dries out and darkens in colour over time. The function of wax is to help to keep the ear canal clean and to moisturise the air. The fine hairs near the entrance of the canal also help to keep the ear clean and to exclude foreign bodies.

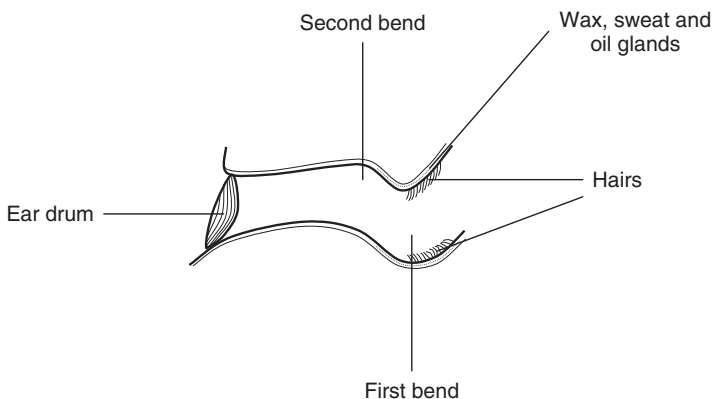


Figure 13.3 The ear canal viewed from above.

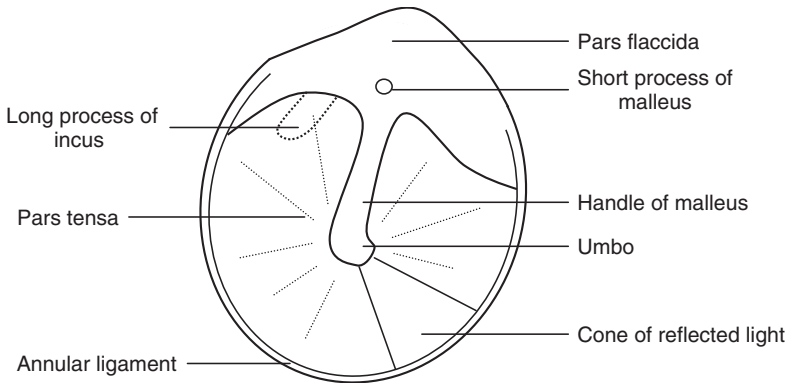


Figure 13.4 The eardrum.

The eardrum

The eardrum or tympanic membrane (Figure 13.4) is an elastic membrane found at the end of the ear canal, separating the outer ear from the middle ear. It is composed of three layers:

1. An outer skin or epithelial layer
2. A middle fibrous layer
3. An inner mucosal layer.

The eardrum lies at an angle of about 55° such that the roof of the ear canal is shorter than its floor, and the drum has a larger surface area than it would have if it were vertical. Across the eardrum runs a branch of the facial nerve known as the chorda tympani. The eardrum is divided into an upper and a lower section. The upper section is the smallest section and here the fibrous layer is deficient; hence the area is flaccid or lacking in elasticity and is known as the pars flaccida. The lower section is tense and elastic and is known as the pars tensa.

The eardrum vibrates in a complex manner in response to sound waves and changes or ‘transduces’ acoustic energy into mechanical vibrations, which are passed on through the bones in the middle ear.

The middle ear

The middle ear (Figure 13.5) is an air-filled cavity beyond the eardrum (ventilated by a tube known as the Eustachian tube) that contains a chain of three bones or ossicles supported by ligaments. The bones are known as the malleus (hammer), incus (anvil) and stapes (stirrup). The handle of the malleus is attached to the eardrum and the footplate of the stapes sits snugly against the oval window. The oval window and the round window are small

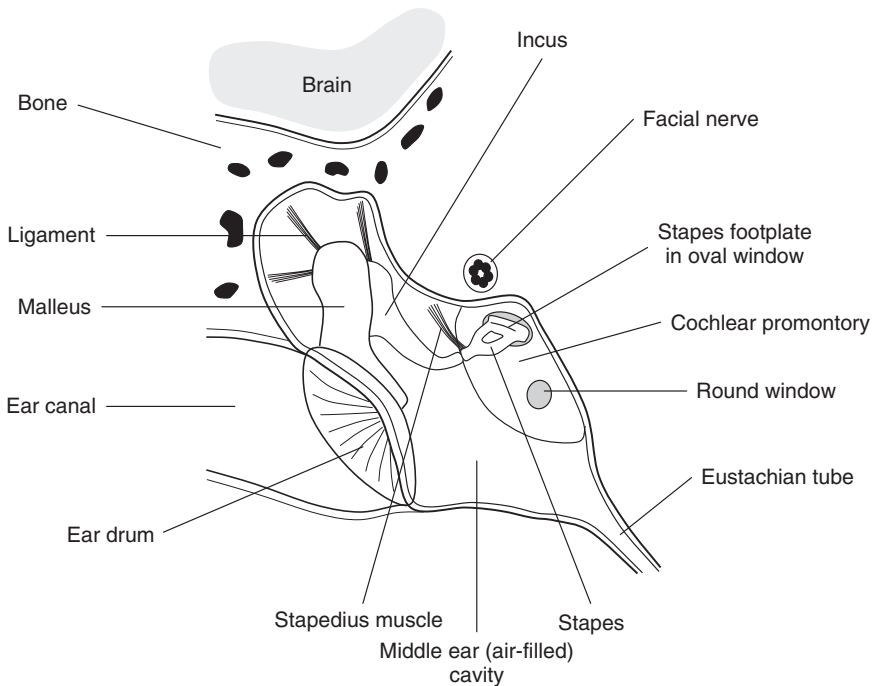


Figure 13.5 The middle ear.

membrane-covered holes situated in the cochlear wall. The middle ear is surrounded by important features, such as the mastoid air cells, the temporomandibular joint, the jugular vein, the carotid artery, the facial nerve and the inner ear; even the brain is only separated from the middle ear by a thin plate of bone. The close proximity of these areas means that they are susceptible to injury or infection in the middle ear and to a lesser extent in the outer ear.

There are two muscles in the middle ear, the tensor tympani muscle and the stapedius muscle. When there is a loud sound, the muscles contract and stiffen the chain of bones so that they are less efficient at passing the sound vibrations. This action is particularly efficient in the low frequencies, below 250 Hz, which may be reduced by as much as 20 dB, but the muscles have little effect with high frequency sounds above 1 kHz. This muscle reflex is also ineffective with transient loud sounds, such as gunfire, because of the small delay between receiving the sound and the action to contract the muscles. The acoustic reflex action is found with loud sounds and, with normal hearing, occurs at about 70 to 90 dBHL.

The cochlea is filled with fluid. Fluid is a denser medium than air and therefore sound will not pass easily from air into fluid. The function of the middle ear is to transfer the sound energy arriving at the eardrum efficiently into the fluids of the

cochlea. In order to achieve this, the middle ear builds up the sound pressure by about 28 dB. Some sound will also reach the cochlea directly via the bones of the skull. The middle ear increases the sound pressure in three ways:

1. The area of the oval window of the cochlea is much smaller than the area of the eardrum. The effective fibrous area of the eardrum is approximately 55 mm², whilst the area of the oval window is only 3 mm². This difference in area enhances the sound pressure at the oval window by a factor of about 18. (The effect of applying a force to a smaller area can be understood by thinking, for example, of the pressure of the body on a stiletto heel.)
2. The three bones act as a series of levers. This gives a mechanical advantage of approximately 1.3.
3. In addition, there is a small enhancement of high frequency sounds due to the characteristics of the eardrum, which concentrate high frequency energy at the centre of the drum, at the umbo.

The middle ear can only work effectively if the pressure inside the middle ear is the same as the pressure in the air outside. Any pressure change will reduce movement of the eardrum. Pressure equalisation is maintained by the Eustachian tube, which runs from the middle ear to the nasopharynx, adjacent to the adenoids. The Eustachian tube is normally closed but it opens to allow air in and out upon swallowing, yawning or blowing the nose. The function of the Eustachian tube is ventilation of, and drainage of mucus from, the middle ear. In young children below the age of about eight years, the Eustachian tube is more horizontal, narrower and less rigid. It is therefore more prone to collapse and also to dysfunction with infection, which can pass up the tube from the upper respiratory tract.

The middle ear also protects the delicate inner ear structures to some extent from potentially damaging noises. When there are loud sounds, especially of low frequency, the stapedius muscle contracts causing the ossicular chain to stiffen and this reduces movement of the stapes in the oval window. Contraction of the stapedius muscle can produce attenuation of over 20 dB at frequencies below 250 Hz but for high frequencies (over 1 kHz) attenuation is negligible.

The inner ear

Hearing and balance

The inner ear consists of all those parts of the auditory system beyond the middle ear, that is the labyrinth, the auditory nerve and the auditory cortex of the brain.

The cochlea and the semi-circular canals make up a fluid-filled cavity, the labyrinth, which is part of the skull. The cochlea is most relevant to hearing, whilst the three semi-circular canals are concerned with balance and are part of the vestibular system. The semi-circular canals lie in three planes at right angles to each other and contain special sensory cells, which send information about

posture and balance along the vestibular branch of the eighth cranial nerve to the brain.

Sound waves pass from the middle ear through the oval window to the cochlea. Here they are converted into electrical signals that travel along the auditory branch of the eighth cranial nerve to the brain.

The cochlea

The cochlea looks rather like a snail's shell and has two and three quarter coils around a central bony core or pillar called the modiolus. The bony labyrinth of the cochlea contains a membranous sac, which effectively divides the cochlea into three chambers (known as the scala vestibuli, the scala media and the scala tympani). The chambers separate the cochlear fluids (endolymph and perilymph). If the cochlea could be unwound, it would appear as in Figure 13.6. The scala vestibuli and the scala tympani connect at a narrow point known as the helicotrema, at the apex of the cochlea, and these chambers are filled with perilymph. The scala media is separated from the scala vestibuli by Reissner's membrane and from the scala tympani by the basilar membrane. The scala media is filled with endolymph. Lining the outside of the cochlear duct is the stria vascularis, which is rich in capillaries and has an important part to play in maintaining the ionic concentration of the endolymphatic fluid in the scala media, without which the cochlea cannot function properly.

The basilar membrane supports the organ of Corti and the tectorial membrane (a jelly-like 'tongue') is positioned above this (Figure 13.7). The sensory cells of the cochlea are contained within the organ of Corti. Between the basilar membrane and the tectorial membrane are rows of hair cells, each served by nerve fibres. The hair cells are so-called because they have fine 'hairs' or stereocilia projecting through a protective layer that lies over the upper surface of the hair cells known as

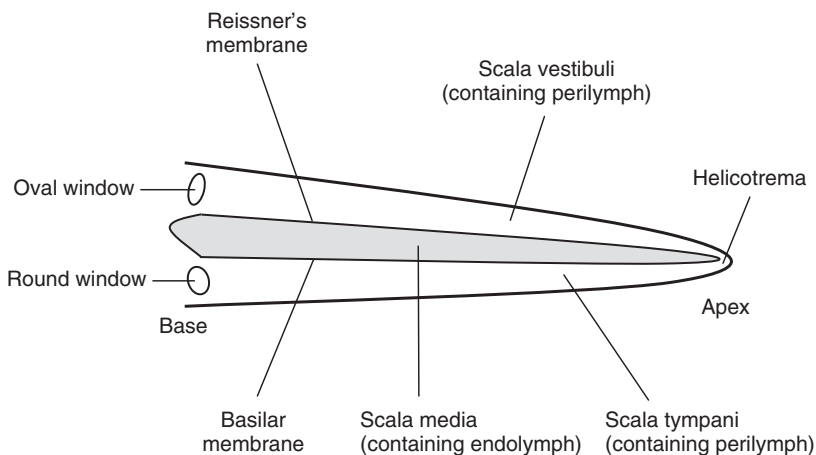


Figure 13.6 The cochlea unwound.

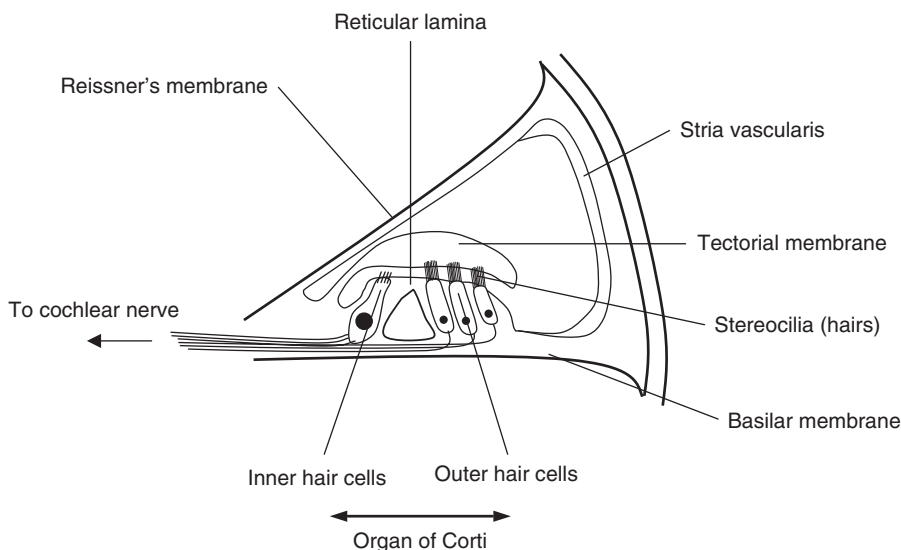


Figure 13.7 The organ of Corti.

the reticular lamina. There are usually three or four rows of outer hair cells (OHC) and one row of inner hair cells. The outer hair cells and inner hair cells are different in shape and function. Outer hair cells are test-tube shaped with many hairs arranged in a wide 'V' or 'W' shape. The hairs gradually reduce in size in a very orderly manner. Inner hair cells are shaped more like a ten pin and have fewer hairs. The nerve supply from the inner hair cells is much richer than that to the outer hair cells and the stereocilia of the outer hair cells are either touching or slightly embedded in the tectorial membrane.

A wave of vibration (sound) travels along the basilar membrane and deforms it, peaking at some point (Figure 13.8). The basilar membrane is stiff and narrow at its base and gradually changes shape along its length to become thick and floppy near the apex. Different parts of the basilar membrane are therefore sensitive to different frequencies. A high frequency sound wave will cause a peak near the base of the cochlea. A low frequency sound wave will cause a peak nearer the apex of the cochlea. This can be imagined to be rather like a piano keyboard. The peak of the travelling wave will cause inner hair cells in the vicinity to 'fire' and send electrical impulses along the auditory nerve to the auditory cortex in the brain.

All sound waves must travel across the base of the cochlea, no matter where they peak. Maximum wear and tear therefore occurs in this high frequency area and high frequency hearing loss is most common.

The function of the outer hair cells is mechanical 'vibration amplification'. When sounds are quiet, the outer hair cells contract and selectively stiffen the basilar membrane causing the inner hair cells in that region fire. The outer hair cell mechanism serves to amplify quiet sounds by a maximum of about 50 dB

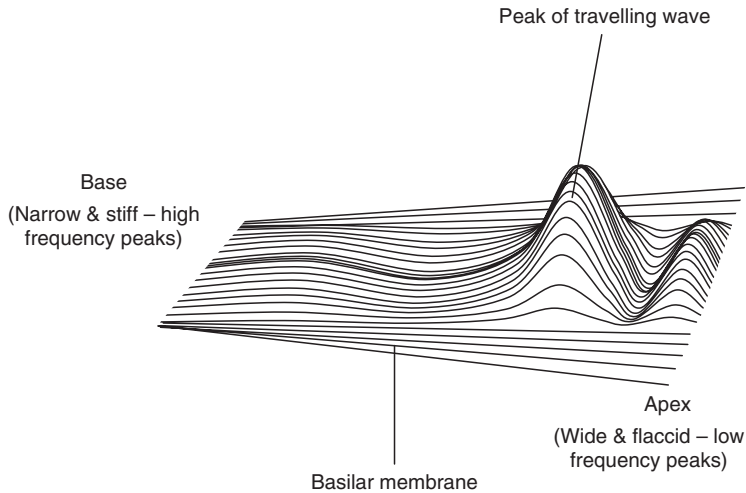


Figure 13.8 The peak of the travelling wave along the basilar membrane.

(Ruggero and Rich, 1991), medium sounds are amplified a little and loud sounds not at all. This action is known as compression and it allows us to hear a much wider range of sounds than would otherwise be possible. However, when the delicate outer hair cells are damaged (by noise or other cause) this will result in weak sounds being inaudible (deafness) whilst there will also be problems of abnormal loudness growth known as recruitment. Someone with this problem may be unable to hear quiet sounds but find loud sounds quickly become too loud. The normal wide difference between quiet and loud is considerably reduced.

The action of the outer hair cells creates a small amount of noise as a by-product of its movement. These sounds can be measured in the ear canal and are often called 'cochlear echoes' or otoacoustic emissions (OAE). Measurement of the OAEs created when a quiet sound is introduced to the ear can be used as an objective screening test of hearing.

The neural and central auditory system

The nerve fibres of the auditory nerve are arranged in an orderly manner, high frequencies on the outside, graduating to low frequencies in the centre. Sound is analysed into its component frequencies in the cochlea and this information is relayed by the auditory nerve to the brain. The auditory nerve passes through the internal auditory canal to enter the brainstem. From here the nerve divides into two and most of the nerve fibres cross over to reach the opposite auditory cortex; thus most of the information from the right ear is received in the left auditory cortex and vice versa.

The auditory cortex of the brain interprets frequency according to the part of the cochlea from which the nerve impulses were sent and interprets intensity largely according to the number of nerve impulses received. A loud signal will result in many impulses being sent, a quieter sound will result in fewer.

Summary

The ear is divided into three parts known as the outer ear, the middle ear and the inner ear. The pinna collects the sound and funnels it into the ear canal and towards the eardrum. The eardrum changes the vibrations in air into mechanical vibrations that pass through the bones of the middle ear to the oval window, which is the entrance to the cochlea. The cochlea is filled with fluid. Sound does not pass efficiently from air to fluid so the function of the middle ear is to build up mechanical vibrations, which results in greater fluid motion in the cochlea. The difference in size between the eardrum and the much smaller oval window creates a 'stiletto heel effect', which is most important in achieving this.

The wave of vibration that enters the cochlea through the oval window reaches a peak at some point along the basilar membrane. The basilar membrane is sensitive to different frequencies along its length, high frequencies near the base and low frequencies near the apex. At the place where the basilar membrane peaks, the inner hair cells fire and send electrical impulses to the brain. The brain recognises the frequency from the place at which the signal arose and the loudness is recognised largely from the number of impulses sent. The function of the outer hair cells is to amplify quiet sounds that would otherwise be too quiet to be heard.

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Basic acoustics

Introduction

Sound, like heat and light, is a form of energy. Sound occurs when a sound source (e.g. a whistle, a drum, a tuning fork or the vocal cords) is set into vibration. The vibrating surface of the sound source moves back and forth and disturbs the particles of air. The vibration is passed across the medium (Figure 14.1) as a series of compressions (areas of high pressure) and rarefactions (areas of low pressure). The sound wave represented as a diagram of the changing sound pressure over distance can be seen in Figure 14.2 (b).

Sound waves are propagated in all directions. Propagation requires the medium to be elastic. Sound waves cannot be transmitted through a vacuum. Sound can be transmitted through a gas, liquid or solid medium. Generally, we are most concerned with the transmission of sound through air.

Air particles in contact with the vibrating body are set into vibration and pass the movement on to other particles with which they come into contact. These pass the movement on to more distant air particles and in this way a flow of sound energy is generated away from the vibrating body. The particles of air vibrate only about their own position; they do not move along with the wave.

The simplest form of vibration is the pure tone. One vibration back and forth is known as one cycle. A pure tone is a single tone formed by simple harmonic motion. In simple harmonic motion, the vibration is repeated back and forth in such a way that the motion repeats itself exactly in equal periods of time (Figure 14.2). Pure tones can be created with a tuning fork or an audiometer but sounds that occur naturally are usually complex and formed from a combination of pure tones.

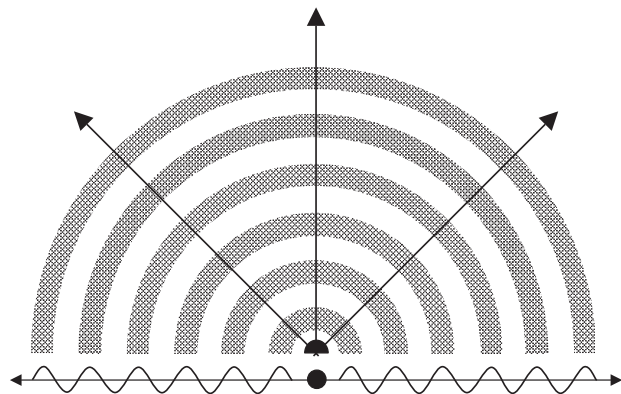
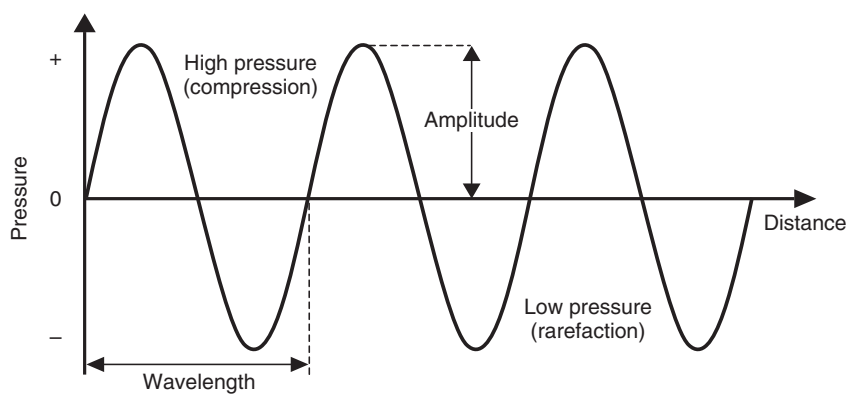
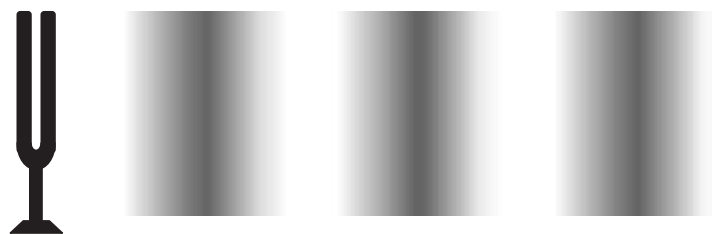


Figure 14.1 Sound moves as areas of high and low pressure across the medium in all directions.



(a) Three cycles shown graphically



(b) Three cycles represented diagrammatically

Figure 14.2 A pure tone sound wave showing the pressure changes through three cycles.

Frequency and pitch

The rate at which the sound source vibrates is called the frequency, which is expressed in Hertz (Hz), named after Heinrich Hertz (1857–1894), who was the first physicist to send and receive radio waves. Frequency is subjectively experienced as pitch. Since pitch is subjective, it cannot be measured directly and frequency is used as an objective form of measurement.

The number of times the vibration repeats itself in a period of one second gives the frequency. For example, if the vibration repeats itself 100 times in one second, it has a frequency of 100 cycles per second or 100 Hz. Many vibrations per second (a thousand or more) produces a high-pitched sound. Less than one thousand vibrations per second produce a low-pitched sound (Figure 14.3). The piano produces its lowest note at 27.5 Hz and its highest at 4186 Hz. Middle C is 261.6 Hz (Music Acoustics, 2005). Human hearing can detect a frequency range from approximately 20 to 20 000 Hz (20 kHz) but sounds that are very low or very high have to have a greater intensity to be heard.

The effect of distance away from the sound source

In an open space, sound becomes weaker with increasing distance from the sound source. As the distance away from the sound source doubles, the sound level falls by 6 dB (Figure 14.4). For example, moving from 1 to 2 m away from the sound source results in a drop in the SPL of 6 dB. Moving 4 m from the sound source causes the SPL to drop by 12 dB and moving 8 m away causes a drop of 18 dB. This phenomenon is known as the inverse square law.

Some surfaces reflect sound, others absorb much of the sound. If the sound is not reflected by walls or objects in its path, it will gradually lose energy and fade away. Reflection of sound takes place when there is a change of medium. The larger the change, the greater is the reflection. Hard surfaces reflect much sound.

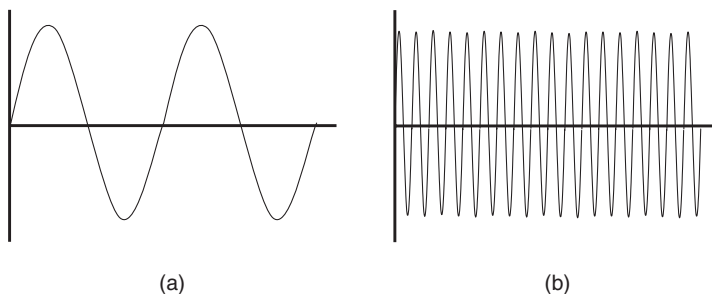


Figure 14.3 (a) A low frequency tone (b) A tone of higher frequency.

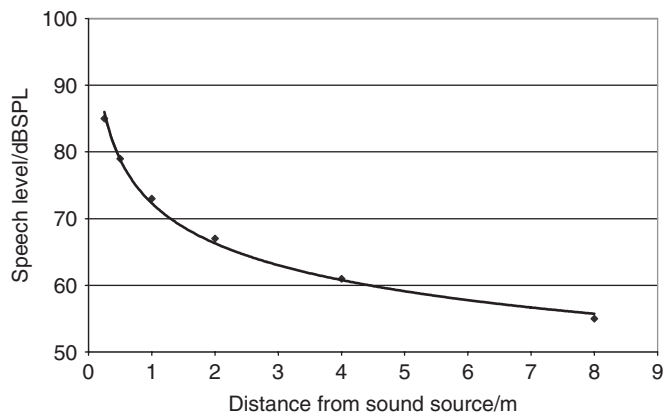


Figure 14.4 Decrease in sound level with increasing distance.

Reflections can have a positive or a negative effect. Reflections which occur very quickly after the original sound may merge together with it such that we hear the sound as louder. This is positive in many normal listening situations. At work these increased sound levels could be damaging to the employee’s hearing. Reflections that are slower may cause interference. Multiple reflections from walls and ceilings within 0.1 seconds of each other cause reverberation, that is the prolonging of a sound. Reflections that occur noticeably after the original may be heard as echoes.

Reverberation that occurs within a confined space can be likened to the sound wave bouncing around the room (Figure 14.5) giving a persistence of sound after the original sound source has ceased, which eventually decays (fades away) or is absorbed by soft materials. When reverberation occurs in an enclosed room or area, the sound pressure will level out, often within about 2 m of the sound source. So, although the sound will be much louder very close to its source, any further distance may not provide much advantage with regard to sound reduction.

Some surfaces reflect sound, others absorb much of the sound. This can create a problem when making noise measurements, as some areas within the room may have unexpectedly high sound levels whilst in other areas the sound waves may meet in such a way that they cancel each other out. Noise measurements taken in two places, a very short distance away from each other (sometimes only a few centimetres), can be significantly different. It is very important to measure noise at the position of the workers’ ears to obtain an accurate picture of noise exposure.

Noisy machinery is supplied accompanied by information giving a value of the sound energy it produces, usually in terms of sound power level in decibels. This refers to the energy produced by the machine at source, rather than in any particular environment.

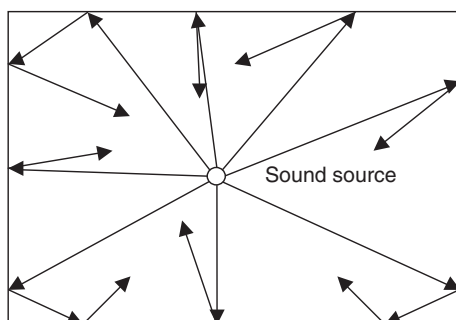


Figure 14.5 Reverberation within a room.

Noise measurement

Sound level and loudness

The greater the energy or force applied to make the body vibrate, the more intense the vibrations and the further the air particles move from their place of rest. Graphically, the distance moved from the equilibrium shows the sound level or amplitude (Figure 14.6). The sound level is subjectively experienced as its volume or loudness.

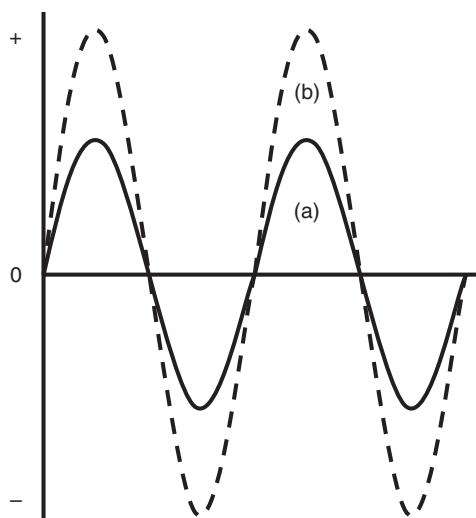


Figure 14.6 (a) A tone of one frequency (b) The same tone with increased sound pressure level, which will be heard as louder.

Table 14.1 The relationship between dBSPL and pascals (Pa)

Sound pressure level (dBSPL)	Sound pressure level (Pa)	Equivalent to
120	20.0	Discomfort
100	2.0	Pneumatic drill
80	0.2	Lathe
60	0.02	Conversation
40	0.002	Loud whispering
20	0.0002	Rustling leaves
0	0.00002	Hearing threshold

Pressure is an amount of force per unit area and can be expressed in pascals (Pa), which is the standard international (SI) unit. Sound pressure level is an objective measure of how much pressure is generated by a sound source. Noise measurements may be expressed in pascals or on a decibel scale. Most commonly sound is measured in decibels. Zero decibels sound pressure level (0 dBSPL) is the same as 0.00002 Pa and one hundred decibels sound pressure level (100 dBSPL) is the same as 2.0 Pa (see Table 14.1).

The term ‘intensity’ is widely, but inaccurately, used to refer to sound pressure level. Intensity really relates to the amount of power flowing across an area, and is correctly expressed in watts per square metre.

The changes in air pressure that create sound are so small they are measured in millionths of a pascal (μ Pa). The human ear can hear sounds from 0.0002 Pa or 20 μ Pa to 20 Pa. Twenty pascals is a million times greater than twenty micro-pascals (20 μ Pa). The wide range of intensities is compressed by transforming it to a logarithmic scale such that a tenfold increase in sound pressure corresponds to 20 dB.

Subjectively, an increase of 10 dB appears twice as loud. The difference between 20 and 40 dB, for example, equates to ten times the sound pressure but is only four times as loud. The gap between increasing sound pressure and loudness widens as we go up the decibel scale. In terms of sound pressure, a noise level of 80 dB is a thousand times (10^3) greater than 20 dB and a noise level of 100 dB is ten thousand times (10^4) greater than a noise level of 20 dB. However, the difference in terms of loudness is far less.

Decibel scales

Decibels are units of relative intensity, that is a ratio between two numbers. This means that the number of decibels describes how much greater is the intensity of a measured sound than a fixed reference level. In other words, the decibels describe the ‘difference’ (e.g. 20 times greater). A ratio must have a reference level for the comparison to be meaningful, that is greater than what? Logarithms are a convenient way of expressing a ratio and tell us how many times the base number is multiplied by itself. Decibels are a logarithmic scale based on the

number 10. There are a number of different reference levels and the one that will be selected to use will vary according to the situation.

The dBSPL and other essentially linear scales

Sound pressure level or SPL can also be written as L_p . The dBSPL scale is one that relates to straightforward pressure measurements and is used widely where we are interested in machines rather than people, for example in the calibration of audiometers. The reference intensity 0 dBSPL is equivalent to a sound pressure of 0.00002 Pa. This reference pressure is fairly arbitrary but is generally accepted as the smallest amount of sound pressure at 1 kHz that may be audible to someone with good hearing. Human hearing is not equal at all frequencies. Very low and very high sounds have to have more energy for us to hear them. The dBSPL scale takes no account of this but has a linear, 'flat' or absolute reference level. This means, for example, that with normal hearing we could hear 20 dBSPL at 1 kHz but we would not hear 20 dBSPL at 125 Hz. The difference in our hearing at low and high frequencies is less marked at high intensity levels and therefore the dBSPL (linear) scale may be used when measuring loud noise.

Other scales are also sometimes used for measuring high noise levels. The dBC scale, for example, is based on the way the human ear responds to sound levels greater than 85 dB and is sometimes used in situations where more importance has to be given to the low frequency content of noise, whilst the dBd scale may be used for measuring aircraft noise. The Z-weighting scale is another essentially linear scale and it is specified in detail in IEC 61672.

Peak sound pressure is the highest noise level encountered during the sound measurement period and often only lasts for a very short time. It is not appropriate to measure short, impulse noises over an 8-hour day and their peak sound pressure is measured and may be expressed as dBSPL, dBC, dBZ or as a simple pressure in pascals. Peak measurements made in dBSPL, dBC or dBZ will not usually vary much unless the peak occurs at the extremes of the frequency range. Two hundred pascals is equivalent to 140 dBSPL. Peak action levels in the Noise at Work Regulations are quoted in pascals, so it will usually be preferable to measure in pascals to ensure easy comparisons with the given peak action levels.

The dBA scale

Most noise assessments are made using the A-weighting scale and all sound level meters provide for measurements in dBA. The ear is not equally sensitive at all frequencies and the dBA-weighting system reflects the way the human ear responds to lower levels of sound. It relates to sound measurements made in an 'open' environment using two ears (binaural hearing). Binaural hearing is slightly better (by about +4 dB) than hearing using one ear only and the A-weighting scale is not an appropriate scale to use when testing hearing under headphones

(one ear at a time). It is also inappropriate to use dBA when making peak level measurements. Almost all sound level meters therefore provide a dBA scale and a flat or linear scale.

The dBHL scale

The dBHL scale is used to reflect the way we hear across the frequency range when our hearing is tested under headphones, one ear at a time and 'HL' stands for hearing level. The reference level used for this scale was found by averaging the hearing of a group of normally hearing young adults. The threshold level (0 dBHL) represents the level at which an average normally hearing young person can just hear. Pure tone audiometry (hearing tests) compares the hearing of the person under test to this average threshold.

The addition of decibels

When we use numbers in everyday life, the difference between each number is exactly the same, that is the difference between 2 and 3 is exactly the same as the difference between 7 and 8. We can easily add and subtract these numbers. When we have to work with logarithms, we can only say that one unit is so many times greater than another. The scale is exponential and the difference between successive numbers becomes larger as one goes higher up the scale. Values in decibels therefore cannot simply be added in the normal way.

If two sound levels such as 80 dB and 80 dB are added together, the resultant sound level will not be 160 dB. In fact, when two or more sound levels are added together, the decibel level is calculated as shown in Figure 14.7.

Measurements are usually taken to the nearest 1/10 dB but after calculations have been carried out the result is usually rounded to the nearest whole decibel. For many purposes, the following information may be adequate:

- The addition of two identical sound sources in dB will give a total sound level that is 3 dB higher than each individual level, for example $80 \text{ dB} + 80 \text{ dB} = 83 \text{ dB}$.
- The addition of four identical sound sources will give a total sound level that is 6 dB higher than each individual level.

$$L_p = 10 \times \log(10^{L_1/10} + 10^{L_2/10} + 10^{L_3/10} \dots)$$

where

L_p = the combined sound level
 L_1 = sound source 1
 L_2 = sound source 2
 L_3 = sound source 3

Figure 14.7 The addition of different sound sources.

Table 14.2 Difference table for addition of decibels from two sound sources

Difference between two levels (dB)	Add to higher level (dB)
0	3
1	2.5
2 or 3	2
4	1.5
5 to 7	1
8 or 9	0.5
10 or more	0

- The addition of ten identical sound sources will give a total sound level that is 10 dB higher than each individual level.
- The addition of twenty identical sound sources will give a total sound level that is 13 dB higher than each individual level.
- If two sound sources differ by 10 dB or more, their combined sound level is essentially the same as the single higher level and the lower sound can be ignored.
- When two different sound levels are added, the difference between the two sound levels can be found by taking one away from the other. This difference can be looked up on Table 14.2, which shows an amount to add to the greatest sound source to obtain the approximate total sound level, for example:

$90\text{ dB} + 94\text{ dB} = 94\text{ dB} + 1.5\text{ dB (from the Table)} = 95.5\text{ dB}.$

Phase and phase cancellation

The phase of a sound wave is expressed in degrees of rotation. The degrees of rotation are used to express the position in the cycle. Thus 0° is the starting point, 90° is a quarter cycle, 180° is a half cycle and 360° is a complete cycle.

In any room, hard surfaces will reflect sound waves; when the reflections meet and combine with the original sound, the sound level will increase if the waves are in phase or decrease if the waves are out of phase (Figure 14.8). The waves caused by combining in phase or out of phase are known as standing waves. If waves are completely in phase, a constructive interaction occurs that increases the amplitude. This is known as resonance. If the waves are completely out of phase, they will cancel each other out. Some hearing protection uses phase cancellation to attenuate noise.

Standing waves create loud and dead spots, which may affect sound level measurements. Measurements of noise exposure should always be taken at, or as close as possible to, the position of *both* ears of the employee. One ear could receive greater noise exposure than the other due to sound reflection.

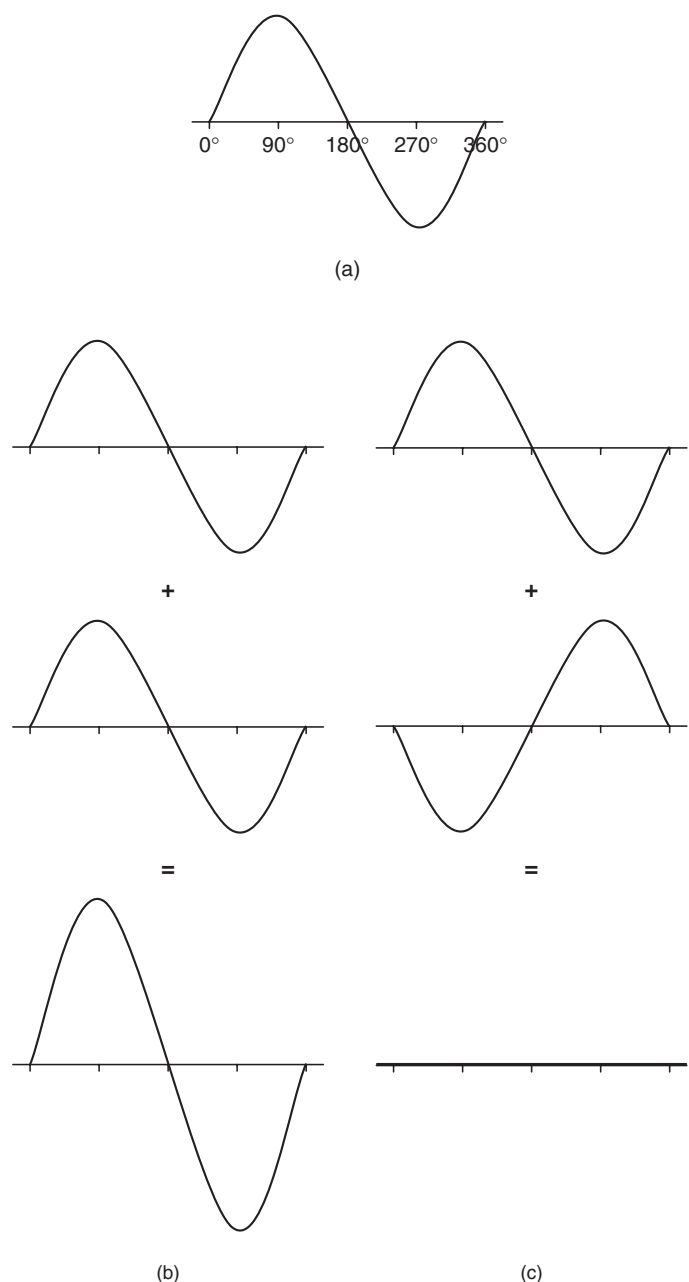


Figure 14.8 (a) The phases of a pure tone sound wave. (b) Pure tones exactly in phase and resultant waveform. (c) Pure tones exactly out of phase and resultant waveform.

Summary

Sound is caused by vibrations which set up a series of compressions (areas of high pressure) and rarefactions (areas of low pressure) that pass across the medium. Important characteristics of a sound wave include its frequency and its amplitude. Frequency equates to the number of cycles in one second and is measured in Hertz (Hz). Amplitude relates to the sound pressure exerted by the sound source and may be measured directly in pascals but is more usually measured in decibels (dB). There are a number of decibel scales, each relating to a somewhat different reference level. The dBA scale is most widely used for measuring noise but cannot be used for measuring peak sound levels. These are usually measured in pascals, dB SPL or dBZ. Hearing is tested under headphones for which a dB HL scale is used. This has a reference level that reflects the average normal hearing of young adults.

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